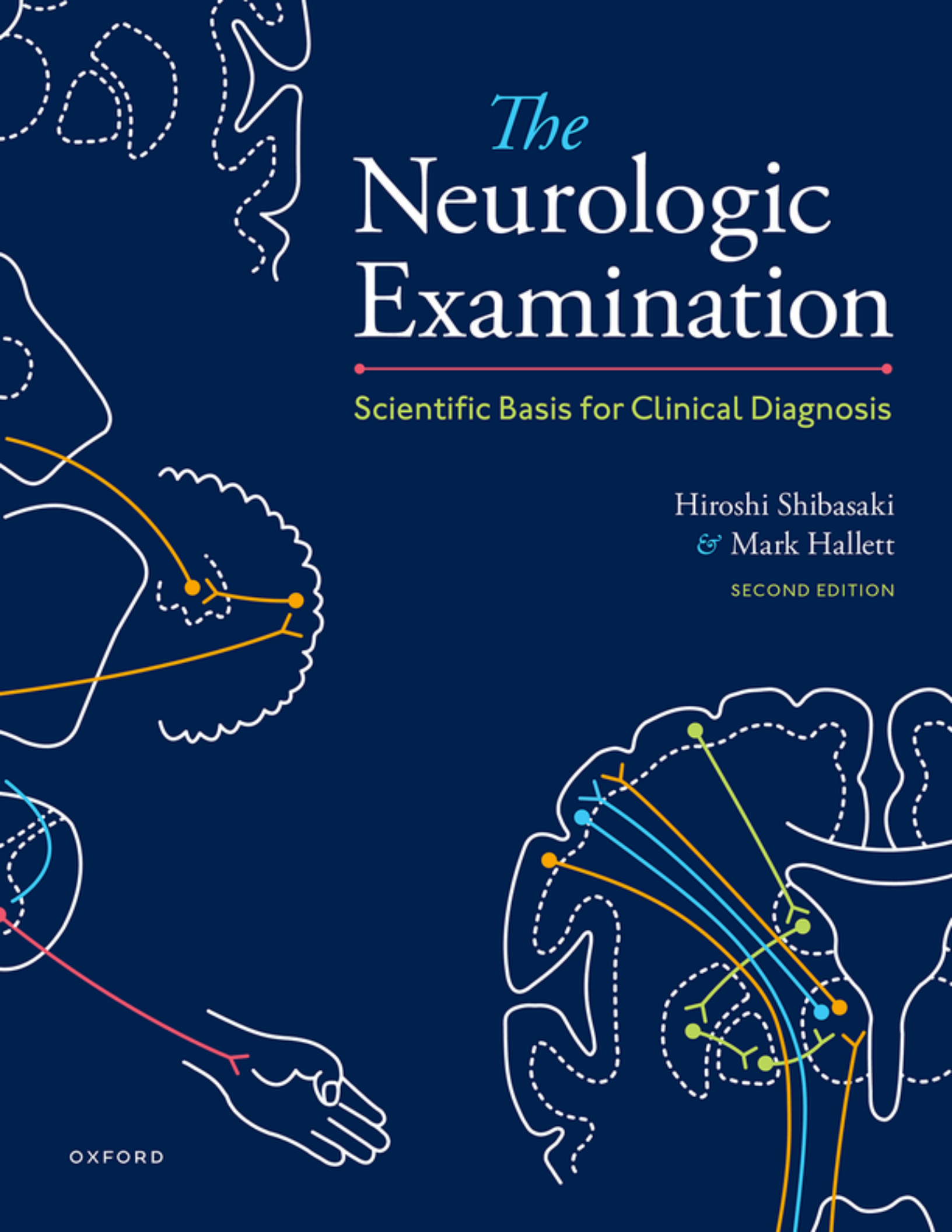




The Neurologic Examination

Scientific Basis for Clinical Diagnosis

Hiroshi Shibasaki
& Mark Hallett

SECOND EDITION

OXFORD

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IN MEMORIAM



Hiroshi Shibasaki, MD, PhD
1939–2022

Hiroshi Shibasaki died as we were finalizing this second edition of *The Neurologic Examination*. He was enthusiastic about the book, pleased with the updates from the first edition, and liked the new cover, a variation on the first edition cover.

I first met Hiroshi at a myoclonus meeting organized by Dr Stanley Fahn in the mid-1980s. Unbeknownst to the attendees, the retreat location, owned by Columbia University, only comprised double rooms, all of which were assigned by Dr Fahn. Hiroshi was my roommate. I had never met him, but we knew each other's work on the physiology of myoclonus. We were fast friends ever since. Over the years, we met at many meetings, and Hiroshi sent some excellent neurologists for post-doctoral fellowships during my tenure at the National Institute of Neurological Disorders and Stroke. When he retired as Chair of Neurology at Kyoto University—mandatory retirement at the early age of 63—he spent two years in my Section at the National Institutes of Health.

Hiroshi made many contributions to the physiology of movement and the pathophysiology of movement disorders. He authored pioneering papers with his deep knowledge of the field and his ability to innovate. Our interests continued to run in tandem, and our mutual interest in myoclonus was a constant theme over the years.

Hiroshi was an excellent teacher who devoted considerable time to teaching even after officially retiring well into his later years. A good example of that is this book. Originally written in Japanese, it was so popular he thought it would be good to have it in English too. He asked me to help, and we had many interesting discussions about the fine points of the neurologic examination. He also had a large number of videotapes of patients with different movement disorders that formed the core of another book on hyperkinetic movement disorders that we wrote together with some other experts in 2020 (also with Oxford University Press).

Hiroshi was a very pleasant person and always a gentleman. Outside of work, he had a large family which he enjoyed. He also had a strong partnership with his devoted wife, Shinobu, also a physician. His family and many neurologists around the world will miss him. I have lost a good friend.

Mark Hallett
June 2022



Dedication

To Shinobu, Judy, and our families for their constant support, and to the many patients we have examined who have taught us about the neurologic examination

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PREFACE TO THE FIRST EDITION

A number of books related to neurology and neurological diagnosis have been published and are currently available in updated editions, but most of them are disease-oriented or anatomy-oriented, and only few are function-oriented. As a consequence of recent advances of laboratory testing including electrophysiology and neuroimaging, the importance of history taking and physical examination in neurological diagnosis tends to be neglected. However, the correct interpretation of symptoms and signs based on modern scientific knowledge is of utmost importance in the diagnosis of neurological diseases. The problems of any patient have to be based on a detailed and thoughtful analysis of the patient; the laboratory testing is just an extension of the examination. In this regard, this book is organized in terms of functional anatomy of the nervous system and aims at providing a bridge from the basic sciences such as anatomy, physiology, pharmacology and molecular biology to neurological symptoms and signs. As one of the unique features of this book, 113 boxes are attached in order to discuss some specific topics of current interest and some clinical issues that the authors have been particularly interested in.

This book is primarily aimed at neurology residents and registrars, but it is hoped that it will be also useful for neurologists in general practice, pediatric neurologists, neurosurgeons, psychiatrists, physical therapists, technicians of clinical neurophysiology and neuroimaging, and medical students.

Hiroshi Shibasaki learned clinical neurology from a number of senior neurologists, including Dr. Yoshigoro Kuwoiwa and Dr. Shukuro Araki of Japan, Dr. A.B. Baker and Dr. John Logothetis of USA, and Dr. A.M. Halliday and Dr. Ian McDonald of UK. He also received a profound impact from Dr. Jun Kimura while they worked together in Kyoto University. He has also collaborated with Dr. Mark Hallett who is a co-author of this book and Dr. Hans O. Lüders of USA for many years. By integrating all the information collected from those distinguished neurologists and based on his own vast clinical experience, he has created his own concept of neurological diagnosis. He published a book entitled *For Those Who Learn Neurological Diagnosis* in Japanese from Igaku-Shoin in Tokyo, the first edition in 2009 (Shibasaki, 2009) and the second edition

in 2013 (Shibasaki, 2013). Since those books have been widely accepted among Japanese neurologists, he decided to publish an updated English version for international readers and asked Mark Hallett to help.

Mark Hallett learned his clinical neurology in Boston from Drs. Raymond Adams, C. Miller Fisher, H. Richard Tyler, and Norman Geschwind. The examination in Boston in those days was heavily influenced by Dr. Derek Denny-Brown, who himself was influenced by Dr. Gordon Holmes. Denny-Brown (1946) authored a very brief booklet on the neurologic examination that was the manual. Like Dr. Shibasaki, Dr. Hallett spent time in London and was influenced there by Dr. C. David Marsden. Dr. Hallett notes that while he and Dr. Shibasaki had the good fortune to be originally trained by giants in the field, the examination does continue to evolve and we appreciate the contributions of all our neurologic colleagues over our years of practice.

Since the two authors have had their main interests in the field of human motor control, movement disorders, involuntary movements and clinical neurophysiology, the number of references cited for each chapter have been influenced, leading to a relatively larger number of literature citations for the chapters related to those specialty fields.

We are grateful to Prof. Per Brodal of Oslo for allowing us to use many of the neuroanatomy figures from his book *The Central Nervous System* (Brodal, 2010).

HS expresses his thanks to Dr. Masao Matsushashi of Kyoto University for his valuable help in computer processing.

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- Shibasaki H. Diagnosis of Neurological Diseases. 2nd ed. Igaku-Shoin, Tokyo, 2013, pp. 1–381 (in Japanese).

PREFACE TO THE SECOND EDITION

Since the first edition of this book appeared in 2016, it has been widely accepted and has developed a favorable reputation. As the scientific information in this specialty field has been rapidly advancing and expanding, we thought it useful to update in this second edition many issues in relation to the phenomenology, pathophysiology, pathology, diagnostic skills, and treatment of disorders of the nervous system. In view of the recent interest in infectious diseases such as Zika virus and the new coronavirus, Chapter 32, “Infectious Diseases of the Nervous System,” stands as an independent chapter in this second edition. We also thought it appropriate to make a new section in the physical examination

on Neurologic Manifestations of the Systemic Medical Diseases, which encompasses collagen vascular diseases, sarcoidosis, endocrine diseases, metabolic diseases, and cancer.

We acknowledge helpful review of this revised edition by Dr. Stephen G. Reich of the University of Maryland School of Medicine and Dr. Yiwen Shi of Johns Hopkins Hospital. H.S. expresses his thanks to Prof. Takashi Nagamine of Sapporo Medical University for his valuable help in computer processing and to Dr. Asako Makino for excellent assistance. Finally, we acknowledge the efficient help of Mr. Craig Allen Panner of Oxford University Press in preparing this book.

EXPLANATORY NOTES

Regarding the use of medical terminology, the following principles are adopted in this book.

A **symptom** is a subjective condition determined from the complaints of the patient during the history taking. A **sign** is an objective finding identified by the physician during the physical examination.

Lesion is the site of the nervous system that is affected by pathology or functionally impaired. **Disturbance** is an abnormality of nervous functions.

Rostral (applied to the brainstem and spinal cord) means superior, or directed toward the cranium. **Caudal**

(applied to the brainstem and spinal cord) means inferior, or directed toward the feet. In the brain, “anterior” and “posterior” are used to describe front and back, respectively. **Ventral** means inferior in the brain, and anterior in the brainstem and spinal cord. **Dorsal** means superior in the brain, and posterior in the brainstem and spinal cord.

Neurologic refers to neurology (e.g., the neurologic examination) whereas **neurological** refers to a disease process (e.g., a neurological symptom).

ACKNOWLEDGMENTS

The following figures and table were reproduced from *Diagnosis of Neurological Diseases*, 2nd edition, Igaku-Shoin Ltd. Tokyo, 2013, with kind permission:

Chapter 1: Figures 1-1, 1-2, 2-1, 4-1, 5-1, 5-2, 5-3, 5-4, 5-5, 7-2, 7-3, 7-5, 7-9, 8-2, 8-3, 8-4, 8-5, 9-1, 9-2, 9-3, 9-4, 9-5, 9-6, 9-8, 10-1, 10-2, 10-3, 11-1, 11-2, 11-3, 12-1, 12-3, 13-1, 14-2, 14-4, 14-6, 14-7, 16-1, 16-2, 16-5, 16-6, 16-7, 16-8, 16-13, 16-14, 16-15, 16-16, 16-19, 16-21, 16-24, 16-25, 16-26, 16-28, 16-29, 16-30, 16-31, 16-32a, 17-2, 17-3, 18-1, 18-2, 18-5, 18-7, 18-10, 18-12, 18-13, 19-1, 19-2, 19-3, 19-4, 19-5, 19-6, 19-7, 19-8, 19-9, 19-10, 19-11, 20-1,

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Chapter 4: Table 4-1, 4-3, 8-1, 9-1, 14-1, 16-1, 16-2, 16-3, 18-1, 18-2, 18-3, 19-1, 19-2, 22-1, 24-2, 25-1, 25-2, 25-3, 29-1, 29-2, 30-1, 30-2, 30-3.

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Chapter 2: Figures 6-1, 7-1, 7-4, 7-6, 7-7, 7-8, 8-1, 9-7, 12-2, 14-3, 14-5, 16-3, 16-4, 16-9, 16-10, 16-11, 16-12, 16-17, 16-18, 16-20, 17-1, 18-3, 18-4, 18-6, 18-11, 22-3, 23-2, 23-4, 23-6, 27-1, 28-1, and Table 24-1.

DIAGNOSIS OF NEUROLOGICAL DISEASES

To diagnose neurological diseases in a systematic way, it is practical and useful to take into account three axes: axis 1, anatomic; axis 2, etiologic; and axis 3, clinical. It is effective to consider these three axes through all steps of neurologic diagnosis, from the history taking to the neurologic examination, before choosing the necessary laboratory tests.

1. AXIS 1: ANATOMIC DIAGNOSIS

While taking the history and carrying out the neurologic examination, it is essential to consider whether the lesion involves the central nervous system, the peripheral nervous system, the muscular system, or more than one of these systems. Furthermore, within each of these systems, the sites of lesion should be estimated as precisely as possible. This is a special feature of neurology because any site of the nervous system, if affected, gives rise to the symptoms and signs that are specific for that particular site. In the cerebrum, for example, whether the lesion involves the superficial gray matter (cortex), the deep gray matter such as the thalamus and basal ganglia, or the intermediate white matter should be considered. Furthermore, it is possible to estimate the precise location of the lesion within each site of the cerebral hemisphere. As for the peripheral nervous system, whether the lesion involves a particular nerve, more than one nerve, or all the nerves diffusely can be estimated. In addition, the precise location of the lesion (the proximal or distal part of the nerve fibers) can be also estimated. Furthermore, careful clinical examination allows us to estimate the kind of nerve fibers affected (motor, sensory, or autonomic) and whether the axon or the myelin sheath is affected.

For example, when the patient complains of muscle weakness in one hand, it is useful to systematically take into account the motor cortex, the corticospinal tract through the cerebral white matter, internal capsule and brainstem to the cervical cord, the anterior horn cells,

and then the α motor fibers, neuromuscular junction, and finally muscles, in this order. Of course, this process can be done in the opposite order from the periphery to the motor cortex. For another example, if the patient complains of numbness in one hand, it is practical to consider the responsible site of lesion first in the peripheral nerve innervating that region of the skin, then the brachial plexus, dorsal root ganglion, dorsal root, gray matter of the cervical cord, the somatosensory fibers in the cervical cord and brainstem, the relay nucleus in the thalamus, the thalamocortical fibers, and finally the hand area of the contralateral somatosensory cortex. This process is often possible even during the step of history taking, if carefully done.

A. DISTRIBUTION OF THE LESION AT THE TISSUE LEVEL

For neurologists, it is not enough to just estimate the location of lesion at the level of gross neuroanatomy; they are expected to be able to consider the more precise distribution of pathology at the tissue level. The mode of distribution of the tissue damage is important not only to make the correct diagnosis but also to predict the severity and prognosis of the symptoms and signs caused by the lesion.

Five forms of representative distribution of tissue damage are illustrated in Figure 1-1, taking the localized lesion in the cervical cord as an example. In a **diffuse** lesion, the tissue is homogeneously damaged throughout. A typical example of this case is transverse myelitis of the cervical cord, in which the patient is expected to show severe quadriplegia at least in the acute stage. In a **disseminated** lesion, the lesion is scattered as if the seeds are spread on the ground. Pathologically it is characterized by perivascular cell infiltration. An example is disseminated myelitis as seen in neuromyelitis optica spectrum disorders (Sato et al, 2014; Takahashi et al, 2007; Wingerchuk et al, 2015), acute disseminated encephalomyelitis (ADEM),

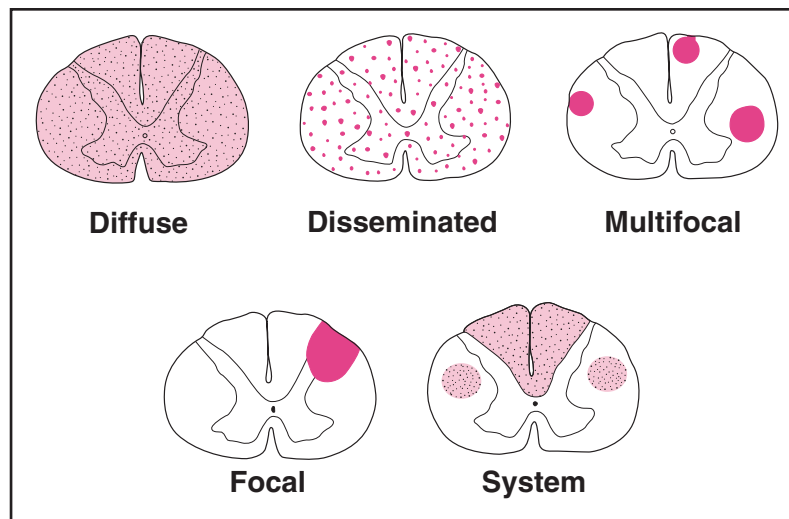


Figure 1-1 Five forms of representative distribution of tissue damage.

neuro-Behçet, and HTLV-I associated myelopathy (HAM/TSP) (Osame, 2002). In this condition, the lesion usually extends also longitudinally over a relatively long distance in the spinal cord. In contrast with acute transverse myelitis, which usually manifests with quadriplegia (paraplegia in case of the thoracic cord lesion) of severe degree below the affected segment, the disseminated cord lesion typically presents with severe spasticity of the lower extremities while muscle strength is relatively spared, and it is especially so in the chronic stage.

A **focal** lesion is localized to an area in the central nervous system and is typically seen in vascular ischemic disease or tumor. Multiple sclerosis and metastatic tumors typically show **multifocal** lesions. **System** involvement is characteristically seen in many neurological diseases, and it affects a group of neurons or a neuronal network that has specific function(s). Typical examples are neurodegenerative diseases such as Parkinson disease, motor neuron disease, and Alzheimer disease. Intoxication due to heavy metals and drug also belongs to this group.

B. SELECTIVE VULNERABILITY

This is the basic concept for explaining system involvement of the nervous system, for example due to heavy metal or drug intoxication. A typical example is organic mercury poisoning (**Minamata disease**), which is clinically characterized by dysarthria and ataxia due to cerebellar damage, constriction of the peripheral visual field due to involvement of the occipital cortex, and hearing difficulty (**Hunter-Russell syndrome**) (see Chapter 7-2B, p. 46).

C. NEGATIVE VERSUS POSITIVE NEUROLOGICAL SYMPTOMS AND SIGNS

Neurological symptoms and signs can be divided into two types: negative and positive. The negative neurological symptoms and signs are neurological deficits caused by impairment or loss of functions of the nervous system, and the positive ones are caused by over-excitation of the nervous system or loss of its inhibition. Examples of negative symptoms and signs are unconsciousness, memory loss, aphasia, visual field defect, motor paralysis, bradykinesia, freezing of gait, sensory deficit, and orthostatic hypotension. Examples of positive ones are convulsion, involuntary movements, spasm/cramp, pain, and dysesthesia. Psychiatric symptoms like hallucinations and delusions may also be interpreted as positive symptoms. As positive phenomena are not always present during the physical examination, it often depends on the patient's description or situations reported by observers. Most of the negative signs are observable by physical examination, but intermittent or paroxysmal negative phenomena like transient ischemic attacks and periodic paralysis have to be judged by careful history taking.

2. AXIS 2: ETIOLOGIC DIAGNOSIS

Information about how the neurological symptoms started (**mode of onset**) and how they have changed after the onset (**clinical course**) provides the most useful clue to the cause or pathogenesis of the disease. The mode of onset is classified into four types: sudden, acute, subacute, and insidious (Figure 1-2 and Box 1-1).

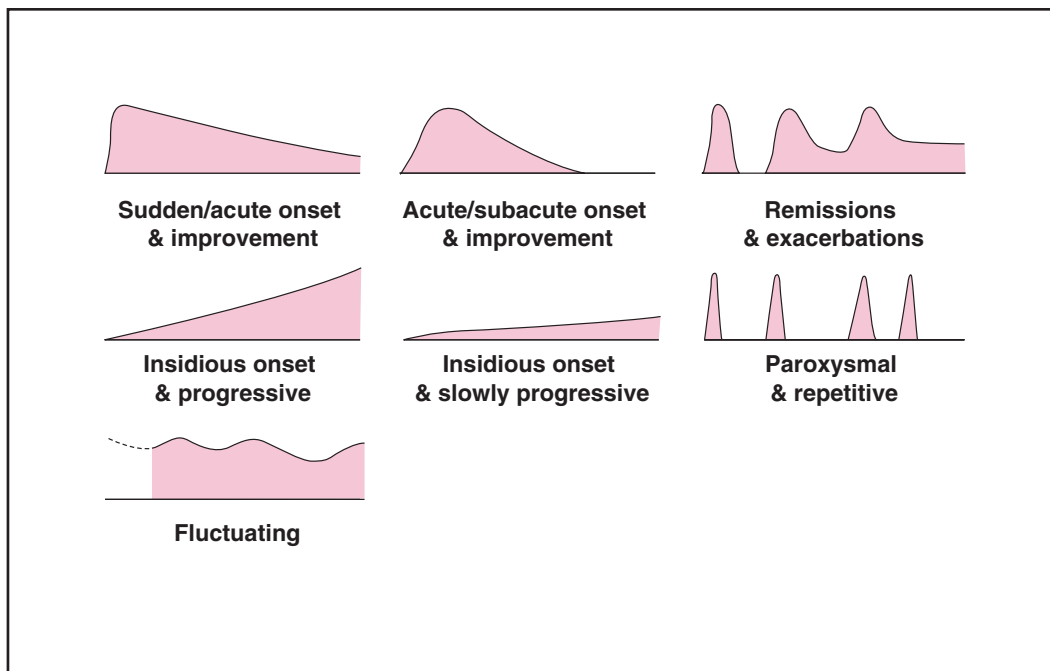


Figure 1-2 Mode of onset.

BOX 1-1 ONSET OF NEUROLOGICAL SYMPTOMS IN CEREBROVASCULAR DISEASES

Cerebral infarction is classified into atherothrombosis, lacunar infarction (see Box 2-2), and cardiogenic embolism. The type of infarction can often be estimated from the clinical symptoms. In **cardiogenic embolism**, the initial neurological symptom is completed suddenly all at once. Moreover, if the functions of cerebral regions supplied by different arteries are simultaneously impaired, it supports the diagnosis of cardiogenic embolism. **Atherothrombosis** is further classified into three types: thrombotic, embolic, and hemodynamic. The **hemodynamic** form tends to occur in the border zone territory supplied by two large arteries, especially when the blood pressure drops (**watershed infarction**). Therefore, this form of infarction tends to be completed over a period of a few hours during sleep at night. The **embolic form of atherothrombosis** occurs when a blood clot is detached from the wall of a large artery and flows into its smaller branch to form an embolic infarction there. A typical example is occlusion of the ophthalmic artery by a blood clot originating from atherothrombosis of the internal carotid artery. In the thrombotic form, thrombus formation in a cerebral artery causes a local occlusion or stenosis of that artery, clinically causing acute neurological deficit of the corresponding functions.

Intracerebral hemorrhage is sudden in onset and the symptoms rapidly progress to cause loss of consciousness and headache. Hemorrhages are common in the putamen, thalamus, pons, and cerebellum, and subcortical hemorrhages are not uncommon. Lobar intracerebral hemorrhage is often caused by **amyloid angiopathy**. In this condition, there are β -amyloid deposits in the wall of small arteries in the cortex and leptomeninges. Its sporadic form is common in the elderly (Biffi & Greenberg, 2011). Cerebral amyloid angiopathy may also be associated with transient focal neurological episodes (Smith et al, 2021) and magnetic resonance imaging (MRI) findings of lobar cerebral microbleeds, cortical superficial siderosis, lobar lacunes, and white matter hyperintensities (Charidimou et al, 2015; Gokcal et al, 2021). **Subarachnoid hemorrhage** is characterized by sudden onset of severe headache rapidly followed by loss of consciousness.

In case of atypical stroke presenting with headache, disturbance of consciousness, and convulsion, **venous sinus thrombosis** has to be kept in mind. MRI with magnetic resonance venography (MRV) may help with the diagnosis (Idiculla et al, 2020). Small juxtacortical hemorrhages on computed tomography (CT) imaging are characteristically seen in cerebral venous thrombosis (Coutinho et al, 2014).

When the symptom reaches the maximal degree within a week, its mode of onset is defined as **acute**. Acute onset is commonly encountered in vascular diseases, inflammatory diseases, and acute poisoning. When the symptom occurs abruptly, its mode of onset is defined as **sudden**, commonly seen in embolism and subarachnoid hemorrhage. The conditions that start acutely tend to be followed by some **improvement** if properly diagnosed and treated. The prognosis of the conditions that start suddenly depends on the severity of the pathology. When the symptom reaches the maximum over a period of more than 1 week but less than 4 weeks after the clinical onset, its mode of onset is classified as **subacute**. It is commonly seen in inflammatory diseases or metabolic/toxic diseases. This definition of categories for indicating the mode of onset is rather arbitrary, and it is actually important to describe the exact time in the medical history.

When the symptom begins gradually and reaches the maximum over a period of more than a month, its mode of onset is called **insidious**. The following clinical course in this case is either **progressive**, as seen in brain tumors, or **slowly progressive**, as seen in neurodegenerative diseases. A **polyphasic** clinical course with **remissions and exacerbations** is typically seen in inflammatory demyelinating diseases such as multiple sclerosis. Repetition of short episodes with intervals of variable length is seen in **paroxysmal** disorders such as epilepsy and migraine. Ion channel disorders such as periodic paralysis and episodic ataxia may also show this pattern of clinical course. A **fluctuating** clinical course is commonly seen in metabolic or endocrine encephalopathy. In the disorders showing a paroxysmal or repetitive clinical course, the onset of each attack or each episode is usually sudden or acute.

When describing the history of the present illness, it is sometimes practical and useful to illustrate the mode of onset and the clinical course in a diagram showing the severity of each symptom in the vertical axis along the horizontally drawn time course, as shown in Figure 1-2. Since more than one symptom may appear in many diseases, illustration of the time course for each individual symptom will help you to understand the temporal relationship among different symptoms.

As described earlier, neurological symptoms are either negative or positive. When indicating the severity of each symptom in the vertical axis of the illustration (see Figure 1-2), a negative symptom or a neurological deficit is usually correlated with the severity of the lesion or the tissue damage, but that is not necessarily the case for positive symptoms. That is because the positive symptoms may result from excessive excitation of the nervous system or loss of inhibition. Furthermore, some positive symptoms, such as pain and involuntary movements, may also result from

plastic changes of the nervous system during the process of functional restoration following an acute insult, such as what is typical following a stroke (see Chapter 33-3, p. 288).

3. AXIS 3: CLINICAL DIAGNOSIS

By considering the anatomic diagnosis and the etiologic diagnosis throughout the process of history taking and neurologic examination as we have described, the most likely sites of lesion and the most likely cause of the lesion can be combined together, which will help you reach the most reasonable clinical diagnosis. A few additional conditions might come out as candidates for the **differential diagnosis**. Of course, a number of conditions may appear in mind as the candidates for differential diagnosis during the history taking, but if an adequate history is obtained, the differential diagnosis is typically expected to be limited to two or three at most at the end of the history taking. An excessive list of differential diagnoses may be difficult to manage and may even make the neurologic examination more difficult since the logic of the examination needs to take the differential diagnosis into account.

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HISTORY TAKING

History taking, if appropriately done, is expected to provide the most important information for reaching the correct neurological diagnosis. Usually the history is obtained from the patient, but if the patient is unconscious, demented, aphasic, or uncooperative, the history must depend on the information obtained from the patient's family or accompanying persons. The information to be obtained when taking the medical history includes the present age, sex, chief complaints, the age of onset of symptoms, history of the present illness, past medical history, social history, and family history. When describing the medical history in the record or when presenting it at a medical conference, it is customary and convenient to describe the information in the above order, but when actually taking the history, it does not always have to follow this sequence.

1. THE PRESENT AGE AND THE AGE AT ONSET

Each neurological disease occurs in the population of a certain age range (Figure 2-1). However, as a consequence of recent advances in the understanding of pathogenesis and with changes in the environment, it is increasingly recognized that there are exceptions to the age predilection. For example, ischemic cerebrovascular diseases are common in the aged population but can also occur in the young population (Box 2-1). Likewise, the paroxysmal disorders or the hereditary degenerative diseases, which were thought to be common in children and young adults, may occur for the first time in adults.

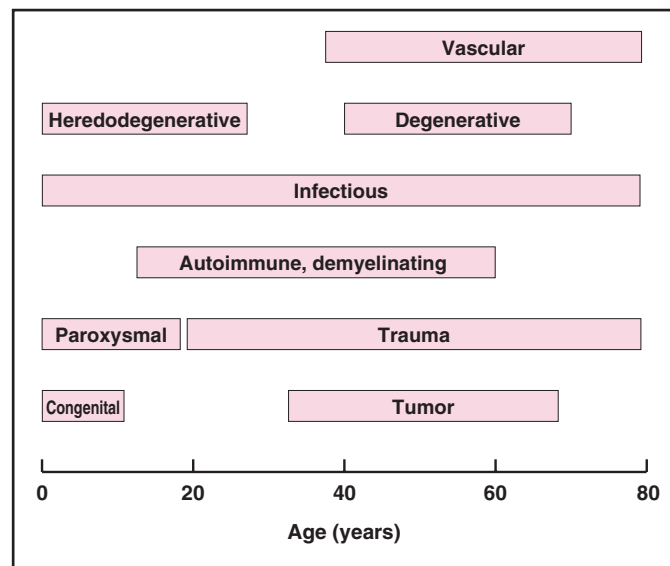


Figure 2-1 Each neurologic disease occurs in the population of a certain age range.

BOX 2-1 RISK FACTORS FOR VASCULAR DISEASES IN THE YOUNG POPULATION

Strokes in young persons have become increasingly common (Aarnio et al, 2014; Larrue et al, 2011). In addition to the common risk factors for vascular diseases such as diabetes mellitus, hypertension, hyperlipidemia, increased blood viscosity, obesity, and heavy smoking, special conditions have to be considered for the young population. Those are congenital heart disease, among others the patent foramen ovale associated with aneurysm of the atrial septum, atrial myxoma, positive antiphospholipid antibody, migraine with aura, use of contraceptives, eclampsia, systemic lupus erythematosus, Behçet disease, malignant atrophic papulosis, Degos disease, fibromuscular dysplasia, homocystinuria, Fabry disease (see Box 20-5, p. 198), pseudoxanthoma elasticum, moyamoya disease, and mitochondrial encephalomyopathy (see Box 19-13, p. 181; see Chapter 3-1, p. 9, for some of the skin abnormalities). **Homocystinuria** is due to lack of cystathionine synthetase and is inherited via autosomal recessive transmission. Clinically it is characterized by dislocation of lens and arachnodactylia similar to Marfan syndrome. **Moyamoya disease** is characterized by an extensive network of abnormal collaterals associated with occlusion of the proximal cerebral arteries (Albrecht et al, 2015; Al-Yassin et al, 2015). It has been commonly reported in East Asian children, but it is also seen in whites (Tho-Calvi et al, 2018). It causes ischemic attacks or convulsion in children and subarachnoid hemorrhage in adults. Physiologically, hyperventilation causes marked, prolonged slowing in the electroencephalogram (EEG). **Mitochondrial encephalomyopathy, lactic acidosis, and stroke-like episodes (MELAS)** is known to cause strokes at a young age (see Box 19-13). This condition also shows convulsions, migraine attacks, and mental retardation. Lactic acid is increased in the serum and cerebrospinal fluid, and ragged red fibers are seen on muscle biopsy. Recent use of drugs such as cocaine has drawn particular attention to this behavior as a risk factor for strokes in the young population (Sordo et al, 2014).

2. SEX

Many diseases are more common in one sex than in the other. The sex of the patient is usually obvious from his or her appearance and from the medical record, but if any uncertainty exists, it should be confirmed during the history taking. Sex is the biological designation of male/female, while gender refers to identification.

3. HANDEDNESS

Since **handedness** is an important factor for higher cortical functions, it can be confirmed when taking the history, although it can also be confirmed while carrying out the physical examination.

4. CHIEF COMPLAINTS

Some physicians might believe that the chief complaints are symptoms the patient mainly complains of, but that is not always correct. Some patients might complain of what they are mainly suffering from at the time of the history taking, but that symptom might not be directly related to the neurological disease that the patient has. It is reasonable to choose the most important complaint(s) of that particular patient during the process of history taking. In this sense, to identify the chief complaint for a patient is to find what the main problem is for that patient, and thus it is a process of **problem-finding**. When writing in the medical record or when presenting in conferences, the chief complaint is usually described or reported before the history of the present illness, but in clinical practice, the most appropriate chief complaint(s) can be chosen after completing the history taking.

It is of utmost importance to find out what the real problem is for the patient and what he or she really means. For this purpose, it is not always advisable to describe exactly what the patient complains of; it is more reasonable to interpret the patient's words in reference to the medical knowledge. For example, a patient who complains of "numbness" in the hand usually means a sensory abnormality such as a tingling or pinprick sensation, but some patients may express weakness by using the word "numb." In this case, it is not appropriate to describe just "numb" in the medical record; rather, it should be described as "weak." Dizziness is a symptom commonly reported, but some patients use "dizzy" when they feel like fainting, others when they feel unsteady, with real vertigo. In this case, these two should be distinguished as much as possible by taking the history carefully (see Chapter 13-2, p. 14).

When describing symptoms when you are taking preliminary notes as well as when you are making the final notes in the medical record, it is advised not to use medical terminology. For example, when a patient complains of difficulty in speaking, it is not advisable to describe it as either "dysarthria" or "aphasia." Instead, it is much better to call it just "speech disturbance." It is because

the first impression might have to be changed later as a result of neurologic examination. It is reasonable to use the medical terminology at the final stage of neurologic evaluation. The chief complaint may not be single, and more than one symptom can be listed as chief complaints as necessary.

5. HISTORY OF THE PRESENT ILLNESS

When taking the history of the present illness, it is important to know how each symptom started and how it has changed since its onset. Patients may certainly have more than one symptom. Regardless of whether those symptoms are caused by a single disease or multiple diseases, mode of onset and clinical course should be carefully described for each individual symptom. In this case, it is convenient to illustrate the time course of each symptom in a diagram as shown in Figure 1-2 (p. 3).

For each symptom, it is also important to obtain the information as to what time of the day, what season of the year, or in what kind of conditions it tends to occur or to worsen (Box 2-2). Furthermore, it is useful to identify any precipitating factor that might bring up or increase the symptom or any factor that may help ameliorate it.

BOX 2-2 LACUNAR INFARCTION

A small infarction (less than 15 mm) in the territory supplied by a **penetrating arterial branch** in the deep cerebral hemisphere is called a lacunar infarction. It mainly occurs in the corona radiata, internal capsule, thalamus, basal ganglia, and brainstem. Typical clinical manifestations are **pure motor hemiplegia** or **hemianesthesia without motor paralysis**, but it can be asymptomatic. An infrequent manifestation caused by lacunar infarction in the internal capsule or pons is known as **dysarthria-clumsy hand syndrome** (Arboix et al, 2004). An underlying pathology of lacunar infarction has been believed to be lipohyalinosis of the penetrating arteries, which is commonly seen in hypertensive patients, but atherothrombosis at the bifurcation of the penetrating arterial branch from a main cerebral artery (**branch atheromatous disease**) has drawn more recent attention (Caplan, 1989). The latter condition in the internal capsule causes repetitive transient hemiparesis (capsular warning syndrome), and a similar condition of the lenticulostriatal artery and anterior pontine artery also causes progressive motor disturbance (Yamamoto et al, 2011).

If a patient reports experiencing a neurological symptom in the past, you should describe in the history of the present illness if you judge it to be related to or a phase of the present illness or in the past medical history if you judge it to be unrelated to the present illness. Likewise, if you judge the present neurological condition to be caused by or part of a systemic condition such as hypertension and diabetes mellitus, you can describe those systemic conditions in the history of the present illness as well. Other systemic conditions can be listed in the past medical history.

6. PAST MEDICAL HISTORY

Regarding the past medical history, the amount of detail to be described in the medical record depends on the clinical situation, but all medical histories must be recorded if they are judged to be related to the present illness. As a patient does not necessarily tell the physician all the past medical history, it may be necessary to ask specific questions to make sure certain topics are covered. For example, when the history of the present illness suggests a possibility of strokes, information about the **risk factors for vascular diseases**, including diabetes mellitus, hypertension, hyperlipidemia, history of cardiovascular diseases, and history of smoking, should be obtained as much as possible (Otite et al, 2017). Among cardiac arrhythmias, atrial fibrillation is known to be a leading cause of recurrent stroke. In particular, paroxysmal atrial fibrillation, which is also an important risk factor, may be detected only by long-term electrocardiographic (ECG) monitoring (Gladstone et al, 2014). Therefore, it is important to know whether such testing was done.

It is noteworthy that some infections might cause neurological disorders a long time afterwards. Subacute sclerosing panencephalitis can occur in adults following childhood measles virus infection, and myelitis can occur years after varicella zoster virus infection (**postherpetic myelitis**). In these cases, confirmation of the past medical history is important for the correct diagnosis.

7. SOCIAL HISTORY

This includes the information about the patient's occupation, allergies, smoking, drinking, hobbies, sports, and environmental situations at home and at work. Just like the past medical history, the information needed depends on the history of the present illness.

8. FAMILY HISTORY

As many neurological diseases are inherited and as the background or precipitating factors might be genetically determined, information about the family is quite important in the history taking. Occurrence of similar diseases among the family members is important, and a history of consanguineous marriage in the patient's parents or grandparents is also important. When there is a possibility of familial conditions, it is useful to draw a family tree (with the patient's permission). It is as important to know the family members who are normal as well as those who are ill.

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PHYSICAL EXAMINATION

1. GENERAL PHYSICAL EXAMINATION AND NEUROLOGIC EXAMINATION

Even in the specialty clinic of neurology, **general physical examination** is as important as **neurologic examination**, mainly because all health aspects are important for the patient and also because the neurological symptoms are often related to systemic diseases in terms of clinical manifestation and pathogenesis (Box 3-1).

The general physical examination and neurologic examination can be carried out separately in this order, but it is often more practical to carry out the two together. Otherwise, a series of examinations

from the top of the head to the toes will need to be repeated, which is not practical particularly when the time spent for each patient is limited in an outpatient clinic. Thus, when examining the eyes for example, we can check the cornea, palpebral conjunctiva, bulbar conjunctiva, and ocular bulb first, and then we can examine the visual acuity, visual fields, pupils, extraocular muscles, ocular movements, and ocular fundi if necessary. When examining the oral cavity, for another example, we check the mucosa of the oral cavity and pharynx, teeth, tongue, and tonsils first, and then check for the presence or absence of muscle atrophy, fasciculation and movement of the tongue, position and movement of the soft palate, gag reflex if necessary, and voice, speech, and swallowing. This way, we do not have to come back to the eyes or mouth after having completed the general physical examination. This approach is more efficient not just for neurologists but also for the patient's convenience.

BOX 3-1 CLINICAL NEUROLOGY FROM A WIDE VIEWPOINT

It is important to take a wide view throughout the diagnostic procedure of neurological diseases. That is because (1) the neurological symptoms in question might be a manifestation of a systemic disease, like Crow–Fukase syndrome (polyneuropathy, organomegaly, endocrinopathy, M-protein, and skin changes; POEMS) (p. 12); (2) the neurological symptom might be a complication of a systemic disease, like diabetic polyneuropathy; (3) the neurological symptom might have resulted from a systemic disease, like cerebrovascular diseases as a result of hypertension, hyperlipidosis, and cardiac diseases; and (4) the neurological disease might have a common pathogenesis with a systemic disease, like paraneoplastic syndrome (see Box 3-2). In bulbospinal muscular atrophy, as another example of this last category, the genetic abnormality of the androgen receptor might be related to the involvement of anterior horn cells (see Box 17-2, p. 137). Approaching neurological conditions with a wide view has contributed to the discovery of some new neurological diseases or to elucidation of their pathogenesis.

A. GENERAL PHYSICAL EXAMINATION AT THE INITIAL CLINICAL EVALUATION

When examining each patient for the first time, the general physical examination usually includes observation of (1) head, hair, and facial skin, (2) eyes (eyelids, cornea, palpebral and bulbar conjunctivae, and iris), (3) ears and nose, (4) oral mucosa, pharynx, and tongue, (5) neck (lymph nodes, thyroid, and vascular bruit if necessary), (6) arterial pulse and blood pressure, (7) chest (skin, respiration, percussion and auscultation of heart, and breast as necessary), (8) abdomen (skin, percussion, palpation and percussion of liver, and bowel sounds as necessary), and (9) extremities (skin, joints, and edema), in this order. Pulsation of arteries should be checked at the radial arteries bilaterally, and the dorsalis pedis arteries should also be palpated as necessary.

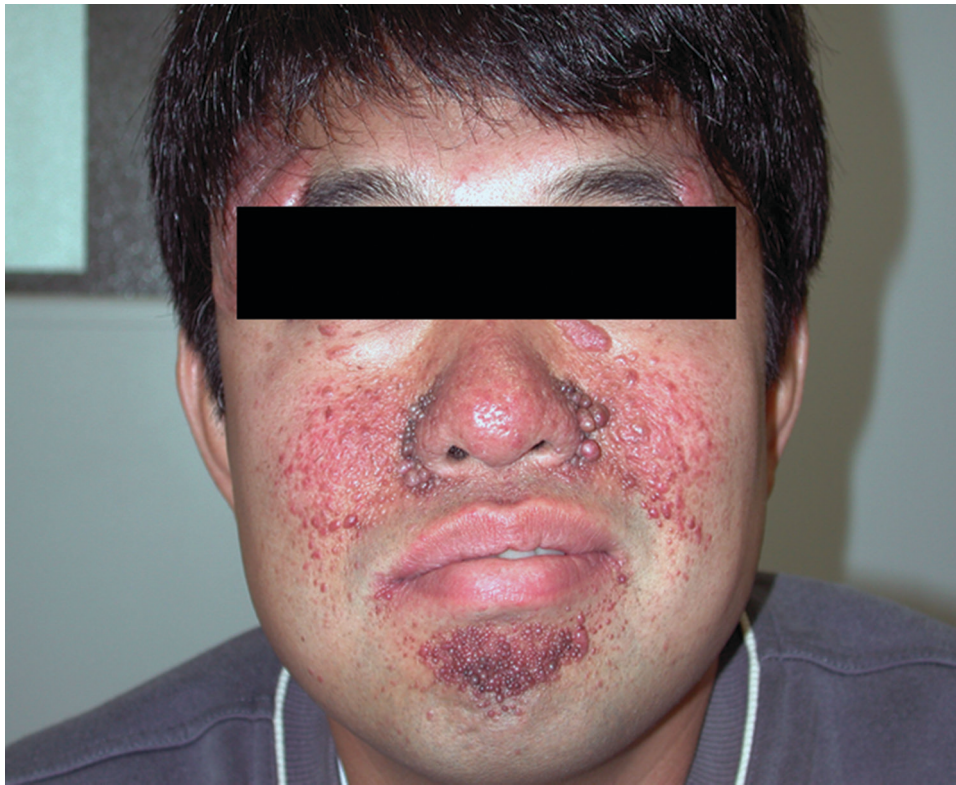


Figure 3-1 Facial angiofibroma seen in a patient with tuberous sclerosis.
By courtesy of Dr. Akio Ikeda, Kyoto University.

B. SKIN ABNORMALITIES SEEN IN CONGENITAL OR HEREDITARY NEUROLOGICAL DISEASES

Observation of skin is especially important when examining patients with possible congenital neurological diseases, because both the nervous system and the skin tissues are embryologically derived from the ectodermal cells. In **tuberous sclerosis**, which is a common cause of epilepsy in children, **facial angiofibroma** (adenoma sebaceum) is characteristic (Figure 3-1), but it is not seen at birth and appears later in infancy. Depigmented nevi, shagreen patch, and periungual fibroma may also be seen. Port wine stain in the territory of the first branch of the trigeminal nerve is seen in **Sturge–Weber syndrome**, which is also called encephalotrigeminal angiomatosis (Figure 3-2). This condition may be associated with atrophy and calcification in the occipital cortex and angioma in the retina and meninges, and clinically it presents with convulsions. Xanthoma in the neck and axilla is seen in **pseudoxanthoma elasticum** (Grönblad–Strandberg syndrome), which may be associated with angioid streaks in the retina and cardiovascular disorders (Figure 3-3). **Ichthyosis** on the anterior aspect of lower legs may be associated with mental retardation and spastic

gait (**Sjögren–Larsson syndrome**) (Nagappa et al, 2017), and it is also a cardinal symptom of **Refsum disease**, which is characterized by sensory ataxia due to polyneuropathy, retinal pigment degeneration, and sensorineural deafness.

Neurofibromatosis is an autosomal dominant hereditary disease that is characterized by neurofibroma and café-au-lait spots (Figure 3-4). Neurofibromatosis is grouped into two types. NF-1 is called **von Recklinghausen disease**, and NF-2 is often associated with acoustic neurofibroma and multiple meningioma. **Fabry disease** is a lipidosis due to X-linked deficiency of α -galactosidase and presents with angiokeratoma, anhidrosis, severe pain induced by exercise and fever, clouding of the cornea and lens, and various kinds of vascular disorders (Lenders et al, 2015; Löhle et al, 2015) (see Box 20-5, p. 198). In this condition, angiokeratoma may be seen only in the scrotum.

Livedo reticularis (livedo racemosa) (Figure 3-5) and acrocyanosis may be seen in patients with positive serum antibody against phospholipid, and it may cause transient ischemic attack and ischemic strokes (**Sneddon syndrome**) (Bottin et al, 2015). **Ehlers–Danlos syndrome** is a hereditary disease characterized by hyperelasticity of

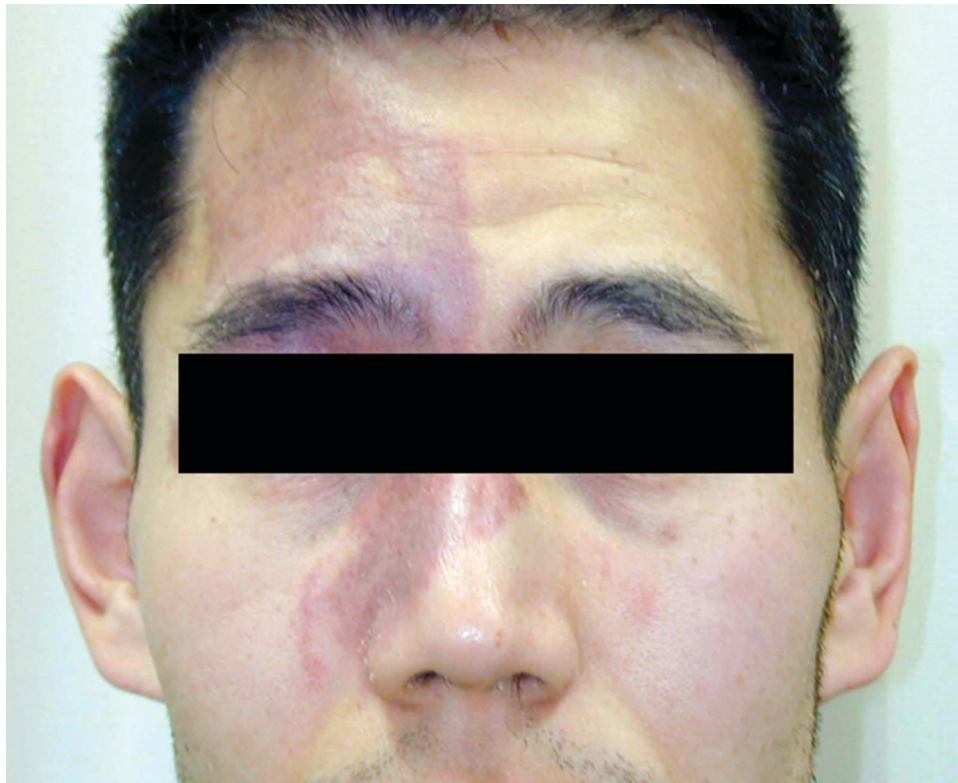


Figure 3-2 Port wine stain in the territory of the first branch of the right trigeminal nerve in a patient with Sturge-Weber syndrome.
By courtesy of Dr. Akio Ikeda, Kyoto University.

skin and hyperextensibility of joints due to abnormality of collagen fibers in the skin, blood vessels, and joint capsules (Figure 3-6). The mode of inheritance is variable, and it may also be associated with abnormalities in the peripheral nerves and muscles (Voermans et al, 2009).

C. SKIN ABNORMALITIES SEEN IN INFLAMMATORY NEUROLOGICAL DISEASES

Some inflammatory diseases of the nervous system are associated with characteristic inflammatory diseases of the skin. Infection of the geniculate ganglion with varicella/zoster virus causes vesicular skin rash in the external auditory meatus and may be associated with facial nerve palsy (**geniculate zoster, Ramsay Hunt syndrome**) (Figure 3-7). Swelling of the ear lobes is also seen in **relapsing polychondritis**, which may be associated with meningoencephalitis (Al-Tabbaa & Habal, 2018). **Behçet disease** is characterized by uveitis, painful oral ulcers (aphthae), and genital ulcers and may also be associated with erythema nodosum (Figure 3-8), and purulent dermatitis at the site of needle injection. This disease is neurologically important because it may cause recurrent meningitis or

encephalitis, or both. Uveitis, meningitis, and hearing loss may also be seen in Vogt–Koyanagi–Harada disease, but there is no skin lesion in this condition. **Sweet disease** is characterized by painful purulent erythema (Figure 3-9), fever, joint pain, leukocytosis, and an elevated erythrocyte sedimentation rate. This condition is also known to cause recurrent meningoencephalitis or cerebral vasculitis (Charlson et al, 2015). About 25% of the patients with this condition have been reported to develop leukemia, lymphoma, or cancer. **Degos disease** (malignant atrophic papulosis) is characterized by red necrotic papules in the trunk and extremities, which will evolve to become white and squamous. This condition can be associated with thrombosis in the skin, gastrointestinal tract, and central nervous system (CNS). Polymyositis (see Chapter 17-3B, p. 139) might be associated with various erythema over the dorsum of the fingers and the bony parts of the shoulders, elbows, and legs (**dermatomyositis**).

Brownish discoloration of the skin, hypertrichosis, and edema are seen in association with osteosclerotic plasmacytoma, and this condition is known as **polyneuropathy, organomegaly, endocrinopathy, M-protein, and skin changes (POEMS syndrome, Crow–Fukase syndrome)**

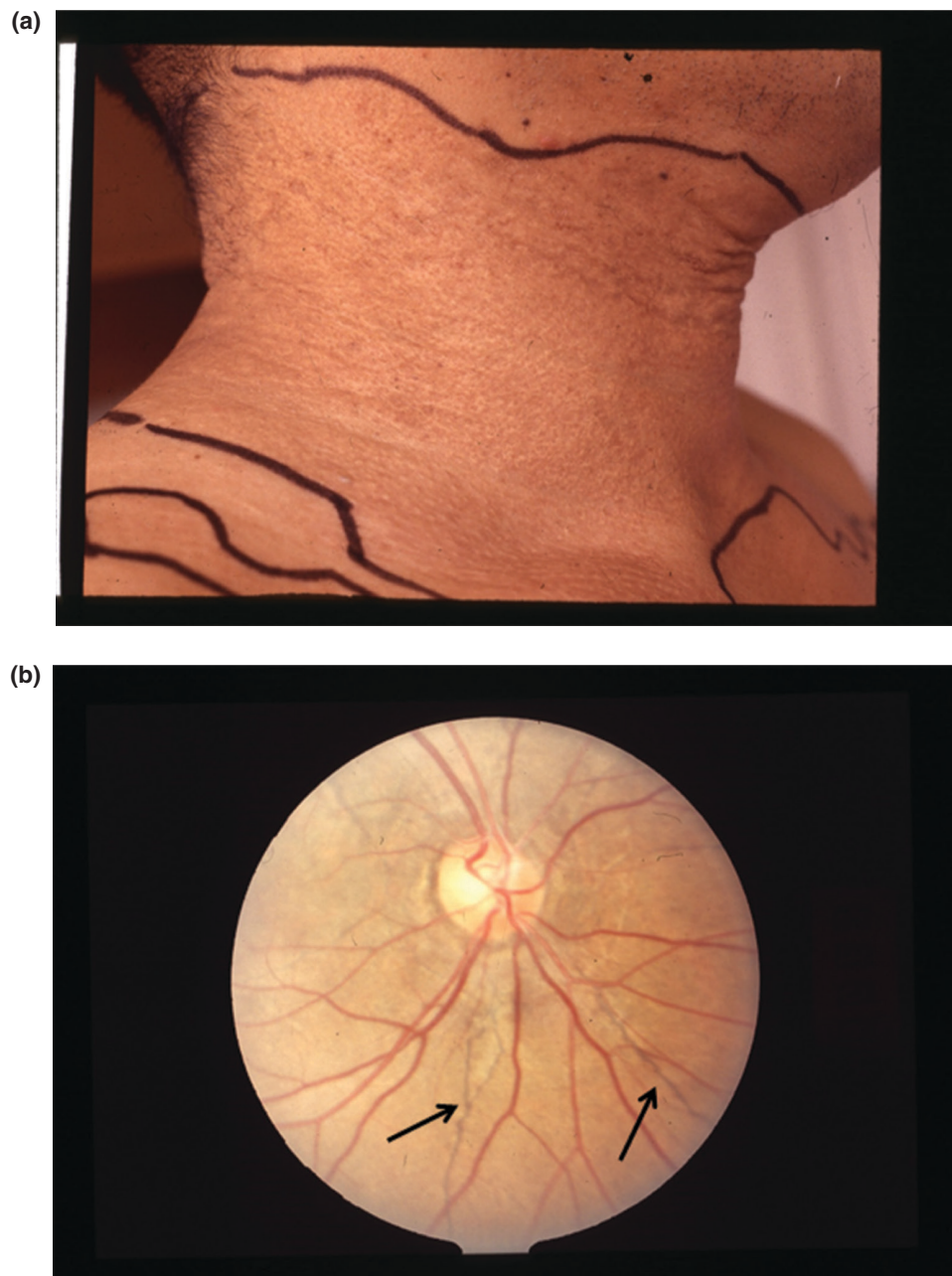


Figure 3-3 Cervical skin lesion (a) and angiod streak of the retina (b) (arrows) seen in a patient with pseudoxanthoma elasticum. By courtesy of Department of Neurology, Kyushu University.

(Figure 3-10). In this condition, vascular endothelial growth factor (VEGF) levels are markedly elevated in the serum. Significant improvement of neurological symptoms by peripheral blood stem cell transplantation has been reported in this condition (Karam et al, 2015; Keddie et al, 2020; Kuwabara et al, 2008). Thalidomide was shown to suppress VEGF production and plasma cell proliferation in patients with POEMS syndrome (Misawa et al, 2016).

D. SKIN ABNORMALITIES SEEN IN TOXIC NEUROLOGIC DISEASES

Intoxication with some heavy metals shows characteristic skin changes. Alopecia is characteristically seen in **thallium poisoning**, which results from oral ingestion of an organophosphate insecticide thallium and causes unconsciousness due to strong inhibition of cholinesterase in the acute phase and polyneuropathy mainly involving the autonomic nervous system in the chronic phase (Figure 3-11). Keratosis of skin,



Figure 3-4 Café-au-lait spots seen in a patient with neurofibromatosis type 1.
By courtesy of Dr. Yumi Matsumura, Department of Dermatology, Kyoto University.

pigmentation, and a white horizontal line on the nail (Mees line) are seen in intoxication with thallium and **arsenic**.

E. SUNLIGHT PHOTSENSITIVITY OF SKIN

In this condition, skin rash is either induced or worsened by exposure to sunlight. It is seen in systemic diseases such as systemic lupus erythematosus (SLE), porphyria, niacin deficiency (pellagra), xeroderma pigmentosum, and Cockayne disease. **Xeroderma pigmentosum** is an autosomal recessive hereditary disease that clinically manifests with marked sunlight photosensitivity of skin from infancy (Figure 3-12). It is associated with progressive decline of intellectual and other neurological functions (**De Sanctis–Cacchione syndrome**). In this condition, abnormality of a repairing mechanism of the damaged DNA has been reported (Anttinen et al, 2008). **Cockayne disease** is also an autosomal recessive hereditary disease, and it is clinically characterized by dwarfism, micrognathia, retinal pigment degeneration, marked sunlight photosensitivity, mental retardation, deafness, and progressive spastic and ataxic gait that start appearing before age 2 years (Cocco et al, 2019).

F. ABNORMALITIES OF ORAL CAVITY SEEN IN NEUROLOGIC DISEASES

In case there is any possibility of **Behçet disease**, the oral cavity should be carefully checked for **aphthae**, typically

white small ulcers (Figure 3-13). In the patients with **pellagra** due to niacin deficiency, ruby erythema of tongue is seen in addition to diarrhea, dermatitis (Figure 3-14), dementia, and polyneuropathy. An abnormally enlarged tongue (**macroglossia**) is seen in Down syndrome, primary amyloidosis, and acromegaly. If there is any scar on the tongue due to a bite, it suggests the presence of vigorous involuntary movements, convulsion involving the mouth, or self-mutilation. **Tongue bite** due to severe choreic movements is seen in **neuro-acanthocytosis** (chorea-acanthocytosis, Levine–Critchley syndrome) (Figure 3-15). A typical example of tongue self-mutilation is **Lesch–Nyhan syndrome**, which is also characterized by hyperuricemia and gout due to the inherited (X-linked) lack of hypoxanthine phosphoribosyltransferase.

G. GENERAL PHYSICAL FINDINGS AND NEUROLOGICAL FINDINGS

As explained at the beginning of this chapter (p. 9), it is practical to carry out the general physical examination and the neurologic examination as a single block of physical examination. However, when describing those findings in the medical record, reporting in case reports, or presenting at medical conferences, it is useful to describe the general physical findings first and then the neurological findings. In this regard, it is important to keep in mind that the general physical findings are not just those dealt with in

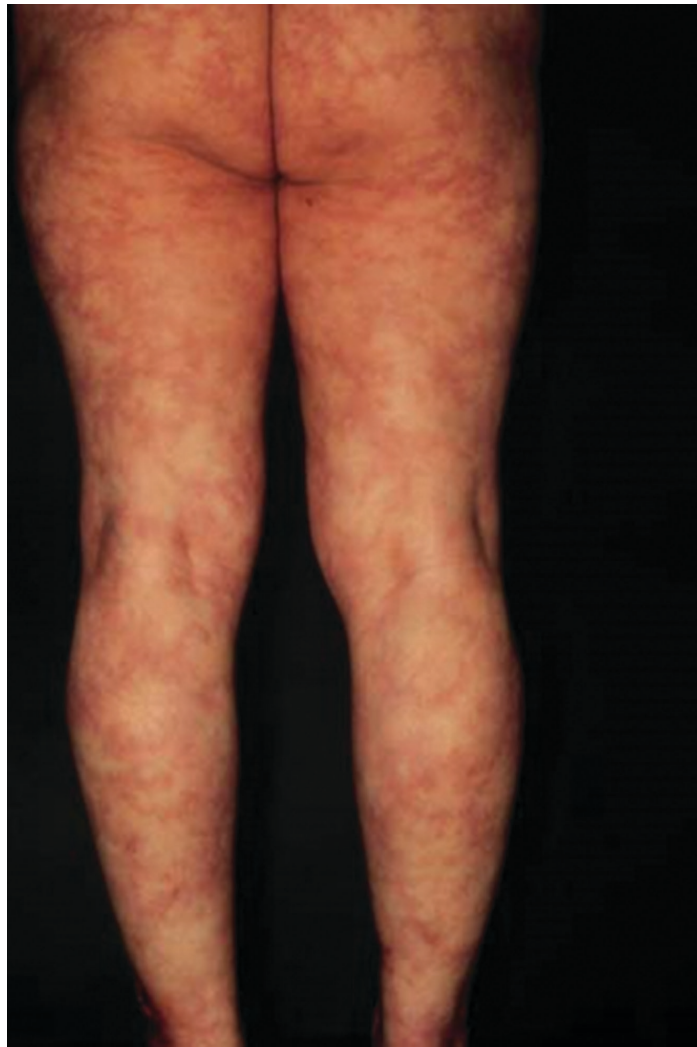


Figure 3-5 Livedo reticularis seen in a patient with Sneddon syndrome.
By courtesy of Dr. Yumi Matsumura, Department of Dermatology, Kyoto University.

internal medicine; rather, it is necessary to include all findings related to other specialties such as ophthalmology, otorhinolaryngology, oral and dental health, dermatology, orthopedics, urology, and gynecology.

2. NEUROLOGICAL MANIFESTATION OF SYSTEMIC MEDICAL DISEASES

A. COLLAGEN VASCULAR DISEASES

This is a group of diseases due to autoimmune inflammatory disorders with antibodies against the collagen tissue, thus causing various forms of systemic vasculitis. These include rheumatoid arteritis (RA), SLE, polyarteritis nodosa (PN), Sjögren syndrome, giant cell arteritis, scleroderma, and mixed connective tissue disease (MCTD). Collagen

diseases often present with neurological manifestations (Lisak et al, 1988) (Table 3-1).

Among these diseases, **RA** is the most common and manifests with a variety of neurological symptoms and signs due to articular inflammation and damage leading to compression of the peripheral nervous system (PNS) or the CNS. Those include carpal tunnel syndrome, atlanto-axial subluxation, compression myelopathy, ischemic neuropathy, and rheumatoid meningitis (DeQuattro & Imboden, 2017).

SLE is characterized by malar flushing (Figure 3-16), multiple arthralgia, photosensitive dermatitis, and fever. It commonly causes neurological manifestations due to involvement of the CNS, the PNS, and the muscles. The PNS involvement in SLE manifests as distal sensory and motor polyneuropathy. Multiple mononeuropathy was also reported in SLE, although it is rare (Rivière et al, 2017).



Figure 3-6 Hyperextensibility of joint and hyperelasticity of skin seen in a patient with Ehlers–Danlos syndrome.
By courtesy of Dr. Yumi Matsumura, Department of Dermatology, Kyoto University.



Figure 3-7 Geniculate zoster seen in a patient with Hunt syndrome.
By courtesy of Dr. Yumi Matsumura, Department of Dermatology, Kyoto University.

3. PHYSICAL EXAMINATION



Figure 3-8 Erythema nodosum seen in a patient with myelodysplastic syndrome.
By courtesy of Dr. Yumi Matsumura, Department of Dermatology, Kyoto University.



Figure 3-9 Purulent erythema seen in a patient with Sweet syndrome.
By courtesy of Dr. Yumi Matsumura, Department of Dermatology, Kyoto University.



Figure 3-10 Hypertrichosis (left) and osteosclerotic plasmacytoma of the spine (right, see arrow) seen in a patient with POEMS syndrome.
By courtesy of Department of Neurology, Kyushu University.

CNS involvement is not rare in SLE (Schwartz et al, 2019), and convulsive seizures may be the first clinical manifestation of SLE.

PN typically causes multiple neuritis (mononeuritis multiplex) (de Boysson & Guillevin, 2019). In this condition, different peripheral nerves may not be involved at the same time, but they may appear at different clinical stages. For example, if the radial wrist flexor muscle (innervated by the median nerve) is paretic whereas the medial wrist flexor muscle (innervated by the ulnar nerve) is intact, it suggests multiple neuritis. Multiple neuritis associated with severe pain and marked eosinophilia is called **eosinophilic granulomatosis with polyangiitis (Churg–Strauss syndrome)** (Furuta et al, 2019; Nishi et al, 2021).

MCTD presents with distinct clinical features, including Raynaud's phenomenon, puffy hands, arthritis, myositis, pleuritis, pericarditis, interstitial lung disease, and pulmonary hypertension. Since MCTD can cause transverse myelopathy and a longitudinally extending spinal cord lesion, it is raised as important differential diagnosis for multiple sclerosis (MS) and

BOX 3-2 INFLAMMATORY DISEASES OF THE NERVOUS SYSTEM

Causes of inflammatory diseases of the nervous system are grouped into **infection** and **autoimmunity** for both the CNS and PNS. The infectious group is caused by direct infection of the nervous system by bacteria or virus. Most of the autoimmune inflammatory disorders of the nervous system are caused by the nervous tissue itself as the antigen, but some of them are caused by the connective tissue as the antigen, like vasculitis. In the former group, the symptoms occur in association with infections (**parainfectious**), or they occur a week or two after the infections (**postinfectious**), like acute disseminated encephalomyelitis (ADEM) and Guillain–Barré syndrome. A special group of autoimmune neurological diseases is the **paraneoplastic syndrome**, which is not due to direct invasion of the nervous system by a tumor but rather due to an autoimmune process against the neoplastic tissue, which is considered to share a common antigen with the nervous system. Clinically a paraneoplastic syndrome is manifested as an inflammation like limbic encephalitis and dermatomyositis, neurodegeneration, or ion channel disorders.



Figure 3-11 Alopecia seen in a patient with thallium poisoning.
By courtesy of Department of Neurology, Kyushu University.

neuromyelitis optica spectrum disorders (NMOSD) (Hao et al, 2019) (see Box 17-4). In MCTD, the sensory symptoms are especially common in the territory of the trigeminal nerve.

Temporal arteritis, which is sometimes included in the collagen vascular diseases, is discussed in Chapter 16.

B. ENDOCRINE DISEASES

Endocrine diseases may cause a variety of neurological presentations, ranging from acute confusional state to slowly progressive gait disturbance (Douglas, 2010).

Thyroid disease is likely the most common. The patient with hyperthyroidism may appear fidgety because of postural tremor and chorea, and hyperactive tendon reflexes. Patients presenting with hypokalemic periodic paralysis might have hyperthyroidism as the trigger. Hypothyroidism is characterized by slowing down and fatigue; cognitive dysfunction and ataxia might be present. Tendon reflex responses are slow.

Autoimmune thyroiditis is associated with other autoimmune neurological diseases such as **steroid-responsive encephalopathy with autoimmune thyroiditis (SRET)**, previously called **Hashimoto encephalopathy**



Figure 3-12 Sunlight photosensitivity of skin in a patient with xeroderma pigmentosum.
By courtesy of Dr. Yumi Matsumura, Department of Dermatology, Kyoto University.



Figure 3-13 Aphthous ulcer of oral mucosa seen in a patient with Behçet disease.
By courtesy of Department of Neurology, Kyushu University.



Figure 3-14 Ruby red dermatitis seen in the hand of a patient with pellagra.
Courtesy of Dr. Soham D. Desai, Department of Neurology, Pramukhswami Medical College, Gujarat, India.

or **thyrotoxic ophthalmoplegia**. Hashimoto encephalopathy may present with cerebellar ataxia (Ercoli et al, 2019). These are not considered to be the outcome of the thyroid dysfunction but rather autoimmune disorders parallel to the autoimmune thyroiditis. Therefore, these neurological abnormalities do not necessarily occur in conjunction with thyroiditis.

ACTH-producing pituitary adenoma causes central obesity, moon face, buffalo hump, striae and plethora of skin, hypertension, proximal muscle weakness, and hypokalemia (**Cushing disease**). A similar clinical picture is seen in patients who are receiving corticosteroid treatment, or in association with an adrenal tumor and an aberrant ACTH-producing tumor (**Cushing syndrome**) (see Chapter 30). Hypoparathyroidism causes tetany due to hypocalcemia (see Chapter 16, p. 151).

C. METABOLIC DISEASES AND SYNDROMES

Diabetes mellitus is complicated by a variety of neurological diseases. It serves as one of the most important risk factors for ischemic cardiac and cerebral vascular diseases. Distal painful dysesthesia with nocturnal burning pain of the feet is characteristic of **diabetic polyneuropathy**. A more proximal mononeuritis multiplex can occur. Excessive insulin

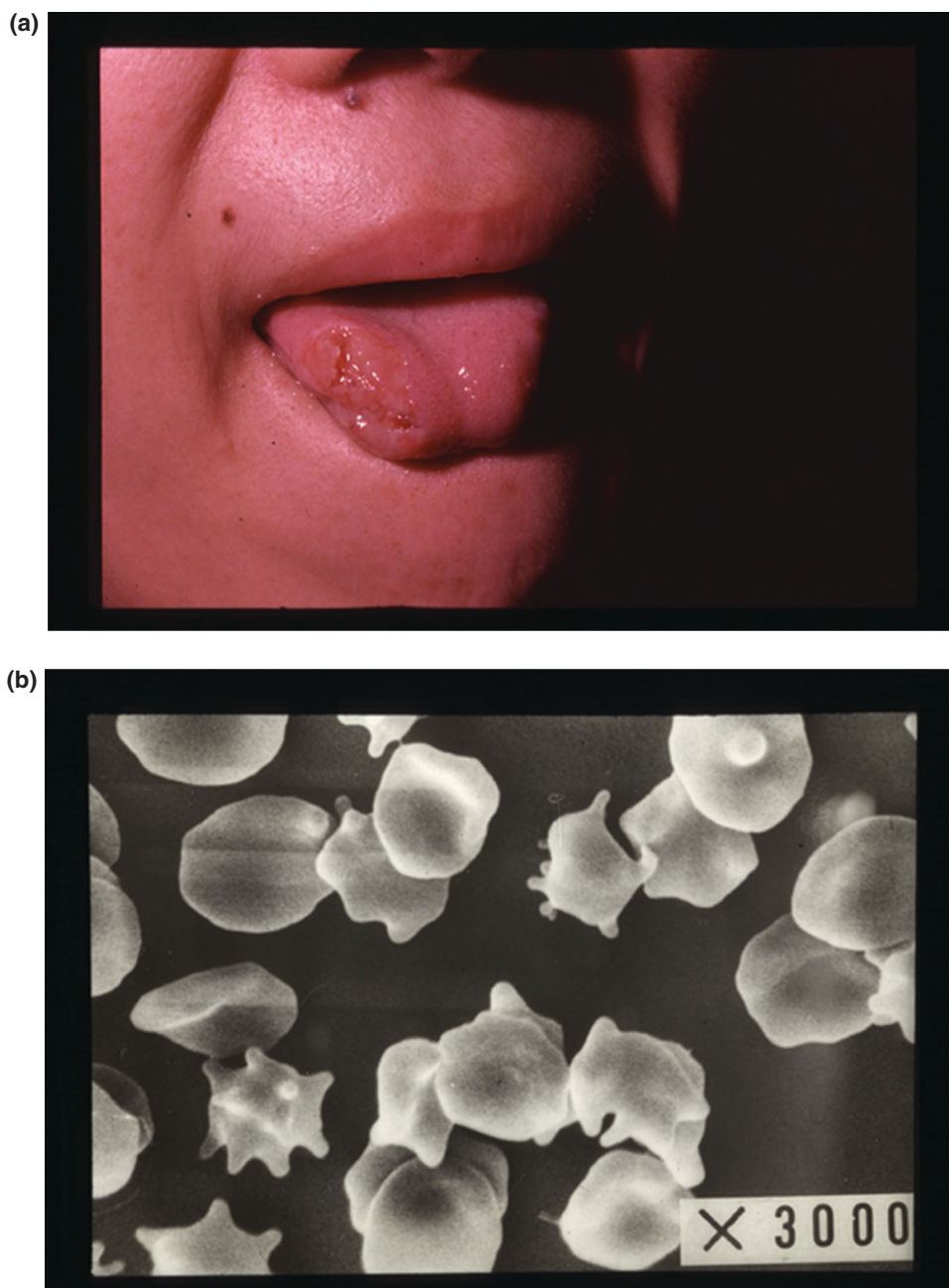


Figure 3-15 Tongue bite (a) and acanthocytes (b) seen in a patient with neuro-acanthocytosis.
By courtesy of Department of Neurology, Kyushu University.

treatment may cause hypoglycemic coma with myoclonus and convulsions. In extreme hyperglycemia there can be damage to the basal ganglia and the acute onset of chorea.

The so-called metabolic syndrome is a collection of risk factors that increase the probability of developing type 2 diabetes. These factors include abdominal obesity, hyperglycemia, hypertriglyceridemia, hypertension, and low levels of high-density lipoprotein cholesterol (see Wang et al, 2020).

Hepatic failure due to cirrhosis may cause consciousness disturbance and asterixis (**hepatic encephalopathy**).

The presence of a shunt between the portal vein and the systemic circulation causes a confusional state after ingesting a protein-rich diet (**portal systemic encephalopathy**) (see Box 23-4). These conditions are often associated with hyperammonemia, and the electroencephalogram (EEG) often shows “triphasic waves.” **Uremic encephalopathy** due to renal failure may show a disturbance in consciousness associated with myoclonus and asterixis.

Acute intermittent porphyria is an autosomal dominant disorder with deficiency of phosphobilinogen

TABLE 3-1 NEUROLOGICAL MANIFESTATION OF COLLAGEN VASCULAR DISEASE

	SLE	Polyarteritis nodosa	MCTD	Rheumatoid arthritis
General	Malar flushing, polyarthritis, fever, dermatitis	Skin nodules	Raynaud phenomenon, arthritis, lung disease	Polyarthritis, interstitial lung disease
CNS	Cognitive impairment, seizures	Cerebral vasculitis	Myelopathy	Atlanto-axial dislocation, meningitis, compression myelopathy
PNS	Polyneuropathy, mononeuropathy	Mononeuritis multiplex, Churg–Strauss syndrome	Trigeminal neuropathy	Entrapment neuropathy, polyneuropathy
Muscles	Proximal myopathy		Myositis	
Antibody	ANA, anti-dsDNA anti-Sm, anti-phospholipid		U1-snRNP	Rheumatoid factor, anti-CCP

deaminase of the heme biosynthesis pathway. It is characterized by recurrent attacks of abdominal pain, polyneuropathy, and mental changes, and an important differential diagnosis for acute abdomen and Guillain–Barré syndrome (Suh et al, 2019).

Disorders of amino acid metabolism with an abnormal urea cycle cause encephalopathy (Wajner, 2019). Among others, the patient with citrullinemia due to deficiency of argininosuccinate synthetase is known to have childhood behavior of beans addiction (see Box 23-4).



Figure 3-16 Malar flushing seen in a patient with systemic lupus erythematosus.
Courtesy of Dr. Tsuneyo Mimori, Ijinkai Takeda General Hospital, Kyoto.

Deficiency of vitamins due to malnutrition causes neurological manifestations specific for each kind of vitamin. Vitamin B₁ deficiency (**beriberi**) is characterized by axonal sensory-dominant polyneuropathy with severe burning dysesthesia of the feet and marked tenderness of the calf muscles, associated with congestive heart failure (see Chapter 20, p. 198). Niacin deficiency causes **pellagra**, which is characterized by ruby erythema of the tongue, diarrhea, dermatitis, dementia, and sensory polyneuropathy (see Chapter 20, p. 198) (see Figure 3-15). Vitamin B₁₂ deficiency may cause subacute combined degeneration of the spinal cord and polyneuropathy (see Chapter 16, p. 131).

Hereditary diseases with known metabolic abnormalities such as Wilson disease (see p. 169 and 176), cerebrotendinous xanthomatosis (see Box 16-10), Bassen–Kornzweig disease (see Box 16-9), and Fabry disease (see Box 20-5) are described in the category of the main clinical feature of each disease.

D. HEMATOLOGIC DISEASES

Various forms of neuropathy can be associated with monoclonal gammopathy (**paraproteinemic neuropathy**) (Vallat et al, 2021).

Systemic amyloidosis as the result of multiple myeloma may cause amyloid polyneuropathy. In this case, just like familial amyloid polyneuropathy (see Box 21-1), small fibers are mainly involved with loss of pain sense and autonomic nervous system failure (Kaku & Berk, 2019).

Osteosclerotic plasmacytoma may manifest with **POEMS (Crow–Fukase) syndrome** (Suichi et al, 2019), as noted earlier (see Figure 3-10). Human T lymphotropic virus-1 (HTLV-1) causes adult T-cell leukemia, and it was also shown to cause myelopathy manifesting with spastic paraparesis, posterior column involvement, and urinary

frequency/urgency (**HTLV-1-associated myelopathy; HAM**) (see Box 17-8). New cases of this condition are no longer seen after breastfeeding of babies by infected mothers and transfusions of blood from infected persons were stopped.

Sickle cell disease is relatively common in Africa, and it commonly presents with stroke, seizure, and headache (Noubiap et al, 2017).

E. BEHÇET DISEASE

Behçet disease is characterized by uveitis, painful oral ulcers (aphthae) (see Figure 3-13), genital ulcers, erythema nodosum, and purulent dermatitis at the site of needle injection. Neurologically, it may cause recurrent meningitis and encephalitis (**neuro-Behçet**). CNS lesions often involve the brainstem and diencephalon with disseminated lesions of perivascular cell infiltration (Kidd, 2017).

F. SARCOIDOSIS

Sarcoidosis is a system disease involving multiple organs and is histologically characterized by non-caseating granuloma. Clinically it is characterized by lymphadenopathy, skin lesions (Figure 3-17), and lung disease with bilateral hilar lymphadenopathy (BHL) (Figure 3-18). Both the CNS and PNS are involved (Kidd, 2020; Krumholz & Stern, 2014). Among the PNS involvement, bilateral facial nerve palsy, either simultaneously or independently, is often seen in this disease (see Box 11-3). **Neurosarcoidosis** of CNS may show clinical manifestation similar to MS, especially when the spinal cord is mainly involved (**sarcoid myelopathy**). In this case, intrathecal synthesis of oligoclonal IgG suggests multiple sclerosis because it is rare in neurosarcoidosis (Arun et al, 2020).

G. NEOPLASM

A young female patient presenting with psychiatric symptoms, confusional state, and dyskinesia might have ovarian teratoma. This condition is called **limbic encephalitis** and is often associated with ovarian teratoma (see Box 28-1). Patients with pulmonary small cell carcinoma may show progressive cerebellar ataxia. In this condition, called **subacute cerebellar degeneration**, antibody against voltage-gated calcium channel (VGCC) is demonstrated (see Chapter 16, p. 131). These patients may also show myasthenic syndrome (Lambert–Eaton syndrome) (see Chapter 25). These conditions are not due to direct or metastatic invasion of the nervous system by the tumor but are



Figure 3-17 Histologically proven skin lesion seen in a patient with sarcoidosis.
Courtesy of Dr. Yumi Matsumura, Kyoto University Hospital.

produced by an autoimmune mechanism (**paraneoplastic syndrome**) (Graus et al, 2021; Rosenfeld & Dalmau, 2018). Paraneoplastic conditions may also manifest as subacute myeloneuropathies (Shar et al, 2021).



Figure 3-18 Chest X-ray showing bilateral hilar lymphadenopathy in a patient with sarcoidosis.
Courtesy of Dr. Michio Ono, Ijinkai Takeda General Hospital, Kyoto.

Cowden syndrome is an autosomal dominant disease with multiple hamartomas involving various organs. This condition is often associated with dysplastic gangliocytoma of the cerebellum, called **Lhermitte–Duclos disease** (Dhamija et al, 2018) (see Figure 16-22).

Erdheim–Chester disease is a histiocytic neoplasm characterized by infiltration and accumulation of inflammatory macrophages in multiple tissues, including the brain (Boyd et al, 2020). The neurological manifestations are cognitive impairment, peripheral neuropathy, pyramidal tract signs, cranial nerve involvement, and cerebellar ataxia.

3. STEPS OF THE NEUROLOGIC EXAMINATION

Neurological findings are systematically described or reported, for example, in the following order: (1) consciousness, (2) mental or intellectual functions, (3) aphasia, apraxia, and agnosia, if any, (4) brainstem and cranial nerve territories, (5) neck, (6) trunk, (7) motor system of extremities, (8) tendon reflexes and pathologic reflexes, (9) somatosensory functions, (10) posture and gait, (11) autonomic nervous functions, and (12) disturbance of daily activities. When we perform the neurologic examination, however, we do not have to follow this sequence. In many cases, it is more convenient and practical to adjust to the actual situation with flexibility. In fact, different systems may be employed in different countries.

In any case, it is useful to start with evaluation of consciousness. In most cases, disturbance of consciousness, mental or intellectual abnormalities, and language abnormality, if any, can be suspected during the history taking. If any abnormality in the higher cortical functions is suspected, they may be tested in detail after completing the physical examination. In the ordinary situation, it is more practical to do the physical examination first and to test the higher cortical functions later in the course of examination, mainly because it can be embarrassing for the patient to go through a series of questions about intelligence and language from the beginning and also because it requires a relatively long time before carrying out the physical examination. Furthermore, abnormalities of the higher cortical functions are usually noticed during the physical examination, because verbal communication with the patient is necessary for the physical examination. Moreover, a more comfortable relationship can be established between the patient and the physician through the course of physical examination. In this book, therefore, the chapters are arranged in the actual order of the physical neurologic

examination, starting from consciousness and immediately moving to the brainstem and cranial nerve functions, and the examination of mental and higher cortical functions will be described after all the chapters related to the physical examination. Furthermore, the examination of patients with coma will be described toward the end of the book because it requires the culmination of knowledge about the whole physical examination.

If the history is effectively taken, responsible sites of the lesion, possible pathogenesis or etiology, and the most likely clinical diagnosis can be postulated before starting the physical examination. Therefore, although the neurologic examination is executed in the order described earlier, emphasis is placed on certain aspects depending on the information obtained from the history taking (**hypothesis-driven neurologic examination**) (Kamel et al, 2011). In other words, a hypothesis is developed after taking the history, which then focuses the examination.

When carrying out the neurologic examination of extremities and trunk, the order of examination can be freely chosen depending on the circumstances. When a patient is examined in the supine position, it is more practical to examine the functions of all four extremities at once. By contrast, when a patient is examined in the sitting position, the upper extremities and the lower extremities may be examined separately. At any rate, stressful burden and uncomfortable maneuvers for the patient should be limited to the minimum necessary.

As structure and function of the nervous system are symmetrically formed between the left and right, with the exception of some higher cortical functions and the peripheral autonomic nervous system, it is important to pay attention to the **symmetry** of neurological signs throughout the neurologic examination. However, there may be some asymmetry in the homologous muscles in volume and strength depending on the occupation of, or sports played, by the patient.

The purpose of the neurologic examination is to confirm the neurological condition that was thought to be most likely based on careful history taking, and to exclude the candidate diagnoses listed in the differential diagnosis. Finally, it is of utmost importance to judge the level of **activities of daily living** and to estimate a possible prognosis of the present condition.

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EVALUATION OF CONSCIOUSNESS

Consciousness is defined as the state of waking, with subjects being aware of self and the environment. Namely, when subjects are fully conscious, they can pay attention to the external environment, accept the information, and respond by language or behavior. Regarding the response by language or behavior, however, patients suffering from neurological deficits such as aphasia and motor paralysis are exceptions. A typical example is the locked-in syndrome, which will be described later in this chapter.

1. ANATOMIC BASIS OF CONSCIOUSNESS

It is generally believed that a critical center for the production of consciousness is in the **brainstem reticular formation**. This is a poorly demarcated gray matter structure situated in the dorsal part of the brainstem, extending

longitudinally through the whole length of the brainstem. Which parts of the reticular formation are critical for consciousness are not known. The brainstem reticular formation is formed by a large number of neurons and their projections. It receives inputs from various ascending afferent fibers and projects ascending fibers to the intralaminar thalamic nucleus and then widely to the cerebral cortex through the **diffuse thalamocortical projection system** (Figure 4-1). This **ascending reticular activating system** activates activities of the cerebral cortex. Therefore, if the reticular formation is damaged, the cerebral cortex cannot be activated, and, as a result, it is unable to pay attention to or receive external information, and to respond by language or behavior (**unconsciousness**). The brainstem reticular formation also sends fiber projections to the spinal cord via several reticulospinal tracts and various nuclei of brainstem and cerebellum for controlling muscle tonus and postural adjustment. Multiple neurotransmitters,

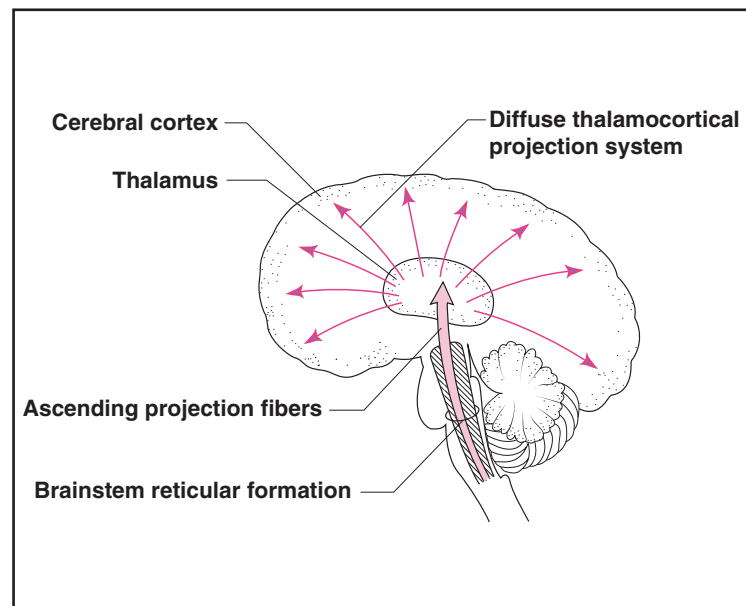


Figure 4-1 Schematic diagram showing the brainstem reticular formation and the thalamocortical activating system.

including acetylcholine, noradrenalin, dopamine, and serotonin, are known to take part in the activities of the reticular formation.

Based on the pathologic study of coma-causing brainstem lesions and by the study of resting-state functional connectivity in healthy subjects, Fischer et al. (2016) reported that injury to a small region in the pontine tegmentum was associated with coma, and that brainstem site was functionally connected to the ventral anterior insula and pregenual anterior cingulate cortex.

2. MECHANISMS OF DISTURBANCES OF CONSCIOUSNESS

Consciousness is impaired as the result of functional loss of neurons of the reticular formation through the following two mechanisms: (1) a loss of excitability of the neurons due to a decrease in or loss of excitatory postsynaptic potentials (EPSP) resulting in inability to generate action potentials and (2) a loss of neuronal activity due to their excessive discharge or hyperexcitability. An example of the first mechanism is tissue damage to the reticular formation due to ischemia, hemorrhage, inflammation, mechanical/traumatic injury, and cerebral herniation. An example of the second mechanism is epilepsy.

As regards ischemic damage of the brainstem reticular formation, unconsciousness may result from two different types of ischemia: (1) a decrease or loss of blood flow due to ischemia of the vertebrobasilar artery territory that directly supplies arterial blood to the reticular formation (Box 4-1) and (2) a decreased supply of blood or oxygen to the whole brain due to cardiac or respiratory failure or a sudden decrease in blood pressure (shock). Loss of consciousness due to the latter mechanism is called **syncope**. Thus, **transient loss of consciousness** is caused by (1) transient ischemic attack in the vertebrobasilar artery territory, (2) syncope due to hypoperfusion or hypoxia, or (3) epileptic attacks.

Syncope is known to occur in at least six specific conditions (Box 4-2). Therefore, precise history taking is critical for the correct diagnosis (Box 4-3). Usually an attack of syncope starts with loss of ability to think clearly, followed by fixation of eyes in the midline, loss of facial expression, constriction of the visual field, blackout, inability to hear, and finally loss of consciousness (Wieling et al, 2009). The whole process may be completed within several seconds. In general, patients with hypoxia due to chronic pulmonary diseases or with anemia due to hematologic diseases tend

BOX 4-1 SUBCLAVIAN STEAL SYNDROME

A **steal phenomenon** is a cause of transient ischemic attack due to “stealing” of blood flow from a large artery supplying the nervous system into the contracting muscles, and the subclavian steal syndrome is its typical example. In this condition, the patient develops vertigo, unsteadiness, and clouding of consciousness when he or she moves an upper limb. In the background of this particular condition, there is stenosis or occlusion of the subclavian artery proximal to the bifurcation of the vertebral artery, and the muscle activities of the upper limb supplied by the subclavian artery steal its blood supply from the vertebral artery territory, causing poor perfusion in the brainstem supplied by the vertebral artery. The subclavian steal syndrome is more common in association with the movement of the left upper limb, because the left vertebral artery bifurcates from the left subclavian artery after the latter bifurcates directly from the aortic arch. In contrast, the right vertebral artery bifurcates from the right subclavian artery soon after it bifurcates from the brachiocephalic artery. Some form of intermittent claudication is also caused by a steal mechanism (see Box 22-5, p. 214).

to suffer from syncope, and this is especially so for elderly patients.

3. OBSERVATION OF THE STATE OF CONSCIOUSNESS

Usually, the state of consciousness can be accurately evaluated while taking the history from the patient. Actually, evaluation of the patients with consciousness disturbance starts from the time the examiner approaches the patient. If the patient pays attention to the examiner and greets

BOX 4-2 ETIOLOGIC CLASSIFICATION OF SYNCOPE

1. Cardiac syncope
2. Orthostatic hypotension
3. Vasovagal syncope
4. Carotid sinus hypersensitivity
5. Micturition syncope
6. Cough syncope
7. Deglutition syncope

BOX 4-3 CAUSES OF SYNCOPE

Syncope is caused by six different conditions (see Box 4-2).

Cardiac syncope, also called **Stokes–Adams syndrome**, is due to transient loss of blood flow due to arrhythmia or cardiac arrest. In this condition, clonic movements might be seen, which necessitates distinction from epileptic attacks. Although cardiac arrest might occur as a result of epilepsy, the distinction is not difficult from the past history and the EEG.

Orthostatic hypotension is a common cause of syncope, which will be explained in relation to the autonomic nervous system (Chapter 20, p. 208).

Vasovagal syncope is caused by a sudden reflex elevation of vagal nerve function resulting in hypotension. This condition may be encountered even in healthy subjects when, for example, exposed to severe pain.

Carotid sinus hypersensitivity is seen in patients whose baroreceptors of the carotid sinus are pathologically sensitive. In this condition, a mechanical stimulus like massage to the anterior neck tends to cause hypotension (Chapter 21, p. 209).

Micturition syncope occurs during or immediately after micturition due to a sudden decrease of intra-abdominal pressure followed by a reflex decrease of cardiac output. This phenomenon may be seen even in healthy subjects after heavy drinking, but the repetitive occurrence of this condition suggests abnormality of the autonomic nervous system.

Cough syncope occurs after vigorous coughing, which causes a sudden increase of the intrathoracic pressure like Valsalva maneuver and reflex excitation of the vagal nerve causing hypotension.

Swallow-induced syncope (**deglutition syncope**) is characterized by lightheadedness, dizziness, and loss of consciousness during or immediately after swallowing (Aydogdu et al, 2017).

In addition, **orthostatic tachycardia** may cause lightheadedness (postural orthostatic tachycardia syndrome; POTS) (Olshansky et al, 2020).

him or her verbally or behaviorally, the consciousness level is considered to be alert or only mildly impaired. If the patient does not pay attention to the examiner, his or her attention should be aroused by calling his or her name or by providing various stimuli (see the next section). Since the consciousness level tends to change from moment to moment, it is important to observe the patient repeatedly along the clinical course. Elderly patients in particular may be more prone to delirious states in which the consciousness level waxes and wanes depending on general conditions like hydration, nutrition, blood count, and respiratory function.

4. MODES OF CONSCIOUSNESS DISTURBANCE AND THE COMA SCALE

Disturbances of consciousness can be classified into two types: (1) a decrease in the level of consciousness (**clouding of consciousness**) and (2) a language or behavioral abnormality (**alteration of consciousness**). Clouding of consciousness is grouped into four categories according to its degree: (1) **somnolence** (the consciousness level is decreased if the patient is left alone); (2) **stupor** (the patient can be aroused by a strong stimulus and moves

for a short period of time); (3) **semi-coma** (the patient can respond to a strong stimulus only with simple movements); and (4) **coma** (no response at all, even to a strong stimulus).

There are various ways for grading the level of consciousness disturbance; the Glasgow Coma Scale (GCS) and the Japan Coma Scale are presented here. In the GCS (Box 4-4), the alert condition (normal) is graded 15 and deep coma is graded 3. In the Japan Coma Scale (Box 4-5), disturbance of consciousness is graded into nine degrees, ranging from score 1 (mildly impaired) to score 300 (deep coma).

When alteration of consciousness is associated with psychomotor agitation, hallucination, and delusion in addition to the clouding of consciousness, the condition is called **delirium**. Delirium of acute onset is often seen in **acute confusional states**. The state of delirium is often associated with tremulous or fumbling movements of hands (**delirium tremens**). This condition is commonly seen in acute ethanol poisoning, drug intoxication, and metabolic encephalopathy. In this regard, a screening test of urine or blood for possible toxins can be done if available; if not, it is important to draw a blood sample from a patient in the acute stage and save it for later laboratory testing to measure the blood level of possible drugs. As the selective serotonin reuptake inhibitors (SSRIs) are often used for the treatment

BOX 4-4 THE GLASGOW COMA SCALE

Eye opening

Open spontaneously	4
Open only to verbal stimuli	3
Open only to pain	2
Never open	1

Best verbal response

Oriented and converses	5
Converses, but disoriented, confused	4
Uses inappropriate words	3
Makes incomprehensible sounds	2
No verbal response	1

Best motor response

Obeys commands	6
Localizes pain	5
Exhibits flexion withdrawal	4
Decorticate rigidity	3
Decerebrate rigidity	2
No motor response	1

of depression, **serotonin syndrome** is not infrequently seen. The serotonin syndrome manifests with shaking of the legs, restlessness, fever, confusion, visual hallucinations, excessive sweating, and tachycardia.

The state of acute clouding or alteration of consciousness without any gross organic lesion in the brain is called **acute encephalopathy**. As this condition is due to a number of different causes (Box 4-6), careful history taking and

BOX 4-5 THE JAPAN COMA SCALE

Grade I: Spontaneously aroused

1. Slightly unclear
2. Poorly oriented in time, place or person
3. Unable to tell the patient's own name and date of birth

Grade II: Aroused by stimulus

10. Opens eyes easily by verbal call
20. Opens eyes in response to loud call or shaking body
30. Barely opens eyes by painful stimulus plus repetitive calling

Grade III: Not aroused even by stimulus

100. Responds to painful stimulus by throw off
200. Responds to painful stimulus by small movements or facial grimacing
300. No response to painful stimulus

BOX 4-6 ETIOLOGIC CLASSIFICATION OF ACUTE ENCEPHALOPATHY

1. Toxic encephalopathy

A. Ethanol and its related disorders

- i. Wernicke encephalopathy (Box 23-3, p. 220)
- ii. Alcoholic pellagra encephalopathy

B. Pharmaceutical drugs

- i. Intoxication due to sleeping drugs
- ii. Serotonin syndrome
- iii. Carbon monoxide intoxication
- iv. Heavy metal intoxication: arsenics, lead
- v. Intoxication due to organic solvents: trichloroethylene
- vi. Insecticide poisoning: organophosphate

2. Metabolic encephalopathy: hyperglycemia, hypoglycemia, hepatic failure, renal failure, electrolyte abnormalities

3. Anoxic encephalopathy

4. Hypertensive encephalopathy

the general physical examination play an important part in its diagnosis.

5. CONDITIONS THAT SHOULD BE DISTINGUISHED FROM CLOUDING OF CONSCIOUSNESS

Some special conditions may resemble disturbance of consciousness and need to be distinguished: (1) deep sleep, (2) advanced stage of dementia, (3) akinetic mutism, and (4) locked-in syndrome. Needless to say, the patient under general anesthesia induced for medical purposes is comatose.

A. DEEP SLEEP

Usually it is not difficult to distinguish sleep from coma, but in some cases, especially after sleep deprivation or after taking sleeping drugs, deep sleep may be confused with disturbance of consciousness. The subject in ordinary sleep can be completely aroused by verbal calling or other stimuli and can maintain the aroused state for some time, but the subject under special conditions listed in the previous sentence may easily fall back to sleep soon after being awakened. On the other hand, even a patient with mild disturbance of consciousness may remain foggy at least for a minute or so after being aroused. This condition is called **lethargy**. The electroencephalogram (EEG) might

help distinguish between sleep and disturbance of consciousness. The EEG activities specific for sleep, like vertex sharp waves (V wave) and sleep spindles, may be seen during sleep, and the subject shows the posterior dominant rhythm of α frequency after being aroused. In contrast, those sleep-specific EEG activities are not seen in the patients with disturbance of consciousness, and the background EEG activity is abnormal even when the patient appears awake. As described earlier (p. 23), observation of daily activities provides useful information for the diagnosis of consciousness disturbance.

B. ADVANCED STAGE OF DEMENTIA

The patient with marked dementia due to diffuse cerebral damage of severe degree cannot pay attention to the external environment and is unable to accept information and respond. This condition corresponds to a state of **inattention** and **apathy** and is encountered (1) in the recovery phase from acute encephalitis or anoxic encephalopathy of severe degree, (2) at the terminal stage of Creutzfeldt–Jakob disease and Alzheimer disease, and (3) in various forms of frontotemporal lobar degeneration (Lansdall et al, 2019). Distinguishing this condition from consciousness disturbance is not difficult from the information about the clinical course and other neurological findings, and the laboratory tests, including EEG and neuroimaging, will provide supplementary information. The sleep–wakefulness cycle is usually maintained in the patient with severe dementia, whereas it is lost or disrupted in the patient with consciousness disturbance. Furthermore, in patients with severe brain lesions, motor abnormalities such as spastic quadriplegia, dystonia, and Gegenhalten (Chapter 17-4C, p. 148) are commonly seen.

C. AKINETIC MUTISM

In this condition, the eyes are open, the patient blinks, and spontaneous ocular movements are seen during the daytime, but the patient cannot direct his or her gaze to, recognize, or respond to even familiar persons. Spontaneous movements are almost absent or very few. However, the patient usually can chew and swallow food once it is put into the mouth. The sleep–wakefulness cycle is relatively well preserved, and functions of the autonomic nervous system and the endocrine system are normally preserved. Thus, the prolonged state of this condition is called **vegetative state**.

Akinetic mutism was originally reported in a patient with an epidermoid cyst of the third ventricle (Cairns et al,

1941), but it is more commonly encountered in association with diffuse cortical lesions or the diffuse white matter lesions that functionally disconnect the cerebral cortex from the deep gray matter. These conditions correspond to the decorticate state or **apallic syndrome** (German: das apallisches Syndrom). The **decorticate state** is often associated with decorticate rigidity and spastic quadriplegia. In **decorticate posture**, the upper limbs are flexed due to increased muscle tonus in the flexors and the lower limbs are extended due to increased muscle tonus in the extensors. In akinetic mutism due to lesions around the third ventricle, the cerebral cortex is considered to be disconnected from the limbic cortex, and physical signs like spastic quadriplegia are few if any. Akinetic mutism might be transiently seen in the recovery phase from anoxic encephalopathy, but this condition should be distinguished from the vegetative state.

It is often claimed that akinetic mutism has been mistaken as brain death, but the distinction is not difficult as far as brain death is correctly defined. Namely, brainstem functions are fully intact in akinetic mutism, whereas all neurons, including the brainstem neurons, are irreversibly damaged in brain death (Chapter 33-7, p. 295).

Catatonic stupor is similar to akinetic mutism in that the patient keeps the eyes open and there are no spontaneous movements or response to verbal call or stimuli. It is characterized by the fact that, once the upper limb is passively lifted, the patient maintains that posture (**catatonia** or waxy flexibility). Furthermore, the patient may recall some of the events that have happened during the unresponsive episode, if asked later. Catatonic stupor usually occurs during the clinical course of psychosis or during the use of neuroleptic medication, but in some cases its cause is undetermined. In contrast with akinetic mutism, catatonic stupor is not associated with any organic lesion in the brain even on conventional neuroimaging. Usually it spontaneously improves over a period of several days, but some patients may respond to benzodiazepine administration. Among the organic brain lesions, catatonic stupor has been reported in patients with multiple sclerosis showing diffuse demyelinating lesions in the cerebral white matter. However, this must be distinguished from akinetic mutism.

D. LOCKED-IN SYNDROME

This condition (Box 4-7) is totally different from the conditions already described, such as clouding of consciousness and akinetic mutism, but it may cause misunderstanding on the part of the patient's family and even the physician.

BOX 4-7 LOCKED-IN SYNDROME

Locked-in syndrome is totally different from clouding of consciousness or akinetic mutism, because patients in this condition, despite intact mental functions, cannot express themselves because of the total motor paralysis. A typical example of this condition is complete tetraplegia due to an infarction of the bilateral pontine basis as a result of the basilar artery occlusion, which interrupts the corticobulbar and corticospinal tracts bilaterally. However, as the vertical gaze center and the oculomotor nucleus in the midbrain are spared in most cases, the patient can still elevate the upper eyelid and move the eyes vertically (see Chapter 9-1, 9-2). Horizontal eye movements are lost because the horizontal gaze center is located in the pontine tegmentum. Furthermore, as the functions of the brainstem tegmentum including the reticular formation and the ascending sensory pathways are preserved in this condition, the patient can understand the environment and spoken language. In fact, locked-in syndrome may also occur in the advanced stage of motor neuron disease and in the acute phase of severe Guillain-Barré syndrome, in which all the voluntary muscles, including the extraocular muscles, may be completely paralyzed.

In this condition, despite involvement of brainstem motor tracks causing quadriplegia and anarthria, consciousness is clear and sensory and cognitive functions are well preserved. Therefore, conversation at the bedside should be done with great caution.

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BRAINSTEM AND CRANIAL NERVE TERRITORIES

1. BRAINSTEM

The brainstem is a cylindrical structure longitudinally situated in the posterior cranial fossa, and is formed of the **midbrain** (mesencephalon), **pons**, and **medulla oblongata** from rostral to caudal.

On transverse section of the midbrain at the level of the superior colliculus (Figure 5-1), a pair of large bulges is seen symmetrically at the anterior aspect (**cerebral peduncle** or **crus cerebri**), and on the posterior aspect there is a pair of small protrusions just lateral to the midline (**superior colliculus**) and another pair below them (**inferior colliculus**). These two pairs of protrusions together are called **quadrigeminal bodies**. This posterior region of the midbrain is called the **tectum**, and the region between the tectum and the cerebral peduncle is called the **mesencephalic tegmentum**, where the substantia nigra, red

nucleus, periaqueductal gray substance, and other structures are located.

The anterior two-thirds of the pons (**pontine basis**) is formed by a large structure containing the descending fibers from the cerebral cortex through the cerebral peduncle and the transverse fibers originating from the pontine nucleus (Figure 5-2). The remaining part of pons is called **pontine tegmentum**, which contains the medial lemniscus anteriorly and a number of nuclei and fiber tracts posteriorly.

The medulla oblongata is located at the caudal end of the brainstem, and anteriorly there is a pair of small protrusions just lateral to the midline (**pyramid**) (Figure 5-3). Through the pyramid, the descending fibers from the cerebral cortex project to the spinal anterior horn cells, and thus this fiber tract is called the **pyramidal tract**. Just lateral to each pyramid, there is another bulge formed by the **inferior**

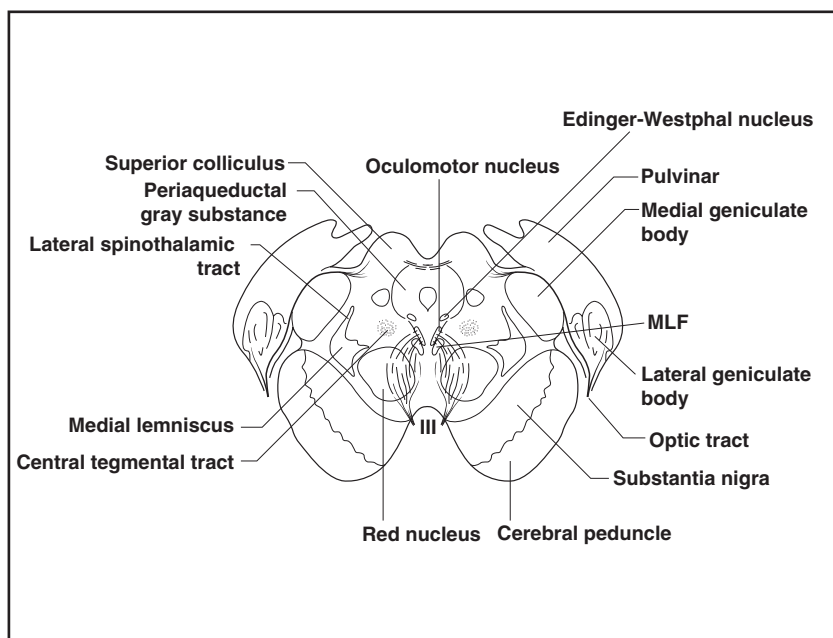


Figure 5-1 Schematic diagram showing a transverse section of midbrain at the level of the superior colliculus. III: oculomotor nerve, MLF: medial longitudinal fasciculus.

Modified from Haymaker, 1969.

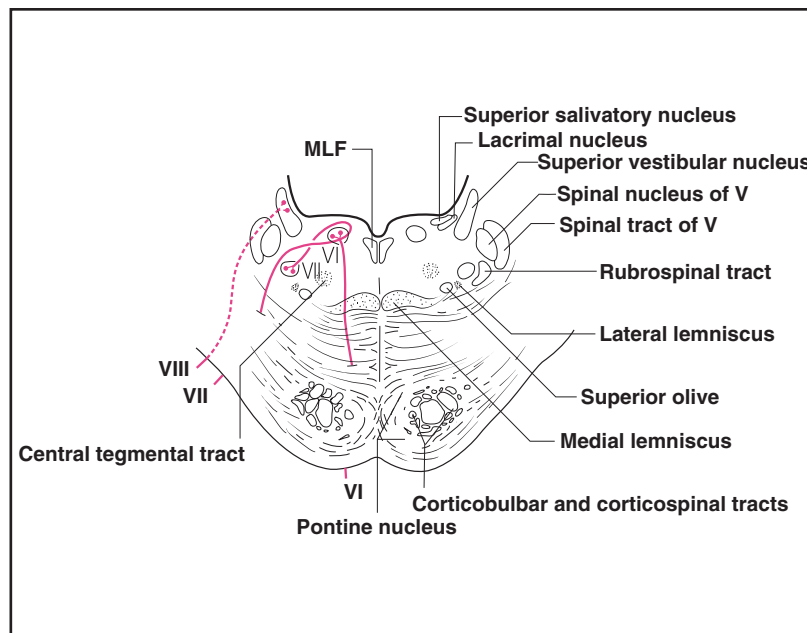


Figure 5-2 Schematic diagram showing a transverse section of pons at the level of the abducens nerve (VI) and the facial nerve (VII). V: trigeminal nerve, VIII: vestibulocochlear nerve, MLF: medial longitudinal fasciculus. Modified from Haymaker, 1969.

olive. Just posterior to the pyramid is the **medial lemniscus**. The posterior rim of medulla oblongata forms the anterior wall of the fourth ventricle, and along the rim are the hypoglossal nucleus, dorsal motor nucleus of the vagus, and vestibular nuclei on each side away from the midline in this order.

2. COMMON STRUCTURES OF CRANIAL NERVES

There are 12 pairs of cranial nerves. Among these, the first, second, and eighth cranial nerves serve special sensory

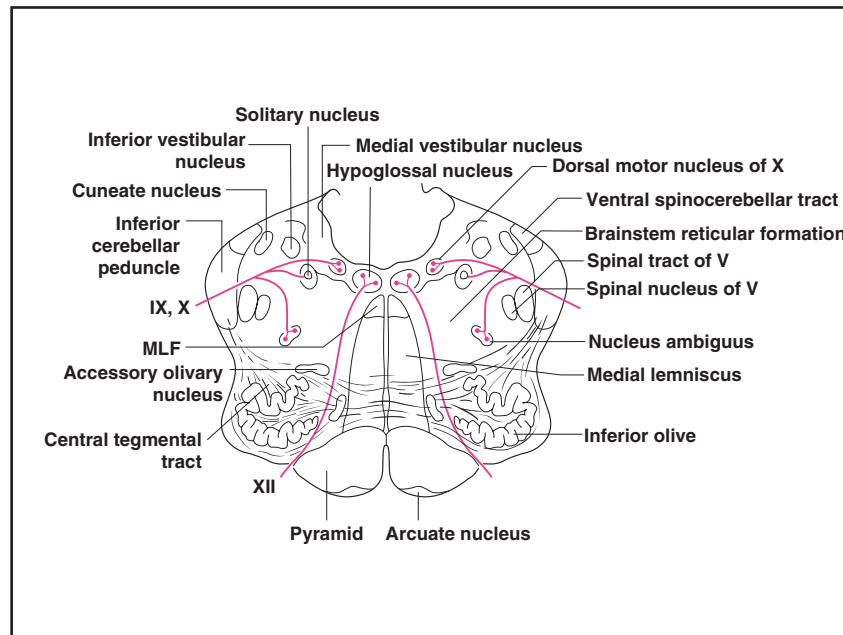


Figure 5-3 Schematic diagram showing a transverse section of medulla oblongata. V: trigeminal nerve, X: vagus nerve, XII: hypoglossal nerve, MLF: medial longitudinal fasciculus. Modified from Haymaker, 1969.

functions (olfaction, vision, and audio-vestibular, respectively). All of the remaining cranial nerves contain motor nerve fibers innervating the skeletal muscles, and the fifth, seventh, ninth, and tenth cranial nerves also contain somatosensory nerve fibers. The third, seventh, ninth, and tenth cranial nerves also contain autonomic nerve fibers.

3. CRANIAL NERVES RELATED TO THE MOTOR FUNCTION

Of the motor fibers contained in the cranial nerves, all of them originate from a corresponding motor nucleus located within the brainstem tegmentum. In contrast, all of the primary somatosensory neurons are located in the somatosensory ganglia, which are located outside of the brainstem. Motor fibers of the cranial nerves do not cross between the sides, at least outside of the brainstem. Within the brainstem, the trochlear nerves cross at the anterior medullary velum (posterior rim of the midbrain at its caudal level), and some of the oculomotor nerve fibers are known to cross within the midbrain (see Chapter 9-1A, p. 57).

Among the sensory fibers contained in the cranial nerves, only the optic nerves decussate partially (see Chapter 7-1B, p. 38), but other sensory nerves do not cross. Therefore, if the sensory fibers are damaged, the symptoms will appear on the same side, corresponding to the innervated territory, with the exception of the optic nerve.

Cytoarchitecturally, the myelin sheath for the cranial nerves originates from the cell membrane of Schwann cells except for the olfactory tract and the optic nerve. Both the olfactory tract and the optic nerve are the secondary neuron of each sensory system, and thus their myelin sheath is formed of the cell membrane of oligodendroglia even though those fibers are situated outside of the brain parenchyma. That is why the optic nerve is often affected in multiple sclerosis, which is a demyelinating disease affecting the central nervous system.

The autonomic nerve fibers contained in the cranial nerves generally innervate the ipsilateral side of the body except for the visceral organs, which are asymmetric in their location (see Chapter 14-5, p. 101).

4. CRANIAL NERVES RELATED TO SOMATOSENSORY FUNCTION

Primary somatosensory neurons of the fifth, seventh, ninth, and tenth cranial nerves are all situated outside of the

brainstem. Like the spinal nerve (see Figure 20-1, p. 192), the primary sensory neuron is a bipolar cell located in a sensory ganglion. The impulses generated in the peripheral sensory receptors are conducted through the peripheral axon to the neuronal soma, and are further conducted through the central axon to reach the brainstem parenchyma. Cytoarchitecturally, therefore, the central axon belongs to the peripheral nervous system until it enters the brainstem parenchyma; once it enters the brainstem, it belongs to the central nervous system. Thus, this single neuron is formed of two different components, peripheral and central, due to this difference in the histologic origin of myelin sheath.

Regarding axonal flow, which is important for the transport of neurotrophic factors from the neuronal soma to the distal axon, the nerve impulse in the motor fibers is conducted in the same direction as the axonal flow. In the peripheral axon of the sensory fibers, in contrast, the nerve impulse is conducted in the opposite direction to the axonal flow. Namely, in the peripheral axon of the sensory nerves, the electric impulse is conducted from the distal terminal to the sensory ganglion cell, whereas the neurotrophic factors are transported from the sensory cell to the distal terminal of the axon. In the central axon of the sensory nerves, the nerve impulse is conducted in the same direction as the axonal flow.

5. CRANIAL NERVES RELATED TO AUTONOMIC NERVOUS FUNCTION

The autonomic nervous system will be described in detail later in this book (Chapter 21, p. 204). The third, seventh, ninth, and tenth cranial nerves contain the parasympathetic efferent fibers. For the sympathetic nervous system, the preganglionic neuron is located in the intermediolateral nucleus, which is located at the lateral column of the spinal cord extending from the first thoracic segment to the first lumbar segment (see Figure 14-5, p. 101). Then the preganglionic fibers leave the cord through the anterior spinal root to reach the sympathetic ganglion, from which the postganglionic fibers originate to innervate the whole body (see Figure 21-1, p. 205). Regarding sympathetic innervation of the head, face, and cranial nerve territories, the preganglionic fibers originate from the intermediolateral nucleus at the first thoracic segment and run through the inferior and middle cervical sympathetic ganglia to reach the superior cervical sympathetic ganglion, where the postganglionic fibers originate (see Figure 8-4, p. 51).

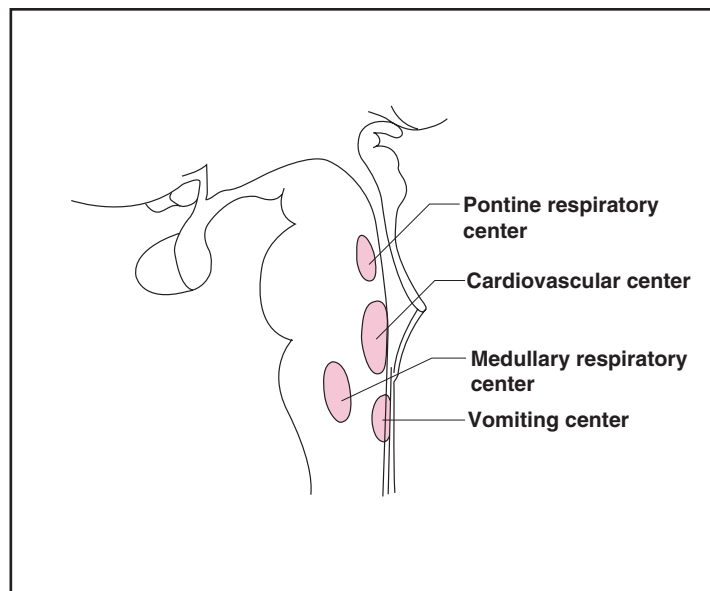


Figure 5-4 Schematic diagram showing the centers for respiration, cardiovascular system, and vomiting in the brainstem.

In the brainstem reticular formation, there are important centers for controlling the vital signs such as respiratory and cardiovascular functions, and vomiting (Figure 5-4). The paramedian dorsal region of the medulla oblongata just ventral to the fourth ventricle is called area postrema, and the syndrome characterized by intractable nausea, vomiting, and hiccups is called **area postrema syndrome**. The lesion in this area is commonly seen in patients with aquaporin-4-IgG-positive neuromyelitis optica spectrum disorder (NMOSD) (see Box 17-4) (Kunchok et al, 2020). Area postrema syndrome was also reported in a patient with Alexander disease (Renaldo et al, 2021). Pathologic yawning was reported in a case of the dorsal pontine lesion (Birca et al, 2020). In addition, the micturition centers also exist in the brainstem, and an isolated lesion involving some areas of the pons or medulla oblongata can cause urinary retention.

6. LOCATION OF THE CRANIAL NERVE NUCLEI IN BRAINSTEM

With the exception of the first cranial nerve (olfactory nerve), the second cranial nerve (optic nerve), and the eleventh cranial nerve (accessory nerve), nuclei of all cranial nerves are located in the tegmentum of the midbrain, pons, or medulla oblongata. As the location of nuclei of these cranial nerves is often indicated on the transverse section

of the brainstem at various levels, it tends to give a wrong impression as to the length of each nucleus. Since some nuclei, in fact, extend for a relatively long distance (Figure 5-5), the topographic interrelationship should be kept in mind when estimating the location of the lesion from the clinical symptoms and signs.

Progressive unilateral involvement of nearly all cranial nerves without sensory or motor long tract signs or intracranial hypertension is called Garcin syndrome (Fukai et al, 2018). It is due to a primary tumor of the skull base, like chondroma and nasopharyngeal tumor, or metastatic carcinoma from lung, breast, or prostate.

7. NEUROLOGIC EXAMINATION OF THE CRANIAL NERVE TERRITORY

Functions of the cranial nerves can be tested in the order of their numbers, but the emphasis can be placed on certain cranial nerves depending on the information obtained from the history taking. In the following chapters, anatomic structures, functions, and the method of neurologic examination of each cranial nerve are described in the order of actual examination. However, the eleventh cranial nerve (accessory nerve) will be dealt with in the chapter on the neck and trunk (Chapter 15, p. 106) because its functions are related to the posture and movement of the neck.

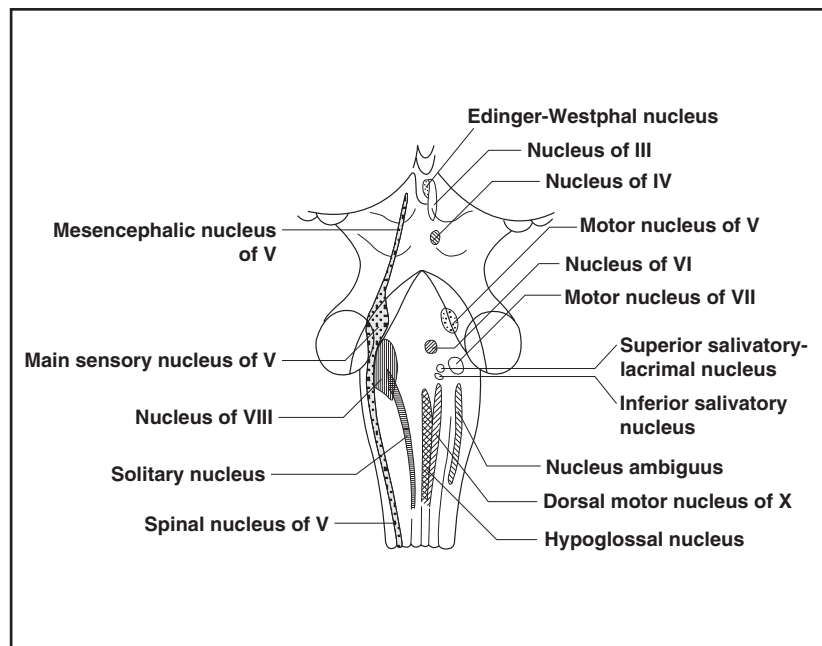


Figure 5-5 Schematic diagram showing the rostrocaudal extension of the cranial nerve nuclei. III: oculomotor nerve, IV: trochlear nerve, V: trigeminal nerve, VI: abducens nerve, VII: facial nerve, VIII: vestibulocochlear nerve, X: vagus nerve.

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OLFACTORY SENSATION

1. STRUCTURES AND NEUROTRANSMITTERS OF THE OLFACTORY SYSTEM

The primary neurons of the olfactory system are the bipolar cells located in the olfactory epithelium of the nasal mucous membrane (Figure 6-1). Their peripheral axonal endings serve as the olfactory receptors, and an electric impulse generated there is conducted to the cell body, from which the central axon originates as the olfactory nerve. The **olfactory nerve** is an unmyelinated fiber. After passing through the lamina cribrosa of the ethmoid bone, it projects to the **olfactory bulb** in the anterior cranial fossa. The secondary neurons originate from the mitral cells in the olfactory bulb, which then run posteriorly in the **olfactory tract** in the olfactory groove and enter the brain parenchyma from the anterior polar region of the temporal lobe. As described in Chapter 5-2 (p. 32), the olfactory tract belongs to the central nervous system because its myelin sheath is derived from oligodendroglia. As the olfactory groove is a common site for a meningioma, unilateral anosmia is a representative sign of such meningioma.

In the olfactory bulb, the neurotransmitter of the mitral cells is glutamate, while dopamine is considered to be involved in the interneurons. In relation to this, in Parkinson disease olfactory sensation is impaired from its early stage, often years prior to onset of motor symptoms (Boesveldt et al, 2008). In fact, Lewy bodies appear in the olfactory bulb in the earliest stage of Parkinson disease (Braak & Del Tredici, 2008). In a 10-year follow-up study of healthy elderly people, poor sense of smell was shown to be associated with a higher risk of Parkinson disease (Chen et al, 2017).

Olfactory sensation is believed to be processed in the limbic system of the brain, which includes, among others, the uncus, hippocampus, and amygdala (see Figure 6-1). A patient with mesial temporal lobe epilepsy might complain of an unusual smell as an aura. This phenomenon is

considered to be due to involvement of the olfactory centers by epileptic discharges, and this type of attack is called an **uncinate fit** according to its anatomic site. According to the results of direct electrical stimulation during the presurgical evaluation in a large number of patients with drug-resistant epilepsy, the mid-dorsal insula seems to play an important role in the olfactory sensation (Mazzola et al, 2017).

2. EXAMINATION OF OLFACTORY SENSATION

The sense of smell is tested by using a non-pungent stimulus such as a mild perfume. It is important to test the left

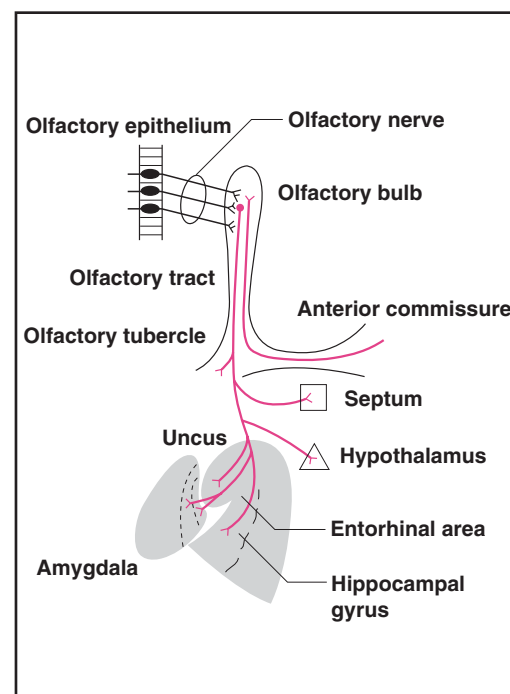


Figure 6-1 Schematic diagram showing the olfactory pathways. From Brodal, 2010, with permission.

and right side of the nose separately. Bilateral disturbance of olfactory sensation is commonly seen in inflammation of the nasal mucosa such as rhinitis or common cold. **Bilateral anosmia** may be a manifestation of some hereditary neurodegenerative diseases. A typical example is an X-linked hereditary disease called **Kallmann syndrome**, which is characterized by anosmia, congenital mirror movements, and hypogonadotropic hypogonadism. Anosmia in this condition is believed to be due to developmental abnormality of the olfactory bulb. It is not difficult to distinguish congenital bilateral anosmia from upper respiratory infection by taking into account the accompanying symptoms. As described earlier, bilateral disturbance of olfactory sensation may be seen in patients with Parkinson disease, but total loss of this sense is rare in this disease. Recently it has been reported that bilateral anosmia is commonly seen in patients with COVID-19 (Laurendon et al, 2020; Pellegrino et al, 2020) (see Box 32-3, p. 282).

Unilateral anosmia is a characteristic sign of **olfactory groove meningioma**, especially when the history of present illness suggests the presence of an intracranial

space-occupying lesion; in fact, it may be the only neurological deficit seen in this slowly progressive tumor.

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VISUAL FUNCTIONS

1. ANATOMY AND FUNCTION OF THE VISUAL SYSTEM

A. RETINA

Visual receptors are **photoreceptor cells** located in the outer layer of the retina (Figure 7-1). There are two kinds of photoreceptor cells: cones and rods. **Cones** are located mainly at the macular area and control photopic vision in the light condition, and **rods** are distributed more toward the periphery of the retina and control scotopic vision in the dimly lit condition. The primary neurons of the visual system are bipolar cells that receive impulses through their peripheral axons and transmit them through their central axons to the retinal ganglion cells. The **retinal ganglion cells** serve as the secondary neurons of the visual system.

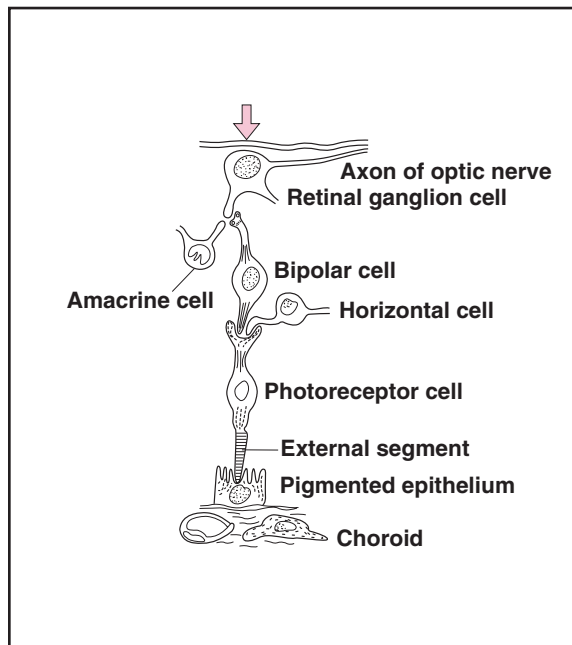


Figure 7-1 Cytoarchitecture of retina. The outer layer is shown below and the inner layer is shown above in the figure. The arrow on top indicates the direction of the incoming light. From Brodal, 2010, with permission.

Axons of ganglion cells from all parts of retina project to the central part of retina and form the **optic disc** (optic papilla) (Figure 7-2). From the optic disc, a bundle of axons runs posteriorly as the **optic nerve** and enters the cranial cavity through the **optic foramen**.

The technique of **optical coherence tomography (OCT)** can be applied to provide an image of retinal layers noninvasively (Aytulun et al, 2021) (Figure 7-3), and **OCT angiography (OCTA)** is used to visualize retinal and choroidal vascular changes in multiple sclerosis and neuromyelitis optica spectrum disorders (Kleerekooper et al, 2020).

As described in Chapter 5-2 (p. 32), the optic nerve histologically belongs to the central nervous system, and it is often involved in multiple sclerosis and neuromyelitis optica (Box 7-1). Isolated demyelination of the optic nerve is called **optic neuritis**, and it may be also called **retrobulbar neuritis** because the lesion commonly involves a part of the optic nerve behind the ocular bulb. It is usually unilateral, but it may be bilateral as is often the case in neuromyelitis optica.

In **Leber hereditary optic neuropathy (atrophy)**, vision is impaired relatively acutely in young adults. Usually it starts unilaterally, but in most cases the opposite eye is also involved within a year. As the visual acuity may occasionally improve to some degree, it may have to be differentiated from bilateral optic neuritis. This condition belongs to the mitochondrial encephalomyopathies (see Box 19-13, p. 181). It is due to a point mutation of mitochondrial DNA transmitted by maternal inheritance. Various treatments are being tried for this disease, including trials of antioxidant agents like idebenone (Newman, 2011).

B. FOVEA CENTRALIS AND OPTIC DISC

Fovea centralis (fovea) is in the center of the macula lutea (macula) and plays an important role in central vision. A bundle of nerve fibers arising from the fovea runs nasally

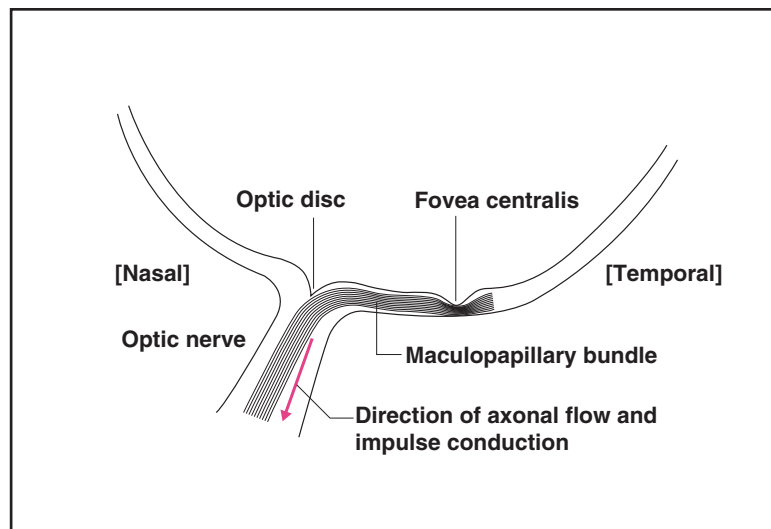


Figure 7-2 Schematic diagram showing the interrelationship of optic disc and fovea. View from top of the right eye. The maculopapillary bundle originating from the fovea centralis enters the optic disc from its temporal margin.

to enter the optic disc from its temporal rim, the **maculopapillary bundle** (see Figure 7-2). This bundle is considered to occupy the central portion of the optic nerve, which is commonly affected by demyelinating lesions in optic neuritis or retrobulbar neuritis. In this condition, therefore, it is common to find pale discoloration at the temporal part of the optic disc (**temporal pallor**). This phenomenon is due to retrograde degeneration of the optic nerve axons as a result of axonal involvement at the demyelinating site (see Figure 7-2). For the same reason, a **central scotoma** is often seen in this condition. Thus, an ophthalmoscopic finding of temporal pallor in this condition suggests that the axons have also been involved secondary to the demyelination at

least several days prior to the time of examination, and thus it suggests poor prognosis for the visual acuity. By contrast, in conditions associated with primary degeneration of the macula, temporal pallor is seen as a result of orthograde degeneration of the axons (Figure 7-4).

C. BLOOD SUPPLY TO THE RETINA

The retina receives its blood supply from the **ophthalmic artery**, which bifurcates from the internal carotid artery soon after it enters the cranial cavity through the carotid canal. The ophthalmic artery enters the orbit along with the optic nerve through the optic foramen, and it supplies

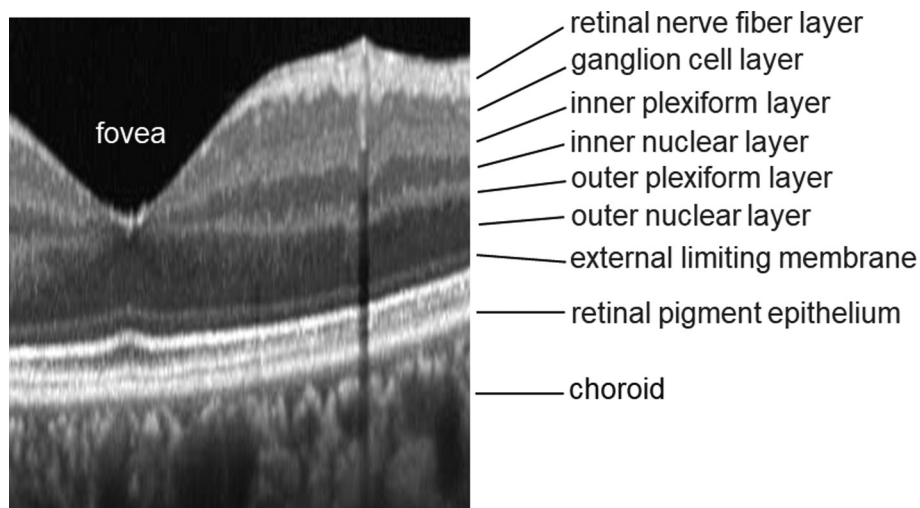


Figure 7-3 Image of normal retina obtained by optical coherence tomography (OCT). Courtesy of Department of Ophthalmology, Kyoto University.

BOX 7-1 CLINICAL FEATURES OF DEMYELINATING DISEASES: IS THE UHTHOFF PHENOMENON DUE TO PHYSICAL EXERCISE OR INCREASE IN THE BODY TEMPERATURE?

It is known that in patients with multiple sclerosis (MS) the symptoms such as motor paresis worsen while taking a bath. In 1890, a German ophthalmologist named Uhthoff reported that, in four among 105 patients with MS, visual loss and abnormality of color vision and visual field were induced by physical exercise. In one of the four patients, in addition to visual loss of the left eye, motor paresis of the right leg and sensory disturbance of the left arm were also induced when standing by the stove (Guthrie & Nelson, 1995). In fact, this phenomenon is considered to be common in patients with MS. Uhthoff initially thought that this phenomenon was due to physical exercise rather than an increase in the body temperature. Based on the following observation by many neurologists, however, it is currently believed that this phenomenon is caused by an increase in body temperature rather than by physical exercise. By using the technique of transcranial magnetic stimulation in MS patients, it was shown that a change in body temperature worsens the conduction block (Humm et al, 2004). Physiologically, it is known that, at the site of incomplete demyelination where the conduction block takes place, the conduction rate is decreased by the temperature increase, whereas, in the intact nerve fibers, the temperature elevation increases the conduction velocity. However, it is also known that, in some cases, the symptoms worsen by a decrease in body temperature. Nerve conduction in demyelinated nerve can be described as an inverted U-shaped function, slowing with both increasing and decreasing temperature. A similar phenomenon can be seen in multifocal motor neuropathy, which is characterized by conduction block in the peripheral motor nerve fibers (see Chapter 17-3H, p. 145). In this condition, fatigue is caused as a result of increased conduction block after strong contraction of the corresponding muscle (activity-dependent conduction block) (Kaji et al, 2000).

blood to retina as the **central retinal artery** (Figure 7-5a). Occlusion of this artery, for example by embolism, causes ipsilateral **blindness**.

Temporal arteritis is an arteritis of the temporal artery affecting elderly people. This special arteritis might cause ipsilateral blindness and optic atrophy due to secondary thrombosis in the ophthalmic artery. It has been reported

that varicella-zoster virus might trigger the immunopathology of this giant cell arteritis (Gilden et al, 2015). Temporal arteritis is often combined with **polymyalgia rheumatica**, and it shows marked elevation of the erythrocyte sedimentation rate and responds to corticosteroid therapy (see Chapter 17-3B, p. 139). If the central retinal artery is transiently occluded by an embolus originating from

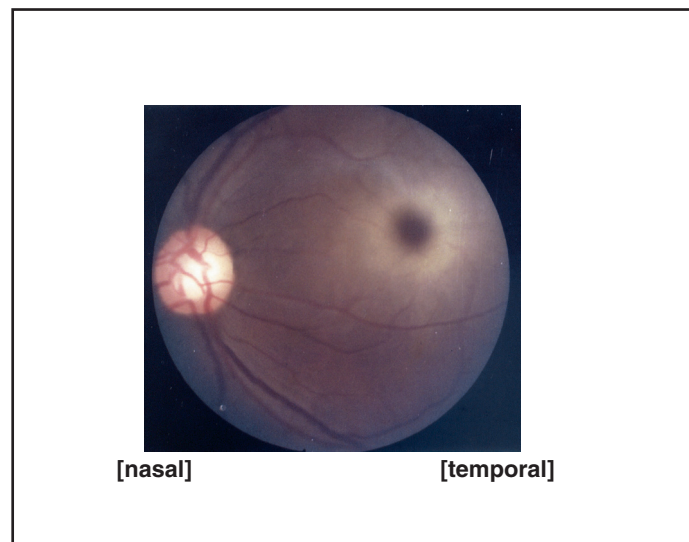


Figure 7-4 Ophthalmoscopic view of the left eye in a case of sialidosis type 2. The fovea centralis appears cherry-red in color (cherry-red spot) and is surrounded by white color due to the lipid deposit. The optic disc shows temporal pallor due to orthograde degeneration of the maculopapillary bundle as a result of macular degeneration. A white spot in the center of the optic disc is due to physiologic cupping.

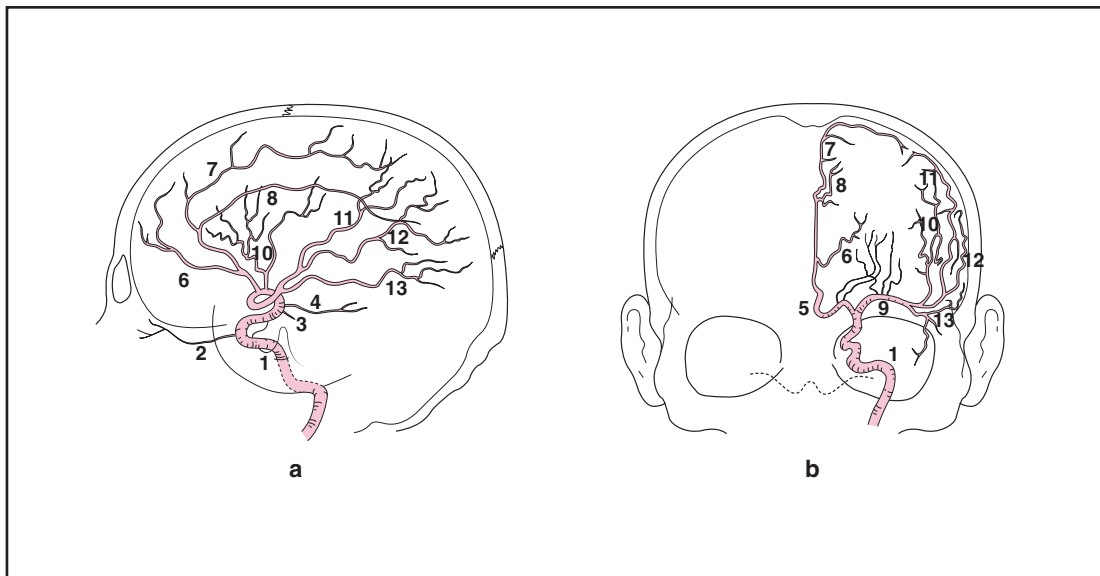


Figure 7-5 Main cerebral arteries (a: lateral view, b: frontal view). 1: internal carotid artery, 2: ophthalmic artery, 3: posterior communicating artery, 4: anterior choroidal artery, 5: anterior cerebral artery, 6: orbitofrontal artery, 7: callosomarginal artery, 8: pericallosal artery, 9: middle cerebral artery, 10: arteries of precentral and central sulci, 11: parietal artery, 12: angular artery, 13: temporal artery. See Figure 9-3 (p. 59) for the vertebrobasilar artery territory.
From List et al., 1945, with permission.

atherothrombotic plaque of the internal carotid artery, it causes transient blindness as a result of transient ischemic attack (**amaurosis fugax**). If this attack is accompanied by contralateral hemiparesis, it suggests the presence of atherothrombosis in the internal carotid artery on the side of blindness.

Transient blindness may be also seen as a prodromal symptom of migraine. In **retinal migraine**, spasm in the ophthalmic artery or in the central retinal artery may cause visual disturbance as a result of ischemia, followed by pulsating pain in the orbit or in the surrounding region. Acute visual loss in optic neuritis or retrobulbar neuritis is characterized by a central scotoma. In this case, ocular pain is induced by upward gaze, because the optic nerve is mechanically stretched by this maneuver, which in turn provokes the inflammatory pain.

D. OPTIC NERVE AND OPTIC CHIASM

The optic nerve runs posteromedially after leaving the orbit, and about half of the nerve fibers cross at the **optic chiasm**. Namely, those fibers coming from the nasal half of the retina, which corresponds to the temporal visual field of that eye, run through the medial half of the optic nerve and cross at the chiasm to form the contralateral optic tract together with the uncrossed fibers coming from the temporal half of the contralateral retina, which corresponds to

the nasal visual field of that eye (Figure 7-6). Thus, the optic tract transmits the visual input arising from the contralateral hemifield of both eyes.

If the optic chiasm is compressed by a pituitary adenoma from below, only the crossed fibers of optic nerves are affected, resulting in **bitemporal hemianopsia** (2 in Figure 7-6). It is known that the crossing fibers originating from the inferior nasal aspect of the retina first enter the contralateral optic nerve, forming an arc for a short distance before crossing at the chiasm. Thus, if the optic nerve is compressed from below just before entering the chiasm, it causes a defect of the superior temporal visual field of the contralateral eye in addition to visual loss of the ipsilateral eye. This phenomenon, called **junctional scotoma**, is extremely rare. Theoretically, compression of the optic chiasm from below on both sides may damage those nerve fibers arising from the temporal half of both retinas and cause **binasal hemianopsia**, but this phenomenon is also extremely rare.

The **optic tract** is formed from the nerve fibers arising from the temporal half of the ipsilateral retina and the nasal half of the contralateral retina, and it runs posterolaterally and projects to the **lateral geniculate body (LGB)**. Thus, if the optic tract is damaged, it causes **homonymous hemianopsia** contralateral to the lesion (3 in Figure 7-6). Since the optic tract is located just below the globus pallidus and internal capsule, deep brain surgery targeting

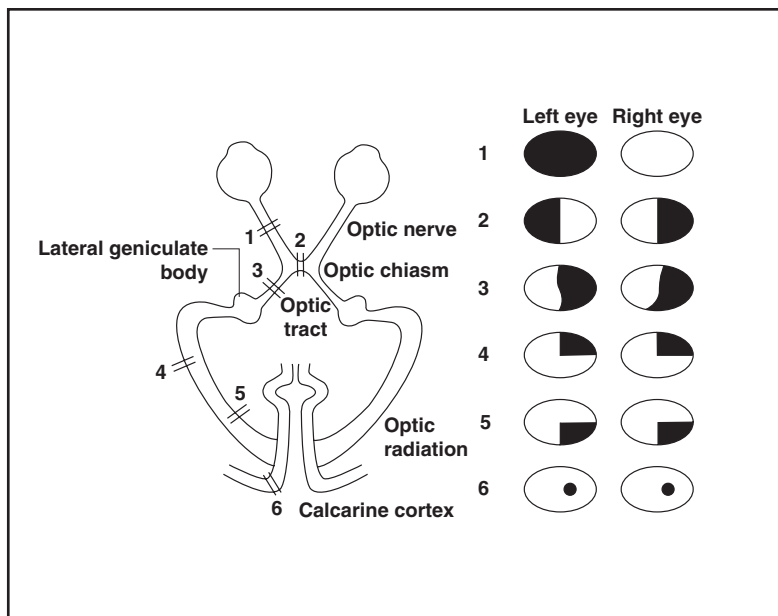


Figure 7-6 Schematic diagram of the visual pathway and defect of visual field caused by lesion of each site. 1. blindness, 2. bitemporal hemianopsia, 3. homonymous hemianopsia, 4. homonymous superior quadrantanopsia, 5. homonymous inferior quadrantanopsia, 6. homonymous visual field defect (homonymous scotoma). Note that hemianopsia due to a lesion of the optic tract is incongruous in shape between the two eyes.

the globus pallidus should be carried out with caution (see Figure 16-8, p. 117).

While most nerve fibers of the optic tract project to LGB, some fibers turn medially just before reaching the LGB and enter the mesencephalic tectum to project to the superior colliculus and pretectal nucleus (Figure 7-7). The **superior colliculus** plays an important role in eye-head coordination (see Chapter 9-4, p. 63), and the **pretectal**

nucleus serves as the afferent arc of the light reflex (see Chapter 8-1, p. 49). Furthermore, some nerve fibers of the optic tract project to the suprachiasmatic nucleus of the hypothalamus, which is located just above the optic chiasm. Melanopsin-containing intrinsically photosensitive retinal ganglion cells (ipRGCs), atypical retinal photoreceptors separate from rod and cone photoreceptors, contribute to this projection (Lax et al, 2019). Neurons in the

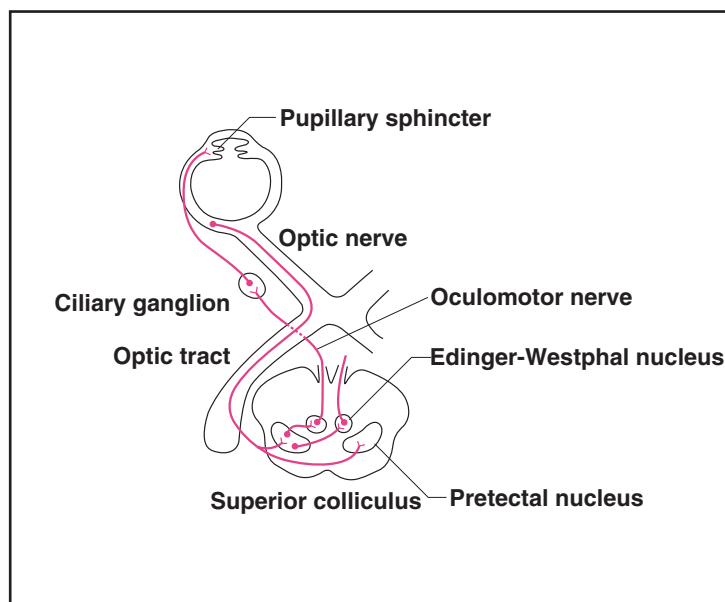


Figure 7-7 Schematic diagram showing the afferent and efferent pathways related to the light reflex.

suprachiasmatic nucleus are activated by bright light and activate the pineal body to produce melatonin, which then activates a part of the hypothalamus involved in the control of **circadian rhythm** (see Chapter 31-3, p. 278). This is why it is believed that exposure to sunlight and the use of melatonin improve jet lag during an overseas trip that involves changing time zones.

E. LATERAL GENICULATE BODY AND OPTIC RADIATION

The tertiary neurons of the visual system send projections from the LGB to the visual cortex (**geniculocalcarine tract**). The tract runs laterally through the posterior limb of the internal capsule and spreads as the **optic radiation**. Therefore, a vascular lesion of that part of the internal capsule causes contralateral hemiplegia and homonymous hemianopsia. The dorsolateral portion of the optic radiation runs posteriorly along the posterior horn of the lateral ventricle and reaches the upper half of the visual cortex through the parietal lobe. The ventral portion of the optic radiation runs laterally just above the inferior horn of the lateral ventricle, forming **Meyer's loop**, after which it reaches the lower half of the visual cortex through the temporal lobe. If Meyer's loop is damaged by surgical resection of the anterior portion of the temporal lobe for treatment of medically intractable mesial temporal lobe epilepsy, contralateral homonymous superior quadrantanopsia may appear.

A lesion of the retrochiasmatic portion of the visual system causes contralateral homonymous hemianopsia, but the shape of visual field defect differs depending on which part of the visual pathway is damaged. In a lesion of the optic tract, the visual field defect is incongruous in its shape between the two hemifields (3 in Figure 7-6). By contrast, the visual field defect caused by a lesion of the optic radiation near the visual cortex is more congruous between the left and right eyes. As the optic radiation is separated into the temporal pathway and the parietal pathway as described earlier, a lesion of the temporal lobe causes **homonymous superior quadrantanopsia** (4 in Figure 7-6), while a lesion of the parietal lobe causes **homonymous inferior quadrantanopsia** (5 in Figure 7-6), both contralaterally.

In hemianopsia caused by a vascular lesion of the visual cortex, the central visual field is often spared (**macular sparing**). The mechanism underlying this phenomenon has not been precisely elucidated, but it has been postulated that a part of the occipital cortex receiving input from the macular portion of retina might receive blood supply from the bilateral posterior cerebral arteries. However, in view of

the fact that this phenomenon is seen not only in ischemic stroke of the posterior cerebral artery territory but also in hemorrhage of the occipital lobe, that hypothesis is unlikely.

F. VISUAL CORTEX

On a histologic section of cortex around the calcarine fissure, a white line called the **line of Gennari** is seen, which is due to abundant myelinated fibers in the fourth layer of the calcarine cortex. This cortex is also called the **striate area** based on this appearance. There is precise topographic organization throughout the whole visual system starting from the retina and terminating at the calcarine cortex. Namely, visual input coming from a certain part of the visual field activates retinal ganglion cells localized to the corresponding part of each retina, and their impulses activate a group of neurons in the corresponding part of the LGB contralateral to the visual field, which then activates neurons in the corresponding part of the striate area (**retinotopic organization**). Thus, if there are blind spots of the congruous shape in the visual fields of two eyes, it may be due to a localized lesion involving a small part of the contralateral striate area (**homonymous scotoma**) (6 in Figure 7-6).

G. BLOOD SUPPLY TO THE VISUAL PATHWAY

As the visual system extends from eyes to the occipital cortex, it receives blood supply from multiple arteries (Figure 7-8). As described earlier (p. 39), the retina and its ganglion cells receive blood supply from the ophthalmic artery. The optic chiasm is supplied by a branch of the anterior cerebral artery. The optic tract and LGB are supplied by the anterior choroidal artery, which bifurcates from the internal carotid artery, and the optic radiation is supplied by branches of the internal carotid artery. Finally, the occipital cortex receives blood supply from the posterior cerebral artery (see Figures 7-5a, 7-5b, and 7-6). Therefore, visual field defects caused by ischemic vascular diseases differ depending on which artery is involved. The responsible artery can be estimated from the mode of visual field defect and other accompanying neurological deficits.

H. CYTOARCHITECTURE OF THE VISUAL SYSTEM AND HIGHER CORTICAL FUNCTIONS

The LGB is divided into six layers, the ventral two layers mainly formed of large neurons (**magnocellular layer**) and the dorsal four layers mainly formed of small neurons (**parvocellular layer**) (Figure 7-9). The magnocellular layer

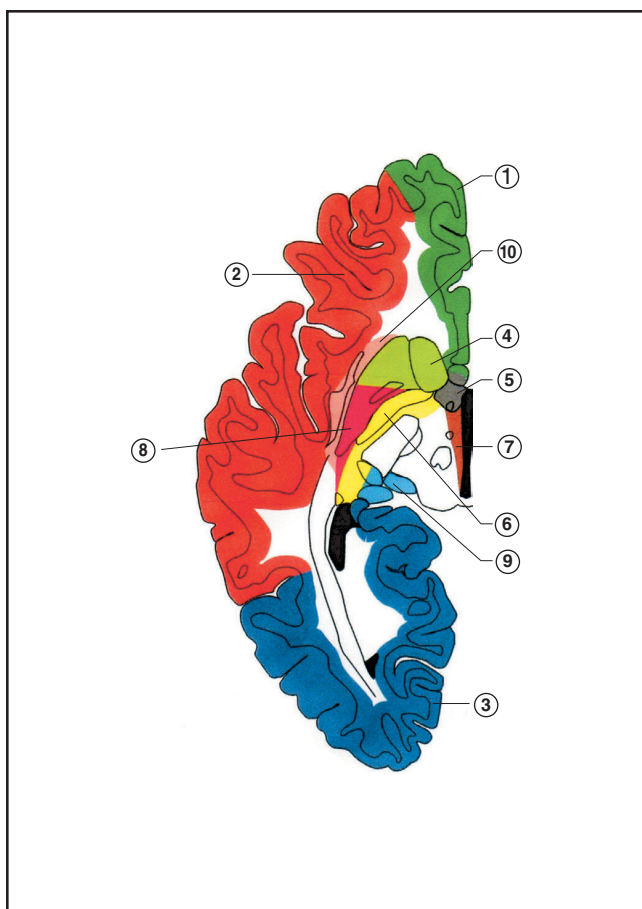


Figure 7-8 A horizontal section of the cerebral hemisphere showing the territories perfused by each cerebral artery. 1. anterior cerebral artery, 2. middle cerebral artery, 3. posterior cerebral artery, 4. perforating branches of the anterior cerebral artery, 5. anterior communicating artery, 6. anterior choroidal artery, 7. perforating branches of the posterior communicating artery, 8. perforating branches of the middle cerebral artery, 9. posterior choroidal artery. From Duvernoy, 1999, with permission.

receives projections from large retinal ganglion cells and the parvocellular layer receives axons originating from small retinal ganglion cells.

The **primary visual cortex** is in the striate area of the occipital cortex (**Brodmann area 17**), and it is connected to the extrastriate area including Brodmann areas 18 and 19 (Figure 7-10) (Brodmann, 1910). The primary visual area is engaged in the processing of basic visual information, and more advanced visual information is processed in the extrastriate area. In general, it is believed that recognition of visual objects is processed in the **ventral pathway** projecting from the occipital cortex to the temporal cortex (“what” pathway) while spatial information and recognition of movement are processed in the **dorsal pathway**

projecting from the occipital cortex to the parietal cortex (“where” pathway). It is also believed that the projection fibers originating from the magnocellular layer of the LGB are engaged in the processing of movement and depth while those originating from the parvocellular layer of the LGB are related to the processing of shape, pattern, and color of objects.

I. SPECIAL VISUAL FUNCTIONS

Several cortical areas are known to be involved in special visual functions. Area V4 (the posterior part of the lingual gyrus and the posterior part of the fusiform gyrus) is important for **color vision**. Area V5 (the posterior part of the middle temporal gyrus, corresponding to MT of monkey) plays a special role in visual recognition of movement (**motor vision**). The fusiform gyrus in the inferior temporal lobe is known to be important for **recognition of faces**, and a bilateral lesion of this area causes **prosopagnosia**. In this condition, the patient cannot recognize a familiar face visually, but once the voice is heard, the person can be easily identified. Information about these higher visual functions has been obtained by observation of the patients with localized cortical lesions and by noninvasive techniques such as magnetoencephalography and functional neuroimaging (see Shibasaki, 2008, for a review).

2. EXAMINATION OF VISUAL FUNCTIONS

A. VISUAL ACUITY

Before examining the visual acuity from a neurologic point of view, it is better to pay attention to the presence or absence of any abnormality in the cornea, lens, and vitreous body. Presence of these ophthalmologic abnormalities can be estimated from the past medical history, but if there is any question, ophthalmologic evaluation will be needed. If a patient usually wears glasses for accommodation failure or astigmatism, it is advisable to examine the visual acuity with the glasses on for the neurologic examination.

How much detail of visual acuity should be measured depends on the tentative diagnosis based on the history taking and on the clinical set-up in the hospital. At any rate, it is important to examine the visual acuity for each eye independently. For quantitative measurement of visual acuity, various charts are available. Visual acuity can be measured for both near and far, relevant for reading and seeing at a

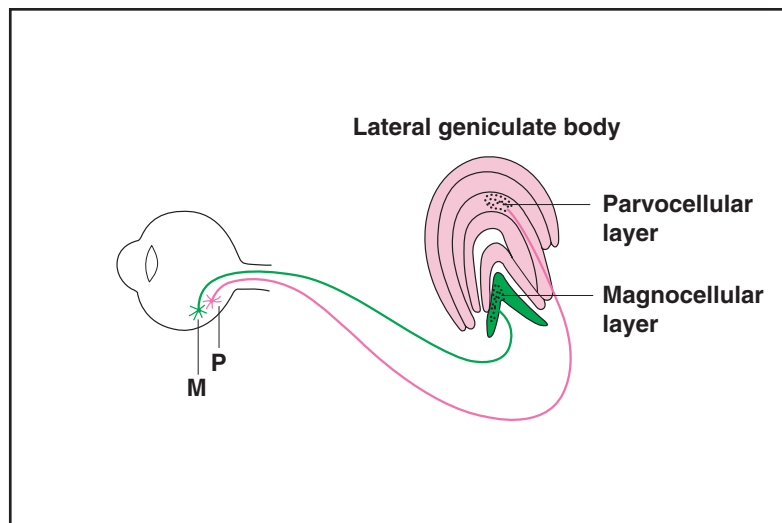


Figure 7-9 Structure of the lateral geniculate body and the projection it receives from the retina. M: large ganglion cells of the retina, P: small ganglion cells of the retina.
From Brodal, 2010, with permission.

distance, respectively. Differences in acuity for near and far are indicative of ophthalmologic causes.

If abnormality of visual acuity is found in one eye, or in both eyes but with significant asymmetry between the two eyes, it may suggest either ophthalmologic abnormalities including the retina or an optic nerve lesion in that eye. In that case, it is important to check whether it is a defect of the central visual field (**central scotoma**) or it involves the whole visual field. As described earlier (p. 39), the presence of a central scotoma suggests a lesion either in the macula of the retina or in the optic nerve.

B. VISUAL FIELDS

Examination of the visual fields may be initiated by a **confrontation test** in which the visual field of two eyes is simultaneously tested by using the examiner's hands. For example, the examiner sits about 50 cm in front of the patient facing each other, and holds his or her hands about 30 degrees apart. Then the examiner wiggles the fingers of the left or right hand and asks the patient to tell which hand was moved.

When doing the confrontation test, it is important to test also simultaneous presentation of visual targets in the

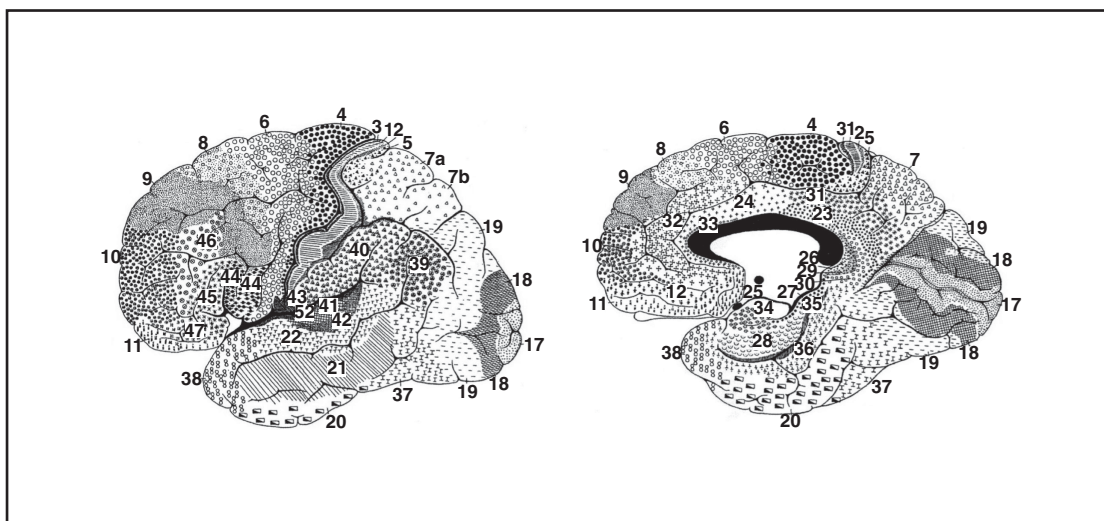


Figure 7-10 Brodmann's cortical areas based on the cytoarchitecture.
From Brodmann, 1910.

left and right visual fields. If the patient constantly neglects one side, it suggests either homonymous hemianopsia or hemispatial neglect (see Chapter 24-2C, p. 241), and in this case the visual field of each eye should be examined in more detail. For testing the visual field of one eye on the confrontation test, for example, the examiner's left eye serves as a control for the patient's right eye by keeping the other eye closed or masked. Then the patient is asked to fix his or her gaze on the examiner's hand held just halfway between the patient and the examiner, while the examiner moves a visual target by another hand in the central as well as peripheral visual field.

If there is any difficulty in testing the upper visual field, it is useful to test it again with the upper eyelid lifted. Furthermore, in the subjects who have a very high nose, the confrontation test might show an apparent defect in the inferior nasal field for both eyes.

As expected, sensitivity of the confrontation test is not very high (Kerr et al, 2010), but gross abnormality of the visual field can be detected if carefully tested. Needless to say, however, if any abnormality of visual system is suggested from the history taking or from the confrontation test, the visual field should be measured precisely by using perimetry.

i. Central scotoma

As described earlier (p. 45), the presence of a central scotoma in one eye suggests that either the macular region of the retina or the optic nerve is damaged. When doing the confrontation test, one may find a blind spot just lateral (temporal) to the central field. This field defect is called the **Mariotte blind spot**, which corresponds to the optic disc where there are no photoreceptor cells (see Figure 7-2). Therefore, this blind spot is enlarged in case of papilledema. Furthermore, in case of central retinitis, a blind spot of racket shape may be formed as a result of fusion of the central scotoma and the Mariotte blind spot.

ii. Other visual field defects

As explained earlier (p. 41), a defect of temporal fields of both eyes (**bitemporal hemianopsia**) is caused by a lesion of the optic chiasm, in particular its compression by a pituitary adenoma from below (2 in Figure 7-6). A hemifield defect of the same side in both eyes (for example, nasal of one eye and temporal of the other eye) is called **homonymous hemianopsia** and is caused by a lesion involving the contralateral visual pathway central to the optic chiasm,

namely the optic tract (3 in Figure 7-6), LGB, optic radiation, or occipital cortex. Likewise, a defect of either the upper half or the lower half of the hemifield of the same side in both eyes is called **homonymous superior quadrantanopsia** (4 in Figure 7-6) or **homonymous inferior quadrantanopsia**, respectively (5 in Figure 7-6). The former is caused by a lesion of the contralateral temporal lobe and the latter is caused by a lesion of the contralateral parietal lobe. A unilateral lesion of the calcarine cortex causes a congruous defect of the contralateral hemifield, often associated with macular sparing, in both eyes (see p. 43).

If the occipital pole is unilaterally affected, by head trauma for example, a localized field defect (scotoma) may be seen in the contralateral half of the central visual field of both eyes (6 in Figure 7-6). This homonymous scotoma may resemble bilateral central scotoma due to bilateral optic neuritis. These two cases, however, can be differentiated from the history of present illness, a precise pattern of the field defect, and results of light reflex testing. In bilateral optic neuritis, the shape of scotoma may not be exactly the same in the two eyes, and the light reflex may be absent or decreased if a light stimulus is projected from the defective visual field, whereas the light reflex is always intact in an occipital lesion (see Chapter 8-2B for examination of the light reflex, p. 54).

iii. Constriction of peripheral visual field

Constriction of the peripheral visual field is encountered in lesions primarily involving the periphery of the retina. A typical example is **retinal pigment degeneration**, which is also called retinitis pigmentosa although it is not an inflammatory disease. This condition is usually seen for both eyes. In this case, the patient complains of **night blindness (nyctalopia)**. The diagnosis is confirmed by ophthalmoscopic observation of pigments in the periphery of the retina.

Constriction of the visual field may be also seen in diffuse lesions of the occipital cortex. A typical example is **Minamata disease**, which is due to organic mercury poisoning. Other neurological symptoms seen in this condition include cerebellar ataxia, dysarthria, and sensorineural hearing deficit (**Hunter–Russell syndrome**) (see Chapter 1-1, p. 2). Pathologically there is marked atrophy in the cerebellar cortex and the calcarine cortex.

In a patient showing constriction of the peripheral visual field due to an organic lesion of the calcarine cortex, the visual field is expected to become increasingly wider as the patient moves away from the visual target. In a patient with

psychogenic visual field defect, however, the field defect does not change in its angle even if the patient moves away from the target. This phenomenon is called **tunnel vision** or **tubular vision** and is considered to be characteristic of psychogenic field constriction.

C. OPHTHALMOSCOPIC EXAMINATION OF OCULAR FUNDUS

For the purpose of neurologic examination, it is usually sufficient to examine the ocular fundus by an ophthalmoscope without using the eye drops for dilating pupils, but turning out the lights in the examining room is often valuable. In case the precise ophthalmoscopic findings are important, eye drops can be used after completing all other neurologic examinations, or at least after examining the pupils (see Chapter 8-2, p. 54). Nowadays, ocular fundus photography facilitates non-ophthalmologists' performance of the ophthalmoscopic examination (Biousse et al, 2018).

i. Optic disc

In the ophthalmoscopic examination as a part of the neurologic examination, it is practical to identify the optic disc

(optic papilla) first. At the initial stage of **papilledema**, the color of the optic disc becomes more red than usual and the surrounding veins look enlarged. At a slightly more advanced stage of papilledema, the optic disc appears poorly demarcated, and the ophthalmoscope can be focused differently between the disc and the nearby blood vessels, which suggests the presence of swelling in the optic disc (Figure 7-11). Papilledema is caused by either intracranial hypertension or a pathologic process directly occluding the central retinal vein.

Pale discoloration of the optic disc is characteristic of **optic atrophy**. It is caused by either orthograde axonal degeneration from a primary pathology of the retina or retrograde axonal degeneration following a pathologic process of the optic nerve. Optic atrophy is commonly found in patients with **multiple sclerosis** or **optic neuritis (retrobulbar neuritis)**, and this finding suggests that axons of the optic nerve have been involved as the result of retrograde degeneration. As explained earlier (p. 38), the nerve fibers originating from the macular region of the retina enter the optic disc from its temporal rim and pass through the central section of the optic nerve. Consequently, a demyelinating lesion involving the central section of the optic nerve causes atrophy in the temporal portion of the optic disc as

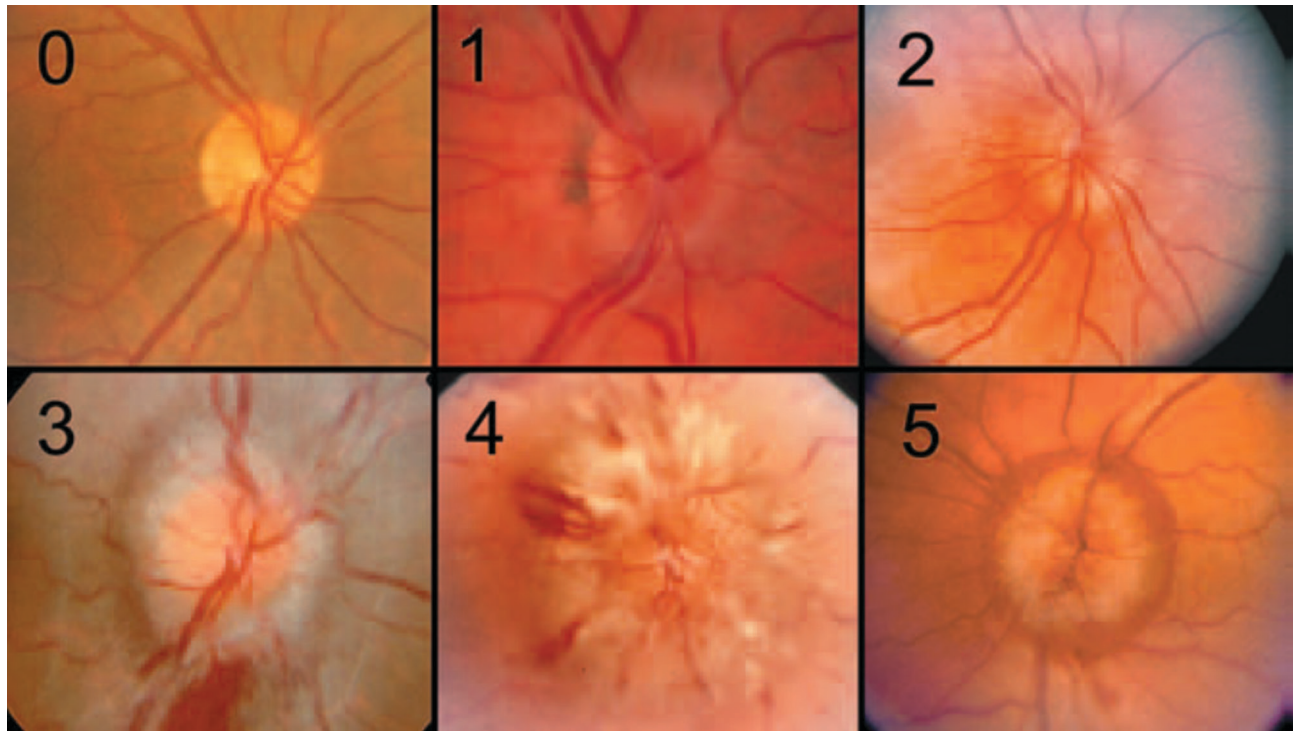


Figure 7-11 Grading of papilledema (Frisen's grading).
Cited from Schiffman et al., 2006, with permission.

a result of retrograde degeneration (**temporal pallor**) (see Figure 7-2).

ii. Physiologic cup of optic disc

The central part of the optic disc looks pale due to a physiologic cup even in healthy subjects. If the optic disc is tilted laterally due to, for example, myopia, the physiologic cup is seen at the temporal portion of the disc, resembling temporal pallor. Distinction from the real temporal pallor is possible based on the history of myopia and correction of visual acuity by the use of glasses.

iii. Fovea centralis

The fovea can be identified by shifting the ophthalmoscope light laterally away from the optic disc or by asking the subject to look at the ophthalmoscope light. In patients with lipidosis, the surrounding area of the fovea looks white due to deposit of lipid, and the fovea looks purple-red. As this color resembles that of a cherry, it is called a **cherry-red spot** (see Figure 7-4). In this condition, examination of the visual field discloses a central scotoma, and temporal pallor is seen in the optic disc due to orthograde axonal degeneration of the maculopapillary bundle.

iv. Retinal pigment degeneration

In this condition, star-shaped black spots are observed mainly in the equatorial (peripheral) region of the retina. The patient complains of night blindness, and the visual field is constricted on the perimetric examination. Retinal pigment degeneration is associated with various kinds of neurodegenerative diseases, among others Kearns–Sayre syndrome (a form of mitochondrial encephalomyopathy) (see Box 19-13, p. 181), Bassen–Kornzweig disease (see Box 16-9, p. 130), Refsum disease (see Chapter 3-1, p. 30), and Cockayne disease (see Chapter 3-1, p. 13).

v. Observation of retinal blood vessels

In patients with hypertension, narrowed vessel diameter and compression of veins at arteriovenous crossings might

be seen. In patients with diabetes mellitus, red spots might be seen in the arterial wall. This had been thought to be due to microaneurysms, but histologically these are small hemorrhages. In addition, attention should be paid to homogeneity of the retinal color and the presence or absence of hemorrhage.

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PUPILS AND ACCOMMODATION

There are two kinds of muscles in the eyes: extraocular muscles, which move the ocular bulb, and intraocular muscles, which are innervated by the autonomic nervous system. The intraocular muscles have two functions; one controls the diameter of pupils and the other controls thickness of the lens (**accommodation**). For examining the ocular muscles, we can start from either muscle, but it is practical to examine the intraocular muscles first in order to avoid overlooking them by examining the extraocular muscles first.

1. NERVE INNERVATION OF INTRAOCULAR MUSCLES

The preganglionic fibers of the parasympathetic nerves innervating pupils originate from the Edinger-Westphal nucleus, which is situated in the midbrain tegmentum (see Figures 5-1 and 7-6). They run anteriorly in the oculomotor nerve and exit the midbrain just medial to the cerebral peduncle (see Figure 5-1). Then they run through the cavernous sinus (Figure 8-1) and enter the orbit through

the superior orbital fissure (Figure 8-2a) to reach the ciliary ganglion. The postganglionic fibers originate from the ciliary ganglion and innervate the pupillary sphincter (see Figure 7-6). As the pupillary sphincter encircles the pupil in the iris, its contraction decreases the pupillary diameter (Figure 8-3). Neurotransmitters are acetylcholine for both the preganglionic and postganglionic fibers, but the mode of transmission is nicotinic for the preganglionic fibers and muscarinic for the postganglionic fibers. When these nerve fibers are involved by a lesion, the pupillary diameter of the affected eye becomes larger compared with the contralateral pupil (**mydriasis**).

A. LIGHT REFLEX AND CONVERGENCE REFLEX

There are two reflexes related to the pupil: light reflex and convergence reflex. The **light reflex** is manifested as shrinkage of the pupillary diameter in response to the light stimulus. Its reflex center is in the midbrain tegmentum, with the afferent arc served by the optic nerve and the efferent arc served by the autonomic fibers of the

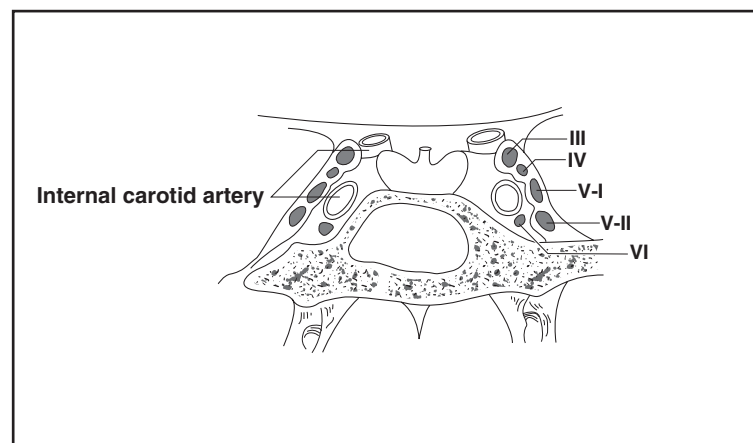


Figure 8-1 Frontal section of the cavernous sinus and the position of cranial nerves that pass through the sinus. III: third cranial nerve (oculomotor nerve), IV: fourth cranial nerve (trochlear nerve), V-I: first branch of fifth (trigeminal) nerve, V-II: second branch of trigeminal nerve, VI: sixth cranial nerve (abducens nerve).
Cited from Haymaker, 1969, with permission.

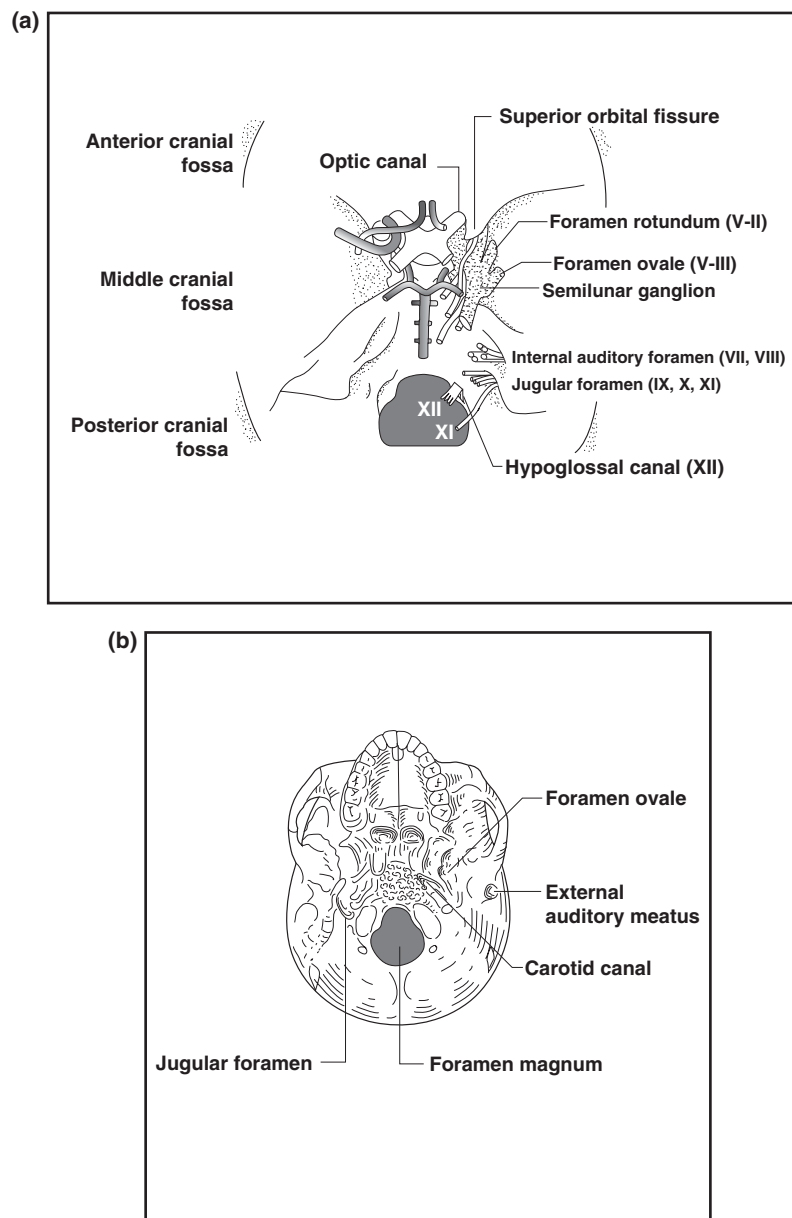


Figure 8-2 Foramina and canals through which the cranial nerves pass. (a) view from inside of the cranial fossa, (b) view from the skull basis. (a) cited from Haymaker, 1969, with permission; (b) modified from Chusid & McDonald, 1964.

oculomotor nerve (see Figure 7-7). The afferent fibers separate from the optic tract just before they reach the lateral geniculate body and project to the pretectal nucleus of mid-brain bilaterally. The fibers originating from the pretectal nucleus project to the Edinger-Westphal nucleus, which is located just anteriorly. The efferent fibers originating from the Edinger-Westphal nucleus run anteriorly as a part of the oculomotor nerve and project to the ipsilateral ciliary ganglion, which then innervates the **pupillary sphincter**.

A light stimulus presented to one eye elicits the light reflex in both eyes because of partial crossing of the optic nerve fibers at the optic chiasm and because of projection

of the afferent fibers from the optic tract to the pretectal nucleus on both sides. The light reflex elicited in the stimulated eye is called the direct pupillary reaction and the reflex elicited in the opposite (non-stimulated) eye is called the indirect or consensual pupillary reaction.

The **convergence reflex** is manifested as shrinkage of the pupillary diameter when one looks at a near object placed just in front of eyes. The efferent pathway of this reflex is the same as that of the light reflex, but the afferent pathway is not precisely known. It used to be thought that the proprioceptive afferent impulse originating from the medial rectus muscle might serve as its afferent pathway, but another

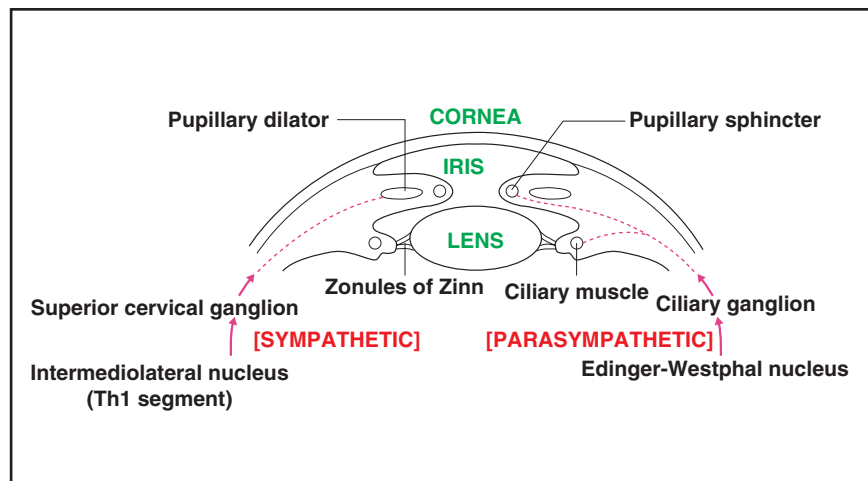


Figure 8-3 Schematic illustration of the nerve tracts controlling the pupil and accommodation. The iris is shown in a relative large size in order to show the pupillary sphincter and dilator.

theory is the near vision itself acts as the afferent because unilateral near vision can elicit this reflex. If the latter theory is correct, this reflex might more properly be called the accommodation reflex rather than the convergence reflex.

B. SYMPATHETIC NERVOUS SYSTEM AND HORNER SYNDROME

Pupils are dilated by a function of the sympathetic nerve fibers (Figure 8-4). The preganglionic fibers originate from

the **intermediolateral nucleus** at the level of the first thoracic segment, which is called the **cilio-spinal center of Budge**. The fibers exit the spinal cord through the anterior root of the first thoracic nerve and enter the sympathetic trunk. After passing through the inferior cervical ganglion (**stellate ganglion**) and the middle cervical ganglion, they make synaptic connection with the postganglionic fibers of the superior cervical ganglion. Then the postganglionic fibers ascend along with the internal carotid artery and the ophthalmic artery and innervate the **pupillary dilator** in the orbit (see Figure 8-3).

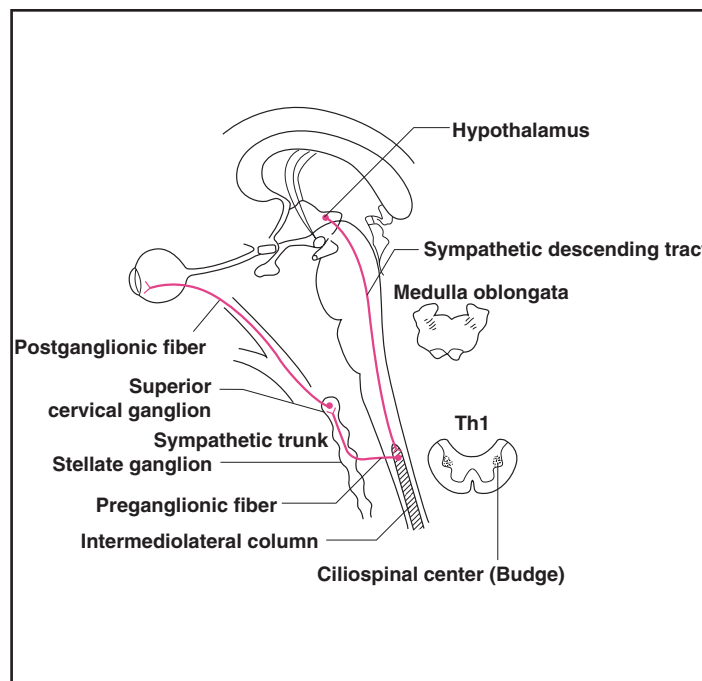


Figure 8-4 Schematic illustration of the sympathetic nervous pathway for controlling the pupils. Th1: first thoracic cord.

The neurotransmitter for the preganglionic fibers is acetylcholine, which acts in a nicotinic way, and that for the postganglionic fibers is noradrenaline. As the pupillary dilator surrounds the pupil radially, its contraction dilates the pupil. Therefore, impairment of unilateral sympathetic pathway results in a smaller pupillary diameter as compared to the intact side (**miosis**). In a stellate ganglion block with an anesthetic agent that is applied for the treatment of pain, miosis is seen on the side of the block because the preganglionic fibers are blocked in the ganglion.

The sympathetic nerve fibers also innervate the superior tarsal muscle (Müller muscle). This muscle does not function in voluntary eyelid opening, but it acts to spontaneously maintain the elevated position of the upper eyelid. Therefore, impairment of those sympathetic nerve fibers causes drooping of the upper eyelid (**ptosis**). In addition, the sympathetic nerve fibers are distributed to the mucous membrane of the eyes, nasal cavity, and oral cavity to inhibit secretion; to the facial skin to increase sweating; and to the arteries inside as well as outside of the cranial cavity to contract them.

A combination of miosis, ptosis, and retraction of the ocular bulb (enophthalmos) forms **Horner syndrome** (Figure 8-5). This condition is caused by involvement of the sympathetic nervous system innervating that eye. In most cases of Horner syndrome, in addition to the above triad, hyperemia of the palpebral and bulbar conjunctiva and decreased sweating and increased temperature of the facial skin are seen on the affected side. In clinical practice, these additional symptoms and signs often serve to

confirm the presence of Horner syndrome. Hyperemia and skin temperature elevation result from dilatation of arterioles and capillaries. The mechanism as to how the enophthalmos occurs has not been elucidated. Decreased sweating and skin temperature elevation are limited to the face if the sympathetic nervous system is involved peripheral to the ciliospinal center of Budge, but when the descending sympathetic tract is involved in the brainstem or cervical cord, these symptoms and signs are seen not just on the face but also on the affected side of the whole body (see Figure 8-4).

The center of the sympathetic nervous system is in the hypothalamus. The central sympathetic fiber tracts originating from the hypothalamus, descend through the medial part of the brainstem tegmentum and cervical cord, and project to the intermediolateral nucleus of the spinal cord, which extends from the first thoracic to the first lumbar segment. As described earlier, those descending fibers related to the pupillary control make synaptic connection at the first thoracic segment of the intermediolateral nucleus (ciliospinal center of Budge). As the central sympathetic fibers run relatively deep in the brainstem and cervical cord, the presence of Horner syndrome in patients with a brainstem or cervical cord lesion suggests the presence of an intramedullary lesion. In cervical spondylosis, for example, the extramedullary compression does not cause Horner syndrome unless the anterior root is compressed at the first thoracic segment.

In the lateral medullary syndrome of Wallenberg, which is caused by infarction of the posterior inferior cerebellar artery, Horner syndrome is a cardinal feature because the descending sympathetic fibers are involved in the medulla oblongata (Table 8-1; see Figures 14-1, 14-2, and 14-3 and Box 14-1).

Contrary to the above, if the sympathetic pathway becomes hyperactive for some reason, it is expected to cause mydriasis, elevation of the upper eyelid, protrusion of the ocular bulb (exophthalmos), anemia of the palpebral conjunctiva, and increased sweating and decreased skin temperature of the ipsilateral face. This rare condition is called **Pourfour du Petit syndrome**. In view of the fact that exophthalmos is also seen in patients with hyperthyroidism along with tachycardia and hyperhidrosis, the sympathetic nervous system is considered to be hyperactive in hyperthyroidism.



Figure 8-5 Horner syndrome due to lateral medullary syndrome. Note ptosis and miosis in the left eye. The lower eyelid is also elevated, which is in conformity with the presence of tarsal muscle also in the lower eyelid.
By courtesy of Dr. Stephen G. Reich, University of Maryland.

C. ACCOMMODATION AND ITS ABNORMALITY

Accommodation is a function of the ciliary muscle, which is innervated by the parasympathetic fibers of the

TABLE 8-1 SYNDROME CAUSED BY LOCALIZED LESION OF THE BRAINSTEM

Symptoms/signs			
Syndrome	Ipsilateral	Contralateral	Sites of lesion
[Crossed]			
Weber*	III nerve palsy	Hemiplegia	Cerebral peduncle
Claude*	III nerve palsy	Motor ataxia	Midbrain medial tegmentum, red nucleus
Benedikt	III nerve palsy	Hemiplegia, tremor, involuntary movement	Midbrain medial basis, red nucleus, substantia nigra
Millard & Gubler*	Facial nerve palsy	Hemiplegia	Caudal pons lateral basis
Foville*	Facial nerve palsy Conjugate gaze palsy	Hemiplegia	Caudal pons basis, paramedian tegmentum
Raymond-Cestan	INO	Dysesthesia, motor ataxia, hemiplegia	Rostral pons tegmentum
Wallenberg	Horner, facial noci-ceptive sensory loss, vertigo, dysphagia, motor ataxia	Nociceptive sensory loss below neck	Lateral medullary tegmentum
Cestan-Chenais	Horner, facial noci-ceptive sensory loss, dysphagia	Nociceptive sensory loss below neck, hemiplegia	Lateral medulla
Babinski & Nageotte*	Horner, facial noci-ceptive sensory loss, vertigo, dysphagia, motor ataxia	Nociceptive sensory loss below neck, hemiplegia	Unilateral medulla except for medial lemniscus and hypo-glossal nucleus
[Uncrossed]			
Parinaud*	Vertical gaze palsy, convergence palsy		Rostral midbrain–thalamus

III nerve: oculomotor nerve, INO: internuclear ophthalmoplegia

* Original reports are listed in the references.

oculomotor nerve. Its action is to focus the visual image on the fovea by controlling the thickness of the lens. As the ciliary muscle surrounds the lens in a ring-like fashion, its contraction loosens the zonules of Zinn, which radially support the lens, and it allows the lens to increase its thickness due to its own elasticity (see Figure 8-3). Owing to this special function of the ciliary muscle, it becomes possible to clearly see a visual target placed as near as 15 cm in front of eyes (**near point**). If accommodation becomes hyperactive, a far object is focused in front of the fovea, causing nearsightedness (myopia). On the contrary, if the refraction becomes hypoactive, a near object is focused behind the fovea, causing hyperopia. Thus, involvement of the parasympathetic nerve fibers of the oculomotor nerve causes accommodation failure, and the near point is extended to longer than 15 cm. Thus, if a patient with oculomotor nerve palsy complains of visual disturbance, accommodation failure should be also considered as its possible cause.

2. EXAMINATION OF INTRAOCULAR MUSCLES

First, the shape and diameter of pupils should be observed, paying special attention to possible asymmetry between the two eyes. Asymmetry of pupillary diameter is called **anisocoria**. A pathologically large pupil is called **mydriasis**, and a pathologically small pupil is called **miosis**. Of course, judgment of this asymmetry is impossible in a condition in which any eye drop, such as a pupillary dilator, has been applied. Furthermore, if the patient underwent any ophthalmologic surgery in the past, the pupil might look irregular in shape.

In general, pupils tend to be small in the aged population. In that case, it is useful to examine the pupils in a dark room in order to dilate the pupils as much as possible. Normal pupils are usually **round** and **isocoric**, meaning the same diameter. When the iris looks dark, as in some races, the pupils can be better observed by lighting the eyes from either side. Projection of the light from the front should be

avoided, because it changes the pupillary diameter as the result of light reflex.

A. OCULOMOTOR NERVE PALSY AND HORNER SYNDROME

If a pupil is extremely large or extremely small as compared with the other side, the pupil of that side is likely abnormal. Sometimes, however, it might be difficult to judge which side is abnormal. If there is any asymmetry in the level of the upper eyelid, then it helps determining the abnormal side. If there is **ptosis** in one eye, that side is often abnormal. If the pupil is larger in the eye with ptosis, it is likely to be oculomotor nerve palsy. In this case, the light reflex and convergence reflex are often lost, insufficient, or sluggish on the affected side.

By contrast, if the pupil is smaller in the eye with ptosis, it is most likely to be Horner syndrome and suggests involvement of the sympathetic nerve fibers on that side. As the tarsal levator muscle is not a skeletal muscle, ptosis due to impairment of the sympathetic nerve fibers is milder in degree compared with paralysis of the levator muscle of upper eyelids due to impairment of the parasympathetic nerve fibers. As described earlier, the accompanying symptoms and signs such as enophthalmos, conjunctival hyperemia, and decreased sweating and elevated temperature of the facial skin support the diagnosis of Horner syndrome. Among these signs, enophthalmos, though it is one of the triad of Horner syndrome, is sometimes difficult to confirm. In that case, other accompanying signs might help in making the diagnosis.

Horner syndrome is often congenital due to a traumatic injury of the sympathetic tract in the neck. Anisocoria due to unilateral miosis is sometimes difficult to detect in a bright room because both pupils react to the light stimulus. In that case, it is useful to examine the patient in a dark room because the intact pupil will dilate more than the affected pupil, making the anisocoria more easily detectable.

B. EXAMINATION OF THE LIGHT REFLEX AND ITS ABNORMALITY

With the eyes kept open, the examiner projects a penlight to either eye and observes the reaction of both pupils simultaneously. In order to avoid overlap of the convergence reflex, the light should be projected to the eye obliquely from the temporal side instead of straight from the front.

If the light reflex is absent or decreased in the stimulated eye (direct reaction absent) but present in the opposite eye (non-stimulated) eye (indirect reaction present), it suggests impairment of the efferent pathway, namely the midbrain

or oculomotor nerve, of the stimulated eye (Figure 8-6; see also Figure 7-6). By contrast, if the light reflex is absent or decreased in the non-stimulated eye (indirect reaction absent) while it is present in the stimulated eye (direct reaction present), it suggests impairment of the efferent pathway of the non-stimulated eye. In other words, if the light reflex is absent in one eye regardless of the side of stimulation, it suggests impairment of the efferent pathway of that eye. By contrast, if the light reflex is absent in both eyes (both direct and indirect reactions absent), it suggests either visual impairment (afferent pathway) of the stimulated eye or bilateral impairment of the efferent pathway. For example, if the light reflex in response to the right eye stimulation is absent in both eyes, it suggests either visual loss of the right eye or bilateral impairment of the autonomic fibers of the oculomotor nerves.

If visual loss is limited to one eye, the light stimulation of the other (intact) eye will elicit the light reflex in both eyes. In this case, if the light stimulus is shifted from the normal eye to the blind eye, it is expected to lead to pupillary dilatation in both eyes, and this phenomenon can be repeated by shifting the light stimulus between the two eyes. As this phenomenon is paradoxical to the ordinary reaction, it is called **paradoxical pupillary reaction**. This paradoxical reaction is often encountered in patients with multiple sclerosis or optic (retrobulbar) neuritis. This phenomenon is called Marcus Gunn pupil, while Marcus Gunn is also attached to jaw-winking as Marcus Gunn phenomenon (see Box 10-1, p. 79).

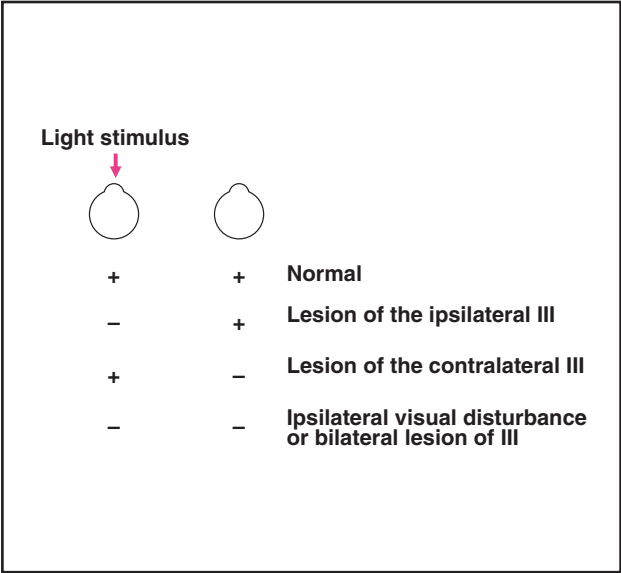


Figure 8-6 Presence (+) or absence (-) of the light reflex and its interpretation. Ipsilateral: ipsilateral to the light stimulus; contralateral: contralateral to the light stimulus.

Bilateral loss of the light reflex is an important finding for the diagnosis of brain death (see Chapter 33-7, p. 295). In relation to this, in a patient lying supine in a bright room like an intensive care unit (ICU), the pupils might already be constricted to the room light before testing. In that case, a very bright light stimulus might be needed to elicit the remaining reflex. An alternative practical way would be to test the light reflex immediately after opening the eyelids or to test the reflex after darkening the room. As a comatose patient usually lies supine in a bright room like ICU, the above caution is important in order to avoid misjudgment.

C. SPECIAL ABNORMAL FINDINGS OF PUPILS

A condition in which the light reflex is lost or decreased in spite of normal vision while the convergence reflex is preserved is called **Argyll Robertson pupil**. In this condition, the pupillary diameter is relatively small. This pupil is seen in patients with neurosyphilis, especially with tabes dorsalis, but it is also seen in diabetic impairment of the autonomic nervous system.

Pupillotonia is characterized by marked mydriasis in which the light reflex is extremely slow, and continuous exposure to the light stimulus in a dark room will shrink the pupil extremely slowly. Then, if the patient is placed in a very dark room immediately afterwards, the pupil will dilate again very slowly. It is, therefore, believed that, in this condition, both the pupillary sphincter and dilator muscles are in a tonic state so that they contract only with extreme slowness. In this condition, the convergence reflex is relatively better preserved than the light reflex.

A typical example of pupillotonia is seen in **Adie syndrome**, which is usually accompanied by loss of Achilles tendon reflex. Pupillotonia associated with segmental anhidrosis and loss of tendon reflexes is called **Ross syndrome** (de Assis Franco et al, 2020). In most cases of pupillotonia, the affected pupil overreacts to eye drops of a cholinesterase inhibitor causing extreme miosis, which is supposed to be due to hypersensitivity of synaptic receptors of the postganglionic fibers. A similar condition is seen following a lesion of the postganglionic fibers and is explained as **denervation supersensitivity**.

Pupils may display small-amplitude fluctuations in size under constant illumination. This phenomenon is called **hippus**, but its clinical significance has not been clarified. Hippus was described as a clinical manifestation of refractory autonomic nonconvulsive status epilepticus (Fernández-Torre et al, 2018).

D. EXAMINATION OF THE CONVERGENCE REFLEX

Convergence reflex is tested by moving a visual target toward the nasion with both eyes kept open or by asking the subject to look at a near object. The reaction is judged in the same way as for the light reflex. If the reaction is absent or decreased in one eye, it suggests impairment of the autonomic fibers of the oculomotor nerve. As the light reflex can be tested and judged more easily than the convergence reflex, the latter is often used as supplementary to the light reflex.

E. EXAMINATION OF ACCOMMODATION

As described earlier, accommodation is a function of the ciliary muscle to control the refraction by changing the thickness of the lens. It is tested by measuring the near point while moving a visual target toward each eye. The near point in healthy subjects is usually 15 cm. In accommodation failure, the near point becomes longer. In other words, the subject cannot see clearly an object placed 15 cm or more in front of the eye. This may be one of the reasons for the visual disturbance reported by patients with disorders of the autonomic nervous system.

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EXTRAOCULAR MUSCLES, GAZE, AND EYE MOVEMENTS

1. NERVE INNERVATION OF EXTRAOCULAR MUSCLES

There are six extraocular muscles attached to each eyeball: medial rectus, lateral rectus, superior rectus, inferior rectus, superior oblique, and inferior oblique muscle (Figure 9-1a). Contraction of the medial rectus and lateral rectus muscles rotates the eyeball medially and laterally, respectively. The exact action of other four muscles depends on the position of the eye (Figure 9-1b). The superior and inferior rectus muscles are attached to the top and the bottom, respectively, of the anterior hemisphere of the eyeball and act most efficiently when the eye is slightly abducted. Therefore, contraction of the superior rectus muscle rotates the eyeball upward diagonally in the lateral direction, and contraction of the inferior rectus muscle rotates the eyeball downward diagonally in the lateral direction. The superior and inferior

oblique muscles are attached to the top and the bottom, respectively, of the posterior hemisphere of the eyeball and act most efficiently when the eye is slightly adducted. Furthermore, the direction of muscle power for these two muscles is reversed by a hook (trochlea), which is attached to the upper and lower medial wall, respectively, of the orbit. Therefore, contraction of the superior and inferior oblique muscles rotates the eyeball downward and upward, respectively, diagonally in the medial direction (see Figure 9-1b).

Thus, simultaneous contraction of the superior rectus and inferior oblique muscle rotates the eyeball upward, and simultaneous contraction of the inferior rectus and superior oblique muscle rotates the eyeball downward. Among these six extraocular muscles, the medial rectus, superior rectus, inferior rectus, and inferior oblique muscles are innervated by the oculomotor nerve. The superior oblique muscle is innervated by the trochlear nerve, the

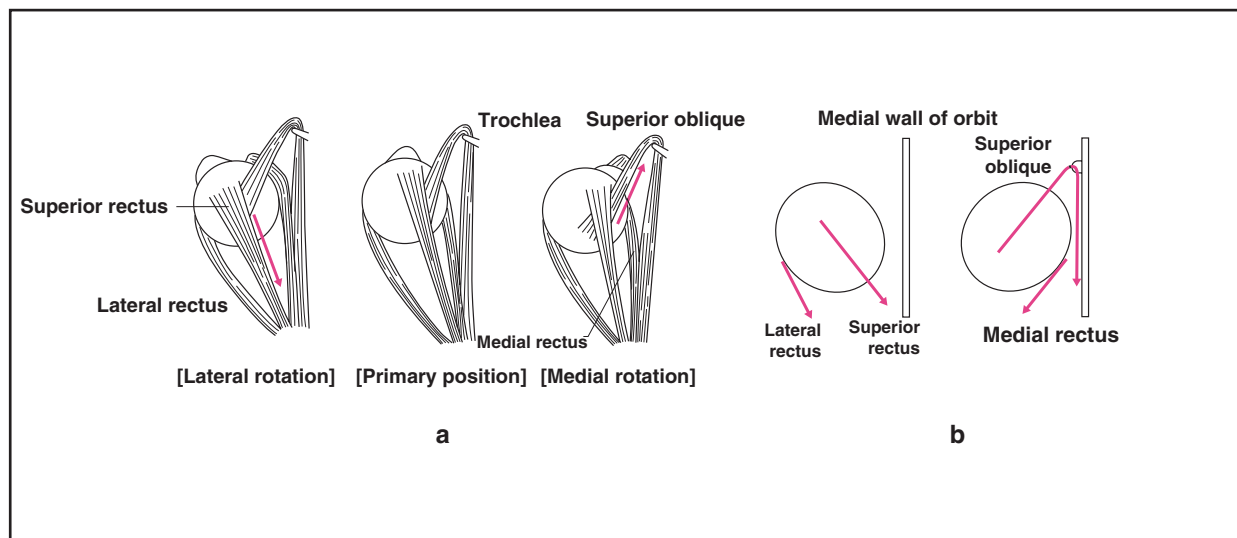


Figure 9-1 Structure of extraocular muscles (a) and direction of action of each muscle shown by arrows (b). View of the left eye from above. The superior rectus muscle rotates the ocular bulb upward in its laterally rotated position, and the superior oblique muscle rotates the ocular bulb downward in its medially rotated position. At the base of the ocular bulb, the inferior rectus muscle and the inferior oblique muscle function according to the same principle.

and the lateral rectus muscle is innervated by the abducens nerve.

A. THIRD CRANIAL NERVE (OCULOMOTOR NERVE)

As just described, the oculomotor nerve innervates four extraocular muscles: medial rectus, superior rectus, inferior rectus, and inferior oblique muscle. Therefore, when the oculomotor nerve is involved by a lesion, medial rotation and upward rotation of the eyeball are paralyzed, but lateral rotation is preserved. Downward rotation is partially possible because the superior oblique muscle is spared. The oculomotor nucleus is located in the midbrain at the level of the superior colliculus, in front of the periaqueductal gray substance and just medial to the medial longitudinal fasciculus (see Figure 5-1, p. 31). The oculomotor nucleus is formed from four groups of neurons, one innervating each extraocular muscle, and a fifth group innervating the levator muscle of the upper eyelid. Within the oculomotor nucleus, there is some topographic arrangement of those neuronal groups in terms of functions. For example, the neuronal groups related to the upward rotation of the eyeball (superior rectus, inferior oblique, and levator muscle of the upper eyelid) are located close each other.

Motor fibers originating from the oculomotor nucleus, together with the autonomic fibers originating from the Edinger-Westphal nucleus, run anteriorly through the medial part of the red nucleus and exit from the midbrain just medial to the cerebral peduncle (see Figure 5-1). As the fibers of the oculomotor nerve are relatively spread apart within the midbrain parenchyma, an intramedullary lesion may affect some fiber groups selectively. In this

case, therefore, weakness of the extraocular muscles might be different in its degree among muscles. As the fiber groups assemble to form a small bundle at the exit from the midbrain, the closer the lesion is to the anterior rim of the midbrain, the more muscles are equally paralyzed to produce complete oculomotor nerve palsy, as seen in Weber syndrome (Figure 9-2).

i. Intramedullary lesion of the oculomotor nerve and crossed hemiplegia

As the oculomotor nerve passes just medial to the red nucleus, its lesion there might have an associated ataxia of the contralateral extremities in addition to the ipsilateral oculomotor nerve palsy (**Claude syndrome**) (see Figure 9-2 and Table 8-1, p. 53). In this case, the ataxia is caused by a lesion of the superior cerebellar peduncle, which originates from the contralateral dentate nucleus of the cerebellum and crosses before reaching the red nucleus (see Chapter 16-2C, p. 127). Claude in 1912 reported a 56-year-old man who presented with complete paralysis of the right oculomotor nerve and probably also paralysis of the trochlear nerve associated with ataxia of the left extremities (Claude, 1912). Later the autopsy findings of that case were reported, which showed a small infarction at the right midbrain involving the oculomotor nerve, the rostral part of the superior cerebellar peduncle, the medial part of the red nucleus, and the medial longitudinal fasciculus (MLF) (Claude & Loyez, 1912). Following this case report, a combination of oculomotor nerve palsy and the contralateral cerebellar ataxia was called Claude syndrome, but it is estimated from the original report that there might have also

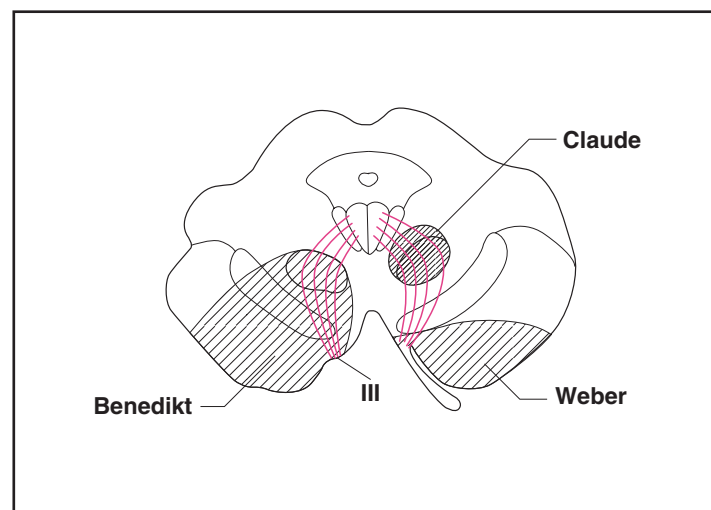


Figure 9-2 Crossed syndrome due to localized lesion of the midbrain.

been internuclear ophthalmoplegia (INO) due to involvement of MLF, although it must have been impossible to clinically confirm INO because of the presence of complete oculomotor nerve palsy (see p. 62).

It is noteworthy that Guillain commented on Claude's report based on his own research that, following a lesion of the red nucleus, retrograde degeneration could occur in the superior cerebellar peduncle down to the dentate nucleus of cerebellum (Claude & Loyez, 1912).

A condition in which oculomotor nerve palsy is associated with contralateral tremulous involuntary movements and hemiplegia or pyramidal signs is called **Benedikt syndrome**. In this condition, the lesion also involves the substantia nigra and cerebral peduncle (see Figure 9-2 and Table 8-1) (Liu et al, 1992). A lesion at the site of exit of the oculomotor nerve from the midbrain causes ipsilateral oculomotor nerve palsy and contralateral hemiplegia or pyramidal signs (**Weber syndrome**) (see Figure 9-2 and Box 9-1). Weber syndrome is caused by a vascular lesion of the midbrain basis, but it is also seen in an extramedullary lesion compressing the midbrain from anteriorly.

The above-described conditions manifesting a cranial nerve lesion associated with contralateral hemiplegia used to be called alternate hemiplegia or alternating hemiplegia. However, as "alternate" or "alternating" is more commonly used in a temporal meaning than in a spatial meaning, **crossed hemiplegia** might be a more appropriate term for these conditions (Liu et al, 1992). Hemiplegia encountered

in crossed hemiplegia is not necessarily severe in its degree, and it may be just a positive Babinski sign due to mild impairment of the corticospinal tract.

It is known in animals that the nerve fibers innervating the superior rectus muscle cross at the level of the oculomotor nucleus and join the contralateral oculomotor nerve, and that the levator muscle of the upper eyelid is bilaterally innervated (Bienfang, 1975). In relation to this, some human cases with a small infarction in the midbrain tegmentum have been reported to manifest isolated paralysis of the contralateral superior rectus muscle (Kwon et al, 2003; Tsukamoto et al, 2005).

ii. Extramedullary lesion of the oculomotor nerve

After exiting from the midbrain, the oculomotor nerve passes between the posterior cerebral artery and the superior cerebellar artery (Figure 9-3), passes through the upper part of the cavernous sinus (see Figure 8-1, p. 49), enters the orbit through the superior orbital fissure (see Figure 8-2a, p. 50), and sends its branches to the four extraocular muscles and the levator muscle of the upper eyelid. Impairment of the oculomotor nerve is an important sign in tentorial herniation (see Chapter 33-6D, p. 294). In this case, the oculomotor nerve is believed to be entrapped between the posterior cerebral artery and the superior cerebellar artery as a result of supratentorial compression rather than being involved intramedullarily in the midbrain.

BOX 9-1 WEBER SYNDROME

A German-born English physician, Hermann Weber, in 1863 reported a male patient who suddenly developed right hemiplegia, left oculomotor nerve palsy, and short-lasting disturbance of consciousness and died of pneumonia 2 months later. The hemiplegia was complete, involving the orbicularis oris muscle and tongue. The oculomotor nerve palsy also involved the pupil and accommodation. The hemiplegia slightly improved and became spastic, but the oculomotor nerve palsy remained complete. At autopsy, a hematoma 15 mm long and 6.3 mm wide was found in the medial inferior part of the left cerebral peduncle. The corticospinal tract was markedly damaged and the oculomotor nerve fibers were totally destroyed (Weber, 1863). The patient also complained of a warm sensation in the right half of the body, and two-point discrimination was decreased in the right extremities. In this report, Weber cited three other reported cases with similar pathology. Andral in 1834 reported a 60-year-old woman who suddenly developed right hemiplegia. At autopsy, a cavity of cherry size filled with green fluid was found in the left cerebral peduncle. There was no oculomotor palsy in that case, and the lesion involved the middle portion of the cerebral peduncle, sparing the medial part. The second case was briefly reported by Hennis Green and involved a girl aged 3.5 years who presented with left hemiplegia and right ophthalmoplegia due to a tuberculoma in the right cerebral peduncle. The third case, reported by Stiebel from Frankfurt, involved an 11-year-old girl who developed severe headache in the left forehead. At autopsy, a large abscess was found in the inferior portion of the left cerebral peduncle, but the neurological deficit consisted of left mydriasis and transient weakness of the left orbicularis oris muscle. Weber concluded that the symptoms and signs due to a lesion of the cerebral peduncle differ depending on which part of the peduncle is affected.

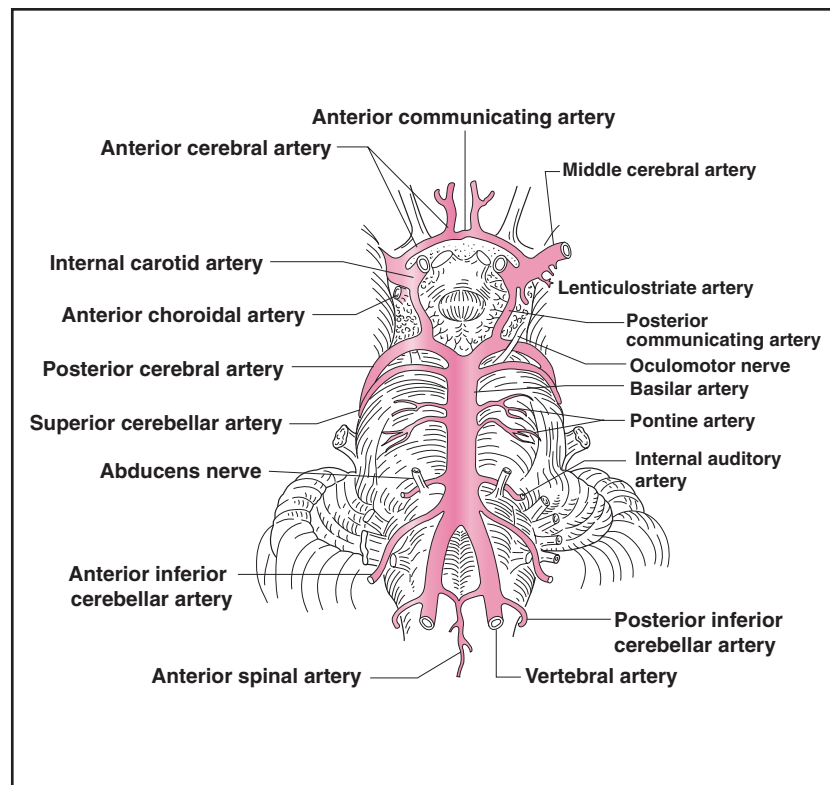


Figure 9-3 Spatial relationship between arteries at the base of the brainstem and cranial nerves.

iii. Cavernous sinus syndrome

As the oculomotor nerve passes through the cavernous sinus, it is easily impaired by inflammation of the cavernous sinus, either by an autoimmune mechanism or as a result of infectious diseases. As it is often associated with severe

ocular pain, this condition is called **painful ophthalmoplegia (Tolosa–Hunt syndrome)**. Similar symptoms and signs can be seen in the **orbital apex syndrome**, which is due to inflammation or tumor of the orbital apex and is characterized by exophthalmos and palpebral swelling in addition to painful ophthalmoplegia.

BOX 9-2 TOP OF THE BASILAR ARTERY SYNDROME

Several branches originate from the distal part of the basilar artery (see Figure 9-3). Therefore, infarction or embolism involving the top of the basilar artery causes ischemic lesions in multiple structures such as thalamus, midbrain, cerebellum, and mesial part of the temporal and occipital lobes. Depending on the anatomy, there can be a variety of clinical symptoms and signs, including bilateral blepharoptosis, extraocular muscle palsy, internuclear ophthalmoplegia, pupillary abnormalities, motor disturbance, impairment of consciousness, confusion, visual field defect, and cerebellar ataxia. The symptom complex is sometimes called “top of the basilar artery syndrome,” but there are no strict criteria for this diagnosis. In some cases of this syndrome, generalized convulsion as the initial symptom or locked-in syndrome has been reported. The diagnosis can be confirmed by magnetic resonance angiography.

B. FOURTH CRANIAL NERVE (TROCHLEAR NERVE)

The nucleus of the trochlear nerve is located in the caudal midbrain at the level of the inferior colliculus just posterior to the MLF. From there the fibers run medio-posteriorly, cross at the anterior medullary velum, and exit from the contralateral midbrain through its posterior rim (Figure 9-4). Then the fibers run anteriorly, pass through the cavernous sinus, and enter the orbit through the superior orbital fissure (see Figures 8-1 and 8-2a). This is the only cranial nerve that crosses intramedullarily except for the fibers of the oculomotor nerve innervating the superior rectus muscle (see p. 58), but it is extremely rare for this nerve to be solely affected by an intramedullary lesion.

C. SIXTH CRANIAL NERVE (ABDUCENS NERVE)

The nucleus of the abducens nerve is located near the anterior wall of the fourth ventricle. Its nerve fibers run

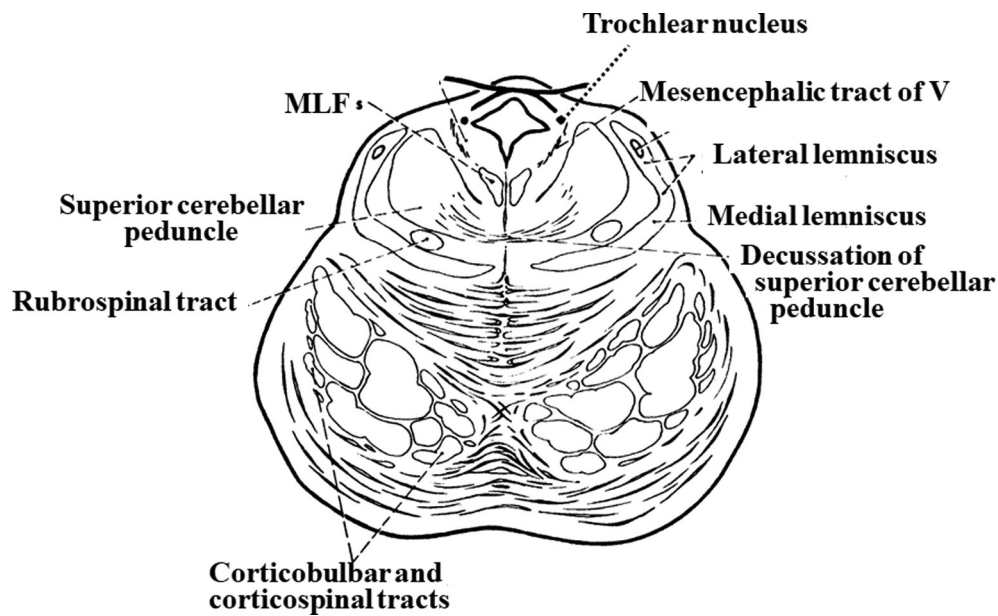


Figure 9-4 Schematic diagram showing the transverse section of the midbrain at the level of crossing of the trochlear nerves.

anteriorly through the medial pons and exit from the pons through the corticospinal tract (see Figure 5-2, p. 32). Then the nerve fibers pass between the internal auditory artery and the anterior inferior cerebellar artery (see Figure 9-3) and through the cavernous sinus, and enter the orbit through the superior orbital fissure to innervate the lateral rectus muscle (see Figures 8-1 and 8-2a). If the abducens nerve is involved together with the corticospinal tract by either an intramedullary lesion or extramedullary compression, an abducens palsy and spastic paralysis of the contralateral extremities are simultaneously seen (**abducens crossed hemiplegia**). This condition, like Weber syndrome, is a representative case of crossed hemiplegia.

The abducens nerve runs in the cranial cavity a longer distance than any other cranial nerves, extending from the posterior cranial fossa along the clivus to the orbit. Therefore, in a state of intracranial hypertension, this nerve tends to be impaired either unilaterally or bilaterally, though to a relatively mild degree. As this is not due to direct compression of the nerve by a mass lesion, this circumstance belongs to the **false localizing signs**.

As will be described later (see Chapter 11-1A, p. 83), the facial nerve first runs posteriorly from its nucleus, turns around the abducens nucleus, and exits from the lateral rim of the pons. Thus, the abducens nerve and the facial nerve are very close together at the pontine tegmentum, but once they exit from the pons, they do not meet again (see Figure 5-2, p. 32). Therefore, when these two nerves

are simultaneously affected, it suggests the presence of an intramedullary lesion in the pontine tegmentum. However, it has to be kept in mind that these two nerves may still be affected simultaneously by an inflammatory process outside of the brainstem, as seen in multiple cranial neuritis.

2. NEURAL CONTROL MECHANISM OF GAZE

A. CENTRAL CONTROL MECHANISM OF GAZE

Voluntary movements of both eyes either horizontally or vertically are called **gaze**. The main cortical gaze centers are located in the frontal lobe. The **frontal eye field (FEF)** is located at the lateral frontal convexity just anterior to the face/hand area of the primary motor cortex, and the **supplementary eye field (SEF)** is located in the mesial frontal cortex just anterior and superior to the supplementary motor area proper (SMA proper) (see Figure 11-3, p. 85). Visual information is transferred from the posterior parietal cortex (area 7) to the FEF. The nerve fibers originating from the FEF and SEF pass through the corona radiata and the centrum semiovale of the cerebral hemisphere, cross at the upper brainstem, and project to the **paramedian pontine reticular formation (PPRF)**, which serves as the lateral gaze center (Figure 9-5). Physiologically, there are saccadic burst neurons in the PPRF. The nerve fibers originating from the PPRF project to the ipsilateral abducens nucleus, in which two different groups of neurons are present. The nerve fibers originating from one group form the abducens

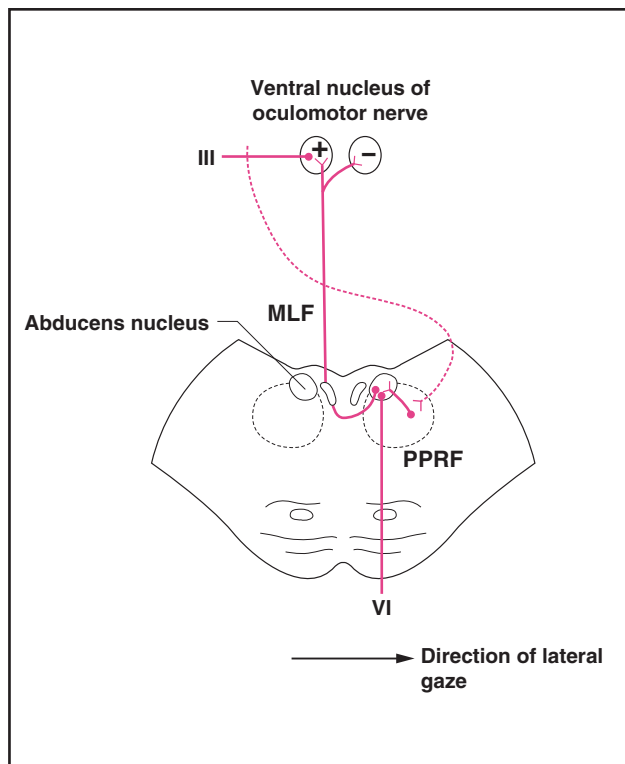


Figure 9-5 Schematic illustration of the nerve pathway to explain the lateral gaze and internuclear ophthalmoplegia. III: oculomotor nerve, VI: abducens nerve, MLF: medial longitudinal fasciculus, PPRF: paramedian pontine reticular formation.

nerve and pass through the medial part of pons anteriorly, and those originating from another group of neurons (internuclear neurons) cross at the same level to form the **medial longitudinal fasciculus (MLF)**. The MLF ascends through the medial part of the pontine tegmentum and projects to the ventral nucleus of the oculomotor nucleus, which innervates the medial rectus muscle. Thus, activation of the PPRF causes contraction of the ipsilateral lateral rectus muscle and the contralateral medial rectus muscle simultaneously, resulting in the lateral gaze to the side ipsilateral to the PPRF or contralateral to the cortical gaze center (see Figure 9-5).

B. LATERAL GAZE AND LATERAL GAZE PALSY

A unilateral lesion of the frontal lobe may cause disturbance of lateral gaze to the other side. This condition may not necessarily present with gaze palsy, but it may just slow down the saccadic movement to the contralateral side. If consciousness of the patient is disturbed in this condition, the eyes are often deviated toward the side of the lesion due to the preserved activity of the contralateral eye fields (**conjugate deviation of eyes**).

A similar phenomenon can also be seen in a parieto-occipital lesion, but in this case the head may be also rotated toward the side of the lesion. This condition is also seen in patients with some disturbance of consciousness, most likely because visual attention is intact toward the side of the lesion and impaired toward the side contralateral to the lesion. It is not completely clear why the conjugate deviation of eyes is more common in a state of impaired consciousness, but it is conceivable that, in conscious patients, visual attention can compensate for the tendency to deviate by intact visual attention.

In the acute phase of thalamic hemorrhage, eyes are often deviated toward the contralateral side. This is considered to be a positive symptom (see Chapter 1-1, p. 2), and the direction of deviation may be reversed later in the clinical course.

As expected from Figure 9-5, a unilateral lesion of the pontine tegmentum causes lateral gaze palsy to the side of the lesion, and the eyes tend to be deviated to the contralateral side. Contrary to the conjugate deviation due to a hemispheric lesion, conjugate deviation due to a pontine lesion can be also seen in patients with intact consciousness.

When the lateral gaze center of the pons (PPRF) is involved by a lesion, the lesion commonly involves the ipsilateral MLF as well. In this condition, lateral gaze may not be conjugate (meaning that the two eyes are not paralyzed to the same degree). In a lesion of the right pontine tegmentum, for example, the left eye is deviated to the left whereas the right eye is not deviated at all or is deviated just a little to the left because of the associated lesion of the right MLF.

Lateral gaze palsy due to a PPRF lesion varies in its severity depending on the nature and degree of the lesion. It could be complete lateral gaze palsy, but in case of relatively mild impairment or during recovery from an acute phase, it may just show either incomplete lateral gaze palsy or it may just decrease the speed of the lateral gaze.

C. VERTICAL GAZE AND VERTICAL GAZE PALSY

In contrast with the lateral gaze center, which is in the pons, the vertical gaze center is located in the midbrain. The **rostral interstitial nucleus of the MLF** plays a main role in the control of vertical gaze, but the interstitial nucleus of Cajal is also known to participate. Like the lateral gaze center, saccadic burst neurons exist in these gaze centers. These structures are believed to receive input from the frontal eye field, vestibular nucleus, and superior

colliculus, but the distinction of upward and downward gaze is not clearly known.

Compression of the quadrigeminal body by a pineal tumor, for example, causes **vertical gaze palsy** and convergence palsy. A combination of vertical gaze palsy and convergence palsy is called **Parinaud syndrome**. This condition is often associated with pupillary abnormalities and up-beat nystagmus (see p. 71). The gaze palsy seen in Parinaud syndrome may be solely upward or solely downward, or may be in both directions (Parinaud, 1886). In four autopsied cases with Parinaud syndrome studied by Pierrot-Deseilligny, infarction was commonly found in the border zone area between the midbrain and the thalamus, involving the rostral interstitial nucleus of the MLF and its efferent fibers (Pierrot-Deseilligny et al, 1982). In this case, the oculocephalic reflex or doll's eye sign (see 5-C of this chapter, p. 69) is usually preserved.

Progressive supranuclear palsy (PSP, Steele–Richardson–Olszewski syndrome) is characterized by vertical gaze palsy, especially downward gaze palsy, in which the oculocephalic reflex is preserved. In this condition, the superior colliculus is pathologically affected, and demonstration of its atrophy on magnetic resonance imaging serves as one of its diagnostic criteria.

Vertical supranuclear saccade palsy is seen in children with neurolipidosis, especially Niemann–Pick disease type C (**ophthalmoplegic lipidosis**) (Patterson et al, 2017).

D. CONVERGENCE AND CONVERGENCE PALSY

In order to see a near object, convergence is produced by bilateral contraction of the medial rectus muscles. The convergence center is called the Perlia nucleus, which is located in the midbrain. Diagnosis of **convergence palsy** is only possible when the medial rectus muscles are not paralyzed. However, as the Perlia nucleus is located near the oculomotor nucleus, the two nuclei are commonly affected by the same lesion. In this case, it is difficult to confirm the presence of convergence palsy. When examining the aged population, it should be kept in mind that convergence and vertical gaze tend to be incomplete even in healthy subjects.

In order to see a far object, the lateral rectus muscles contract bilaterally. This is called **divergence**. If double vision is a complaint when looking at a far object, it may be due to bilateral paralysis of lateral rectus muscles or convergence palsy, and the distinction between the two conditions is sometimes difficult. If the lateral rectus muscle is

paralyzed, it is expected to worsen the double vision when looking laterally toward the impaired side whereas, in convergence palsy, double vision is expected to disappear or decrease when looking toward either side.

3. MLF AND INTERNUCLEAR OPHTHALMOPLÉGIA

As the MLF connects the abducens nucleus with the contralateral oculomotor nucleus (see Figure 9-5), its interruption causes disconjugate lateral gaze palsy to the contralateral side. Namely, the contralateral eye can be abducted but the ipsilateral eye cannot be adducted (**internuclear ophthalmoplegia, INO**). In this case, horizontal nystagmus is seen on lateral gaze to the side contralateral to the lesion with the rapid phase directed toward the side of gaze. Characteristically, this nystagmus is seen solely or predominantly in the abducting eye (**dissociated nystagmus**). As the same phenomenon can be seen in patients with horizontal nystagmus associated with oculomotor nerve palsy, it has to be proved that the medial rectus muscle is intact before making the diagnosis of INO. In this case, the presence of convergence is used for proving that it is intact. Thus, the diagnostic criteria for INO is composed of (1) impairment of adduction of the eye ipsilateral to the lesion, (2) dissociated nystagmus, and (3) relative preservation of convergence.

Since the MLF lies close to the midline in the brainstem, demonstration of an INO always suggests the presence of an intramedullary lesion. When the MLF is only mildly affected or during the recovery phase from an acute phase, the adduction of the ipsilateral eye may appear normal, and instead only the dissociated nystagmus might be seen. In that case, careful observation of the lateral saccade toward the side contralateral to the lesion might disclose slower speed of the adducting eye as compared with the speed of the abducting eye. INO is commonly seen in ischemic vascular disease or multiple sclerosis. The most common cause of bilateral INO in young subjects is multiple sclerosis.

In a unilateral lesion of the pontine tegmentum involving both the PPRF and MLF, the lateral gaze to the side ipsilateral to the lesion is impaired while the lateral gaze to the contralateral side shows INO. In this condition, therefore, the eye ipsilateral to the lesion does not move to either side while the contralateral eye moves only laterally but not medially. Thus, this condition is called the **one-and-a-half syndrome** (Figure 9-6).

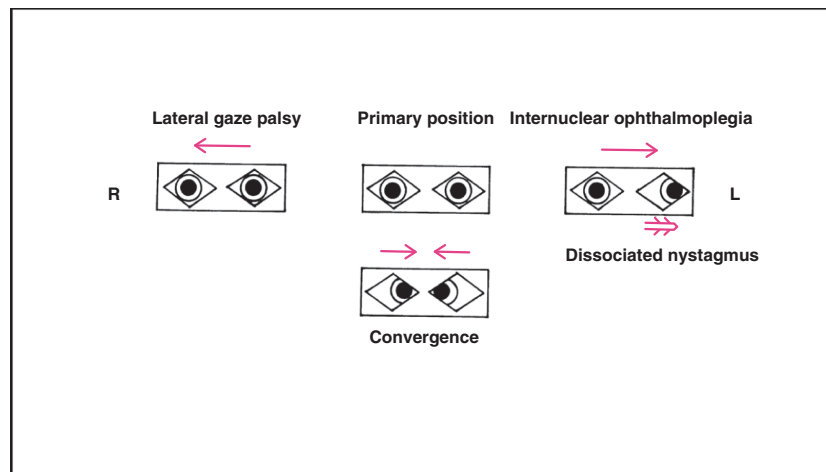


Figure 9-6 Schematic illustration showing the one-and-a-half syndrome. In this case, the lesion is in the right pontine tegmentum. Right lateral gaze is paralyzed due to the lesion of the right PPRF, and internuclear palsy is seen on the left lateral gaze due to the lesion of the right MLF. Convergence is normal. R: right, L: left. *Single arrows* indicate the direction of gaze.

4. CENTRAL CONTROL MECHANISM OF EYE MOVEMENTS

Eye movements are categorized into two kinds; **saccadic eye movement (saccades)** and **smooth pursuit eye movement (smooth pursuit)** (Box 9-3). Saccades occur when the eyes rapidly shift their fixation from one visual target to the other, and smooth pursuit occurs when the eyes pursue

a smoothly moving visual target. For these two, the final common pathway from the brainstem nuclei to the extraocular muscles is the same, but the central control mechanism is different.

A. CENTRAL CONTROL MECHANISM OF SACCADES

Physiologically, saccades are grouped into two kinds: memory-guided and visually guided. **Memory-guided saccades (MGS)** are generated when a subject is requested to look at a visual target after it has been removed. **Visually guided saccades (VGS)** are generated by asking a subject to look at a visual target. Thus MGS are more voluntary while VGS are more reflexive if done immediately upon presentation of the visual stimulus. Results of experimental studies in monkeys, clinical observations in patients, and neuroimaging studies in healthy subjects all indicate that MGS are mainly controlled by the frontal cortex including the SEF, FEF, dorsolateral prefrontal cortex, and anterior cingulate, and that VGS are mainly controlled by the posterior parietal cortex (Girard & Berthoz, 2005; Ramat et al, 2007) (Figure 9-7).

It is believed that the nerve fibers originating from the frontal cortex project directly to the saccadic burst neurons in the gaze centers of midbrain and pons, but there is also a role of the **superior colliculus** in the control of saccades (see Figure 9-7) (Girard & Berthoz, 2005; Ramat et al, 2007; Terao et al, 2011). In the superior colliculus, a group of neurons is retinotopically organized and related to visual function, and another group of neurons is engaged in the motor control. The superior colliculus receives input directly from the frontal and parietal cortices and the striatum, and its

BOX 9-3 SACCADIC EYE MOVEMENT AND SMOOTH PURSUIT EYE MOVEMENT

It is generally believed that a saccadic eye movement (saccade) is a rapid eye movement whereas a smooth pursuit eye movement (smooth pursuit) is a slow movement, but from the clinical point of view, the distinction between the two should be made based on not just the speed of ocular movements but rather on the way the eye movement was induced in the examination. A saccade is induced by shifting the visual fixation from one visual target rapidly to another target. In this case, therefore, the fixation point jumps from a visual target to another target. Consequently, the movement is rapid in healthy subjects, but in some patients the saccade may become very slow. By contrast, smooth pursuit is induced when the eyes maintain fixation to a slowly moving visual target, and its speed depends on the speed of the target movement. As the visual target is usually moved slowly for inducing the smooth pursuit, the induced eye movement is slow and smooth in healthy subjects. However, if the visual target is moved rapidly, the induced ocular movement can be as fast as the saccade.

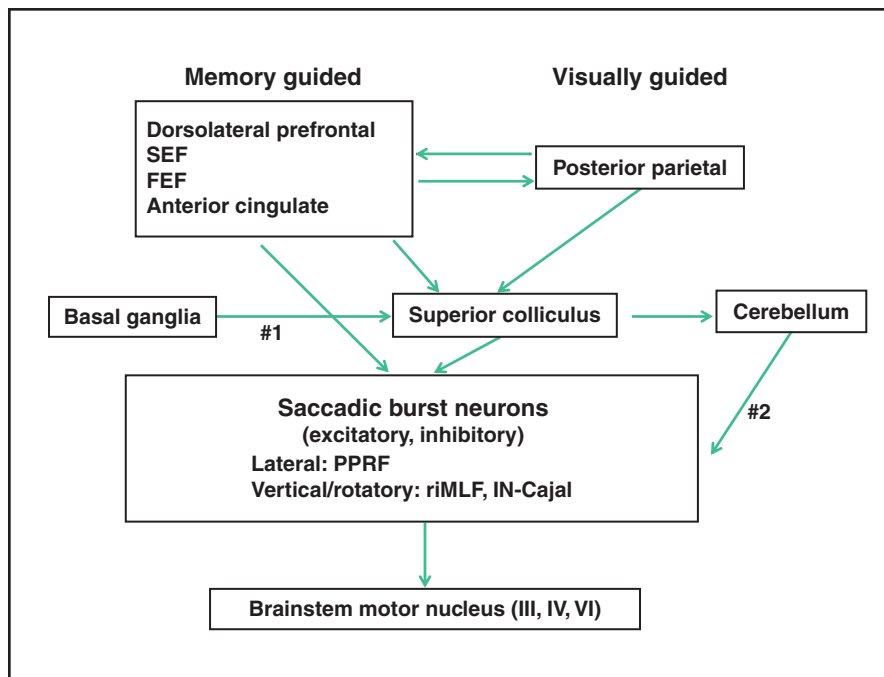


Figure 9-7 Diagram illustrating the central control mechanism of saccadic eye movement. The difference between a memory-guided saccade and a visually guided saccade is not absolute but relative. The *arrows* show the main flow of information between different structures, and in fact most connections are believed to be bidirectional. Furthermore, the ascending pathway from the subcortical structures to the cerebral cortex through the thalamic relay nuclei is not shown here. #1 is inhibitory and mainly related to motor selection. #2 controls accurate performance of the saccades. SEF: supplementary eye field, FEF: frontal eye field, PPRF: paramedian pontine reticular formation, riMLF: rostral interstitial nucleus of the medial longitudinal fasciculus, IN-Cajal: interstitial nucleus of Cajal. Prepared by the authors based on Girard & Berthoz, 2005, and Ramat et al., 2007.

output projects to the saccadic burst neurons in the gaze centers and to the cerebellum. However, as an experimental lesion of the superior colliculus in animals does not alter the amplitude of saccades or smooth pursuit, it is believed that the superior colliculus is not a direct relay center from the cortex to the burst neurons but rather it controls selection of visual target, and initiation and speed of eye movements (Ramat et al, 2007).

B. CENTRAL CONTROL MECHANISM OF SMOOTH PURSUIT

There is some overlap of the central control system of smooth pursuit with that of VGS, but **area V5**, which corresponds to the middle temporal visual area (MT) and the medial superior temporal visual area (MST) of monkeys, plays an important role in motion vision and thus in smooth pursuit (Figure 9-8) (Lencer & Trillenber, 2008). Area V5 in humans is located in the posterior part of the superior temporal sulcus at the site where its ascending branch bifurcates (Matsumoto et al, 2004) (see Chapter 24-1K, p. 238). Area V5 receives input from the primary visual

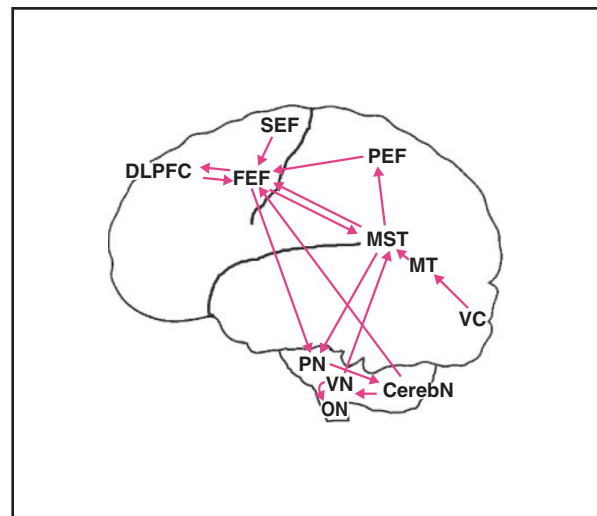


Figure 9-8 Schematic diagram showing the central pathway related to the control of smooth pursuit eye movement. CerebN: cerebellar nucleus, DLPFC: dorsolateral prefrontal cortex, FEF: frontal eye field, MST: medial superior temporal visual area, MT: middle temporal visual area, ON: motor nuclei innervating extraocular muscles, PEF: parietal eye field, PN: smooth pursuit-related nucleus in the pontine reticular formation, SEF: supplementary eye field, VC: primary visual cortex, VN: vestibular nuclei. Cited from Lencer & Trillenber, 2008, with permission.

cortex directly or indirectly, and also from the vestibular nuclei, and it projects to the FEF and the parietal eye field. Furthermore, the FEF receives input from SEF, dorsolateral prefrontal cortex, and parietal cortex, each of which controls the function of FEF. The smooth pursuit-related neurons in the brainstem are considered to be in the reticular formation, which corresponds to the dorsolateral pontine nucleus and nucleus reticularis tegmenti pontis in monkeys. The neuronal group receives input from area V5 and FEF and projects indirectly via the cerebellum and vestibular nuclei to the motor nuclei innervating the extraocular muscles (see Figure 9-8) (Mustari et al, 2009).

C. ROLES OF CEREBELLUM AND BASAL GANGLIA IN EYE MOVEMENTS

The cerebellum and basal ganglia play an important role in the control of eye movements. The cerebellum receives input from the superior colliculus and projects output through the cerebellar nuclei to the saccadic burst neurons in the midbrain and pons (see Figures 9-7 and 9-8). The cerebellum acts to accurately control the amplitude of saccades, and its dysfunction causes instability of the saccadic amplitude and impairment of VGS. Within the cerebellum, among others, the flocculus and paraflocculus play an important role in stabilizing the visual image on the fovea of the retina.

Output from the basal ganglia is inhibitory and acts to select necessary movements by suppressing unnecessary movements (see Chapter 16-2B, p. 116). In Parkinson disease, the reaction time with saccades is delayed and the speed of saccade becomes slow in proportion to the reaction time of the hand ipsilateral to the side of gaze (Shibasaki et al, 1979). As the striatum is functionally connected to the supplementary motor area, it is considered to be especially important for the control of MGS. In Parkinson disease, MGS are mainly impaired, but VGS were also shown to be impaired, though to a relatively mild degree, from the early clinical stage (Terao et al, 2011). Saccadic bradykinesia was documented especially in self-paced saccades in patients with Parkinson disease (Koohi et al, 2021).

5. EXAMINATION OF EXTRAOCULAR MUSCLES AND GAZE

As gaze and eye movements cannot be tested if the extraocular muscles are paralyzed, it is reasonable to test the

extraocular muscles first before testing the gaze and eye movements.

A. PTOSIS

Although the levator muscle of the superior eyelid is not directly related to eye movement, it is useful to observe the presence or absence of **ptosis (blepharoptosis)** to start with. This can be done by comparing the palpebral fissure between the two eyes.

i. Bilateral ptosis

Bilateral ptosis is seen in disorders of muscles or the neuromuscular junction. In particular, it is important to think about the possibility of **myasthenia gravis (MG)**. The cardinal feature of this condition is **easy fatigability**, which is usually suggested by the history. When examining the patient, ptosis may be noticeable several seconds after the patient arises from the lying position and it may return to normal once the patient lies down again.

When there is a possibility of MG, the diagnosis can be confirmed according to the following tests. For ocular MG, the **ice pack test** was shown to have similar diagnostic accuracy to single-fiber electromyography (EMG; Giannocaro et al, 2020). Another test that is specific and sensitive for MG is the **Cogan lid twitch** (Singman et al, 2011). To elicit this sign, the patient is instructed to maintain downward gaze for 15 seconds and then gaze upward, finally returning to primary gaze. In patients with a positive sign, the upper lid produces an upward twitch or overshoot soon after the patient has returned to primary gaze.

Then, the effect of cholinesterase inhibitor should be tested (**Tensilon test**) (see Chapter 33-1B, p. 285 for details). Then, the response of a skeletal muscle to a 3-Hz repetitive electrical stimulation of the corresponding motor nerve should be examined. In case of ocular MG, either the orbicularis oculi or oris muscle can be studied by stimulating the facial nerve. Decrement of the muscle evoked potentials by more than 10% is judged to be positive (**waning**). Single-fiber EMG may be also applied. Then, the serum should be tested for the antibody against acetylcholine receptor. This test is positive in the majority of patients with MG, but in a small proportion of patients, antibodies against muscle-specific kinase (MuSK) and low-density receptor-related protein 4 (Lrp4) might be positive instead (Ha & Richman, 2015; Vincent, 2014). Furthermore, as thymoma is present in about 15% of MG

patients, X-ray and computed tomography (CT) of the chest should be obtained.

A bilateral lesion of the brainstem tegmentum may cause bilateral ptosis as a part of Horner syndrome. In this case, however, it is extremely rare to show ptosis of equal degree in both eyes. In bilateral Horner syndrome, the ptosis is more conspicuous on the side of the smaller pupil. By contrast, in case of bilateral oculomotor nerve palsy, the ptosis is more conspicuous on the side of the larger pupil. Bilateral lesion of the brainstem tegmentum can be judged from other accompanying signs.

In aged subjects, the tissue of upper eyelids may become loose and result in apparent ptosis. In this case, lifting of the slackened upper eyelid may disclose its real position.

ii. Unilateral ptosis

Unilateral ptosis of marked degree is usually caused by a lesion of the ipsilateral oculomotor nerve, because ptosis in Horner syndrome is usually mild in degree. In unilateral ptosis of mild degree, its cause could be either oculomotor nerve palsy or Horner syndrome. Observation of pupils is of utmost importance to distinguish between the two conditions (see Chapter 8-2, p. 54).

iii. Blepharospasm

When a patient complains of difficulty in seeing because of the eyelid drop, it may be due to blepharospasm instead of ptosis. Since blepharospasm is caused by excessive contraction of the orbicularis oculi muscle (see Chapter 11-1A, p. 83), the position of the lower eyelid is elevated in blepharospasm, which is not the case in ptosis. Furthermore, in blepharospasm, contraction of the orbicularis oculi muscle is enhanced either continuously (tonic contraction) or repetitively (phasic contraction), and blinks might be increased in intensity and frequency.

If intermittent eye closure is limited to one eye, it could be a part of **hemifacial spasm**, which is due to hyperexcitation of the facial nerve. In this case, however, contraction may also be seen in the ipsilateral orbicularis oris muscle, either continuously or repetitively in synchrony with the contraction of the orbicularis oculi muscle (see Chapter 11-1A). On the other hand, blepharospasm may look asymmetric in some cases. In this case, observation of the orbicularis oris muscle is again important to make the distinction.

Blepharospasm is considered a kind of focal dystonia (see Chapter 19-6, p. 172). The condition in which

blepharospasm is accompanied by dystonia of the jaw and pharynx used to be called **Meige syndrome**.

B. EXAMINATION OF EXTRAOCULAR MUSCLES

Examination of the extraocular muscles can be effectively started by observing the eyes in the primary position when the subject spontaneously looks straight ahead. If an eye is deviated either medially or laterally, it may not necessarily be paralysis of extraocular muscles but rather congenital strabismus (squint) or due to other ophthalmologic causes. In congenital strabismus, the patient rarely complains of double vision (diplopia), and paralysis of extraocular muscles can be excluded by the examination as described in the following.

In paralysis of an extraocular muscle, the involved eye is deviated to the side of the opposing muscle. For example, in paralysis of the medial rectus muscle, the affected eye is deviated laterally in the primary position, and in paralysis of the lateral rectus muscle, the eye is deviated medially. If one eye is deviated upward or downward, a pair of muscles rotating the eye either downward or upward may be affected, or one muscle of each pair may be affected.

i. Vertical paralysis of one eye

If one eye is rotated upward in the primary position, it suggests paralysis of either the superior oblique muscle or the inferior rectus muscle, or both, of that eye. In **unilateral paralysis of the superior oblique muscle**, the patient complains of diplopia (double vision) when looking downward, and the degree of diplopia (distance between the real image from the intact eye and the false image from the paralyzed eye) is increased when the visual target placed downward is moved laterally away from the side of the affected eye (see Figure 9-1b). In unilateral paralysis of the inferior rectus muscle, by contrast, the diplopia is increased in its degree when the visual target is moved toward the side of the affected eye. Furthermore, patients with unilateral superior oblique muscle palsy tend to tilt their head toward the side of the intact eye. This is because they are spontaneously (involuntarily) trying to compensate for the relative position of the affected eye, which is otherwise rotated upward due to the downward paralysis of that eye. This phenomenon can be tested by tilting the patient's head toward the side of the affected eye which will elevate the relative position of the eye further, resulting in an increase in the degree of diplopia. This phenomenon is called the **Bielschowsky sign**.

If one eye is rotated downward in the primary eye position, it suggests paralysis of either the superior rectus muscle or the inferior oblique muscle, or both, of that eye. The patient complains of diplopia when looking at an object placed upward. If the diplopia is increased by shifting the visual target from there to the side of the affected eye, it suggests impairment of the superior rectus muscle. Likewise, if the diplopia is increased by shifting the target to the side away from the affected eye, it suggests impairment of the inferior oblique muscle. In fact, as these two muscles are innervated by the same (oculomotor) nerve, they are often paralyzed together.

If one eye is rotated either upward or downward in the primary position, it is important to observe the position of the other eye, because it might be skew deviation and not extraocular muscle palsy. In **skew deviation**, one eye is rotated upward while the other eye is rotated downward, producing a diagonal position. This phenomenon is seen in some lesions of either the cerebellum or brainstem.

ii. Congenital disorders of extraocular muscles

Congenital paralysis of the bilateral abducens nerves and/or the bilateral facial nerves is called **Möbius syndrome** (see Box 11-3, p. 88). These conditions are due to hypoplasia or aplasia of neurons of the abducens nucleus or the facial nucleus. Therefore, the signs are present from birth and not progressive. Möbius syndrome may also present with lateral gaze palsy due to hypoplasia of neurons in the lateral gaze center.

In another congenital condition, **Duane syndrome**, the affected eye is medially rotated because of disturbance of lateral rotation, and an attempt to medially rotate the eye produces retraction of the eyeball associated with narrowing of the palpebral fissure. This phenomenon can be seen bilaterally. By using magnetic resonance imaging (MRI) with high spatial resolution, marked hypoplasia of the extramedullary fibers of the abducens nerve was demonstrated in this condition (Denis et al, 2008). It was thought that this condition was due to the histologic reorganization of the abducens muscle itself, but it is now believed to be secondary to abducens nerve palsy.

iii. Causes of diplopia or extraocular muscle paralysis

Causes of diplopia or extraocular muscle paralysis can be grouped into four main conditions (Box 9-4). Namely, diplopia may be caused by a lesion involving any part of the final common pathway extending from the relevant

BOX 9-4 CAUSES OF EXTRAOCULAR MUSCLE PARALYSIS

1. Muscle disease
 - Ocular myopathy
 - Mitochondrial encephalomyopathy (among others, Kearns–Sayre syndrome)
 - Myopathy associated with thyroiditis (so-called thyrotoxic ophthalmoplegia)
2. Disorders of neuromuscular junction
 - Myasthenia gravis
3. Cranial nerve lesion (III, IV, VI)
 - Cranial neuritis
 - Miller Fisher syndrome (Box 9-5)
 - Painful ophthalmoplegia (Tolosa–Hunt syndrome)
 - Diabetic neuropathy
 - Sarcoidosis
 - Aneurysm (internal carotid artery, basilar artery)
 - Metastatic brain tumor
 - Sphenoidal ridge meningioma
 - Tumor of the paranasal sinus (especially mucocoele)
 - Trauma
4. Brainstem lesion
 - Multiple sclerosis
 - Angioma
 - Infarction
 - Hemorrhage
 - Tumor
 - Wernicke encephalopathy
 - Brainstem encephalitis

brainstem motor nuclei to the extraocular muscles. For the patients complaining of diplopia, therefore, it is useful to consider each possibility systematically throughout the history taking and the physical examination. The tests for MG, one of the most common causes of diplopia, were explained earlier (see p. 65). Diplopia can occur in a lesion involving one or more of the three cranial nerves (oculomotor, trochlear, abducens) or those nuclei or fiber pathways within the brainstem.

In relation to this, some patients with a lesion in the occipital lobe may complain that a single object looks like more than one to them. In this case, it is not just double vision but more often multiple vision (**polyopia**), and the degree of polyopia does not change by moving eyes or by changing the position of the object. Furthermore, it is often associated with disturbance of other visual cognitive functions (see Chapter 24-1K, p. 237). Therefore, this condition

BOX 9-5 MILLER FISHER SYNDROME

A combination of ophthalmoplegia, trunk ataxia, and loss of tendon reflexes of acute onset following the symptoms of upper respiratory or gastrointestinal infection is called Miller Fisher syndrome, named for C. Miller Fisher. Despite severe involvement of the oculomotor nerve, trochlear nerve, and abducens nerve, consciousness is preserved and there is no evidence of long tract signs, which suggests extramedullary, inflammatory involvement of those cranial nerves. In this condition, levels of the antibody against GQ1b are often elevated in the serum (Kusunoki et al, 2008). Some patients may have a combination of clinical features of Guillain-Barré syndrome, brainstem encephalitis of Bickerstaff, and Miller Fisher syndrome. Those cases might be regarded as a syndrome of anti-GQ1b antibody.

can be easily distinguished from diplopia caused by a lesion of the final common pathway.

iv. Ocular myopathy

Ocular myopathy associated with thyrotoxicosis used to be called **thyrotoxic myopathy**, but this condition may not be directly related to the thyroid hormone. It is more likely that the extraocular muscles and the thyroid gland happen to be affected independently by an autoimmune process. Temporally, therefore, these two conditions do not always occur or change in parallel.

A group of conditions due to abnormality of mitochondria-related genes and manifesting progressive external ophthalmoplegia in young adults is called **ophthalmoplegia plus** or **progressive external ophthalmoplegia** (Copeland, 2014) (see Box 19-13, p. 181). The most representative form of this condition is **Kearns-Sayre syndrome**, which is characterized by age at onset before 20 years, progressive external ophthalmoplegia, retinal pigment degeneration, cardiac conduction block, cerebellar ataxia, and increased protein in the cerebrospinal fluid. It is usually sporadic in its occurrence and is due to deletion of mitochondrial DNA (mtDNA). Another condition in this group is **mitochondrial neurogastrointestinal encephalomyopathy** (MNGIE), which is characterized by progressive external ophthalmoplegia, motor and sensory polyneuropathy, chronic diarrhea, recurrent intestinal obstruction, and malnutrition (Filosto et al, 2018). This condition is inherited via an autosomal recessive trait and is due to multiple deletion of mtDNA.

C. EXAMINATION OF FIXATION AND GAZE

After having confirmed that the extraocular muscles are intact, or at least not markedly paralyzed, it is possible to test the gaze and eye movements. First, the primary position of the eyes is observed while the patient is looking straight ahead. If the eyes are unstable, associated with small rapid horizontal movements, the electronystagmogram (ENG) may show small square waves. This phenomenon, called **square-wave jerks**, is seen in patients with cerebellar lesions. Then, to test fixation, the patient is asked to look at an object placed in the central visual field and saccade to small shifts of the target. If the eyes miss the target and show several small oscillations before fixating on the target, it is called **ocular dysmetria**. In this case, the ENG shows a short burst of several rapid decrescendo beats before catching the target. **Flutter-like oscillations** may look like ocular dysmetria, but in this condition there is no decline of the ENG amplitude in the burst.

i. Lateral gaze palsy and conjugate deviation of eyes

Deviation of the two eyes to either left or right to equal angle is called **conjugate deviation of eyes**. This is caused by a lesion involving either the cerebral hemisphere, especially the frontal lobe, ipsilateral to the side of deviation or the pontine tegmentum contralateral to the side of deviation. In the former case, it is usually seen in association with disturbance of consciousness, as in acute stroke. If the head is also rotated to the same direction as the eyes, it is usually due to a lesion of the parieto-occipital lobe ipsilateral to the side of deviation. This condition is associated with either homonymous hemianopsia or hemispatial neglect on the side contralateral to the lesion. As visual spatial attention is mainly controlled by the right cerebral hemisphere in right-handed subjects, this conjugate deviation of eyes and head is commonly seen in patients with right hemispheric lesions. Upward or downward deviation of the eyes suggests the presence of downward gaze palsy or upward gaze palsy, respectively.

When there is lateral or vertical gaze palsy, it is important to examine the presence or absence of **oculocephalic reflex**. For this purpose, the head is passively rotated, and if the eyes change their relative position toward the opposite side of passive rotation, then the oculocephalic reflex is judged to be present. The presence of the oculocephalic reflex when there is a gaze palsy suggests **supranuclear paralysis**, and its absence suggests nuclear or infranuclear paralysis. In the latter case, the lateral gaze center in the pons (PPRF) or the vertical gaze center in the midbrain is directly involved by

a lesion, and in the former case, the lesion is present central to the gaze centers. Progressive supranuclear palsy (see Box 22-2, p. 213) is a typical example of supranuclear gaze palsy. The oculocephalic reflex is also called **doll's eye sign**, and it is an important test for examining a patient with possible brain death (see Chapter 33-7, p. 295).

ii. Oculogyric crisis

Paroxysmal, conjugate, tonic upward, or oblique upward deviation of the eyes is called **oculogyric crisis**. Oculogyric crisis was first described in patients with postencephalitic parkinsonism. Nowadays this phenomenon can be seen in a residual state of acute encephalitis or in the chronic phase of anoxic encephalopathy. It is also seen as complications of dopamine-receptor blocking medications (Slow & Lang, 2017). Similar phenomena might be encountered during epileptic attacks, but the diagnosis of epileptic attacks can be easily made by taking into account other manifestations associated with epilepsy (see Chapter 25-1, p. 244).

In healthy subjects, closure of eyelids is accompanied by upward rotation (elevation) of eyes, which is called the **Bell phenomenon**. In patients with a unilateral cerebral hemispheric lesion, elevation of the eyes produced by strong eyelid closure is deviated to either left or right. In 70% of cases, the eyes are deviated away from the side of the hemispheric lesion (**Cogan sign**), and in 20% of cases, they are deviated toward the side of the lesion (Sullivan et al, 1991).

iii. Gaze paralysis and ocular motor apraxia

Examination of gaze can be done while examining the extraocular muscles, but as patients with intact extraocular muscles may still have gaze disturbance, it is better to pay attention to the extraocular muscles and gaze separately. Gaze can be tested, with the eyes placed in the primary position, by asking the patient to look at either side or up and down without showing any visual target. If a patient is unable to look at any direction, it may be due to either gaze palsy or disturbance of even higher cortical functions. To distinguish these two conditions, it is useful to observe spontaneous eye movements while the patient is talking, for example. In **ocular motor apraxia**, eye movements may occur randomly in all directions in the spontaneous condition, but once the patient is asked to fixate on an object placed in the lateral visual field, then he or she is unable to move the eyes to that direction or finds it difficult to do so. In this case, the head first rotates promptly and overshoots the target, and then it returns so that both the head and eyes

catch the object. This phenomenon is called **head thrust**. Actually, both ocular motor apraxia and supranuclear gaze palsy are overcome by the vestibulo-ocular reflex (Schweyer et al, 2018).

In **gaze palsy**, in contrast to the ocular motor apraxia, eye movements to the paralyzed direction are absent even in the spontaneous condition. In fact, ocular motor apraxia is much less common than gaze palsy. Gaze palsy is divided into two types: supranuclear and infranuclear. The infranuclear gaze palsy is caused by a lesion involving the gaze center in the midbrain or the pons or its efferent nerve fibers innervating the extraocular muscles. In contrast to the other skeletal muscles, neither the muscle volume nor the tendon reflex can be examined in gaze palsy. Therefore, the oculocephalic reflex (see p. 68) and other accompanying signs provide important clues to distinguishing supranuclear and infranuclear ophthalmoplegia. The presence of the oculocephalic reflex (doll's eye sign) suggests supranuclear palsy. In incomplete gaze paresis or during the recovery phase from the acute stage, gaze may be no longer paralyzed, but it may manifest as a slow speed of saccades. When testing the gaze, involuntary movements of eyes (to be described in the next section) can be also observed, if any.

iv. Eye-tracking test

It is important to test saccades and smooth pursuit separately for the horizontal as well as vertical gaze. To test the horizontal saccades, the examiner sits about 50 cm in front of the patient and holds visual targets (usually the examiner's hands) horizontally with the visual angle of about 30 degrees while the patient is requested to fixate at each target alternately. By this maneuver, approximate information is obtained as to the speed of saccades and how accurately the eyes can fixate on the target. To obtain more detailed information, the patient is asked to look at a target placed in the central visual field and to promptly look at another target placed in the lateral field. If the eyes cannot fixate their gaze to the target accurately, it is called **dysmetria**. There are two types of dysmetria. In **hypermetria** or **hypermetric saccades**, the eyes overshoot and then return to the target, which is considered to be characteristic of a cerebellar lesion, especially one involving the dorsal vermis and fastigial nucleus (Milenkovic et al, 2021). In **hypometria** or **hypometric saccades**, the eyes stop before reaching the target and then proceed further to catch the target. This condition is less lesion-specific than hypermetria and is seen in disorders of the cerebellum or basal ganglia, or the supranuclear ocular motor system.

Smooth pursuit can be tested by moving a visual target smoothly over the visual angle of about 30 degrees and by asking the patient to follow the moving target. If small saccades are superimposed on the smooth movement, the phenomenon is called **saccadic pursuit** or **catch-up saccades**. This phenomenon is also seen in cerebellar lesions, but its specificity to any site or any nervous system is relatively low.

v. Optokinetic nystagmus

Optokinetic nystagmus (OKN) is produced by asking patients to watch a tape with a series of alternating stripes (or other objects). The tape can be moved horizontally or vertically. The eyes should move at the speed of the tape in the direction of the tape and then have periodic saccades in the opposite direction. OKN is a nice way of assessing whether saccades are present and symmetric.

6. INVOLUNTARY MOVEMENTS OF EYES

There are several different kinds of involuntary movements involving the eyes (Table 9-1). The most common one is nystagmus. In most cases of nystagmus, both eyes move to the same direction to the same degree and with the same speed (**conjugate nystagmus**). An exception to this is the dissociated nystagmus seen in internuclear ophthalmoplegia (see p. 62) and seesaw nystagmus. Nystagmus may be seen in the primary position of the eyes (**spontaneous nystagmus**), but it is often induced or increased by gaze (**gaze nystagmus**). Depending on the direction of involuntary movements, it is classified into **horizontal nystagmus**, **vertical nystagmus**, and **rotatory nystagmus**. Patients having nystagmus often complain of **oscillopsia**, but they may not

complain of any symptoms. To diagnose ocular oscillations accurately, it is important to take a careful history and to perform a systematic examination of vision and each functional class of eye movement (Thurtell & Leigh, 2011).

A. SPONTANEOUS NYSTAGMUS

Nystagmus is commonly formed by a quick phase and a slow phase, but in some cases of nystagmus the eyes may move at the same speed to both directions. This is called **pendular nystagmus** and is commonly seen in patients with congenital nystagmus. **Congenital nystagmus** is characterized by horizontal pendular nystagmus of large amplitude, its suppression by eyelid closure, and a reversed direction of OKN. As noted earlier, in healthy subjects, the quick phase of OKN is induced in the direction away from the side of the target movement, but in congenital nystagmus, it is induced toward the side of target movement. Moreover, congenital nystagmus is often associated with amblyopia from childhood, and the patient may not be aware of any oscillopsia or unsteadiness despite the presence of conspicuous nystagmus. Suppression of congenital nystagmus by eyelid closure may not be caused by masking of visual stimulus but may be related to the voluntary attempt to close the eyes (Shibasaki et al, 1978).

In some cases of spontaneous nystagmus, its quick phase might change its direction from one to the other periodically. This phenomenon is called **periodic alternating nystagmus** (PAN) and is typically seen in a lesion of the cerebellum, especially the nodulus. PAN was reported in a patient on long-term lithium medication (Rust et al, 2016).

B. GAZE NYSTAGMUS

Horizontal nystagmus is commonly induced or enhanced by lateral gaze. The quick phase is induced toward the side of gaze in most cases, but in some cases it may be to the opposite direction. Lateral gaze may also induce vertical nystagmus or rotatory nystagmus. In patients with cerebellar lesions, the quick phase of nystagmus induced by lateral gaze may change its direction when the eyes return to the primary position. For example, the gaze nystagmus that is induced by lateral gaze to the right with its quick phase directed to the right changes its quick phase to the left as soon as the eyes are returned to the primary position. This phenomenon is called **rebound nystagmus** and is seen in a lesion of the cerebellum, especially the flocculus and paraflocculus.

TABLE 9-1 INVOLUNTARY MOVEMENTS OF EYES

Primary position	On fixation	On gaze
Square-wave jerks	Ocular dysmetria	Opsoclonus
Spontaneous nystagmus		Ocular flutter
Ocular tremor		Gaze nystagmus
Ocular bobbing		Periodic alternating nystagmus
Ocular dipping		Rebound nystagmus
		Dissociated nystagmus

Vertical gaze nystagmus is commonly induced to the direction of vertical gaze, but it may be induced to the reverse direction. Namely, upward gaze commonly induces **up-beat nystagmus**, but it may induce **down-beat nystagmus** and vice versa. Down-beat nystagmus is seen in patients with Arnold–Chiari malformation, paraneoplastic syndrome, and intoxication with lithium, carbamazepine, or amiodarone. This is also believed to be caused by involvement of the flocculus and parafocculus. **Arnold–Chiari malformation (Chiari malformation)** is a congenital condition in which the cerebellar tonsils, and in severe cases the inferior vermis and lower medulla as well, are displaced below the foramen magnum.

In patients with the **sylian aqueduct syndrome**, **convergence–retraction nystagmus** may be induced by vertical gaze, especially by upward saccade provoked by downward movement of the OKN stimulus. In this condition, pupillary abnormality may be also seen.

Superior oblique myokymia is a burst of rotatory pendular rapid small movements seen in one eye (Joosten & Hoeijmakers, 2018). This condition can be only recognized with a slip-lamp examination because each movement is quite small. The patient with this condition may complain of rotatory oscillopsia when looking at an object monocularly.

C. OCULAR MYOCLONUS

Rhythmic vertical oscillation of the eyes at a frequency of about 3 Hz used to be called **ocular myoclonus**, but its mechanism seems to be similar to that of tremor. In most cases, the ocular movements are associated with synchronous vertical movements of the soft palate. The latter phenomenon used to be called palatal myoclonus but is now more often called **palatal tremor**. **Symptomatic oculopalatal tremor** is caused by a lesion involving any part of the Guillain–Mollaret triangle, but most commonly by a lesion of the central tegmental tract in the pons, and then by a cerebellar lesion (see Box 19-16, p. 182).

D. OPSOCLONUS

Opsoclonus is seen in the spontaneous eye position and is characterized by irregular gross quick conjugate movements of the eyes to various directions. Each movement is shock-like and resembles myoclonus of the extremities (Figure 9-9). In fact, opsoclonus is often associated with myoclonus in the trunk or extremities. However, unlike myoclonus of cortical origin, opsoclonus is not preceded by electroencephalogram (EEG) spikes. It is postulated that the saccadic burst neurons in the brainstem gaze centers generate bursts of discharges as a result of a cerebellar lesion.

Saccadic lateral gaze to right by 20 degrees

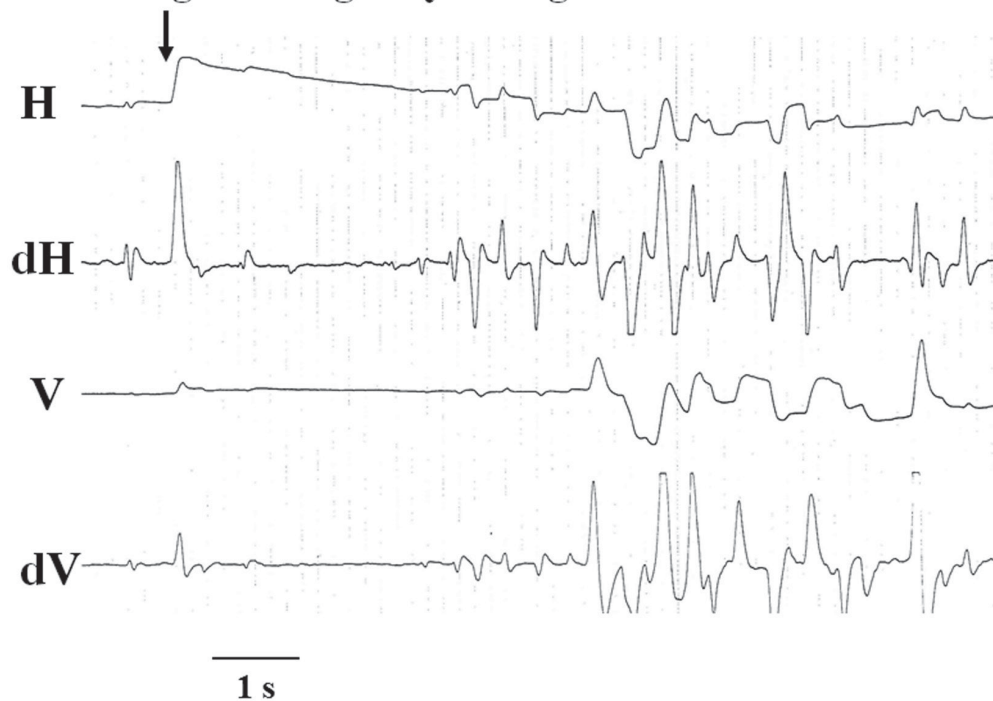


Figure 9-9 Electronystagmogram of opsoclonus. H: original waveform (horizontal), V: original waveform (vertical), dH: differential waveform (horizontal), dV: differential waveform (vertical).

BOX 9-6 COMPREHENSIVE EXAMINATION OF EYE MOVEMENTS

Examine eyes at rest.

Look for involuntary movements such as square-wave jerks.

Look for alignment of the eyes.

Include cover/uncover test for strabismus.

Examine for convergence (can check for pupillary constriction at same time).

Examine smooth pursuit, horizontal and vertical, including looking for movement paresis or paralysis.
nystagmus at end gaze
Add oculocephalic reflex if appropriate.

Examine saccadic movements, horizontal and vertical.
Look for speed and accuracy.

Examine optokinetic nystagmus, horizontal and vertical

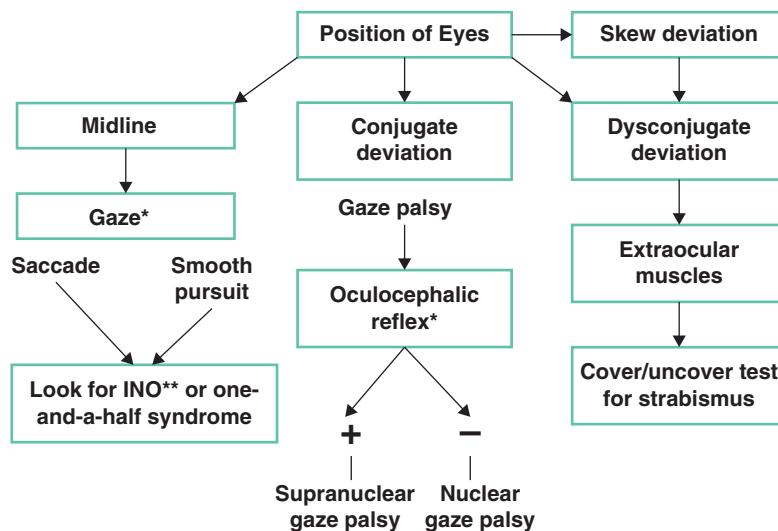
Opsoclonus–myoclonus syndrome is seen in children with cerebellar neuroblastoma and is also seen in adults with autoimmune or infectious cerebellitis. It is also known to occur as a paraneoplastic syndrome. Opsoclonus has also been reported in cases of neuroborreliosis due to infection with *Borrelia burgdorferi*.

In some cases, opsoclonus may need to be distinguished from ocular flutter. **Ocular flutter**, also called **flutter-like oscillation**, is a short burst of horizontal rhythmic quick movements and is enhanced by gaze. It is distinguished from opsoclonus in that flutter is a burst of rhythmic beats whereas opsoclonus is irregular, but the two conditions can be seen in the same patient at the same time or at different stages of the clinical course.

E. OCULAR BOBBING AND OCULAR DIPPING

Quick sinking of both eyes followed by slow return to the primary position is called **ocular bobbing**, and slow sinking of both eyes followed by quick return to the primary

Examination of Eye Movements



* Horizontal and Vertical ** Internuclear ophthalmoplegia

Figure 9-10 Flow chart illustrating the examination eye movements. First, the position of eyes should be examined as to whether there is any conjugate or dysconjugate deviation of eyes. The presence of dysconjugate deviation suggests paresis of extraocular muscles or strabismus. The presence of conjugate deviation suggests gaze palsy, and the question as to whether it is nuclear or supranuclear should be answered by examining the oculocephalic reflex. In the absence of deviation, the saccade and smooth pursuit should be examined by the eye-tracking test in the horizontal and vertical directions. Possible involuntary movements should be kept in mind throughout those examinations.

position is called **ocular dipping**. Both conditions are seen in bilateral massive lesions of the pontine tegmentum.

7. COMPREHENSIVE EXAMINATION OF EYE MOVEMENTS

It is practical to consider two axes for the comprehensive examination of eye movements: extraocular muscles and gaze (Axis 1) and eye movements (Axis 2) (Box 9-6). For example, saccades and smooth pursuit can be fully examined only when there is no extraocular muscle palsy or no gaze palsy. Therefore, it is practical to first pay attention to whether there is any deviation of the eyes in the primary forward-looking position. If there is any dysconjugate deviation, it suggests the presence of extraocular muscle palsy or skew deviation (Figure 9-10). When one eye is rotated upward while the other eye is rotated downward, producing a diagonal position, it suggests the presence of skew deviation (refer to p. 67). By contrast, conjugate deviation of the eyes suggests the presence of gaze palsy, either horizontal or vertical. Then the question as to whether the gaze palsy is nuclear or supranuclear can be addressed by checking the oculocephalic reflex (refer to Chapter 9-2). When each eye is on the midline position, saccade and smooth pursuit should be examined separately (refer to Chapter 9-4). Internuclear ophthalmoplegia or one-and-a-half syndrome may be demonstrated by examining saccades or smooth pursuit, even if there is no deviation in the primary position. While performing these steps, the presence or absence of ocular involuntary movements should be always kept in mind, because involuntary movements might occur during any of these examinations.

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TRIGEMINAL NERVE

The fifth cranial nerve (trigeminal nerve) innervates facial sensation and jaw movement. Anatomically the nerve consists of three branches: the first branch (ophthalmic nerve), the second branch (maxillary nerve), and the third branch (mandibular nerve).

1. STRUCTURES AND FUNCTIONS

A. SOMATOSENSORY PATHWAY

Sensory nerves of the trigeminal nerve innervate the skin of the face and forehead, a part of the oral mucosa, and a large part of the meninges. The primary neurons are in the **semilunar ganglion (gasserian ganglion)**, which is located in the middle cranial fossa (Figure 10-1; see also Figure 8-2a, p. 50).

The **ophthalmic nerve** transmits sensory impulses from the skin of the frontal scalp (supraorbital nerve) and

the nasal ala. The nerve fibers arising from the sensory receptors of those areas pass posteriorly below the superior wall of the orbit and, together with the fibers coming from the orbital mucosa, exit from the orbit through the superior orbital fissure (see Figure 8-2a, p. 50) to enter the anterior cranial fossa. Then they pass further posteriorly through the cavernous sinus (see Figure 8-1, p. 49) to enter the middle cranial fossa, where they project to the semilunar ganglion. Intracranially, the ophthalmic nerve transmits nociceptive impulses from the meninges of the anterior cranial fossa and the cerebellar tent. In relation to this tentorial branch, the frontal headache occasionally complained of by a patient with a posterior fossa lesion is explained as a referred pain. The patient feels as if the pain is coming from the frontal region because the tentorial branch enters the ophthalmic nerve.

The **maxillary nerve** transmits sensory impulses from the skin of the upper lip and cheek, and from the

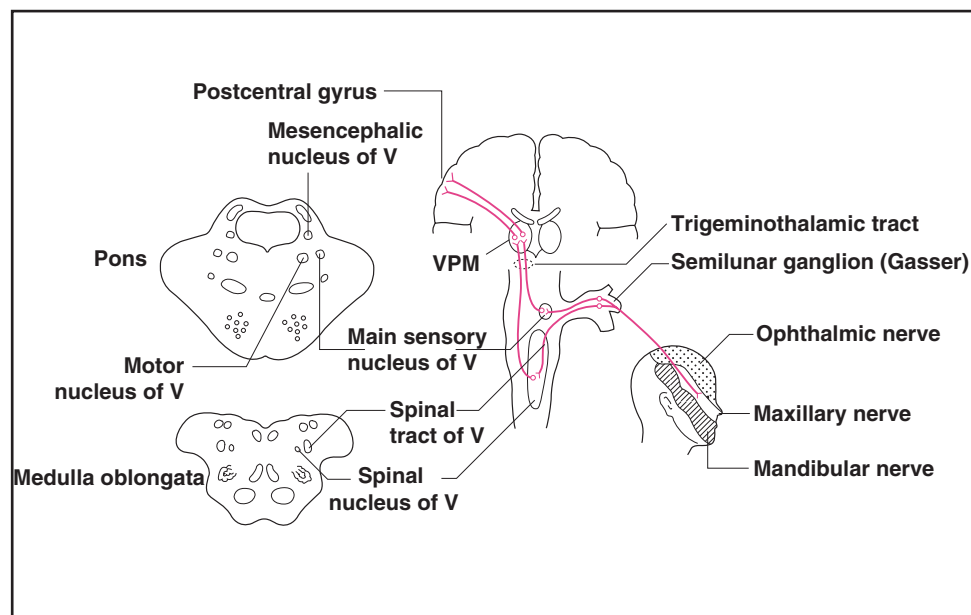


Figure 10-1 Schematic illustration of the composition of the trigeminal nerve and its branches.

mucosa of the nose and the upper oral cavity. It enters the middle cranial fossa through the foramen rotundum (see Figure 8-2a, p. 50), and after being joined by the nociceptive input from the middle cranial fossa, it projects to the semilunar ganglion. The **mandibular nerve** transmits sensory impulses from the skin of the jaw and temple and the mucosa of the lower oral cavity. It enters the middle cranial fossa through the foramen ovale (see Figure 8-2a) and projects to the semilunar ganglion.

The primary sensory neuron of the trigeminal nerve, like other primary somatosensory neurons, is a bipolar cell (see Figure 20-1, p. 192). The impulse generated at the peripheral receptors is conducted through the peripheral axon to the semilunar ganglion and enters the lateral pons through its central axon. Soon after entering the pontine tegmentum, the fibers separate into two groups, as follows.

Small nociceptive fibers form the **spinal tract of the trigeminal nerve** and descend through the lateral tegmentum of the pons and medulla oblongata to reach the upper cervical segments (see Figures 5-2, 5-3, and 5-5, p. 32, 35). As the impulses descend through the spinal tract, they are transmitted to the secondary neurons at each level in the **spinal nucleus of the trigeminal nerve**, which is located just medial to the spinal tract (see Figure 10-1). The secondary sensory fibers cross to the other side at each level and join the medial part of the contralateral spinothalamic tract to form the **trigeminothalamic tract**. This tract ascends to reach the **ventral posteromedial nucleus (VPM) of the thalamus**, where the third sensory neurons originate. As described in Chapter 5-6 (p. 34), the spinal nucleus of the trigeminal nerve extends from the middle pons to the upper cervical segments, and thus is the longest nucleus among the cranial nerve nuclei.

Large sensory fibers related to tactile sensation enter the **main sensory nucleus of the trigeminal nerve**, which is located in the middle part of the pontine tegmentum (see Figure 10-1). The secondary fibers originating from the main sensory nucleus cross to the other side to join the medial lemniscus and project to the VPM of the thalamus.

As expected from the previous discussion, if one of the three branches is affected, the patient suffers from superficial sensory loss and/or complains of pain or numbness localized to the area of the skin and mucosa innervated by that branch. If a lesion involves the main trunk of the trigeminal nerve (central to the semilunar ganglion), then the sensory impairment occurs in the entire area innervated by the whole nerve.

An extramedullary lesion of the trigeminal nerve usually affects both nociceptive and tactile sensation. As for

intramedullary lesions, a lesion located at the entrance area is expected to disturb all kinds of sensation, but after the fibers are separated into the nociceptive group and the tactile group, either of the two senses might be selectively affected (**dissociated sensory loss**). Furthermore, as the spinal nucleus of the trigeminal tract extends for a long distance, the sensory impairment caused by the lesion may not involve the whole face. However, correlation between the site of lesion and the distribution of sensory impairment has not been clearly established.

In lateral medullary syndrome (Wallenberg syndrome) (Box 14-1, p. 97), the spinal tract is damaged at the upper medullary level, but the pain and temperature senses are usually lost over the whole area of the ipsilateral face. On the other hand, it is expected theoretically that a lesion of the upper cervical cord might cause sensory symptoms on the face, but this phenomenon is extremely rare.

B. CORTICAL RECEPTION OF SOMATOSENSORY INPUT FROM FACE

The third sensory fibers originating from the VPM of thalamus ascend through the internal capsule and the corona radiata and project to the face area of the **primary somatosensory cortex (S1)** in the postcentral gyrus. In the postcentral gyrus, the receptive area of impulses arising from the face is located over the lateral convexity whereas that from the foot is located over the mesial area, forming an upside-down homunculus (Figure 10-2). Regarding specific parts of the face, however, there is no upside-down somatotopy, and the area for the eyes is located dorsal to that for the jaw. Functional localization in the S1 will be described in the section of Chapter 20 on the somatosensory system (Chapter 20-1) (see Figure 20-3, p. 194).

Nerve fibers originating from the S1 project bilaterally to the **secondary somatosensory cortex (S2)** located in the superior bank of the sylvian fissure, and also to the ipsilateral areas 5 and 7 located posterior to the S1. There is some somatotopic organization also in the S2, with the face area being located more anteriorly and the foot area more posteriorly.

C. NERVE INNERVATION OF MASTICATORY MUSCLES

The **motor nucleus of the trigeminal nerve** is located in the middle of the pontine tegmentum just medial to the main sensory nucleus (Figures 10-1, 10-3). Its efferent fibers exit the pons from its lateral rim and pass through the semilunar

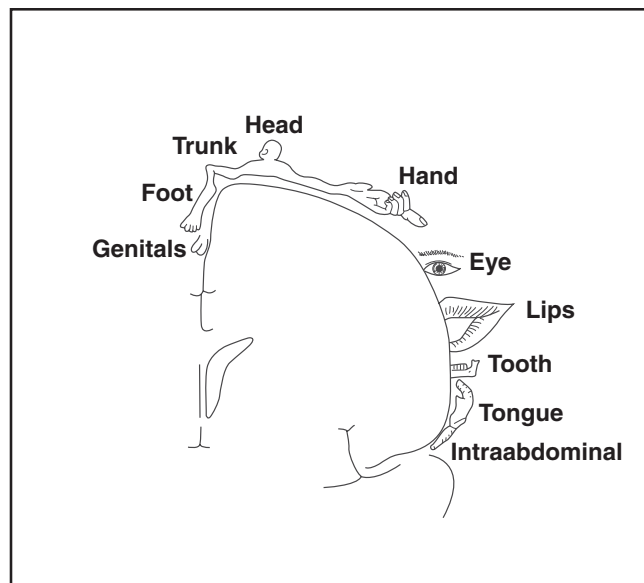


Figure 10-2 Schematic illustration of somatotopic organization of the primary somatosensory cortex. Frontal view of the left hemisphere.

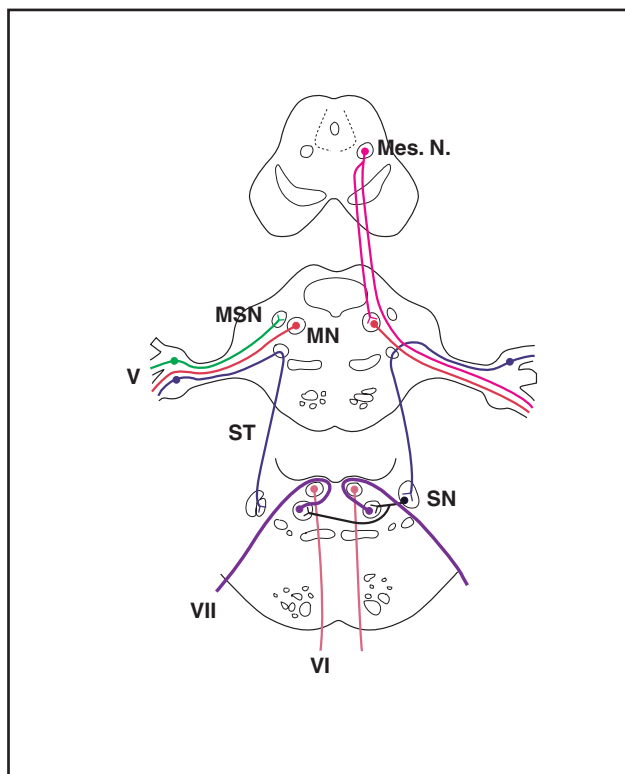


Figure 10-3 Schematic diagram showing nuclei of the trigeminal nerve (V) and neuronal circuits of the corneal reflex and the masseter reflex (jaw jerk). The polysynaptic reflex arc of the corneal reflex is shown in purple with the spinal tract (ST) projecting to the motor nucleus of the facial nerve (VII) via the spinal nucleus (SN). The monosynaptic reflex arc of the masseter reflex is shown in pink with the mesencephalic nucleus (Mes.N.) sending its efferent nerve to the ipsilateral motor nucleus of the trigeminal nerve (MN). MSN: main sensory nucleus, VI: abducens nerve.

ganglion to form the mandibular nerve. After exiting from the middle cranial fossa through the foramen ovale, the motor fibers separate into three branches to innervate the temporal muscle, the masseter, and the pterygoid muscles. In a lesion of the mandibular nerve, atrophy and weakness can be seen or palpated in the temporal muscle and masseter. As the lateral pterygoid muscle acts to open the mouth by protruding the mandible forward and downward, its unilateral paralysis pushes the jaw to the paralyzed side when opening the mouth, causing deviation of the jaw to the affected side. As the lateral pterygoid muscle also acts to push the jaw to the other side, its weakness can be felt by palpation by asking the patient to push the jaw to the side contralateral to the weakened muscle. As the motor nucleus of the trigeminal nerve is bilaterally innervated by the corticobulbar tract, a unilateral lesion of the corticobulbar tract does not cause paralysis of the masticatory muscles.

The motor fibers of the mandibular nerve also innervate the tensor tympani muscle. As this muscle acts to control the tension of the tympanic membrane, its paralysis is theoretically considered to decrease the tension of the tympanic membrane. However, hearing deficit due to a trigeminal nerve lesion is not encountered in the clinical setting.

D. REFLEX PATHWAY OF JAW JERK

As the masseter is bilaterally innervated by the corticobulbar tracts (as already mentioned), its tendon reflex is not enhanced with a unilateral lesion. The **masseter reflex**

(**jaw jerk**) is elicited by tapping the jaw downward with the mouth gently opened. Its afferent neuron is in the **mesencephalic nucleus of the trigeminal nerve**, which is located in the midbrain (see Figures 10-1, 10-3). Unlike other tendon reflexes, which are monosynaptic with the afferent neuron situated in the sensory ganglion, the afferent neuron of this particular monosynaptic reflex is situated in the brainstem. An afferent impulse generated by stretch of the masseter muscle spindle is conducted through the afferent axon of the mandibular nerve and is projected to the bipolar neuron in the mesencephalic nucleus, where the impulse is reflected through its efferent axon and monosynaptically transmitted to the ipsilateral motor nucleus of the trigeminal nerve, which then causes contraction of the masseter.

In patients with dysphagia and dysphonia or dysarthria, the jaw jerk is used to distinguish between bulbar palsy and pseudobulbar palsy (see Chapter 14-7, p. 103, for definition), because it is impossible to test the tendon reflex in the bulbar muscles such as the soft palate and tongue (see Chapter 14-1, p. 97). Although the reflex center is located in the pons, enhancement of jaw jerk in those patients suggests pseudobulbar palsy due to bilateral lesions of the corticobulbar tracts. However, in a condition in which upper and lower motor neurons are both affected, as in amyotrophic lateral sclerosis, the jaw jerk might be enhanced with the presence of muscle atrophy of the tongue.

E. NEURAL PATHWAY OF CORNEAL REFLEX

The **corneal reflex** is a polysynaptic reflex with the trigeminal nerve forming its afferent arc and the facial nerve forming its efferent arc (see Chapter 11-1A, p. 83). An impulse generated in the nociceptive receptors of the cornea is conducted through the ophthalmic nerve to the spinal tract of the trigeminal nerve (see p. 80). Then an impulse arising from the spinal nucleus of the trigeminal nerve is conducted to the ipsilateral as well as contralateral facial nucleus (see Figure 10-3). Thus, a nociceptive stimulus given to the cornea of one eye causes eyelid closure bilaterally.

The blink reflex, a reflex closely associated with the corneal reflex, can be tested electrophysiologically by presenting an electric shock to the ophthalmic nerve (usually the supraorbital branch) and by recording the surface electromyogram (EMG) from the orbicularis oculi muscles (**blink reflex**). The initial response (R1), oligosynaptic in nature, is recorded from the muscle ipsilateral to the stimulus at a latency of about 10 ms, and the second response (R2) is recorded bilaterally at a latency of about 30 ms (see

Kimura, 2013). As only R2 can be recognized on clinical observation, the corneal reflex is seen bilaterally in healthy subjects.

2. EXAMINATION OF FUNCTIONS RELATED TO THE TRIGEMINAL NERVE

A. MOTOR NERVE

First, the volume of the masticatory muscles (masseter and temporal) should be observed by visual inspection and palpation. In the case of atrophy of these muscles, the zygomatic bone of the affected side appears protruded. Even when the muscle atrophy is not visible, palpation of those muscles while clenching the teeth might disclose the atrophy on the affected side.

Next, the position of the jaw can be observed while the patient opens the mouth. In unilateral paralysis of the lateral pterygoid muscle, the jaw is deviated to the affected side upon the mouth opening because the muscle of the intact side pushes the jaw forward and downward. Furthermore, palpation of the jaw on both sides while the patient tries to shift the jaw laterally might disclose weakness of the lateral pterygoid muscle contralateral to the direction of shifting.

The temporal muscle and masseter are commonly affected in **myotonic dystrophy**. Because these muscles are bilaterally affected in this condition, the zygomatic bones appear bilaterally protruded (**hatchet face**). Early frontal baldness is also commonly seen in this condition (see Chapter 17-6, p. 152).

i. Trismus (lockjaw)

When a patient complains of difficulty in opening the mouth, it could be the initial symptom of **tetanus**. Therefore, **trismus (lockjaw)** of acute onset should be considered with caution. Tetanus is caused by a toxin produced by the anaerobic bacteria *Clostridium tetani*. Infection occurs when the toxin produced at an injury site is transported through the motor nerve axon to the spinal cord and brainstem. There, inhibition is blocked by the toxin, leading to excessive activity of the motor neuron and severe spasm of the corresponding muscles. The motor nucleus of the trigeminal nerve is commonly affected in this condition, resulting in trismus. The trismus may continue and look like a smiling face (**risus sardonicus**). In the patient with tetanus, any external stimulus causes generalized muscle spasm and **opisthotonos**, an abnormal posture in which

the trunk is markedly bent backward due to excessive spasm of the spinal muscles. It could be fatal and requires intensive treatment (see Chapter 33-4B, p. 289).

Difficulty in jaw opening can be caused by local inflammatory as well as non-inflammatory diseases of the temporomandibular joints and peritonsillar or retropharyngeal abscess, but tetanus is the most important differential diagnosis from a neurological point of view.

ii. Jaw-winking phenomenon

In this condition, chewing movements are associated with opening and closure of one eye (**Marcus Gunn phenomenon**) (Box 10-1). Marcus Gunn in 1883 reported a 15-year-old girl who had drop of the left upper eyelid that quickly elevated when she opened the mouth or moved her jaw to the right (Marcus Gunn, 1883). The left upper eyelid remained elevated as long as she kept the new position of the jaw. However, voluntary opening of the left eye did not cause either opening of the mouth or deviation of the jaw.

BOX 10-1 MARCUS GUNN PHENOMENON

Regarding the hypothesis of possible misconnection between the motor fibers of the mandibular nerve targeting the lateral pterygoid muscle and those of the oculomotor nerve targeting the levator muscle of the upper eyelid, these two nerves or their nuclei are never located in any proximity. Moreover, if that theory is correct, voluntary elevation of the upper eyelid is also expected to open the mouth—which is not the case.

In view of the fact that the mesencephalic nucleus is the only component of the trigeminal nerve located in the mid-brain, Shimono et al. (1975) studied possible participation of the proprioceptive afferents from either the masseter or the lateral pterygoid muscle (see p. 78). Namely, the isometric contraction of the lateral pterygoid muscle induced by asking the patient to open the mouth while the jaw was fixed from below also elevated the upper eyelid. This finding was interpreted to exclude participation of the stretch afferent from the masseter and to suggest involvement of proprioceptive afferent from the lateral pterygoid muscle through the group Ib fibers. It was thus postulated that fibers of the mesencephalic tract of the trigeminal nerve and fibers of the oculomotor nerve targeting the levator muscle of the upper eyelid might be misconnected. However, a question remains as to why, among other fibers of the oculomotor nerve, only the nerve fibers targeting the levator muscle of the upper eyelid are affected.

This phenomenon was noted about 5 weeks after birth and slightly improved at age 12 years. Regarding the mechanism underlying this phenomenon, Marcus Gunn postulated the presence of hypoplasia in the motor fibers of the left oculomotor nerve targeting the levator muscle of upper eyelid, and its compensatory innervation by fibers of the left trigeminal nerve targeting the lateral pterygoid muscle, thus causing elevation of the upper eyelid upon voluntary contraction of the lateral pterygoid muscle.

The Marcus Gunn phenomenon can be either congenital or acquired, and familial occurrence has also been reported. Marin Amat, a Spanish ophthalmologist, reported in 1919, in addition to a typical case of Marcus Gunn phenomenon, a 56-year-old man with inverse Marcus Gunn phenomenon (**Marin Amat syndrome**) (Marin Amat, 1919). That patient suffered from severe trauma on the left face resulting in atrophy of the left optic disc and right facial palsy and deafness, and 45 days later he was treated with palpebral suture for the right lagophthalmos (failure to close eyelids) (see Chapter 11-2A, p. 86). About a year and half later, it was noted that opening the mouth or shifting the jaw to the left closed the right eye. The right eye remained closed as long as the right lateral pterygoid muscle was kept contracted, and it was also accompanied by lacrimation and nasal discharge on the right. It was postulated that the motor fibers of the mandibular nerve and those of the facial nerve might have been misconnected.

B. SENSORY NERVE

Impairment of the sensory fibers of the trigeminal nerve causes numbness of the face and the anterior half of the scalp on the affected side. Pain, temperature, and tactile senses can be tested by using sterile pins, flasks filled with hot or cold water, and tissue paper or cotton, respectively. It is important to compare the two sides for each test. If any sensory abnormality is found, it is important to know the affected area of the skin precisely, as to whether the abnormality is restricted to a region corresponding to any of the three branches or involves the whole unilateral face. It is useful to note that the ophthalmic nerve shares a border with the greater occipital nerve of cervical origin at the vertex of the head and that the mandibular nerve shares a border with the third cervical nerve at the chin. As the trigeminal nerve also innervates the mucosa of the oral cavity, including the tongue (namely the maxillary nerve for the upper oral mucosa and the mandibular nerve for the lower oral mucosa), it is important to pay attention to the oral mucosa as well, including the teeth. Numbness of either half of the tongue suggests impairment of the corresponding branch

of the trigeminal nerve. Toothache is felt through either the maxillary nerve or the mandibular nerve.

Facial-onset sensory and motor neuronopathy (FOSMN) is characterized by asymmetric facial numbness or paresthesia, bulbar palsy, and facial weakness. Recently an autopsied case of this syndrome was shown to have TDP43 pathology (Rossor et al, 2019).

Shock-like shooting pain in the face is known as **trigeminal neuralgia (TN, tic douloureux)**. A territory of the maxillary nerve is most commonly involved, but **post-herpetic neuralgia** is more common in the territory of the ophthalmic nerve. In neuralgia of the maxillary nerve, pain shoots up laterally from the maxilla toward the temple, and several beats of radiating pain tend to occur as a cluster. It is often triggered by stimulating the skin or mucosa (**trigger zone**). TN is mostly considered to result from compression of the trigeminal nerve by tortuous branches of the vertebralbasilar artery. This type of TN is classified as classical TN and is common in the aged population (Cruccu et al, 2016). It is postulated that the pain is not directly caused by the mechanical compression but is due to the secondary demyelination of the nerve trunk. However, in view of the fact that surgical decompression of the nerve can remove the pain immediately, demyelination may not be the sole mechanism.

TN may also occur in a tumor at the cerebellopontine angle or an intramedullary brainstem lesion like multiple sclerosis (secondary TN) (Cruccu et al, 2016). In fact, multiple sclerosis is one of the common causes of TN in young subjects. It has been reported based on a magnetic resonance imaging (MRI) study that, among patients with TN, an organic lesion is found in about 15% (Gronseth et al, 2008). The association of TN with Horner syndrome is known as **paratrigeminal syndrome (Raeder syndrome)** (Box 10-2).

Persistent idiopathic facial pain or atypical facial pain is a syndrome of constant facial pain without any neurological abnormality or clear explanation. It could be a disorder of central pain processing (Ziegeler et al, 2021).

C. JAW JERK (MASSETER REFLEX)

The jaw jerk is the only tendon reflex that can be tested for the muscles innervated by the cranial nerves. To elicit this reflex, the jaw is tapped downward while the mouth is kept gently open (see Figure 18-2a, p. 159). The response can be seen as jaw elevation or masseter contraction. This reflex may not be very conspicuous even in healthy subjects, and therefore loss of this reflex cannot be confirmed clinically.

Enhancement of this reflex suggests hyperexcitability of motor neurons in the motor nucleus of the trigeminal nerve in the pons as a result of a bilateral lesion of the

BOX 10-2 RAEDER (PARATRIGEMINAL) SYNDROME

Raeder in 1924 reported five cases of neuralgia or sensory disturbance in a territory of the trigeminal nerve associated with ipsilateral Horner syndrome (Raeder, 1924). Etiology was trauma in two cases, tumor of the middle cranial fossa in one, and undetermined in two. Motor fibers of the trigeminal nerve were also affected in one case, and optic neuropathy and external ophthalmoplegia were also seen each in two cases. The neuralgia most commonly affected the ophthalmic nerve. In all cases, responsible lesions were localized at or around the semilunar ganglion in the middle cranial fossa and extended to the internal carotid artery to involve the sympathetic nerve fibers (see Chapter 8-1, p. 49). Regarding Horner syndrome in those cases, only blepharoptosis and miosis were seen, but enophthalmos, conjunctival hyperemia, hypohidrosis (decreased sweating), and increased temperature of the facial skin were not present. This is considered to be due to the fact that the sympathetic fibers innervating those functions other than the eyelid elevation and pupillary dilatation bifurcate before the internal carotid artery enters the cranial fossa.

corticobulbar tracts or of impairment of the upper motor neurons innervating those motor nuclei. In patients with bulbar symptoms such as dysphagia and dysphonia/dysarthria, enhancement of jaw jerk serves as supplementary evidence for **pseudobulbar palsy** (see Chapter 14-7F, p. 204).

D. CORNEAL REFLEX

This reflex is generally tested only when it is judged necessary from the history and other neurological findings. To elicit this reflex, a twisted string of tissue paper or strand of cotton can be used to gently touch the cornea from the lateral visual field, and the response can be judged by observing closure of eyelids. Coming in from the lateral side, rather than from directly in front, reduces the chances of a visual response to the stimulus. Touching the sclera is not effective because there are no nociceptive receptors in the sclera. In patients who have too frequent blinks or who are unable to keep their eyes open, the eye to be stimulated is held open by the examiner and the response of the stimulus eye can be judged by observing contraction of the orbicularis oculi muscle in the other eye. The same technique can be used to examine comatose patients.

A similar reflex can be elicited by touching the cilia (eyelash) and may be called the ciliary reflex, but this term

is more properly used for the accommodation reflex of the ciliary muscle (see Chapter 8-1, p. 52). As the corneal reflex is more sensitive than the ciliary reflex, it should be tested even when the ciliary reflex is absent.

Loss of the corneal reflex suggests either impairment of the ophthalmic nerve of the stimulated side or bilateral paralysis of the orbicularis oculi muscles. When the corneal reflex is lost in one eye regardless of the side of stimulation, it suggests paralysis of the facial nerve on that side. In this case, spontaneous blinking is either incomplete or lost in the affected eye. Likewise, the blink reflex elicited by tapping the glabella is either lost or slower on the side of the affected facial nerve.

As the trigeminothalamic tract ascends through the pontine tegmentum along the lateral edge, it can be affected by extramedullary compression (see Figure 5-5). For example, in a large cerebellopontine angle tumor like acoustic neurinoma compressing the pons, loss of the corneal reflex is an important sign in addition to sensorineural deafness and cerebellar ataxia.

In some cases of degenerative diseases involving the basal ganglia, a blink reflex (glabellar tap reflex) may be exaggerated or may respond to repetitive stimuli continuously. This phenomenon is called the **Myerson sign**.

E. PRIMITIVE REFLEXES

In patients with cerebral hemispheric lesions, especially with bilateral frontal lobe lesions, stimuli given to the skin or mucosa innervated by the trigeminal nerve may elicit reflexes that are seen only in infants (primitive reflex) (Schott & Rossor, 2003). Gentle stroking of the upper lip will elicit a sucking movement in the lips (**sucking reflexes**), and gentle tapping of the upper lip will elicit pouting of the lips (**snout reflex**). Stimuli presented to the cornea might elicit contraction in the mental muscles or even might move the jaw (**corneo-mandibular reflex**). A patient with predominant right pontine hemorrhage who showed jaw deviation to the left in response to right corneal stimulation was reported (Caballero-Avila et al, 2019).

Other primitive reflexes may be elicited by stimulating the skin of the upper extremity. Stroking the thenar aspect of the palm might cause contraction in the mental muscles (**palmomental reflex**). In an extreme case, this reflex might be induced by stroking the forearm. The apparently curious distant response to palmar stimulation is due to the proximity of these body parts in the primary sensory cortex. The frequency of a palmomental response in normal persons means that the presence of the reflex in isolation cannot be considered abnormal. **Forced grasping**

BOX 10-3 FORCED GRASPING

In infants, stroking the palm or sole toward the tips causes flexion of the fingers or toes, which is called **palmar grasp** and **plantar grasp**, respectively. The grasp reflex disappears in adults but may reappear in special pathologic conditions. **Forced grasping** is considered to be an exaggerated state of this grasp reflex, but it might be overlooked unless specifically examined. In forced grasping of severe degree, the patient might forcefully grasp the caregiver's hand or object and cannot let it go, which will cause significant disability in daily living (see Schott & Rossor, 2003, and Mestre & Lang, 2010, for reviews). This phenomenon is especially common in neurodegenerative diseases presenting with parkinsonism and dementia, such as corticobasal degeneration and progressive supranuclear palsy (see Mestre & Lang, 2010, for a review). Forced grasping might be also seen in vascular lesions of the frontal lobe, in the hand contralateral to the lesion. The structures of the frontal lobe that are important for this phenomenon have not been precisely elucidated, but the supplementary motor area and the cingulate gyrus contralateral to the grasp have been incriminated. Forced grasping might be also seen in response to visual stimulus (**forced groping**).

(Box 10-3) and **utilization behavior** (see Chapter 24-1F, p. 235) are also regarded as primitive reflexes. Primitive reflexes may be also induced by a visual stimulus. For example, reaching out the examiner's hand toward the patient's

BOX 10-4 HEAD RETRACTION REFLEX AND OPISTHOTONOS

The **head retraction reflex** is a primitive reflex that can reappear later in life with loss of inhibition of various causes. Head retraction is commonly elicited by tapping the upper lip (Wartenberg, 1949). The head retraction reflex might be seen in neurodegenerative diseases presenting with parkinsonism. Among others, this sign is especially common in progressive supranuclear palsy and corticobasal degeneration, but it has been reported also in patients with stiff-person syndrome and other related conditions (Berger & Meinck, 2003). Head retraction after tapping the nose is a feature of hereditary startle syndrome (hyperekplexia). **Opisthotonos** seen in patients with stiff-person syndrome and tetanus may appear similar to the head retraction reflex, but in opisthotonos, not only the head but also the trunk is retracted. In opisthotonos, therefore, the lower motor neurons of the whole body are considered to be hyperexcitable (see Chapter 33-4, p. 289), while the neck extensors are selectively involved in the head retraction reflex.

face might make the patient open the mouth, or touching the cheek or just showing an object visually might make the patient reach out his or her face and mouth toward the object (**rooting reflex**).

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FACIAL NERVE

The seventh cranial nerve (facial nerve) contains four different groups of nerve fibers: motor, somatosensory, gustatory, and autonomic (parasympathetic).

1. STRUCTURES AND FUNCTIONS

A. MOTOR NERVE

The facial motor nucleus is located in the pontine tegmentum, anterior and lateral to the abducens nucleus on each side. Its nerve fibers first run posteromedially toward the fourth ventricle and, after having encircled the abducens nucleus, they run anterolaterally and exit the pons through its lateral edge (see Figure 5-2, p. 32). In the cerebellopontine angle of the posterior cranial fossa, they enter the internal auditory meatus of the temporal bone, pass through the **geniculate ganglion** in the facial canal,

and exit from the stylomastoid foramen (Figure 11-1a). After passing through the parotid gland, they separate to project to the frontal, orbicularis oculi, and orbicularis oris muscles and platysma. If the facial nerve is affected by infectious or autoimmune inflammation, these muscles are paralyzed to various degrees (**Bell palsy**).

When the facial nerve is in a hyperexcitable state, the orbicularis oculi muscle and/or the orbicularis oris muscle twitches repetitively or continuously, resulting in eyelid closure or lateral upward pulling of the oral angle. This condition is called **hemifacial spasm**, which is, like trigeminal neuralgia (see Chapter 10-2B, p. 79), considered to be due to demyelination of the nerve fibers secondary to its compression by tortuous branches of the verte-brobasilar artery. However, as this phenomenon is not due to direct compression by the arteries, the twitches seen in those muscles are not synchronous with cardiac pulsation. As the result of electrophysiologic studies by stimulating

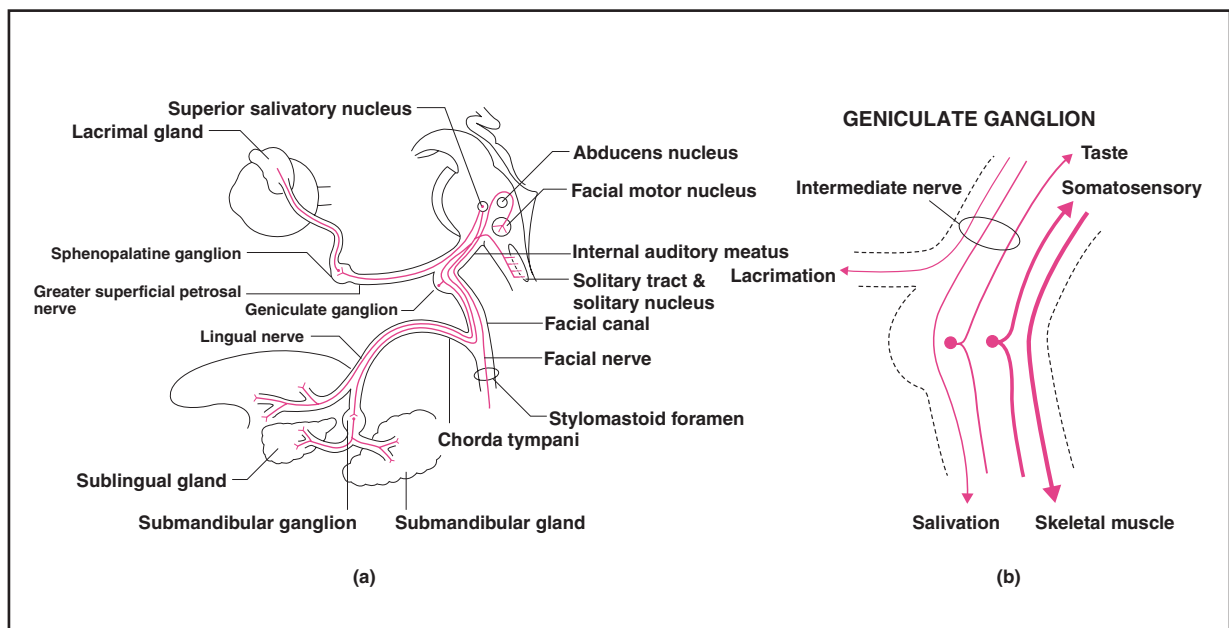


Figure 11-1 Structures of the facial nerve (a) and magnified view of the geniculate ganglion to show fiber compositions in it (b).

the facial nerve and by recording the evoked electromyogram (EMG) from the corresponding muscle (F-wave), motor neurons of the facial nucleus were shown to be hyperexcitable (Ishikawa et al, 2010). The spasms, however, may arise from ectopic activity in the demyelinated segment of the nerve. The synchronous activity of muscles in the face is likely due to ephaptic transmission between nerve fibers under the compression.

In the facial canal, the facial nerve sends a branch also to the stapedius muscle, which acts to relieve tension of the tympanic membrane. In Bell palsy, therefore, the tympanic membrane becomes tense, causing **hyperacusis**.

The face area of the primary motor cortex (M1) is located at the lateral convexity of the precentral gyrus, and like the primary somatosensory cortex (S1), there is somatotopic organization in the M1, showing the upside-down homunculus (Figure 11-2). However, just like the S1, the organization within the face area is not upside-down, and the eye area is located dorsal to the mouth area. The M1 is mostly located in the anterior wall (area 4) of the central sulcus, and at least for the face and hand areas, the crown of the precentral gyrus is occupied by the premotor area (area 6) (Figure 11-3). For the upper part of the face, in addition to the M1, the cingulate area sends a strong projection to the facial nucleus (Hanakawa et al, 2008).

Corticospinal tracts and corticobulbar tracts are made of nerve fibers directly projecting from the pyramidal tract cells of the M1 and the premotor area. As for the nerve fibers originating from the supplementary motor area (SMA), at



Figure 11-2 Somatotopic organization of the primary motor cortex.

least as far as those projecting to the anterior horn cells related to the distal extremity muscles are concerned, they join the corticospinal tract via the M1 or the premotor area. Electrophysiologically, a direct cortical stimulation presented to the hand area of the M1 or the premotor area evokes a phasic contraction in the contralateral hand muscle at a latency of 20 ms, while an electric shock given to the SMA evokes slow muscle contraction bilaterally. Thus, for the extremity muscles, the M1 and the premotor area innervate contralaterally whereas the SMA innervates bilaterally.

B. SOMATOSENSORY NERVE

Somatosensory fibers of the facial nerve transmit impulses arising from the skin of the inner aspect of the earlobe and the external auditory meatus. The primary sensory neuron is a bipolar cell of the geniculate ganglion (see Figure 11-1b). Its peripheral axon conducts impulses from that skin area, and through the central axon, the nociceptive input is conducted through the spinal tract of the trigeminal nerve to its nucleus while the tactile input projects to the main sensory nucleus of the trigeminal nerve. Nerve fibers originating from the secondary neuron cross and ascend through the trigeminothalamic tract to reach the VPM of the contralateral thalamus (see Chapter 10-1, p. 75).

When the geniculate ganglion is infected with **varicella zoster virus**, the somatosensory fibers are primarily affected because varicella zoster virus has a strong affinity for the sensory ganglion. In this condition, a vesicular skin rash is observed in the inner earlobe and the external auditory meatus associated with neuralgia, numbness, and sensory loss in that region (see Figure 3-7, p. 15). Motor fibers are also affected by inflammation in the geniculate ganglion, causing motor paralysis (**Ramsay Hunt syndrome** or **Hunt syndrome**). Although other conditions, including deep palmar branch ulnar neuropathy, dyssynergia cerebellaris myoclonica, juvenile Parkinson disease, and dentatorubropallidoluysian atrophy, might be also referred to as Ramsay Hunt syndrome, only the geniculate zoster is called Ramsay Hunt syndrome in this book to avoid confusion.

C. TASTE SENSE

Special nerve fibers for taste sense conduct impulses from the anterior two-thirds of the tongue. The primary neuron is a bipolar cell in the geniculate ganglion. Impulses arising from the taste buds are conducted through the peripheral axon as the lingual nerve and chorda tympani into the geniculate ganglion. The central axon conducts impulses

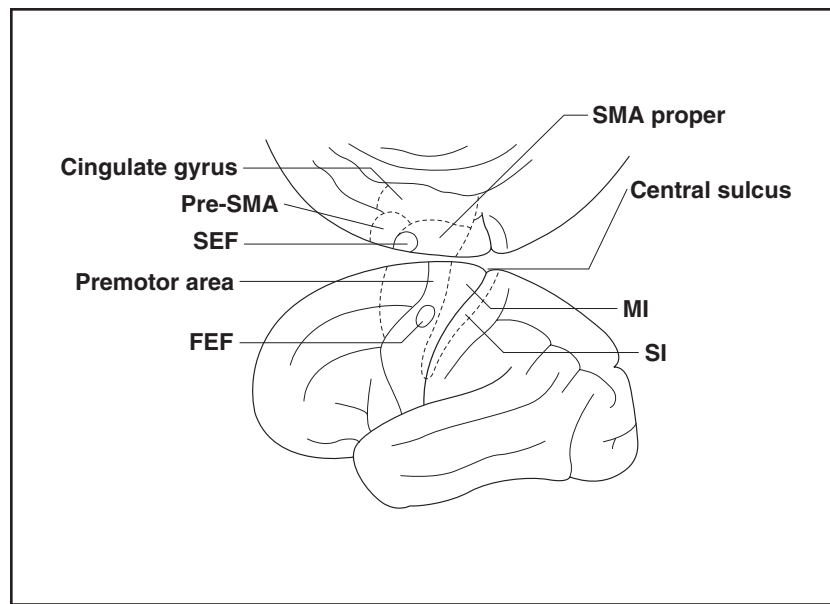


Figure 11-3 Schematic illustration of the precentral gyrus to show the subareas of the motor cortices. MI: primary motor cortex, SI: primary somatosensory cortex, SMA: supplementary motor area, SEF: supplementary eye field, FEF: frontal eye field.

to the **solitary nucleus** in the pontine tegmentum (see Figures 11-1a, 5-3, and 5-5). In Bell palsy, the facial nerve is usually affected central to the bifurcation of the chorda tympani from the facial nerve, thus involving the gustatory fibers, and therefore taste sense is lost in the anterior two-thirds of the tongue unilaterally.

The secondary gustatory fibers originating from the solitary nucleus project to the VPM of the thalamus, and the third fibers project to the tongue area of the S1 and the anterior insula. In relation to this, some patients with mesial temporal lobe epilepsy complain of an unusual taste as an aura. According to the results of direct electrical stimulation during presurgical evaluation in a large number of patients with drug-resistant epilepsy, the insular posterior short gyrus seems to play an important role in the gustatory sensation (Mazzola et al, 2017). In addition, some fibers originating from the solitary nucleus project to the salivatory nucleus, the dorsal motor nucleus of the vagus nerve, and the hypothalamus, and they control digestion in a reflexive way. Taste sense of the posterior third of tongue is innervated by the glossopharyngeal nerve (see Chapter 14-3, p. 101).

D. LACRIMATION AND SALIVATION

Parasympathetic efferent fibers of the facial nerve control secretion of the lacrimal gland and the salivary gland. The nucleus innervating salivation is the **superior salivatory nucleus** situated in the pontine tegmentum, and that for

lacrimation is the **lacrimal nucleus**. These two nuclei are attached together and may be considered as a single nuclear group (see Figure 5-2, p. 32). The preganglionic fibers emerging from the nucleus form the **intermediate nerve** and enter the facial canal as a part of the facial nerve (see Figure 11-1b).

The preganglionic fibers from the lacrimal nucleus bifurcate from the facial nerve just before they reach the geniculate ganglion and form the greater superficial petrosal nerve to project to the sphenopalatine ganglion. The postganglionic fibers emerging from the ganglion innervate the lacrimal gland (see Figure 11-1a). The neurotransmitter of the preganglionic fibers is acetylcholine for nicotinic receptors and that of the postganglionic fibers is acetylcholine for muscarinic receptors. This system acts to facilitate lacrimation.

The preganglionic fibers from the salivatory nucleus bifurcate from the facial nerve after it emerges from the geniculate ganglion and form the chorda tympani to project to the submandibular ganglion (see Figure 11-1a). The postganglionic fibers emerging from the ganglion project to the sublingual gland and submandibular gland to facilitate salivation. The synaptic transmitters for this system are the same as for lacrimation.

In Bell palsy, if the inflammation involves the facial nerve before (proximal to) the bifurcation of the greater superficial petrosal nerve, the preganglionic fibers related to lacrimation are involved, which will cause loss of lacrimation. Furthermore, as the orbicularis oculi muscle is

paralyzed to various degrees, it worsens the dryness of the eyes and necessitates protection of the cornea and conjunctivae from excessive dryness. Theoretically, unilateral involvement of the facial nerve before the bifurcation of the chorda tympani will cause loss of salivation on the affected side, but this is compensated for by the intact salivation of the opposite salivary glands.

E. PARASYMPATHETIC INNERVATION OF FACIAL SKIN AND MUCOSA

In contrast with the sympathetic nervous system, the parasympathetic innervation of the facial skin and mucosa is not precisely known. However, it is most likely that the parasympathetic pathway goes through the facial nerve. If that is the case, a lesion of the facial nerve is expected to show, contrary to Horner syndrome (see Chapter 8-1, p. 51), anemic palpebral conjunctiva, decreased skin temperature, and decreased nasal discharge; however, these findings are not detected on routine clinical examination.

2. EXAMINATION OF FUNCTIONS RELATED TO THE FACIAL NERVE

A. MOTOR FUNCTION

First, it is important to observe the face closely, paying special attention to possible asymmetry in the forehead furrows, palpebral fissure, nasolabial folds, and the level of oral angles. In peripheral facial paralysis, the furrows in the forehead look shallower on the affected side, and weakness of the frontal muscle can be confirmed by asking the patient to raise the eyebrows. The palpebral fissure on the affected side may be larger due to paralysis of the orbicularis oculi muscle. Complete failure to close the eyelids is called **lagophthalmos**. It should be kept in mind, however, that, in the chronic stage of Bell palsy, the palpebral fissure may be even smaller on the affected side due to abnormal electric discharge of the facial nerve causing repetitive or continuous contraction of the orbicularis oculi muscle. Furthermore, attempting to contract the orbicularis oris muscle may narrow the palpebral fissure of the affected side due to synkinesis of the orbicularis oculi muscle (Box 11-1).

It is also important to pay attention to blinks. In a unilateral facial nerve palsy, blinks might decrease in amplitude and speed, and may even be completely absent on the affected side. When the patient is asked to close the eyes

BOX 11-1 ORBICULARIS ORIS MUSCLE AND ORBICULARIS OCULI MUSCLE

The orbicularis oris muscle is innervated by the contralateral motor cortex. By contrast, the orbicularis oculi muscle receives bilateral innervation though with the contralateral predominance, and the frontalis muscle is innervated equally by both hemispheres. Moreover, for the upper part of the face, the cingulate gyrus sends a strong projection to the facial nucleus (Hanakawa et al, 2008). Therefore, in facial palsy due to a cerebral hemispheric lesion, the orbicularis oculi muscle is involved much more mildly, if at all, than the orbicularis oris muscle (**central facial paralysis**). This is in strong contrast to a lesion of the facial nerve that affects both muscles equally (**peripheral facial paralysis**). For the same reasons, if the face area of the M1 is involved by an epileptogenic event, convulsion will involve the orbicularis oris muscle only contralaterally but the orbicularis oculi muscle bilaterally. By contrast, when the facial nerve is involved by abnormal discharge as in **hemifacial spasm**, both the orbicularis oris and orbicularis oculi muscles of the affected side show contractions.

In patients having a past history of Bell palsy, voluntary contraction of the orbicularis oris muscle is often associated with co-contraction of the orbicularis oculi muscle (**synkinesis**). This is considered to result from misconnection between the nerve fibers innervating the orbicularis oris muscle and those innervating the orbicularis oculi muscle during recovery from the inflammatory demyelination. In patients with persistent synkinesis following Bell palsy, the oral angle of the involved side may appear pulled up laterally, and it may look as if the other side might be paralyzed.

tight, cilia on the affected side might protrude forward (**ciliary sign**). Furthermore, the blink reflex elicited by tapping the glabella may be incomplete, slower, or even absent on the affected side.

When the orbicularis oris muscle is paralyzed, the nasolabial fold is shallower and the oral angle might be lower on the affected side compared with the intact side. However, since even healthy subjects might show some asymmetry in the nasolabial folds and oral angles, their asymmetry has to be confirmed by muscle testing and other accompanying signs. Strength of the orbicularis oris muscle can be tested by asking the patient to move the mouth to various directions, whistle, and blow. If the patient is asked to pronounce a long “eee” sound, contraction of the platysma can be observed in the neck.

Paralysis of the stapedius muscle can be tested by checking hearing with a tuning fork. As described earlier in this chapter, the hearing threshold is decreased because the paralyzed stapedius muscle cannot relieve the tension of the tympanic membrane (**hyperacusis**).

i. Central facial paralysis and peripheral facial paralysis

In most cases, it is not difficult to distinguish central from peripheral facial paralysis based on the history and the accompanying symptoms and signs. If the motor paralysis is restricted to unilateral face without involving the extremities, relative impairment of the orbicularis oris and oculi muscles provides a clue for the distinction. Paralysis of the orbicularis oris muscle with no associated paralysis, or only mild if any, of the orbicularis oculi muscle usually suggests **central facial paralysis**, because the upper facial muscles are bilaterally innervated from the motor cortex and probably also because the facial subnucleus corresponding to the upper part of the face receives projection from the cingulate motor area (Hanakawa et al, 2008) (see Box 11-1).

Unilateral involvement of all facial muscles suggests a **peripheral facial paralysis**. Theoretically, there is no stretch reflex in the orbicularis oris muscle, because muscle spindles are not known to exist in this muscle. However, tapping the lips occasionally causes a twitch in the muscle. Like the blink reflex (see Chapter 10-1, p. 78), this reflex is considered to be a polysynaptic reflex in response to activation of the skin receptors. On some occasions, it is also difficult to distinguish it from the snout reflex. In peripheral facial paralysis, all these reflexes are expected to be lost.

Central facial paralysis is often associated with deviation of the tongue to the side of facial paralysis when voluntarily protruded, because the mouth and the tongue areas are represented close each other in both the motor cortex and the corticobulbar tracts (see Box 14-2, p. 99).

ii. Crossed hemiplegia of facial muscle

A combination of unilateral peripheral facial paralysis with contralateral hemiplegia is called **Millard–Gubler syndrome**, and further addition of lateral gaze palsy to the side of facial palsy is called **Foville syndrome** (see Table 8-1, p. 53). Both conditions are crossed syndromes due to an intramedullary lesion in the pons (Silverman et al, 1995). Millard in 1856 reported autopsy findings of one of two cases reported by Senac and a case reported by Poisson in the *Bulletin of the Anatomical Society of Paris*,

and ascribed the former combination to a pontine hemorrhage (Millard, 1856). Foville in 1858 reported a 43-year-old man with possible pontine hemorrhage manifesting the latter combination (Foville, 1858). He postulated the presence of some mechanism in the pons that controls abduction of one eye and adduction of the other eye. There was no concept of lateral gaze center at that time.

iii. Mimetic muscles

Mimetic muscles employed for laughing and crying are also involved in peripheral facial paralysis but are not much impaired in central facial paralysis. Facial muscles related to emotional expression might also be controlled by the basal ganglia, hypothalamus, and cingulate gyrus, either directly or indirectly. In fact, those muscles are affected in basal ganglia disorders, and facial expression is lost contralateral to the lesion. A **mask-like face** is often seen in patients with Parkinson disease.

B. SOMATOSENSORY FUNCTION

If there is any possibility of herpes zoster infection as a cause of peripheral facial paralysis, it is important to look for vesicular rash on the skin of the inner earlobe and the external auditory meatus (**Ramsay Hunt syndrome**) (see p. 84). It is usually associated with sensory impairment in that region.

C. GUSTATORY SENSE

Loss of gustatory sense (**ageusia**) is noticed by the patient only when the taste buds are totally damaged, as may occur as side effect of drugs and lead deficiency. Unilateral loss of taste sense due to a facial nerve lesion can be detected by comparing the taste sense between two sides of the anterior two-thirds of the tongue. Taste sense can be tested by touching each side of the tongue with a wet swab soaked in salt or sugar, while the tongue is gently protruded. The patient is asked to show by hand if the taste is felt and to tell later what kind of taste it was. Then the other half of the tongue is tested after washing out the mouth.

D. LACRIMATION AND SALIVATION

In unilateral peripheral facial paralysis, the patient often complains of dryness in the affected eye. This is tested by measuring the volume of tears with a piece of filter paper placed on the lower eyelid (**Schirmer test**). In Bell palsy, as described earlier in this chapter (see p. 85), dryness of

BOX 11-2 CROCODILE TEARS SYNDROME

Some patients who have recovered from Bell palsy might complain of tearing from the affected eye just when looking at food. This is considered to result from abnormal connections between the preganglionic fibers related to lacrimation and those related to salivation during recovery from acute inflammation. Physiologically, it is considered to be due to nonsynaptic transmission of impulses between the different groups of nerve fibers (**ephaptic transmission**). It is natural for presentation of food to induce salivation, but in crocodile tears syndrome, lacrimation is also induced as a result of the erroneous transmission of impulses. The reason why this phenomenon is named “crocodile tears” remains a mystery, but there is some tradition that crocodiles distract other animals by showing tears to catch them. Recently, it has been demonstrated that crocodiles actually do tear when eating.

the affected eye becomes even worse due to paralysis of the orbicularis oculi muscle. In the recovery phase from Bell palsy, the crocodile tears syndrome (Box 11-2) may be confirmed during the history taking or by actual observation while the patient is eating.

BOX 11-3 BILATERAL FACIAL PARALYSIS

Bilateral peripheral facial paralysis is not very common, but if it is seen, the following causes should be considered. Bilateral peripheral facial paralysis of acute or subacute onset suggests systemic conditions, including sarcoidosis, Lyme disease, and leprosy. In **sarcoidosis**, the facial nerves are involved by granulomatous inflammation. Both facial nerves may be affected simultaneously or independently one after the other. Association of facial nerve palsy with uveitis and parotid swelling is known as **Heerfordt disease**. **Lyme disease** is caused by infection with the spirochete *Borrelia burgdorferi*, which is mediated by ticks. In facial palsy due to tuberculoid leprosy, the frontal and orbicularis oculi muscles are commonly involved. As was described in Chapter 9-5B (p. 67), congenital bilateral facial nerve paralysis (Möbius syndrome) is due to hypoplasia of the corresponding neurons in the facial motor nucleus.

Dryness in the mouth is a complaint of a patient with a **sicca syndrome** like Sjögren syndrome and Mikulicz syndrome. In this case, the eyes are also found to be dry. In a unilateral lesion of the facial nerve, unilateral loss of salivation is never noticed by the patient because of normal salivation from the intact side and also because of the intact glossopharyngeal nerve, which innervates salivation of the parotid gland (see Chapter 14-4, p. 101).

E. FACIAL SKIN AND MUCOSA

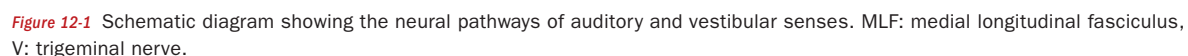
In relation to the autonomic nervous functions, attention might be paid to color of the facial skin, palpebral conjunctiva, and oral mucosa and the temperature and moisture of the facial skin. In Horner syndrome, conjunctival hyperemia, decreased sweating, and elevated temperature might be observed on the affected side (see Chapter 8-1, p. 51). In peripheral facial paralysis, although the parasympathetic fibers are considered to be contained in the facial nerve (see p. 86), abnormalities of skin and mucosa are not usually observed. Occasionally, peripheral facial paralysis might be accompanied by orofacial edema and furrowing of the tongue (scrotal tongue) (**Melkersson–Rosenthal syndrome**) (Dhawan et al, 2020; Hallett & Mitchell, 1968).

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Receptors of auditory sense are hair cells in the organ of Corti of the cochlea. The primary neuron is a bipolar cell in the **spiral ganglion**, and its peripheral axon is very short. Its central axon (**cochlear nerve** or **acoustic nerve**) forms, together with the vestibular nerve, the eighth cranial nerve (**vestibulocochlear nerve**), which passes medially through the internal auditory meatus to enter the cerebellopontine angle in the posterior cranial fossa and enters the lateral posterior part of the medulla oblongata (Figure 12-1). The secondary neuron is in the **dorsal and ventral cochlear nucleus** located in the upper lateral part of the medullary tegmentum.

contralateral **superior olive** to form the **lateral lemniscus** (Figure 12-2). Some fibers form synaptic connections to the tertiary neurons in the superior olive and ascend through the lateral lemniscus. Some fibers from the ventral cochlear nucleus also project to the ipsilateral superior olive, and the tertiary fibers from there join the ipsilateral lateral lemniscus. The **lateral lemniscus** projects to the **inferior colliculus** in the midbrain. The secondary fibers originating from the dorsal cochlear nucleus cross in the tegmentum and join the contralateral lateral lemniscus. Some of the third- or fourth-order neurons originating from the inferior colliculus send their axons to the ipsilateral **medial geniculate body**, and the fibers of the other group cross and project to the contralateral medial geniculate body. The fibers originating from the medial geniculate body project through the acoustic radiation of the cerebral white matter to the ipsilateral **auditory cortex**.



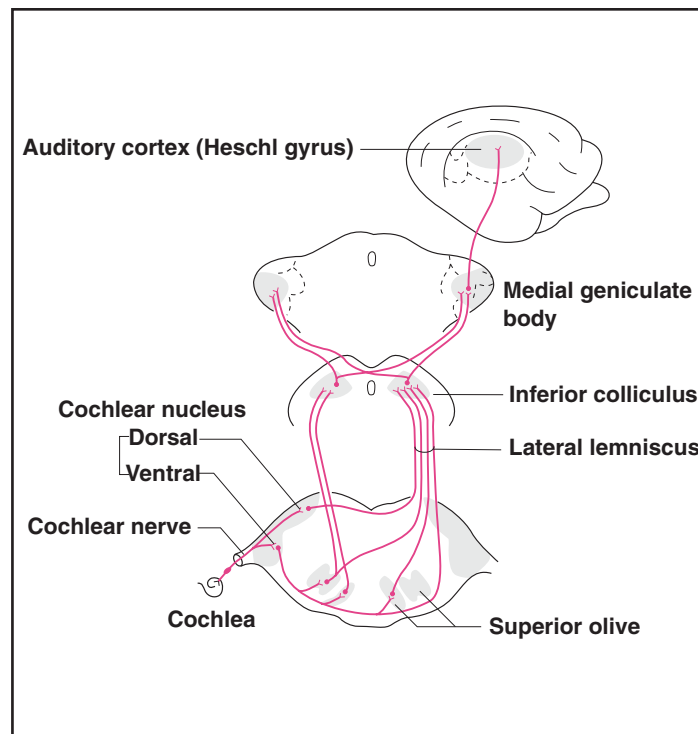


Figure 12-2 Central pathway of the auditory sense.
Cited from Brodal, 2010, with permission.

(**Heschl gyrus**) located in the inferior bank of the sylvian fissure (see Figure 12-2).

As the auditory pathways cross partially at the upper part of the medulla oblongata, and again partially after the inferior colliculus, each auditory cortex receives the input from both ears. However, based on studies using magnetoencephalography and neuroimaging techniques, it is known that the input to the auditory cortex from the contralateral ear is greater compared to the ipsilateral ear. In sensory systems in general, it is the tertiary neurons that project to the cortical receptive areas, but in the auditory system, it is mediated by more than three steps.

A. BRAINSTEM AUDITORY EVOKED POTENTIALS

Based on the concept of far-field potentials, which can be recorded from the electrodes placed on the head surface, it is possible to follow the conduction of auditory impulses through the brainstem pathways noninvasively. In the **brainstem auditory evoked potentials (BAEPs)**, seven small surface-positive peaks can be identified within 10 ms after presentation of a click stimulus (Figure 12-3). The generator sources of those peaks are postulated to be the acoustic nerve for peak I, acoustic nerve or cochlear nucleus for II, superior olive for III, lateral lemniscus for IV, inferior

colliculus for V, medial geniculate body for VI, and auditory radiation for VII. BAEPs are used as a method to assess the integrity of parts of the brainstem, including confirmation of loss of brainstem functions in possible cases of brain death (see Chapter 33-7, p. 295).

B. COGNITIVE FUNCTIONS RELATED TO THE AUDITORY SENSE

By magnetoencephalographic studies, **tonotopic organization** was demonstrated in the primary auditory cortex. Namely, the auditory input is received in sequentially different areas of the cortex depending on its frequency. **Spatial perception of auditory input**, or the information as to where a sound has come from, is achieved based on a subtle difference in the arrival time of the sound to both ears and consequently on a subtle difference in the arrival time of impulses to the cortices on both sides. Regarding **selective attention**, if special attention is paid to a particular voice among others at a party, that voice can be heard even in noisy circumstances. This so-called **cocktail party effect** has been shown to occur also in the primary auditory cortex (see Box 24-8, p. 237). Thus, the primary auditory cortex plays an important role in the cognitive auditory functions.

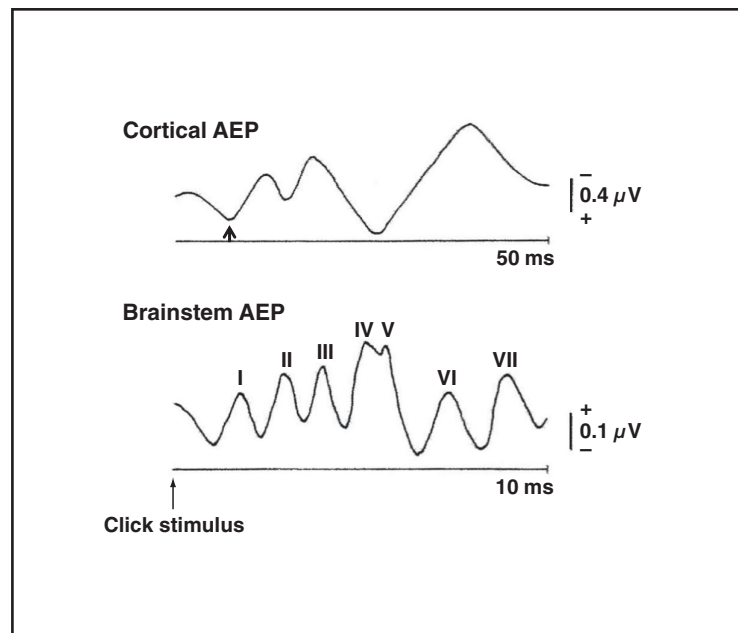


Figure 12-3 Auditory evoked potentials (AEPs) in response to click stimulus. In the cortical AEP shown on the top, the initial impulse arrives at the cortex at a latency of about 17 ms. The brainstem AEP shown on the bottom is generated in deep structures but can be recorded from the electrodes placed on the scalp as far-field potentials due to volume conduction. See text for the generator source of each peak.

2. EXAMINATION OF AUDITORY FUNCTION

Although bedside testing of hearing has low sensitivity, hearing of each ear can be tested by using, for example, a vibrating tuning fork (Boatman et al, 2007). Neurologists commonly use a 128-Hz tuning fork since that is used to assess sensation and is readily available, but most hearing loss in patients will be at higher frequencies. If hearing is decreased in one ear, the **Weber test** can be done by placing a vibrating tuning fork on the midline forehead and asking the patient to judge on which side the sound of the tuning fork is heard. For this test, it is important to specify in which ear the patient hears, but not on which side he or she feels the vibration of, the tuning fork. If the patient hears on the side of hearing loss, it suggests **conductive deafness**. In that case, the tympanic membrane should be checked. If the patient hears on the intact side, it suggests **sensorineural deafness**. As some patients might predict that they should be able to hear on the intact side, if there is any question as to the results of Weber test, they should be well informed that they might hear more on the affected side. If some question still remains in the interpretation of the results, it is useful to mask one ear and confirm that it is heard on the masked side. This maneuver might help obtain some clue to a possible diagnosis of functional (psychogenic) disorders. In case of bilateral hearing loss, the Weber test can be still applied. If the patient hears the tuning fork better on the

more severely affected side, it suggests conductive deafness, and vice versa.

Alternatively, a vibrating tuning fork can be placed on the retroauricular part of the temporal bone to compare bone conduction to air conduction (**Rinne test**). The bone conduction is preserved in conductive deafness and therefore is better than air conduction. In sensorineural deafness, both air conduction and bone conduction are diminished, and air conduction will remain more than bone conduction.

As was described in Chapter 10-2D (p. 81), if unilateral sensorineural deafness is associated with cerebellar ataxia and loss of the ipsilateral corneal reflex, it suggests acoustic neurinoma occupying the cerebellopontine angle, compressing the pons.

A. TINNITUS

Tinnitus is a common complaint related to hearing disturbance. Sound produced intracranially can be heard continuously or intermittently in one ear or in both ears. Usually it is heard only in the patient's own ears, but occasionally it can be heard by other nearby people. Causes of the latter condition include disturbance of intracranial blood circulation, some types of palatal tremor, or abnormalities of temporomandibular joints. Subjective tinnitus occurs in organic disorders of any of the outer, middle, and inner ears, and any part of the auditory pathway, but

it is particularly common in association with sensorineural deafness. However, tinnitus is extremely rare in disorders of the central nervous system even if the auditory pathway is involved. Unilateral tinnitus might be an initial symptom of acoustic neurinoma.

B. SENSORINEURAL DEAFNESS

Sensorineural deafness is caused by a number of conditions. It may be seen as a sequela of inflammatory diseases such as meningitis, mumps, and syphilis. It may be also seen in drug intoxication, organic mercury poisoning (Minamata disease), and Ménière disease. A typical case of progressive unilateral sensorineural deafness is acoustic neurinoma. A gradual bilateral loss of high tones is common with aging.

In some neurodegenerative diseases, sensorineural deafness may be seen as one of its cardinal symptoms. Among others, **Refsum disease** (heredopathia atactica polyneuritiformis due to phytanic acid oxidase deficiency) and **Kearns–Sayre syndrome** (see Box 19-13, p. 181) are well known. **Susac syndrome** occurs in women aged 20 to 40 years and is characterized by acute encephalopathy with headache and psychiatric symptoms, occlusion of the retinal arteries, and sensorineural deafness. In some cases of Susac syndrome, cranial magnetic resonance imaging (MRI) shows a white matter change like acute disseminated encephalitis, probably suggesting an autoimmune disorder (Rennebohm et al, 2008). Ischemic lesions known as snowballs are seen in the corpus callosum (Marrodan et al, 2017). In Susac syndrome, retinal pathology can be detected by spectral-domain optical coherence tomography

(Ringelstein et al, 2015). **Superficial siderosis** occurs in the aged population and is characterized by progressive bilateral sensorineural deafness, cerebellar ataxia, and pyramidal signs (Kumar et al, 2009; Vernooij et al, 2009). This condition is thought to be due to repetitive subarachnoid hemorrhage of mild degree or amyloid angiopathy.

Auditory pathways may be involved by intramedullary lesions of the brainstem, but as the input arising from each ear ascends bilaterally in the brainstem, it does not manifest hearing deficit in the clinical setting. As the lateral lemniscus ascends through the lateral part of the pons, its bilateral involvement is rarely encountered. Bilateral involvement of the lateral lemniscus may occur in a massive pontine lesion, but in that case the patient may not be in a sufficiently alert condition for the hearing test.

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SENSE OF EQUILIBRIUM

1. STRUCTURES AND FUNCTIONS OF THE PATHWAYS RELATED TO EQUILIBRIUM

Receptors of the sense of equilibrium are hair cells located in the semicircular canals, saccule, and utricle of the labyrinth. The **semicircular canals** receive input related to rotatory acceleration of head, and the saccule and utricle receive input related to the position and rectilinear acceleration of the head. Among the latter two receptors, the **saccule** is sensitive to the acceleration of gravity while the **utricle** is sensitive to horizontal acceleration. For example, when riding a roller coaster, sudden acceleration forward is received by the utricle, and when taking an elevator, sudden vertical acceleration is received by the saccule.

The primary neuron of the vestibular system is a bipolar cell located in the **vestibular ganglion (Scarpa's ganglion)**. Its peripheral axon conducts impulses from the receptors to the central axon (**vestibular nerve**), which, along with the cochlear nerve, passes through the internal auditory meatus. In the posterior cranial fossa, they enter the dorsolateral tegmentum of the medulla oblongata and project to the vestibular nuclei (vestibular complex) (see Figure 12-1, p. 89).

The **vestibular complex** is made up of four nuclei: medial, lateral, superior, and inferior. Most nerve fibers originating in the semicircular canals project to the superior vestibular nucleus and the rostral part of the medial vestibular nucleus. Nerve fibers originating in the saccule and utricle mainly project to the lateral vestibular nucleus (Deiters nucleus). In addition, the vestibular complex receives input from the spinal cord, brainstem reticular formation, superior colliculus, and flocculus, nodulus, and anterior lobe of the cerebellum.

The secondary nerves originating in the vestibular complex ascend through the ipsilateral vestibulothalamic tract in the medial brainstem (Yang et al, 2014) and project to the ventral posterolateral nucleus of the thalamus (VPL), from which the tertiary nerves project to the

cortex. Based on the results of evoked potential recording and direct stimulation study, the cortical receptive areas of the vestibular input are postulated to be a region posterior to the face area of the postcentral gyrus and the parietal insular vestibular cortex located in the junction between the parietal lobe and insula (Mazzola et al, 2014). In relation to this, some patients with temporal lobe epilepsy complain of vertigo as an aura (**vertiginous seizure, epileptic vertigo**) (Tarnutzer et al, 2015).

In addition to these pathways directly related to the sense of equilibrium, nerve fibers originating in the vestibular complex also project to motor nuclei targeting the extraocular muscles, cerebellum, and anterior horn of the spinal cord. Those fibers targeting the spinal cord originate mainly in the lateral nucleus and form the **lateral vestibulospinal tract**, which projects to the α and γ motor neurons at the anterior horn. The vestibulospinal tract is considered to have a facilitatory effect on the muscle tone of extensors. This pathway is considered to be significant in the pathophysiology of the **decerebrate posture** and **decerebrate rigidity** seen in tentorial herniation. Namely, as the inhibitory pathway projecting to the vestibular complex is impaired due to the midbrain compression, it disinhibits the facilitatory vestibulospinal input to the extensor muscles, causing the extensor posture in all extremities (see Chapter 33-6D, p. 294). If the herniation progresses to compress the lower brainstem, the vestibular complex itself is also damaged, resulting in total loss of muscle tone.

Nerve fibers related to eye movements originate mainly in the superior and medial vestibular nuclei. Most of them ascend through the ipsilateral medial longitudinal fasciculus (MLF), but some fibers cross and ascend through the contralateral MLF (see Chapter 9-3, p. 62). In addition, some other fibers form synaptic connections to the tertiary neurons in the ipsilateral as well as contralateral paramedian pontine reticular formation (PPRF) and then join the MLF. The latter pathway controls conjugate lateral gaze.

The vestibulocerebellar tract originates from the medial and inferior (descending) vestibular nuclei and projects via the inferior cerebellar peduncle to the vestibulocerebellum (see Figure 16-19, p. 128). This pathway plays an important role in the control of equilibrium. Some patients with a localized cerebellar lesion may complain of vertigo, most likely due to involvement of the vestibulocerebellum.

2. EXAMINATION OF THE SENSE OF EQUILIBRIUM

Whenever a patient complains of “dizziness,” it is important to judge whether it is vertigo in the narrow sense or dizziness in general (Figure 13-1). Dizziness is a common symptom in patients having syncope. It is usually described as a feeling of fainting or even blackout in a severe case. A variety of systemic conditions can cause dizziness, including cardiovascular insufficiency, postural hypotension, pulmonary diseases, and severe anemia (see Box 4-2, p. 27).

By contrast, vertigo is commonly associated with oscillopsia because of the presence of nystagmus. **Oscillopsia** is a phenomenon in which the patient sees the visual scene moving, and with vertigo, it might be rotating. If it is accompanied by tinnitus or hearing disturbance, it suggests vertigo rather than presyncope. It is because the vestibular pathway and the cochlear pathway are anatomically close to each other all the way from the inner ear to the brainstem nuclei. A typical example is **Ménière disease**, which presents with attacks of vertigo and slowly progressive low-tone

sensorineural hearing disturbance due to dysfunction of the labyrinth.

Benign paroxysmal positional vertigo (BPPV) is a common disease that is characterized by vertigo of sudden onset triggered by changing the head position, especially upon awakening in the morning. Individual episodes provoked by head motion are short in duration, lasting seconds. Vertigo in this condition lasts for several hours to several days and tends to relapse frequently. Its most common cause is considered to be the fall of an otolith (small stone) from the utricular macula into the space of the posterior semicircular canal, which produces the flow of endolymph and pathologically excites the cupula ampularis. This syndrome can often be diagnosed clinically via the Dix–Halpike maneuver, which tests for canalithiasis of the posterior semicircular canal. Involvement of the horizontal canal can be tested via the log roll test (Yu et al, 2020). A positive test is suggested by provoked vertigo and nystagmus usually appearing after a latency of a few seconds. As it often resists drug treatment, various maneuvers have been applied to restore the otolith position (Epley maneuver) (Fife et al, 2008; Kerber et al, 2012; Kim et al, 2012).

Vertigo is a common symptom encountered in an ischemic condition of the **vertebrobasilar artery territory**, especially in the **lateral medullary syndrome (Wallenberg syndrome)** (see Box 14-1, p. 97). Wallenberg syndrome is commonly caused by infarction in the territory of the posterior inferior cerebellar artery, but vertigo is also seen in infarction of the territory of the anterior inferior cerebellar artery. By contrast with Wallenberg syndrome, the latter infarction is characterized by peripheral facial paralysis and

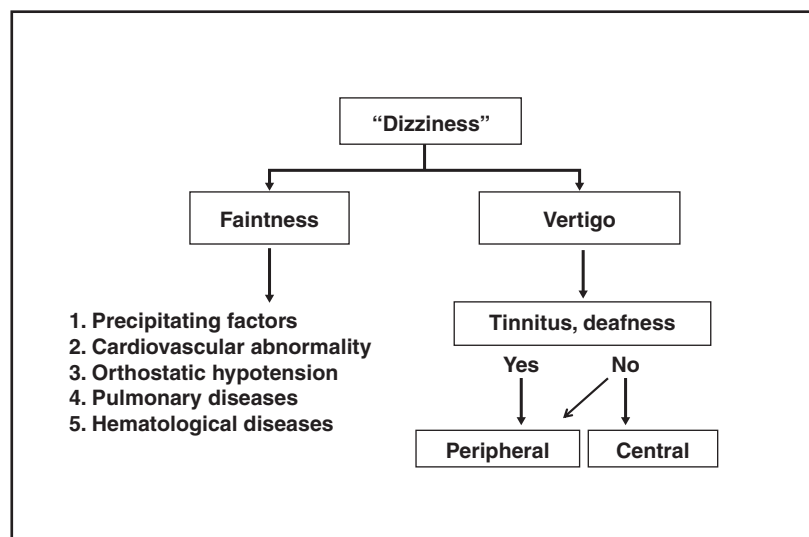


Figure 13-1 Flow chart used for the diagnosis of “dizziness.” It is practical to distinguish faintness and vertigo.

hearing deficit of the affected side in addition to vertigo. As the fibers targeting at the solitary nucleus are not affected in this case, peripheral facial paralysis is not associated with loss of taste.

Examination of vestibular functions depends on the history and other accompanying signs. If the vestibular system is expected to be impaired, a detailed examination is needed. The most important test of vestibular functions is observation of the standing posture and gait, which may be done after other neurologic examinations are completed (see Chapter 22, p. 211).

In the standing position, the trunk may be tilted, usually toward the side of the affected vestibular system. Upon walking or on tandem gait, the body tends to fall to the affected side (**body lateropulsion**, **axial lateropulsion**). In a medullary lesion like Wallenberg syndrome, the lateropulsion occurs to the affected side, but in a lesion of the paramedian region of the rostral midbrain, it occurs to the side contralateral to the lesion. In Wallenberg syndrome, lateropulsion may be also observed in the eye movement (**saccadic lateropulsion**). Namely, lateral saccadic movement of the eyes to the affected side causes their overshoot, and upon the vertical saccadic movement, the eyes may be pulled toward the affected side during the saccade. The stepping test of Unterberger and Fukuda may be applied in combination with other electrophysiologic tests of vestibular function (Fornos et al, 2019). Patients step in place with their eyes closed; in a positive test, they will rotate (most commonly) to the side of the vestibular deficit.

If stability of the trunk position could be confirmed even with the eyes closed, then the past pointing test may be done. With the eyes closed in the standing position, the patient is asked to hold his or her arms straight ahead so that each hand can touch the examiner's opposing hand, and after raising both arms above, to return them to touch the examiner's hands again. Deviation of the patient's arms to either side is called **past pointing**, and usually it deviates to the side of the affected vestibular system. If a Bårany chair (a chair used for turning the patient for testing vestibular function) is available, it might be used to induce vertigo and nystagmus. If a Bårany chair is not available, it may be possible to ask the patient to turn around in the standing position, but this maneuver should be avoided in order to prevent falling.

An autoimmune inflammation of the audiovestibular system is called **Cogan syndrome**. This condition presents with positional vertigo and nystagmus resembling benign paroxysmal positional vertigo initially, followed by hearing loss, keratitis, fever, and myalgia. It is considered to be

caused by autoantibodies against the peptide expressed in the sensory epithelia of the inner ear, and thus is treatable (Lee et al, 2019).

A. VESTIBULAR NYSTAGMUS

Close observation of nystagmus is very important for testing vestibular functions. **Rotatory nystagmus** and **positional nystagmus** are suggestive of vestibular dysfunction.

On the bedside testing of vestibular functions, the first thing to do is to see if there is any spontaneous involuntary movement in the eyes in the primary position. Then, in the supine position, with the head tilted either up by 30 degrees or down by 60 degrees, the presence or absence of nystagmus is checked. If no nystagmus is induced in these positions, then the patient is requested to do lateral gaze to either side in each of the head positions. Then, with the head returned to the original flat position, the patient is asked to take the recumbent position on each side to see if vertigo or nystagmus is induced.

Some patients with cervical spondylosis may complain of vertigo when rotating the head on either side. In this case, the vertebral artery might be compressed during its passage through the transverse foramen at the upper cervical level, and this may cause vertigo as a result of ischemia. Vertigo and/or consciousness disturbance induced by the head rotation is known as **bow hunter's stroke**. Vestibular nystagmus is usually enhanced by blocking fixation, such as closing eyes or applying Frenzel glasses (lenses for inhibiting fixation), which is contrary to congenital nystagmus (see Chapter 9-6, p. 70).

B. THE VESTIBULO-OCULAR REFLEX

When the head moves, vestibular signals can maintain fixation of the eyes. A simple test of the vestibulo-ocular reflex (VOR) is the head thrust test. Patients are asked to fixate straight ahead, and the examiner quickly turns the head to the left or right. With an intact VOR the eyes will remain on target rather than moving with the head.

C. CALORIC TEST

The most objective test of vestibular function available at the bedside is the caloric test, although its execution requires some expertise. With the patient placed in the supine position with the head extended by 60 degrees, 100 to 200 ml of cold water (or 5 to 10 ml of ice water) is injected into the external auditory meatus while the patient

is asked to indicate as soon as vertigo or nausea is felt, and the induced nystagmus is observed. In healthy subjects, this procedure induces nausea and horizontal nystagmus with the quick phase directed toward the side contralateral to the injection. If the vestibular system is totally damaged, neither nausea nor nystagmus is induced. In case of mild impairment, the response to stimulation of the affected side is weaker compared with the intact side. If only nausea is induced while no nystagmus is induced, it might be due to external ophthalmoplegia or lateral gaze palsy.

In case of abnormally enhanced excitability of the vestibular system, the response to the water injection might be earlier and stronger than on the intact side. The caloric test is one of the most valuable tests in the diagnosis of brain death as it is a very good probe for brainstem function (see Chapter 33-7, p. 295).

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SWALLOWING, PHONATION, AND ARTICULATION

1. INNERVATION OF SWALLOWING, PHONATION, AND ARTICULATION

The final common pathway related to swallowing (deglutition), phonation (vocalization), and articulation is the ninth cranial nerve (glossopharyngeal nerve), the 10th cranial nerve (vagus nerve), and the 12th cranial nerve (hypoglossal nerve).

A. GLOSSOPHARYNGEAL NERVE AND VAGUS NERVE

The motor nucleus of the glossopharyngeal and vagus nerves is the **nucleus ambiguus** located in the center of the tegmentum of medulla oblongata on each side (see

Figure 5-3, p. 32). These two nerves pass behind the spinal tract of the trigeminal nerve and exit the medulla oblongata from its lateral edge. Then they exit the posterior cranial fossa through the **jugular foramen** just lateral to the foramen magnum (see Figure 8-2, p. 50).

Because the motor fibers of the glossopharyngeal nerve control swallowing and phonation by innervating the soft palate and pharyngeal muscles, their bilateral lesion causes marked **nasal voice (rhinolalia)** and **dysphagia**. In a unilateral lesion of the glossopharyngeal nerve, the soft palate of the affected side appears as a curtain because of its paralysis (**curtain sign**), and the uvula is pulled toward the intact side. Nasal voice and dysphagia can be quite severe even in the unilateral bulbar palsy, like in **Wallenberg syndrome** (Box 14-1).

BOX 14-1 WALLENBERG SYNDROME AND RELATED CONDITIONS

This is a good example of how many symptoms and signs can be produced by even a small lesion in the brainstem depending on its anatomic compactness.

The **lateral medullary syndrome** (Wallenberg syndrome) is systematically composed of (1) **bulbar palsy** due to a lesion of the nucleus ambiguus, (2) **crossed analgesia** on the ipsilateral face due to a lesion of the spinal tract and nucleus of the trigeminal nerve and on the contralateral limbs and trunk due to a lesion of the spinothalamic tract, (3) **vertigo and loss of equilibrium** due to a lesion of the vestibular nucleus, (4) **cerebellar ataxia** of the ipsilateral limbs due to a lesion of the cerebellar hemisphere, and (5) ipsilateral **Horner syndrome** due to a lesion of the descending sympathetic tract (Figures 14-1 and 14-2; see Table 8-1). This syndrome is caused by an infarction of the lateral part of the medullary tegmentum due to thrombosis of the posterior inferior cerebellar artery. Regarding (2) in the list earlier in the paragraph, if a lesion also involves the trigeminothalamic tract that crossed from the contralateral spinal nucleus of the trigeminal nerve, the hemianalgesia contralateral to the lesion involves the face as well.

Babinski and Nageotte in 1902 reported a 50-year-old man who suddenly presented with vertigo, ataxic gait, dysphagia, and dysphonia and showed soft palate paralysis and loss of gag reflex, Horner syndrome and cerebellar ataxia on the left, hemiparesis and decreased tactile and pain sense on the right, and sensory loss on bilateral face. At autopsy, four lesions were found in the left medulla oblongata, the largest occupying nearly the whole area of the left rostral medulla (Babinski & Nageotte, 1902). In the same paper, they also reported two other clinical cases in which the crossed sensory disturbance was seen only for pain but sparing touch and proprioception. Thus, Babinski–Nageotte syndrome consists of symptoms and signs seen in Horner syndrome plus hemiplegia (Figure 14-3). A condition similar to Babinski–Nageotte syndrome but without vertigo and cerebellar ataxia is called Cestan–Chenais syndrome (see Figure 14-3) (Krasnianski et al, 2006).

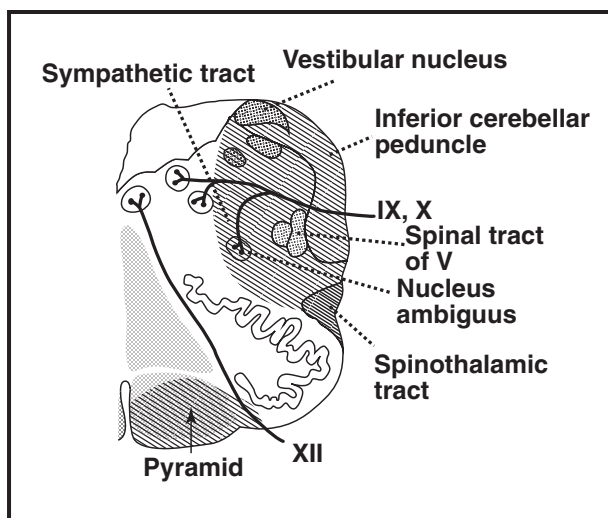


Figure 14-1 Schematic diagram of medulla oblongata indicating a lesion of the lateral medullary syndrome (Wallenberg syndrome). The lesion is indicated by striped lines on the left lateral medulla. V: trigeminal nerve, IX: glossopharyngeal nerve, X: vagus nerve, XII: hypoglossal nerve.

Motor fibers of the vagus nerve innervate not only pharyngeal muscles but also laryngeal muscles, including the vocal cord. In a severe unilateral lesion of the vagus nerve, both abductor and adductor muscles of the vocal cord are paralyzed so that the affected vocal cord becomes

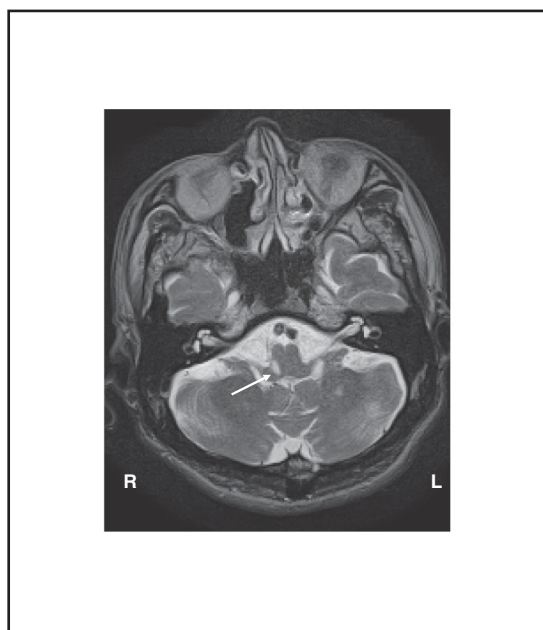


Figure 14-2 T2-weighted magnetic resonance imaging (MRI) of a patient who acutely presented with dysphagia, vertigo, nystagmus, left hemianalgesia including the left face, and Horner syndrome and cerebellar ataxia on the right (Wallenberg syndrome). A localized area of increased signal intensity is seen in the lateral part of the right medullary tegmentum (arrow).

By courtesy of Dr. Masutaro Kanda, Ijinkai Takeda General Hospital, Kyoto.

immobile and fixed in the intermediate position, causing **hoarseness** or **aphonia**, and even difficulty in breathing. As the **recurrent laryngeal nerve** innervates all laryngeal muscles other than the cricothyroid muscle, its unilateral lesion causes hoarseness, but it could well be asymptomatic since movement of the unaffected contralateral laryngeal muscles could be adequate. Bilateral lesions of the recurrent laryngeal nerve are seen in lead poisoning and may cause stridor and dyspnea. Bilateral complete lesions of the vagus nerve impair all the parasympathetic nerve supply to the visceral organs in addition to dyspnea.

As the glossopharyngeal nerve and the vagus nerve take the same course within and outside of the brainstem, they are often impaired together. When hoarseness and dyspnea are seen in addition to nasal voice and dysphagia, it suggests an impairment of not only the glossopharyngeal nerve but also the vagus nerve.

B. HYPOGLOSSAL NERVE

All the tongue muscles are innervated by the hypoglossal nerve, which is purely made of motor fibers. The **hypoglossal nucleus** is located in the medial part of the medullary tegmentum just anterior to the fourth ventricle (see Figure 5-3, p. 32). Fibers emerging from the nucleus pass anteriorly all the way through the medial part of the medulla and exit the medulla just lateral to the pyramid. Then they turn laterally and exit the posterior cranial fossa through the **hypoglossal canal** just lateral to the foramen magnum (see Figure 8-2a, p. 50).

In a unilateral lesion of the hypoglossal nerve, muscle atrophy is seen in the affected half of the tongue. In an acute lesion like hypoglossal neuritis, however, muscle atrophy may not be recognized until about a week after the clinical onset. Unilateral weakness of the tongue muscle can be detected by asking the patient to push the tongue against the cheek and by palpating its force from the cheek skin, but it can be confirmed more efficiently by demonstrating deviation of the protruded tongue to the affected side. It is due to the fact that the genioglossus muscle acts to protrude the tongue forward, and thus the muscle on the intact side overcomes that on the affected side, causing the tongue to deviate to the affected side on protrusion (Figure 14-4).

If the hypoglossal nerve is affected at the pyramid together with the corticospinal tract, the hypoglossal nerve palsy might be associated with the contralateral spastic hemiplegia (**hypoglossal crossed hemiplegia, medial medullary syndrome**).

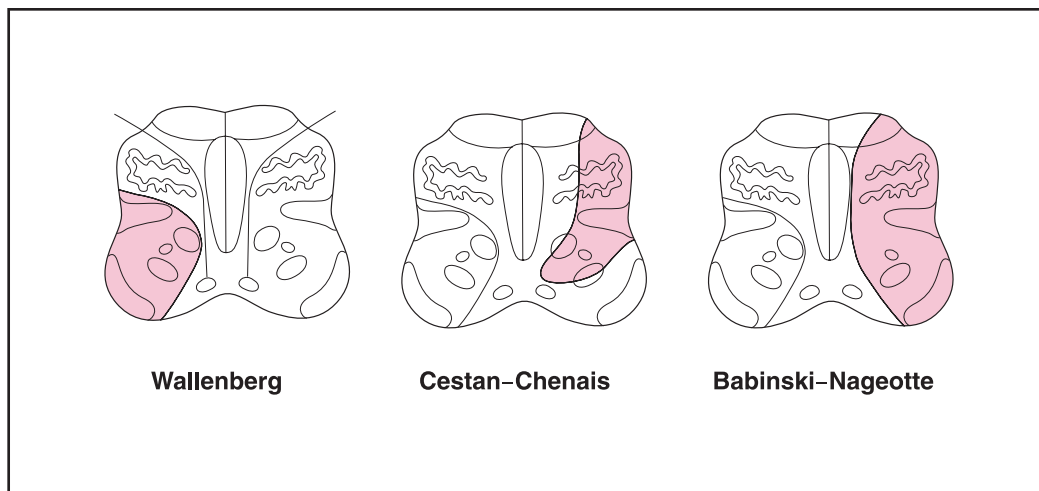


Figure 14-3 Schematic diagrams of medulla oblongata indicating three different distributions of lesions causing a crossed syndrome. Cited from Krasnianski et al., 2006, with permission.

C. CORTICAL INNERVATION OF BULBAR MUSCLES

The cortical area related to movements of the pharynx, larynx, and tongue is located in the ventral part of the face area of the primary motor cortex (see Figure 11-2, p. 84). The nucleus ambiguus and the hypoglossal nucleus are bilaterally innervated by motor cortices, with one exception (Box 14-2). Therefore, only bilateral lesions of the corticobulbar tracts cause functional disturbance such as dysphagia and dysarthria. This condition is called **pseudobulbar palsy** (see p. 104).

Even in a unilateral vascular lesion of the frontal lobe, mild dysphagia might be occasionally seen. This is explained by the hypothesis that there is some hemispheric

dominance in terms of swallowing, although the dominant side seems to differ among subjects. Furthermore, if there was a latent ischemic lesion in one cerebral hemisphere in the past, the new vascular lesion in the other hemisphere might cause pseudobulbar palsy.

D. INVOLVEMENT OF THE CRANIAL NERVES INNERVATING BULBAR MUSCLES

As the jugular foramen and the hypoglossal canal are closely located at the exit from the posterior cranial fossa, the last

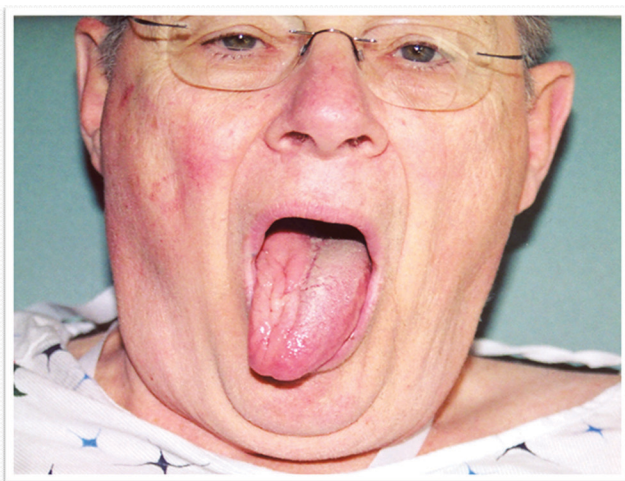


Figure 14-4 Right hypoglossal nerve palsy as a consequence of endarterectomy. Note atrophy of the right tongue and deviation to the right on protrusion. By courtesy of Dr. Stephen G. Reich of University of Maryland.

BOX 14-2 WHY DOES THE TONGUE DEVIATE ON PROTRUSION EVEN IN A UNILATERAL HEMISPHERIC LESION?

In patients with hemiplegia due to a unilateral vascular lesion of the cerebral hemispheric, the tongue often deviates to the hemiplegic side on protrusion despite bilateral cortical innervation of the bulbar muscles. This is because, unlike other bulbar muscles, the genioglossus muscle is unilaterally innervated by the contralateral motor cortex. As the genioglossus muscle acts to protrude the tongue forward, a lesion of the left cerebral hemisphere, for example, causes paralysis of the right genioglossus muscle so that the left genioglossus muscle overcomes the right to push the tongue toward the right. Thus, this phenomenon is similar to the peripheral hypoglossal paralysis, but muscle atrophy does not appear in the supranuclear lesion. Moreover, the central hypoglossal palsy is commonly associated with central paralysis of the orbicularis oris muscle, because of the anatomic proximity of the central pathways related to the two muscles.

TABLE 14-1 SYNDROMES DUE TO LESIONS OF MULTIPLE CRANIAL NERVES

Syndrome	Affected nerves	Sites of lesion
Gradenigo	V, VI	Tip of petrous pyramid
Vernet	IX, X, XI	Jugular foramen
Collet-Sicard	IX, X, XI, XII	Jugular foramen, hypoglossal canal
Raeder*	V, sympathetic nerve	Semilunar ganglion
Villaret	IX, X, XII, sympathetic trunk	Retroparotid skull base

* See Box 10-2 (p. 80).

four cranial nerves (IX to XII) tend to be affected together, a condition called **Collet-Sicard syndrome** (Table 14-1). Furthermore, the IXth and Xth cranial nerves may be affected together with the XIth cranial nerve (accessory nerve) at the exit from the jugular foramen (**jugular foramen syndrome**, **Vernet syndrome**) (see Chapter 15-1 for accessory nerve, p. 106). These syndromes can be caused by a mechanical injury or tumor but can be also due to inflammation (**multiple cranial neuritis**). Jackson in 1886 reported a patient with syphilis who presented with tongue muscle atrophy and paralysis of the soft palate and vocal cord on the right and a left brachial plexus lesion (Jackson, 1886). Consequently, similar conditions have been reported with various names such as Schmidt syndrome, resulting in some confusion.

Villaret in 1916 reported two cases of trauma in the mastoid process, manifesting with ipsilateral tongue atrophy, Horner syndrome, and paralysis of the vocal cord and soft palate; additional facial palsy occurred in one of the cases. Villaret postulated an extracranial involvement of the IXth, Xth, and XIIth cranial nerves and the sympathetic trunk, and named this condition “le syndrome nerveux de l’espace rétro-parotidien postérieur” (Villaret, 1916).

2. SOMATOSENSORY INNERVATION OF PHARYNX AND LARYNX

As described in relation to the trigeminal nerve (Chapter 10-2B, p. 79), the somatic sense of the oral cavity and tongue is mostly innervated by the trigeminal nerve, but the pharynx, uvula, tonsils, posterior two-thirds of the tongue, eustachian tube, and mucosa of the epiglottis are innervated by the sensory fibers of the glossopharyngeal nerve. Somatosensory impulses arising from the larynx are conveyed through the vagus nerve, and those from the inner aspect of the tympanic membrane and the tympanic cavity are conveyed through the tympanic nerve (Jacobson’s nerve), which joins the vagus nerve.

The primary sensory neurons are in the two small ganglia located just outside the jugular foramen, called the superior ganglion and the petrous (petrosal) ganglion (see Figure 14-5). Their central axons enter the spinal nucleus and the main sensory nucleus of the trigeminal nerve in the medulla

BOX 14-3 NAMED SYNDROMES RELATED TO BRAINSTEM AND CRANIAL NERVES

A number of combinations of symptoms and signs due to intramedullary lesions of the brainstem or extramedullary lesions of multiple cranial nerves have been described from the 19th century to the beginning of the 20th century. Based on subsequent clinical and pathologic studies of similar cases, those syndromes are called by the authors’ names (see Table 8-1, p. 53, and Table 14-1). For a complex combination of various symptoms and signs like some crossed syndromes, it is more convenient to use the names than to list the symptoms and signs. In some syndromes, however, later neurologists might not have correctly followed the initial description of the original authors, and some syndromes might have been modified. Moreover, not all the original descriptions were very precise. For example, an autopsy case reported by Jackson, who showed lesions of four (IX to XII) cranial nerves, had additional multiple ischemic lesions in the brain that made the interpretation of symptoms complex (see Pearce, 2006, for a review). A combination of lesions involving the vagus nerve and the hypoglossal nerve tends to be called Tapia syndrome, but the case reported by Tapia in 1904 was not precisely described (see Schoenberg & Massey, 1979, for a review). In those days, traumatic lesions due to combat were common in addition to cerebrovascular lesions and tumors. Moreover, similar conditions might have been called by different names. Furthermore, if some authors happen to have erroneously cited the original paper, then the subsequent authors might not have noticed those errors. In some cases, later descriptions were found to fit the relevant clinical cases better than the original report. Anyway, it is important to use the original author’s name(s) correctly. One of the authors (H.S.) has reviewed as many of those original papers as possible with the help of Dr. Kerry Mills in the medical library of Oxford University, and it was judged that the syndromes listed in Tables 8-1 and 14-1 were clearly described.

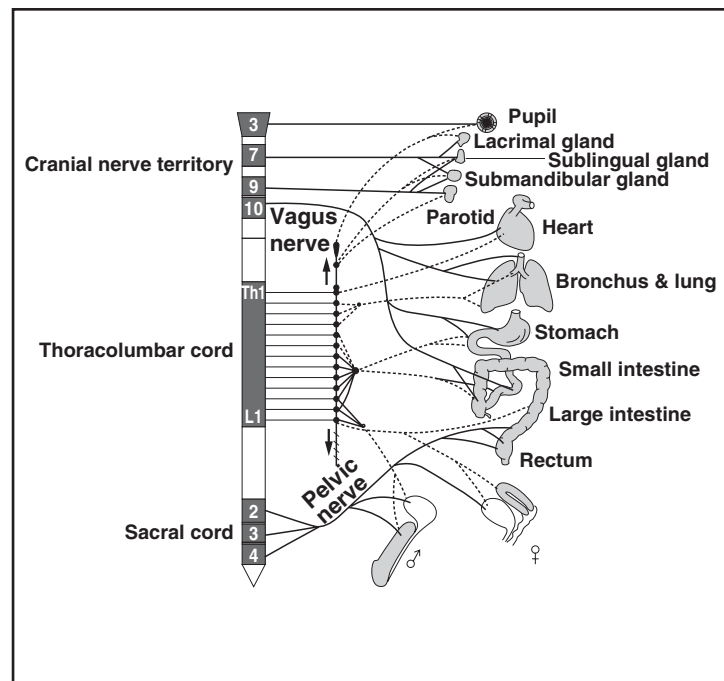


Figure 14-5 Autonomic efferent and afferent fibers, and gustatory fibers of the glossopharyngeal nerve.

oblongata. Neuralgia of these sensory nerves is called **glossopharyngeal neuralgia**. In an attack of this neuralgia, lightning pain starts from the root of tongue or the lateral wall of the pharynx and radiates toward the inner ear.

This sensory pathway also acts as an afferent arc of the **pharyngeal reflex (gag reflex)**, with the reflex center being in the solitary nucleus and with its efferent arc formed by motor fibers of the glossopharyngeal and vagus nerves. A typical example of this reflex is a vomiting reflex elicited when a toothbrush happens to touch the pharynx. The presence or absence of this reflex is also important for testing the integrity of medullary function in the examination of comatose patients (see Chapter 33-6C, p. 293).

Recently, transcutaneous stimulation of the auricular branch of the vagus nerve has been explored as a management option in some clinical disorders, including heart failure, migraine, and inflammatory bowel disease (Butt et al, 2020; Colzato & Beste, 2020; Wang et al, 2021).

3. TASTE SENSE

Gustatory impulses from the posterior third of the tongue are conducted through the glossopharyngeal nerve (see Figure 14-5), while those from the anterior two-thirds are conveyed through the facial nerve (see Chapter 11-1C, p. 84). The primary neurons are in the superior or petrous ganglion, and

their central axons enter the solitary tract and solitary nucleus in the medulla oblongata along with the gustatory fibers of the facial nerve. The central pathway from the solitary nucleus is the same as that originating from the facial nerve.

4. SALIVATION

Parasympathetic efferent fibers of the glossopharyngeal nerve act to facilitate salivation of the parotid gland. Its nucleus is the inferior salivatory nucleus located in the medullary tegmentum, and the preganglionic fibers bifurcate from the glossopharyngeal nerve as the lesser petrosal nerve to project to the otic ganglion (see Figure 14-5). The postganglionic fibers arising from the otic ganglion pass through the auriculotemporal nerve, which is a branch of the trigeminal nerve, and innervate the parotid gland (Box 14-4). The synaptic transmitter for the preganglionic fibers is acetylcholine (nicotinic) and that for the postganglionic fibers is acetylcholine (muscarinic).

5. AUTONOMIC INNERVATION OF VISCERAL ORGANS

Parasympathetic efferent fibers of the vagus nerve innervate the visceral organs such as the cardiovascular system,

BOX 14-4 FREY SYNDROME

A combination of sweating and flushing seen in the earlobe and the temporal skin while eating is called **auriculotemporal syndrome**. This phenomenon has been known as a complication of parotid gland surgery for many years. Frey in 1923 postulated the pathogenesis of this syndrome, and subsequently it has been called Frey syndrome. The syndrome is considered to be due to a posttraumatic misconnection between the parasympathetic efferent fibers facilitating salivation of the parotid gland and those of the auriculotemporal nerve targeting the skin. This condition used to be referred to as “gustatory sweating,” but it is reasonable to call it “salivary sweating” because the primary pathophysiology is a misconnection between the two groups of parasympathetic efferent fibers, just like the “crocodile tear phenomenon” (see Box 11-2, p. 88). Subcutaneous injection of botulinum toxin is used to treat this condition (see de Bree et al, 2007, for a review).

bronchus and lung, esophagus, small intestine, large intestine proximal to the middle of the transverse colon, liver, gallbladder, pancreas, and others (Figure 14-6). The preganglionic fibers arise from the **dorsal motor nucleus of the vagus**, which is located in the medullary tegmentum

anterior to the fourth ventricle and just lateral to the hypoglossal nucleus (see Figure 5-3, p. 32). Recently, this particular nucleus has drawn increasing attention as one of the earliest nervous systems to be affected by Lewy bodies in Parkinson disease (see Box 16-2, p. 122). This autonomic pathway is also called the **visceral efferent** or **visceromotor nerve**. Postganglionic neurons are located in close proximity to their target organs, and their fibers innervate each target organ by acetylcholine (muscarinic) as the neurotransmitter (see Figure 21-2, p. 205). Their action for the cardiovascular system is to slow down the pulse rate, to dilate the vessels, and to lower blood pressure, but for the gastrointestinal system, they act to facilitate intestinal peristalsis.

6. AFFERENT PATHWAY FROM THE VISCERAL ORGANS

The glossopharyngeal and vagus nerves play an important role in conveying the afferent input from visceral organs. Visceral afferents from the **baroreceptor of the carotid sinus (carotid body)**, which is situated in the wall of the internal carotid artery, pass through the glossopharyngeal nerve and convey impulses to the visceral afferent bipolar neurons in the superior and petrous ganglia (Figure 14-7).

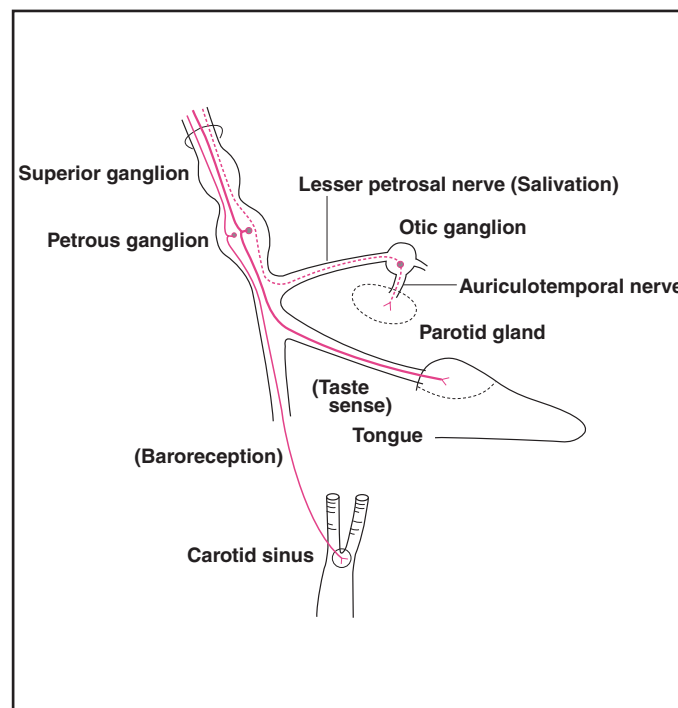


Figure 14-6 Parasympathetic efferent innervation of visceral organs.
Cited from Haymaker, 1969, with permission.

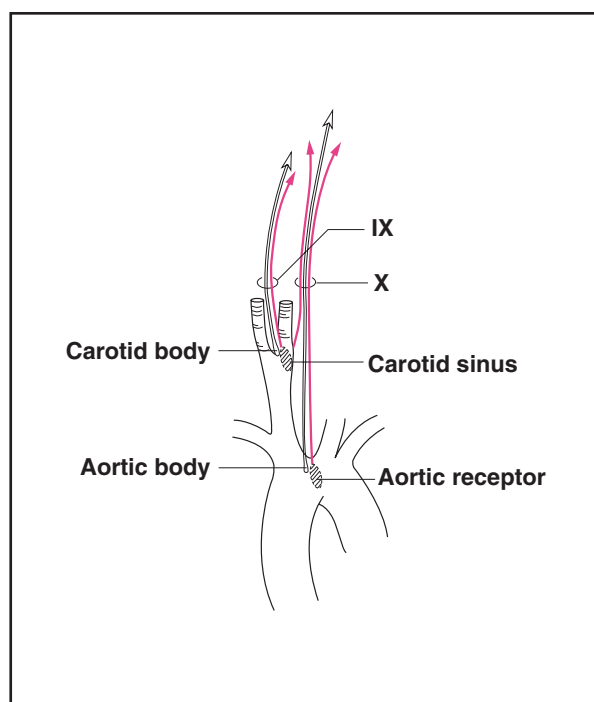


Figure 14-7 Afferent fibers related to baroreception. IX: glossopharyngeal nerve, X: vagus nerve.

The central axons of those neurons enter the solitary nucleus in the medulla oblongata and form a reflex arc with the dorsal motor nucleus of the vagus. On the other hand, the afferents from the **aortic receptors (aortic body)** in the aortic arch pass through the vagus nerve and project to the visceral afferent bipolar neurons in the nodose ganglion, and the central axons also project to the solitary nucleus to form a reflex arc like the baroreceptors from the carotid sinus (see Figure 14-7). Thus, the solitary nucleus plays an important role in various somatic as well as autonomic medullary reflexes (Cutsforth-Gregory & Benarroch, 2017).

These two pathways play an important role in the regulation of blood pressure. For example, if blood pressure changes rapidly, the baroreceptors sense the change and, via the medullary reflex arc, regulate the blood pressure automatically. If this reflex pathway is impaired, it becomes difficult to control the blood pressure upon sudden arising (**orthostatic hypotension**).

Visceral sensation from the pharynx, larynx, bronchus, esophagus, and abdominal visceral organs is conveyed to the central nervous system through the vagus nerve. Typical examples of visceral sensation are feelings of satiety and hunger. These visceral afferent neurons are also in the nodose ganglion, and the central axons convey impulses to the solitary nucleus. The efferents from the solitary nucleus are conducted to the cerebral cortex via the thalamus on

one hand and form a reflex arc with the autonomic efferent pathway in the brainstem on the other hand (see Chapter 21-2, p. 207). As visceral pain is conveyed through the sympathetic fibers, it will be discussed in Chapter 20-2.

7. EXAMINATION OF SWALLOWING, PHONATION, AND ARTICULATION

Swallowing is a highly complex neuromuscular process that requires precise coordination of multiple muscles, intact pharyngeal sensation, and central control, and its disturbance (dysphagia) can be caused by a variety of pathophysiologic mechanisms (Warnecke et al, 2021).

A. EXAMINATION OF THE TONGUE

If there is muscle atrophy and/or fasciculation (see Chapter 17-2B for fasciculation, p. 138) on one side of the tongue, it suggests involvement of either the ipsilateral medulla oblongata or the hypoglossal nerve. As the genioglossus muscle acts to protrude the tongue forward, the tongue will deviate to the affected side upon protrusion (see p. 146). By contrast, bilateral involvement of the tongue with muscle atrophy or fasciculation suggests degenerative diseases of the motor neurons (Figure 14-8). As was described in Box 14-2, among other tongue muscles, only the genioglossus muscle receives unilateral innervation from the contralateral motor cortex. Therefore, even in a unilateral hemispheric lesion, the tongue deviates to the side contralateral to the lesion upon protrusion.

B. EXAMINATION OF THE SOFT PALATE

In a unilateral lesion of the medullary tegmentum or the IX/X cranial nerves, the affected soft palate drops and moves incompletely (**curtain sign**). If the patient is asked to pronounce an “ah” sound loudly with the mouth kept open, the intact soft palate is seen to rise whereas the affected side rises only incompletely, if any. The **gag reflex** elicited by touching the lateral pharyngeal mucosa with a swab may be lost on the affected side. As the gag reflex causes bilateral contraction of the relevant muscles, loss of gag reflex on only one side suggests impairment of the afferent reflex arc on the affected side. Bilateral loss of the gag reflex suggests bilateral impairment of the IX/X cranial nerves, but it should be kept in mind that this reflex may not be clearly observable in normal aged subjects and even in some young subjects.



Figure 14-8 Muscle atrophy of the tongue in a patient with bulbospinal muscular atrophy (Kennedy disease).
By courtesy of Dr. Masutaro Kanda, Ijinkai Takeda General Hospital, Kyoto.

C. SPASM OF THE DEGLUTITION MUSCLES

Dysphagia in some patients might be due to excessive contraction or spasm of the deglutition muscles instead of their paralysis. This condition is called **spasmodic dysphagia**. In patients with dysphagia, therefore, it is useful to observe how they actually swallow a cup of water during the physical examination, and it is especially so for a possible case of spasmodic dysphagia.

D. DYSPHONIA

For testing phonation, it is useful to ask the patient to pronounce the same sound repeatedly, including a labial sound (mi-mi-mi), a palatal sound (ga-ga-ga), and a lingual sound (la-la-la). Paralysis of the soft palate is manifested as **nasal voice (rhinolalia)** whereas that of the vocal cord appears as **hoarseness**. When sound is lost completely, it is called **aphonia**. Although nasal voice and hoarseness are commonly seen in the common cold, the distinction is not difficult from the history and other accompanying symptoms. Furthermore, like dysphagia, spasm of the articulatory muscles may cause dysphonia (**spasmodic dysphonia**). Spasmodic dysphonia is most commonly an adult-onset focal dystonia but may be seen as a manifestation of autosomal dominant hereditary dystonia (“whispering dysphonia,” DYT4 dystonia) (Lohmann et al, 2013). Spasmodic dysphonia may be due to spasm of not just the vocal cord

but also the respiratory muscles as a whole (**laryngeal dys-tonia**) (Payne et al, 2014; Simonyan et al, 2021).

E. DYSARTHRIA

A condition in which a patient cannot speak properly or articulately is called **dysarthria**. Dysarthria characterized by prolonged and irregular intervals and variable loudness among syllables is called **scanning speech** or **slurred speech**. This kind of speech is also characterized by an abrupt increase in the loudness (**explosive speech**). Dysarthria with these characteristics is commonly seen in patients with cerebellar ataxia. In disorders of the basal ganglia like Parkinson disease, speech is characterized by low-pitched and monotonous voice, and intervals between syllables become increasingly shorter while speaking. This speech is compared to the propulsion or festination seen in parkinsonian gait (see Chapter 16-2B, p. 121) and may be called **oral festination** (Moreau et al, 2007). It is important to know that dysarthria due to pseudobulbar palsy is usually accompanied by severe dysphagia.

F. BULBAR PALSY AND PSEUDOBULBAR PALSY

When muscle atrophy and/or fasciculations are seen in the tongue in patients with dysphagia, dysphonia, and/or dysarthria, it suggests **bulbar palsy**, and if the jaw jerk

is clearly enhanced, it suggests **pseudobulbar palsy**. The term “pseudobulbar palsy” comes about because it looks like a lesion of the medulla, but it is not. However, bulbar palsy may not necessarily be accompanied by tongue atrophy, and the jaw jerk may not necessarily be enhanced even in pseudobulbar palsy. If there is a clear asymmetry in paralysis of the soft palate or if there are nasal voice and/or hoarseness of marked degree, then it more likely suggests the presence of bulbar palsy. **Amyotrophic lateral sclerosis** is characterized by a coexistence of bulbar palsy and pseudobulbar palsy. In this condition, therefore, there is muscle atrophy in the tongue and yet the jaw jerk is hyperactive.

Patients with pseudobulbar palsy occasionally tend to cry or laugh for no reasonable cause (emotional incontinence). These are called **forced crying** and **forced laughing (forced laughter)**, respectively. The pseudobulbar affect is commonly seen in motor neuron diseases (Floeter et al, 2014). In most cases, the facial expressions are not accompanied by emotional changes (mirthless crying, mirthless laughter), and they are considered to be due to paroxysmal involuntary contractions of emotional muscles. **Frequent yawning** has been reported in patients with amyotrophic lateral sclerosis as another sign of pseudobulbar palsy (Williams, 2000).

Somatic sense of the soft palate and pharynx can be tested by using swabs only when judged necessary based on the history and other physical signs. The gag reflex can also be tested by the same maneuver. Taste sense of the posterior third of the tongue may also be tested, but this is more difficult to test than the anterior two-thirds of the tongue. As was described in relation to the facial nerve, salivation of the parotid gland is not tested in the ordinary clinical setting (see Chapter 11-2D, p. 87). Autonomic reflexes of the cardiovascular system will be explained in the chapter of the autonomic nervous system (see Chapter 21-3D, p. 208). On the bedside examination, we can test whether the pulse rate changes with deep breathing or not and whether the blood pressure drops when the patient stand up from the supine position, and we can also perform auscultation of bowel sounds.

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NECK AND TRUNK

1. EXAMINATION OF THE NECK

A. POSTURAL ABNORMALITY OF THE NECK

Posture of the neck can be observed in the sitting or standing position. Stooped posture of the neck may be physiologically seen in the aged population. Two abnormal postures of the neck are **flexed posture**, seen in patients with Parkinson disease or parkinsonism, and dropped head syndrome. In Parkinson disease and parkinsonism, not only the neck but also the whole upper trunk is commonly flexed. Moreover, the tone of the neck muscle is often increased, and there is resistance against passive lateral flexion and rotation. Occasionally, cogwheel rigidity might also be found in the neck muscles (see Chapter 16-2B, p. 121).

In the **dropped head syndrome**, flexed posture is almost restricted to the neck muscles. Two conditions are conceivable for this syndrome: weakness of the neck extensors and excessive contraction (dystonia) of the neck flexors. To distinguish between these, it is useful to test the strength of the neck extensor muscles by asking the patient to lift up the head in a prone position. Weakness of the neck extensors in this condition is thought to be myogenic rather than neurogenic, and is seen in parkinsonism, especially multiple system atrophy (see Box 21-2, p. 209).

Extended posture of the neck (**retrocollis**) is commonly seen in patients with progressive supranuclear palsy. However, absence of retrocollis does not completely rule out this disease. It is also seen in tardive dyskinesia associated with extension of the trunk.

Rotated posture of the neck to either side is called **torticollis**. As this condition is considered to be caused by a mechanism of focal dystonia, it is also called **cervical dystonia**. It is often a partial manifestation of generalized dystonia, and it could be its initial manifestation, although it more commonly begins in the leg, particularly

with childhood onset. As the postural abnormality of neck may almost disappear in the supine position, it is important to observe the neck in the upright position. In the early stage of torticollis, it may not stay with the rotated posture, but it may present with repetitive rotatory phasic movements of the neck to one side, resembling tremor. In the continuously rotated state of the neck, examination of the neck tone might show increased resistance to passive rotation to the normal position. This is in contrast with the neck tone in generalized dystonia, because muscle tone is not increased in most cases of generalized dystonia. In torticollis of long duration, the sternocleidomastoid muscle (SCM) contralateral to the side of neck rotation is often hypertrophic. Usually the patient cannot return the neck to normal position, and even if it is possible to do so to some degree, it tends to return to the rotated posture soon.

Sensory trick is a characteristic sign of focal dystonia, especially torticollis (see Chapter 19-6, p. 172). Typically, the neck returns to the normal position as soon as the cheek or other part of the head is touched by the patient's own hand or, less commonly, by another person's hand. In standing posture against a wall, the neck may return to the normal position as soon as the posterior head touches the wall, but it rotates back again as soon as the patient steps away from the wall. In this case, the sensory trick seems to occur even before the posterior head is actually touched if the patient could anticipate the touch in advance. A mechanism underlying the sensory trick has not been fully elucidated, but this is an example of sensorimotor integration that serves favorably for the patient.

B. EXAMINATION OF MUSCLES INNERVATED BY THE ACCESSORY NERVE

The muscles innervated by the 11th cranial nerve (accessory nerve), namely the **sternocleidomastoid muscle** (SCM) and the **trapezius muscle**, can be tested when

examining the neck. The SCM acts to rotate the neck to the opposite side. Therefore, when testing the left SCM, for example, that muscle should be observed as well as palpated by asking the patient to turn the neck to the right. The two SCM muscles are activated together to flex the neck. The SCM is considered to be innervated by the ipsilateral motor cortex, because, in a focal motor seizure, the neck rotates to the side contralateral to the epileptogenic focus. The trapezius muscle can be observed and palpated while the patient is asked to raise the shoulders. In unilateral trapezius palsy, the shoulder of the affected side is lower compared to the intact side when the patient in a standing position is observed from behind.

C. CERVICAL SPONDYLOSIS

In radiculopathy (disorders of nerve roots) due to cervical spondylosis, radiating pain might be induced in the arm by tilting the neck posteriorly and laterally to the affected side (**Spurling sign**). This maneuver makes the intervertebral foramen even narrower so that the nerve root is compressed to produce “root pain.”

Other causes of **neck pain** include whiplash injury, intervertebral disc protrusion, ossification of the posterior longitudinal ligament, atlantoaxial dislocation, and Klippel–Feil syndrome. Severe neck pain of acute onset associated with limited range of motion in aged subjects is seen in the **crowned dens syndrome**. In this case, computed tomography (CT) of the neck shows calcification around the odontoid process (dens), and levels of C-reactive protein (CRP) are elevated in the serum. It is important to consider this condition because it sometimes resembles meningitis and also because it responds well to nonsteroidal anti-inflammatory agents (Taniguchi et al, 2010).

In addition to radiculopathies due to abnormalities of the cervical spine as listed earlier, it should be kept in mind that systemic diseases such as thyroiditis, lymphadenitis, and vascular diseases may also cause neck pain.

D. LHERMITTE SIGN

Lhermitte sign is a radiating pain starting from the upper back down to the dorsal aspects of lower limbs upon neck flexion. This phenomenon is commonly encountered with an intramedullary lesion of the cervical cord, especially in multiple sclerosis or neuromyelitis optica. The underlying mechanism is thought to be due to abnormal impulse transmission within the posterior column.

E. MENINGEAL IRRITATION

When the history suggests a possibility of meningitis or subarachnoid hemorrhage with severe headache and nausea, the presence or absence of **nuchal stiffness** is important. This is tested by observing the range of neck flexion by raising the head passively in supine position. If one of the examiner's hands is placed behind the neck during this maneuver, it may be possible to feel a reflex contraction of the nuchal muscle. This reflex spasm is considered to be due to inflammatory involvement of the posterior cervical roots as they pass through the meninges. It is important to keep in mind that, in subarachnoid hemorrhage, the nuchal stiffness is not detectable until about 24 hours after the onset.

Kernig sign is also well known as a sign of meningeal irritation. This is tested by passively raising a lower limb with the knee flexed in the supine position and then by extending the knee joint. If the angle of knee extension is limited or if the patient complains of pain on the posterior aspect of the thigh, Kernig sign is judged positive. Sometimes, this test might be confused with **Lasègue sign**, which is seen in patients with lumbar radiculopathy upon the straight leg raising test (see the next paragraph). Moreover, Kernig sign may be negative while nuchal stiffness is present, and thus nuchal stiffness is more sensitive to the state of meningeal irritation. Similarly, **Brudzinski sign** is judged positive when neck flexion causes a patient's hips and knees to flex.

For the **straight leg raising test**, the relaxed and extended lower extremity is slowly lifted passively in a supine position, and the patient is instructed to report whether pain occurs or not. If this maneuver causes radiating pain downward along the back of the leg (Lasègue sign), it suggests the presence of lumbar radiculopathy.

2. EXAMINATION OF THE TRUNK

Observation of posture in the standing position provides important information. Flexed posture of the neck and upper trunk is commonly seen in patients with Parkinson disease or parkinsonism. Marked flexion of the trunk to 45 degrees or even more in the standing position is called **camptocormia (bent spine syndrome)** (Ali et al, 2018). This postural abnormality becomes less pronounced in the recumbent position. There are a variety of etiologies, both peripheral and central, for this condition (see Srivanthapoom & Hallett, 2016, for a review), but there are two main hypotheses: (1) dystonia or rigidity of the truncal muscles and (2) myopathy of the paraspinal muscles (Jankovic, 2010; von Coelln et al, 2008). Lateral bending

of the trunk is called **Pisa syndrome** (Barone et al, 2016). In addition, scoliosis is also common in Parkinson disease.

In patients with Parkinson disease or parkinsonism, movement in general is slower than normal, especially in turning around or sitting up on bed, and standing on the floor (**bradykinesia**).

Muscle tone of the truncal muscles is difficult to test, but tone of the paraspinal muscles can be estimated by the following maneuver. With a patient placed in the supine position with the knees flexed and with the examiner's hands holding the lateral aspect of the knees, the examiner rotates the patient's trunk to each side and feels the resistance. In healthy subjects, there is little, if any, resistance against this maneuver, but in patients with Parkinson disease, resistance can be felt, often with significant asymmetry. In patients with rigidity of limbs more on one side than on the other, the resistance against the trunk rotation away from the more rigid side is greater than that toward the more rigid side. In this case, the trunk often shows scoliosis concave to the more rigid side.

This postural abnormality of the cervical or thoracic spine is seen in disorders of the spine itself, such as kyphosis, lordosis, and scoliosis of the spine. Usually, differentiating neurological disorders from these spine disorders

is not difficult from the history and other accompanying signs, but it is helpful to compare the postural abnormality during wakefulness and sleep. Abnormalities of the spine persist during sleep while abnormality of muscle tone tends to decrease during sleep. Furthermore, abnormality of muscle tone may change depending on the posture of the patient.

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MOTOR FUNCTIONS AND MOVEMENT DISORDERS

1. FINAL COMMON PATHWAY OF THE MOTOR SYSTEM

The final common pathway for movement consists of the central motor pathway, which projects from the motor cortex through the corticobulbar tract to the brainstem motor nuclei or through the corticospinal tract to the anterior horn of the spinal cord, and the peripheral motor pathway, which projects from the brainstem motor nuclei or the anterior horn cells by means of the α motor fibers to skeletal muscles. Neurons of the cerebral cortex forming the corticobulbar and corticospinal tracts are called **upper motor neurons**, and neurons of the brainstem motor nuclei and the spinal anterior horn forming the α motor fibers are called **lower motor neurons**. A representative degenerative disease characterized by systematic involvement of the upper and lower, or either, motor neurons is **motor neuron disease**.

A. MOTOR CORTEX AND CORTICOSPINAL TRACTS

The motor cortex is formed of the **primary motor area** (Brodmann area 4; M1), **supplementary motor area** (SMA), **lateral premotor area**, and **cingulate gyrus**. SMA is further divided into **SMA proper** and **pre-SMA** (see Figure 11-3, p. 85). Electrophysiologic and neuroimaging studies of cortical activities in humans suggest that voluntary unilateral simple movement of the hand is preceded by increasing activity first in the pre-SMA and SMA proper bilaterally, then in the lateral premotor area and M1 bilaterally but with contralateral predominance, and finally neurons primarily of the M1 are activated so that their impulses descend through the fast-conducting corticospinal tract to activate the corresponding anterior horn cells (Shibasaki, 2012). For the corticospinal tracts, all fibers originating from the M1 and most fibers from the lateral premotor area form the **direct corticospinal tract**, while the fibers originating from the SMA proper and some fibers from the lateral premotor area project to

the spinal cord via the M1. In addition, some fibers from the SMA proper and the lateral premotor area descend to the spinal cord indirectly via brainstem nuclei such as the reticular formation and vestibular nuclei (**indirect corticospinal tract**). This indirect pathway via the reticular formation is called the **corticoreticulospinal tract** (Jang & Lee, 2019).

i. Pyramidal tract and upper motor neuron

The corticospinal tract in the narrow sense indicates a pathway directly projecting to the spinal cord, but in the wider sense it does include fibers that arise not in the cortex. The **pyramidal tract** was originally named to indicate fiber tracts passing through the medullary pyramid, and the term is used to mean the direct corticospinal tract. Fibers of the pyramidal tract originate not only from large neurons in the M1 (pyramidal tract cell, Betz cell) but also from a much larger number of other neurons in the M1, neurons in the SMA proper, and some neurons in the lateral premotor area and in the postcentral gyrus. Clinically, signs due to involvement of the pyramidal tract are called **upper motor neuron signs** or **pyramidal tract signs** (pyramidal signs).

ii. Corticospinal tract

Impulses conducted through the direct corticospinal tract are transmitted to the anterior horn cells either directly (monosynaptically) or via spinal interneurons. Neurotransmission at the nerve terminal is mostly glutamergic. In particular, most impulses targeting intrinsic hand muscles are transmitted monosynaptically, which is thought to be effective for the quick and precise movement of fingers. In fact, in a relatively mild or early lesion of the motor cortex or corticospinal tract, the initial sign is disturbance of fine finger movement rather than weakness.

About 85% of fibers constituting the pyramidal tract cross at the **pyramidal decussation** of the medulla and

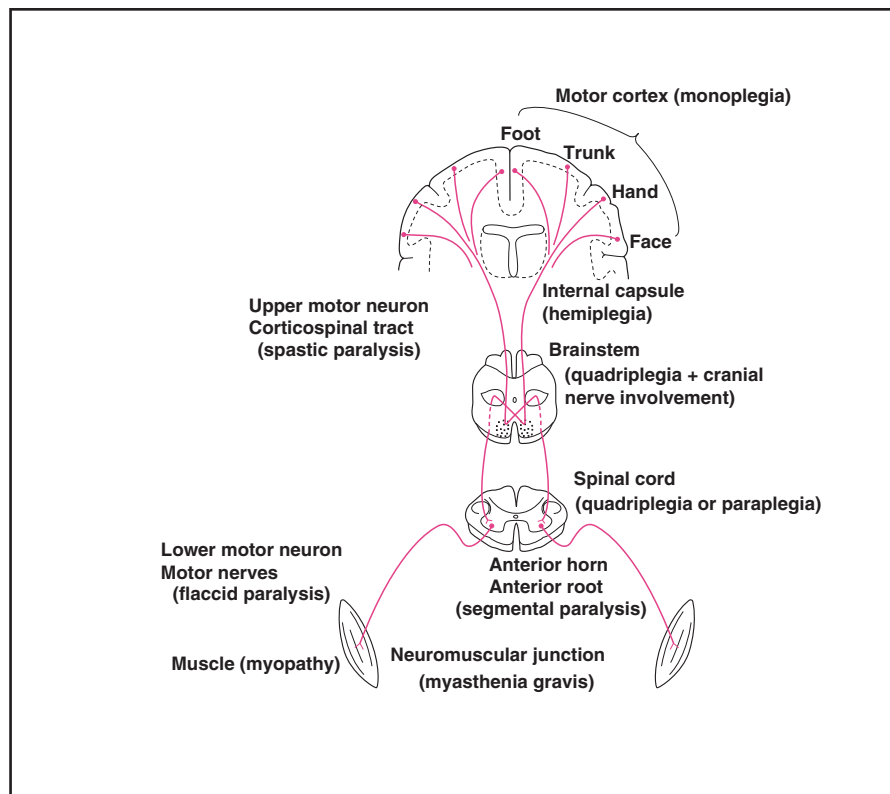


Figure 16-1 Schematic illustration of motor pathways from the motor cortex to the muscles. Distribution of motor paralysis is shown in the parenthesis for each site of lesion.

descend through the contralateral **lateral corticospinal tract**. The remaining 15% of fibers descend through the ipsilateral **anterior corticospinal tract** (Figures 16-1, 16-2, and 16-3). However, the ratio of the crossing fibers is known to vary considerably among subjects, and it is known that, in some exceptional cases, all fibers cross. The anterior corticospinal tract terminates at the lower cervical level. There are also reports of cases with ipsilateral hemiparesis due to a hemispheric lesion as a result of nondecussating corticospinal tract (Zhou et al, 2021).

In a unilateral infarction of the medulla, ipsilateral hemiplegia may be seen occasionally (Opalski syndrome). This is explained by an involvement of the corticospinal tract after crossing (distal to the pyramidal decussation) in the medulla (Nakamura et al, 2010). Furthermore, in mechanical compression of the craniocervical junction, spastic paralysis of unusual distribution may be seen (Yayama et al, 2006). For example, bilateral weakness of the upper limbs is called cruciate paralysis, and a combination of spastic paralysis of the ipsilateral upper limb and pyramidal signs of the contralateral lower limb may be called hemiplegia cruciata. Structural or functional misinnervation of the corticospinal tracts may present itself as mirror movements (Box 16-1).

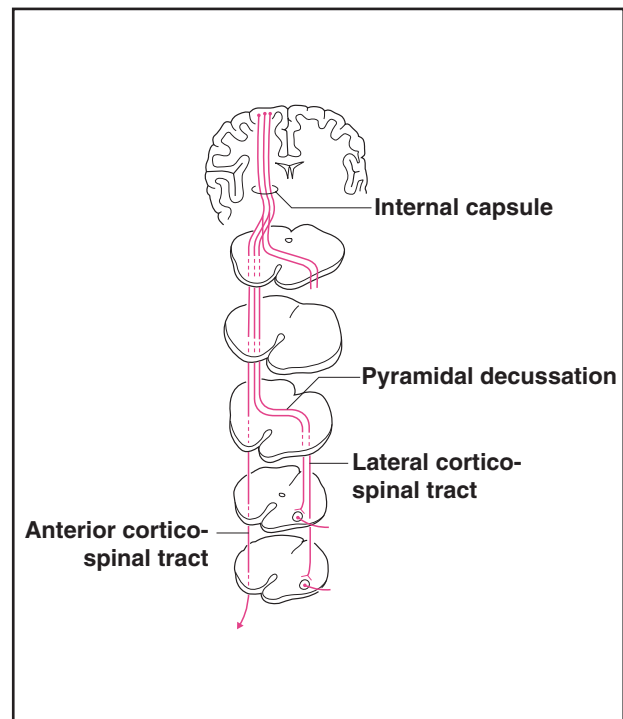


Figure 16-2 Schematic diagram showing the lateral and anterior corticospinal tracts.

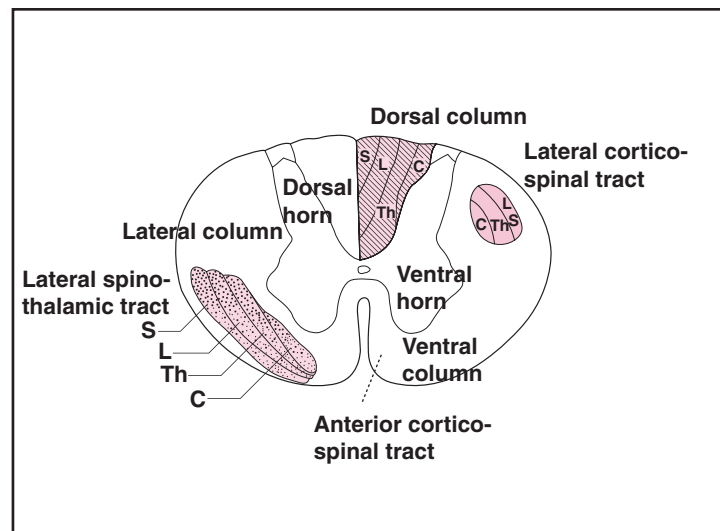


Figure 16-3 Laminar structures of the lateral corticospinal tract and somatosensory pathway in the cervical cord. C (cervical), Th (thoracic), L (lumbar), and S (sacral) in the descending lateral corticospinal tract indicate the target segments and those in the ascending lateral spinothalamic tract and dorsal column indicate the segments of origin. Cited from Haymaker, 1969, with permission.

iii. Lesion of the corticospinal tract and distribution of motor paralysis

As described in Chapter 11 (see Figure 11-2, p. 84), there is somatotopic organization in the M1. Different body parts are quite separate, but within a body part the muscle representations are intermixed. The corticospinal tract descends

medially through the cerebral white matter and converges at the posterior limb of the internal capsule like the pivot of a fan (see Figure 16-1). Thus, if the M1 is affected focally, or if the corresponding group of the corticospinal fibers is subcortically affected, the resulting paralysis is limited to a part of the contralateral extremity or face innervated by the

BOX 16-1 MIRROR MOVEMENT

In this condition, intended movement of one hand is accompanied by the same movement in the opposite hand. This phenomenon may cause significant difficulty in daily life, especially in actions requiring movement of bilateral fingers like piano playing. Mechanisms underlying **congenital mirror movement** differ among patients. In some cases, the uncrossed corticospinal tract is excessively developed, and activation of unilateral motor cortex may cause simultaneous activation of both the crossed and uncrossed corticospinal tracts. In other cases, the motor cortex ipsilateral to the intended movement might be co-activated with the contralateral motor cortex. By using the triple stimulation technique of transcranial magnetic stimulation (TMS) in a case of congenital mirror movement, Ueki et al. (2005) reported evidence for predominant ipsilateral innervation. In a patient with **Kallmann syndrome**, which is an X-linked hereditary disease characterized by anosmia, hypogonadotropic hypogonadism, and mirror movement (see Chapter 6-2, p. 37), the late phase of Bereitschaftspotential was distributed over the motor area not only contralateral but also ipsilateral to the intended unilateral hand movement, suggesting activation of bilateral motor cortices (Shibasaki & Nagae, 1984).

Mirror movement can be also seen in acquired conditions (**acquired mirror movement**). When a patient during the recovery phase from stroke moves the intact hand, similar movement may be seen in the paretic hand (Etoh et al, 2010). Additionally, an attempt to move the paretic hand may be associated with similar movement in the intact hand. The latter phenomenon might be related to the fact that, even in healthy subjects, an attempt to make a complex movement by one hand is associated with bilateral activation of motor cortices (Kitamura et al, 1993b; Shibasaki et al, 1993). Therefore, it is postulated that, in patients with severe motor disability irrespective of its etiology, even a simple movement of one hand might cause bilateral activation of motor cortices, thus resulting in mirror movement.

In relation to this, when interpreting data of Bereitschaftspotential and neuroimaging studies for investigating a mechanism of motor recovery from stroke, increased activation of the motor cortex ipsilateral to the paretic limb might be due to subjective complexity of the motor task and may have little to do with compensatory activation of the nonaffected cerebral hemisphere.

damaged area (**monoplegia**). For example, if the hand area of the left precentral gyrus is focally affected, motor paralysis or disturbance of fine finger movement may appear only in the right hand.

When the posterior limb of the internal capsule is affected, all fibers of the corticospinal tract are involved, causing weakness of all contralateral skeletal muscles, including the lower face (**hemiplegia**). A typical example is lacunar infarction in the internal capsule. In this case, even a small lesion localized to the posterior limb of the internal capsule can cause complete hemiplegia involving the lower face and the upper as well as lower extremities. As the somatosensory system is not involved in this case, it causes **pure motor hemiplegia** (see Box 2-2, p. 7). Clinically, pure motor hemiplegia involving the lower face most likely suggests the presence of a small lesion in the posterior limb of the internal capsule. A lesion in the pontine base may also cause pure motor hemiparesis sparing the lower face. An ischemic lesion in the corona radiata may cause hemiparesis predominantly involving the contralateral upper and lower limbs.

By contrast with these two examples, if the corticospinal tract is involved between the motor cortex and the internal capsule in the cerebral white matter, the resulting hemiplegia is incomplete in its distribution. Namely, if an infarction occurs in the deep cerebral white matter, weakness is often seen in the contralateral upper and lower limbs, sparing the face. In this case, somatosensory impairment of various degrees may be also present.

In the brainstem, the corticospinal tract is located most anteriorly from the midbrain to the medulla. In the cerebral peduncle, fibers projecting to the brainstem motor nuclei are located medially and those projecting to the caudal spinal cord are located laterally, with those projecting to the rostral spinal cord in between. This arrangement is reasonable because the fibers targeting the more rostral motor nuclei can easily cross from the medial part of the tract. Most of the fibers project to the brainstem motor nuclei bilaterally, but those fibers innervating the lower facial muscles and the genioglossus muscle of the tongue project only to the contralateral facial motor nucleus and hypoglossal nucleus, respectively (see Box 11-1, p. 86, and Box 14-2, p. 99).

B. STRUCTURE AND FUNCTION OF SPINAL CORD

The spinal cord is formed of 30 segments: eight cervical, 12 thoracic, five lumbar, and five sacral. There is no structural

boundary dividing the segments, and the segments are numbered according to the nerve root emerging from each segment. By contrast, there are 29 vertebrae: seven cervical, 12 thoracic, five lumbar, and five sacral. In the cervical cord, therefore, the number of spinal segments is not the same as that of vertebrae. The first cervical nerve exits the spinal canal rostral to the first cervical vertebra, and the eighth cervical nerve exits the canal between the seventh cervical vertebra and the first thoracic vertebra. Thus, the cervical nerves emerge from the spinal canal rostral to each corresponding vertebra with the exception of the eighth cervical nerve. All other spinal nerves from the first thoracic nerve down emerge from the spinal canal caudal to each respective vertebra (Figure 16-4).

The caudal end of the spinal cord (**medullary conus, conus medullaris**) is located between the first and second lumbar vertebrae, and thus the spinal cord is shorter than the spinal canal. The lumbar cord is actually located at the level of the caudal end of the thoracic vertebrae, and the nerve roots originating in the lumbar cord (**cauda equina**) descend through the spinal canal for some distance and then emerge from each respective intervertebral foramen (see Figure 16-4).

i. Laminar structure of the lateral corticospinal tract

Fibers of the lateral corticospinal tract and those of the anterior corticospinal tract targeting each segment of the spinal cord leave the tracts and turn anteromedially and posteriorly, respectively, so that they project to the anterior horn at each level. As seen in the transverse section of the lateral corticospinal tract (see Figure 16-3), fibers projecting to the cervical segments are located medially and those projecting to the sacral segments are located laterally, so that the fibers targeting the more rostral segments can leave the tract easily. For the same anatomic reason, when the cervical cord is compressed from the lateral side, the pyramidal signs first appear in the ipsilateral lower extremity. When the lateral compression is strong, pyramidal signs also appear in the contralateral lower extremity because of a contrecoup effect against the opposite wall of the spinal canal and partly because the spinal cord is fixed to the lateral wall of the spinal canal by the denticulate ligament, which causes mechanical distortion of the cord. For the same reason, if the pyramidal signs are clearly limited to the ipsilateral extremities, it more likely suggests the presence of an intramedullary lesion.

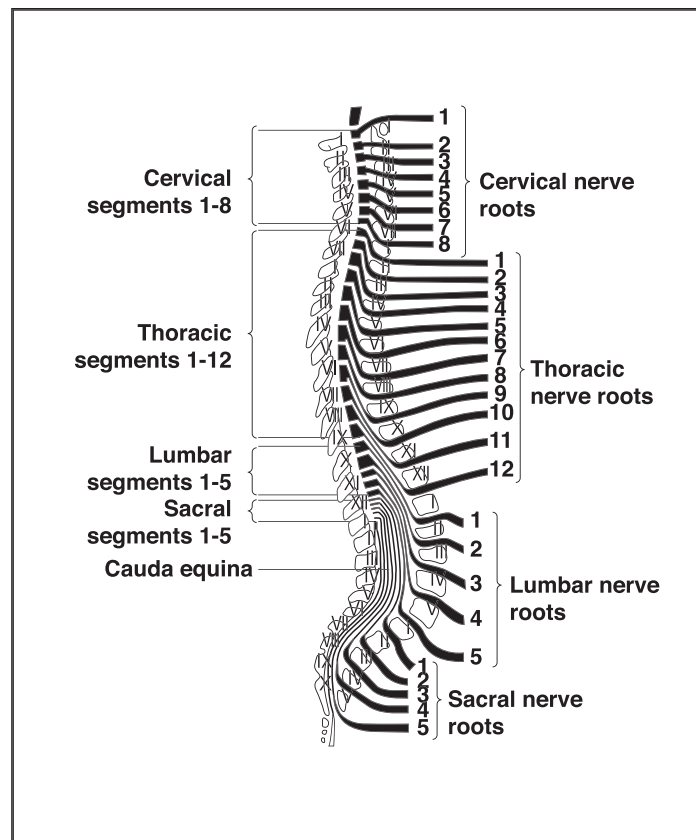


Figure 16-4 Anatomic interrelationship among the spinal cord segments, spinal nerve roots, and vertebrae. Cited from Haymaker, 1969, with permission.

ii. Blood perfusion of the spinal cord

The cervical cord receives blood perfusion mostly from the **anterior spinal artery**, which is formed by branches of the vertebral arteries (see Figure 9-3, p. 59). The thoracic cord is supplied by **radicular arteries**, which bifurcate from the aorta. Therefore, narrowing of the intervertebral foramen may compress not only the nerve roots but also the radicular arteries, causing ischemic damage to the cord. In relation to this, as surgical treatment of aortic diseases may cause an ischemic lesion in the cord, intraoperative electrophysiologic monitoring of spinal functions can be used to detect any functional abnormality, which otherwise may not be detected until the effect of anesthesia wears off. Caudal segments of the thoracic cord and the lumbar cord are supplied by the **Adamkiewicz artery**, which bifurcates from the aorta and enters the upper lumbar cord from the left.

The anterior spinal artery descends along the anterior aspect of the spinal cord at the midline, and its branches perfuse the anterior two-thirds of the cord (Figure 16-5).

If the anterior spinal artery is occluded, the corticospinal tracts, spinothalamic tracts, and sympathetic nervous pathways are bilaterally affected, causing paralysis and loss of pain/temperature sense below the affected segment and sphincter disturbance, but sparing proprioceptive sensation because of intact posterior columns (see Chapter 16-1, p. 112). This condition is called **anterior spinal artery syndrome**. Posterior spinal arteries are located at the lateral posterior aspect of the spinal cord and send their branches to the posterior column on each side. However, some patients who presented with bilateral impairment of proprioceptive sensation, pyramidal signs, and sphincter disturbance due to unilateral occlusion of the posterior spinal artery have been reported (Moriya et al, 2011).

Veins from the spinal cord exit the spinal canal along with the nerve roots and enter the corresponding venous plexus. Therefore, the radicular veins might be also compressed by the narrowed intervertebral foramen, thus causing a congestive lesion in the corresponding segments of the spinal cord.

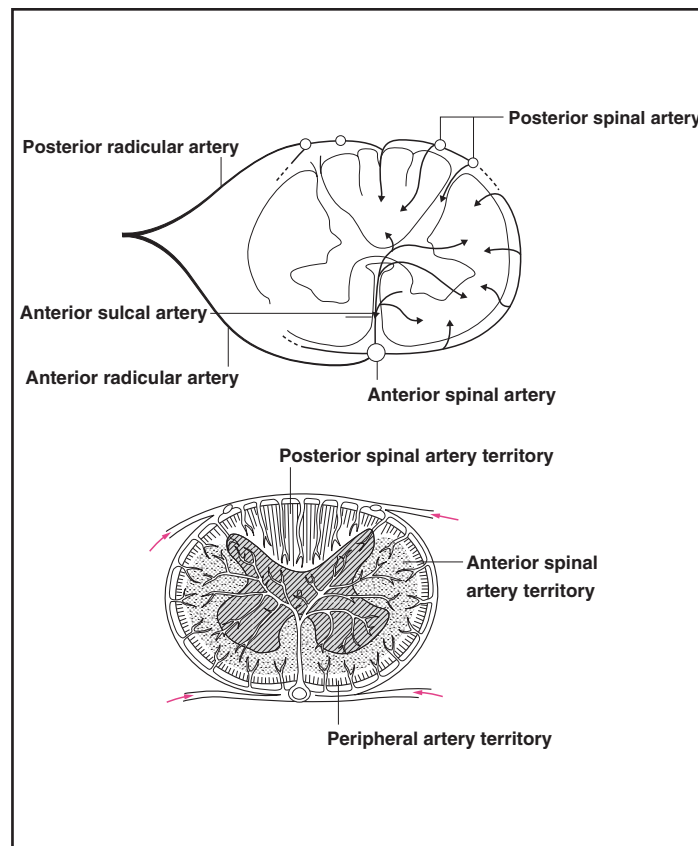


Figure 16-5 Transverse section of spinal cord showing its arterial supply.

C. ANTERIOR HORN CELLS AND PERIPHERAL MOTOR NERVES

i. α motor fibers

For voluntary movements of the extremities, the anterior horn cells receive excitatory impulses from the motor cortex through the corticospinal tract and through their axons (α motor fibers) activate their innervated muscles. The intramedullary part of the α motor fibers histologically belongs to the central nervous system because their myelin sheath is derived from the cell membrane of oligodendroglia. The extramedullary axons are covered by myelin sheath derived from the cell membrane of Schwann cells, and thus they histologically belong to the peripheral nervous system (Figure 16-6). The α motor fiber is a large myelinated fiber with diameter of 12 to 21 μm , and its conduction velocity is about 50 m/s. These fibers exit the spinal cord at each segment as the **anterior root**, and along with the posterior root, they exit the spinal canal through the **intervertebral foramen** and innervate the corresponding muscle via the neuromuscular junction. For the α motor fibers of cranial nerves, they

exit the cranium through the foramen corresponding to each nerve (see Chapter 5-2, p. 32).

In the α motor fibers, an electric impulse is conducted in the same direction as the axonal flow. Therefore, if its axon is damaged, the innervated muscle not only becomes paralyzed but also rapidly falls into atrophy due to deprivation of neurotrophic factors. As the α motor fibers innervating the distal extremities are as long as 1 m, they are vulnerable to axonal damage.

ii. Motor unit and reinnervation

The combination of a single anterior horn cell and all the skeletal muscle fibers innervated by that cell is called a **motor unit**, and its size is expressed by the number of muscle fibers innervated by that single anterior horn cell. The size ranges from less than 10 to more than 1,000, with the mean being about 150. Motor units of a muscle engaged in fine finger movement are small. In chronic neurogenic muscle atrophy, when a single anterior horn cell is degenerated and lost, the surviving anterior horn cells reinnervate those muscle fibers deprived of innervation, and this

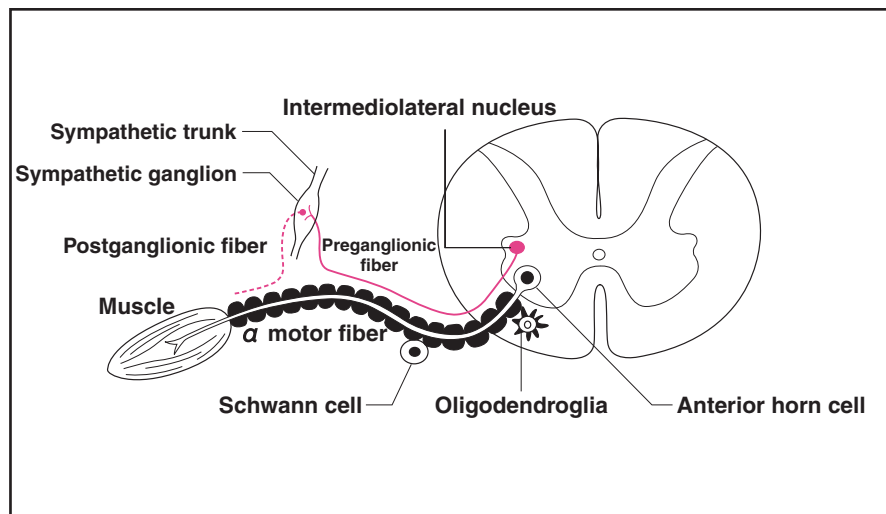


Figure 16-6 Schematic diagram of a transverse section of the thoracic cord to illustrate the anterior horn cell and anterior spinal root. Sympathetic preganglionic fiber exits the cord through the anterior spinal root.

process is repeated so that the motor unit of the reinnervating cell keeps increasing. If this condition is recorded by needle electromyography (EMG), it shows **giant motor unit potentials**.

iii. γ motor fibers

There is another kind of motor neuron in the anterior horn, a small neuron called the γ motor neuron, and its fiber runs along with the α motor fiber. While the α motor fiber innervates the main muscle fibers of a muscle, the extrafusal fibers, the γ motor fiber innervates the intrafusal fibers of the muscle spindle. The γ motor system plays an important role in the spinal reflexes (see Chapter 18-1, p. 157). There are also β motor fibers that innervate both extrafusal and intrafusal fibers.

D. NEUROMUSCULAR JUNCTION AND MUSCLE

An electric impulse of an α motor fiber is transmitted to a muscle fiber via the neuromuscular junction. Each muscle fiber has a single neuromuscular junction regardless of its length, and the junction is generally located in the middle of the fiber. Extraocular muscle fibers are the exception in this regard, because each fiber is known to have multiple junctions. The synaptic terminal of the α motor fiber is called the **motor end-plate**, from which synaptic vesicles release acetylcholine into the synaptic cleft. Then acetylcholine activates the acetylcholine receptor located in the junctional fold of the muscle fiber and opens sodium channels. Electrophysiologically, a **miniature end-plate potential** is

recorded corresponding to each synaptic vesicle, but such events are generally only appreciated due to spontaneous release when the nerve is not active. An action potential in the nerve will lead to the simultaneous release of many vesicles and is called the **end-plate potential**. Typically, end-plate potentials are larger than necessary (**safety factor**) to trigger an action potential in the muscle cell membrane, which causes contraction of the muscle fiber (**excitation-contraction coupling**; **EC coupling**).

2. CENTRAL CONTROL OF VOLUNTARY MOVEMENT

As described in the preceding section, the final common pathway for voluntary movement of extremities is formed of the motor cortices, corticospinal tract, anterior horn cells, α motor fibers, and muscles. However, stimulus-induced simple activation of the upper motor neuron in the primary motor cortex (M1) will produce just a simple twitch of the corresponding muscles, but no purposeful movement. To produce purposeful movement, the secondary motor cortices such as the SMA and lateral premotor area, the basal ganglia, and the cerebellum participate in controlling the excitability of the motor cortex neurons (Figure 16-7), and these structures are known to start functioning before the beginning of the movement and keep working throughout the motor execution. Control of the motor cortex neurons in advance of the actual movement is known as **feedforward control** as opposed to feedback control.

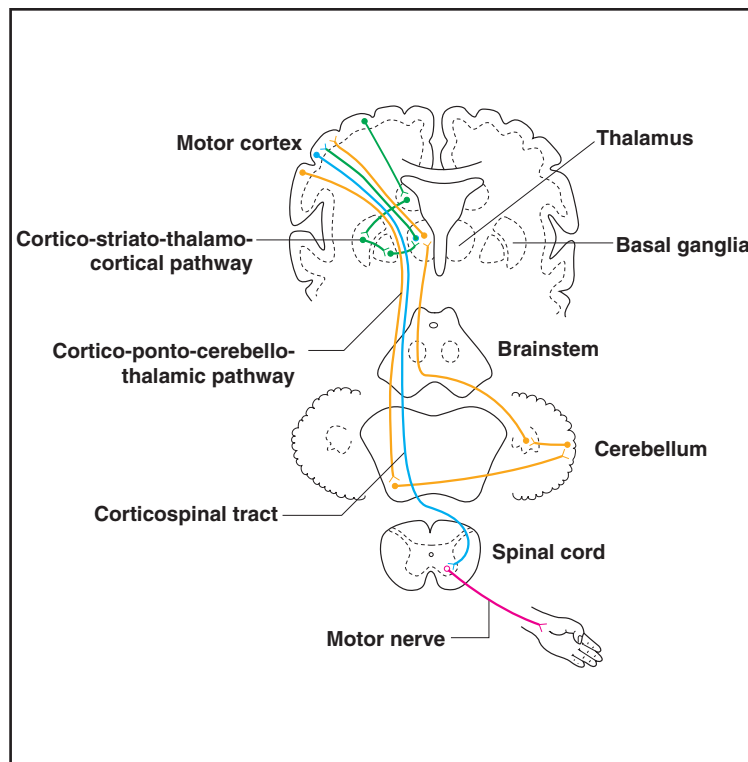


Figure 16-7 Schematic illustration of neural circuits involved in the central control of voluntary movement. The circuits involving the basal ganglia and cerebellum are shown in green and orange, respectively, and the corticospinal tract is shown in light blue.

A. HIGHER FUNCTIONS OF MOTOR CORTICES

As described earlier (p. 109), voluntary unilateral simple movement of the hand is preceded by increasing activities first in the pre-SMA and the SMA proper bilaterally, then in the lateral premotor area and the M1 bilaterally but with contralateral predominance, and finally in the contralateral M1 (Shibasaki, 2012). There is somatotopic organization in the SMA proper to some extent and in the lateral premotor area and M1 more precisely. In unilateral complex movement of the hand, SMA is more activated than in simple movement, and the M1 tends to be activated bilaterally just before the movement onset (Kitamura et al, 1993b; Shibasaki et al, 1993). The SMA is also known to be activated when the subject just imagines movement without actual motor execution. Furthermore, an isolated discrete finger movement is preceded by larger activity of the contralateral M1 compared with the simultaneous movement of more than one fingers (Kitamura et al, 1993a). This is in conformity with the clinical notion that the essential motor deficit of the M1 lesion is a disturbance of fine finger movement.

In those movements employed for daily activities such as tool use and gesture, the parietal cortex of the dominant hemisphere is activated first, then the frontal motor cortices

(Wheaton et al, 2005). In this case, the functional connectivity from the left inferior parietal cortex to the left premotor area plays an important role (Hattori et al, 2009). This is related to the clinical concept of ideational and ideomotor apraxia (see Chapter 24-1F, p. 234).

B. BASAL GANGLIA

The basal ganglia are situated in the deep cerebral hemisphere and are formed of multiple nuclei (Figure 16-8). The main input from the cerebral cortex to the basal ganglia enters the **putamen** and **caudate nucleus**, and its synaptic transmitter is glutamate. The putamen and caudate nucleus together are called the **corpus striatum** or **striatum**.

i. Cytoarchitecture of striatum and neural circuits of basal ganglia

In the striatum, more than 90% of neurons are **medium spiny neurons**, whose neurotransmitter is gamma-aminobutyric acid (GABA). There are two classes of medium spiny neurons with two outputs: the **direct pathway** with GABA and substance P (SP) as the transmitter and the

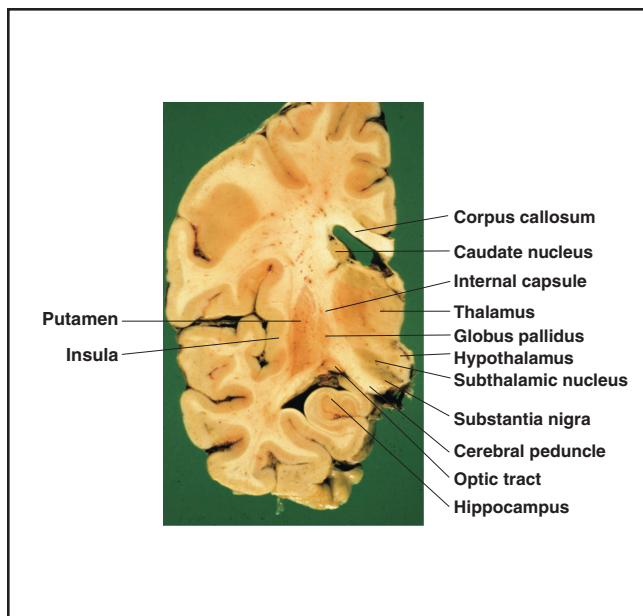


Figure 16-8 Frontal section of the right cerebral hemisphere showing deep structures.
By courtesy of Dr. Jun Tateishi, Emeritus Professor of Kyushu University.

indirect pathway with GABA and enkephalin as the transmitter (Figures 16-9 and 16-10). The direct pathway projects directly to the internal segment of the globus pallidus (GPi) while the indirect pathway first projects to the external segment of the globus pallidus (GPe) and from there sends GABAergic fibers to the **subthalamic nucleus** (STN, also called Luys body). The STN sends its glutamatergic output to the GPi. In effect, the direct pathway inhibits the GPi while the indirect pathway excites the GPi so that they maintain balance. This concept is known as the parallel pathway hypothesis. In fact, however, this neural circuit does not seem that straightforward; there are multiple other connections and pathways. One critical additional pathway is the excitatory hyperdirect pathway from the cortex to the STN; its action supplements that of the indirect pathway. It is noteworthy that GABA is produced from glutamate by glutamic acid decarboxylase (GAD), but the action of the glutamate pathway is excitatory whereas that of GABA is inhibitory to the target receptors.

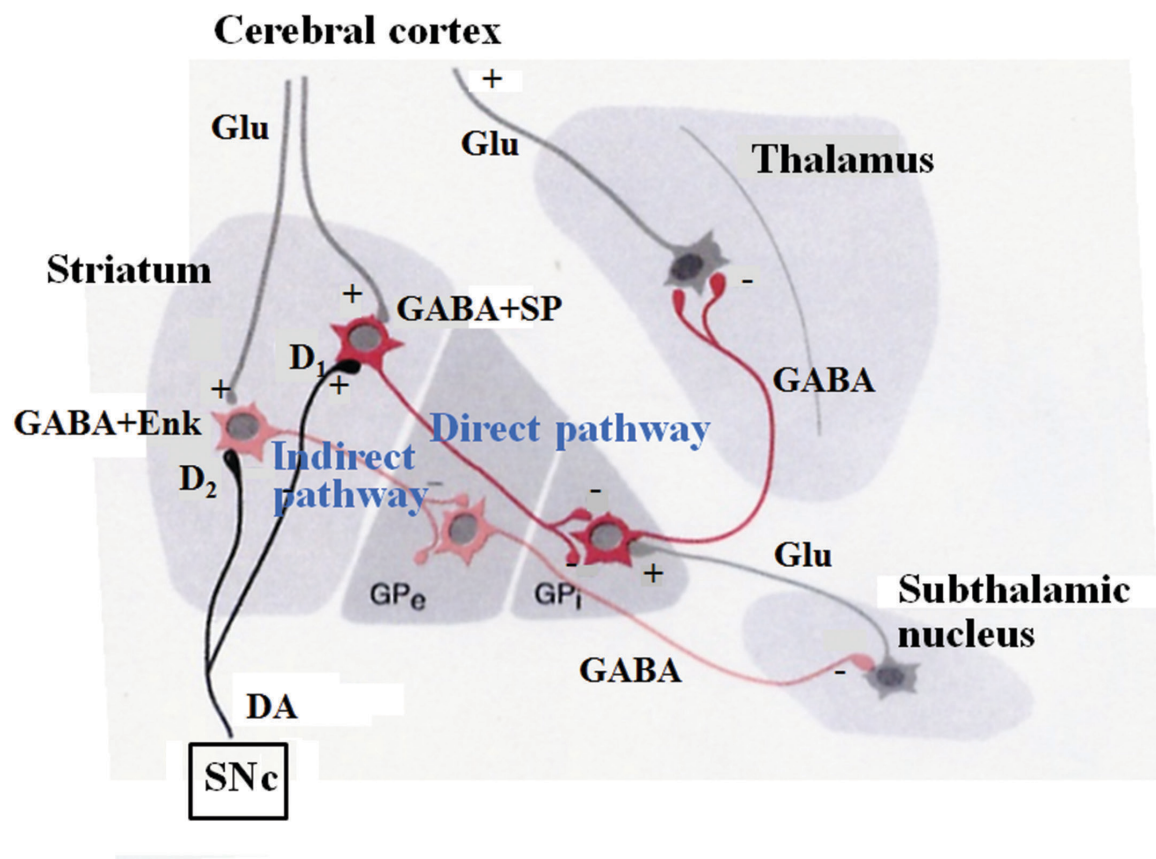


Figure 16-9 Cytoarchitecture of the striatum and neural circuits of basal ganglia with neurotransmitters. D1: D1 receptor, D2: D2 receptor, DA: dopamine, Enk: enkephalin, GABA: gamma-aminobutyric acid, Glu: glutamate, GPe: external segment of globus pallidus, GPi: internal segment of globus pallidus, SNc: substantia nigra pars compacta, SP: substance P, STN: subthalamic nucleus.
Cited from Brodal, 2004, with permission.

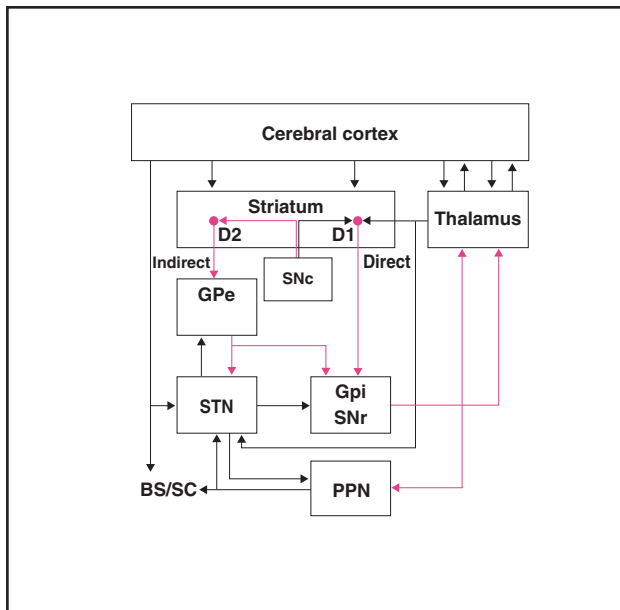


Figure 16-10 Neural circuit of basal ganglia and cerebral cortex. Excitatory pathways are shown in black arrows and inhibitory pathways are shown in red arrows. BS: brainstem, PPN: pedunculo pontine nucleus, SC: spinal cord, SNc: substantia nigra pars compacta, SNr: substantia nigra pars reticulata, STN: subthalamic nucleus; otherwise the same abbreviations as in Figure 16-9.

Cholinergic interneurons represent a small percentage of all striatal neurons, but acetylcholine is believed to regulate the function of the striatal microcircuit and striatal output (Girasole & Nelson, 2015).

Output from the basal ganglia is mostly through the GPi and projects the GABAergic pathway mainly to the **ventral anterior nucleus (VA)** and **ventral lateral nucleus (ventrolateral nucleus, VL)** of the thalamus (see Figure 30-1, p. 271). The receiving areas of the VL are considered to correspond to the **anterior oral ventral nucleus (Voa)** and **posterior oral ventral nucleus (Vop)** by Hassler's terminology (Krack et al, 2002; Molnar et al, 2005).

Figure 16-11 introduces the center surround hypothesis, which takes into account the striatal function of motor selection (Mink, 2007). According to this theory, there is some somatotopic organization in the striatum and GPi, and the GABAergic fibers from the striatum give rise to focal inhibition of the GPi. As the result, the GABAergic output from the GPi to the thalamic relay nuclei is focally disinhibited, and consequently the glutamatergic pathway from the thalamus to the cortex is focally facilitated. By contrast, as the glutamatergic excitatory projection from the STN to the GPi has no somatotopic organization, it excites the GPi rather diffusely. As a result, the thalamic relay nuclei are diffusely inhibited and serve to suppress unintended movement. The balance between these two systems is postulated to play a role in selecting the intended movement and suppressing other movements (**motor selection**). In this hypothesis, the direct glutamatergic projection from the frontal cortex to the STN (**hyperdirect pathway**) also plays a significant role (Mink, 2007).

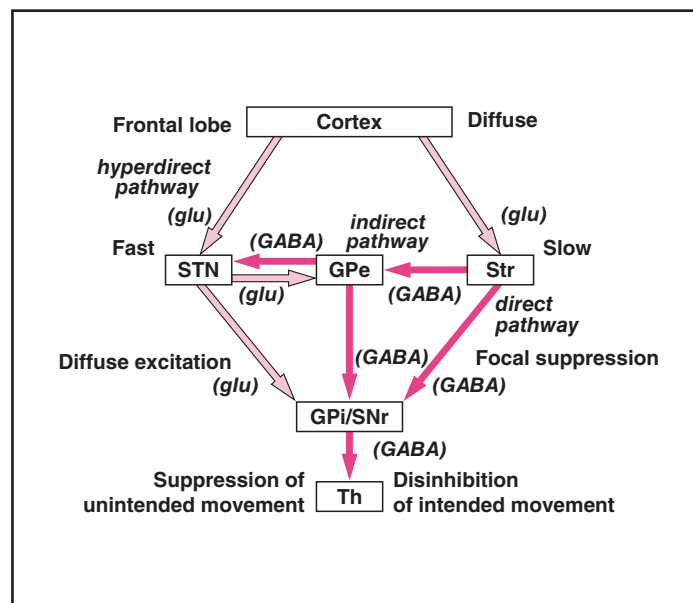


Figure 16-11 Schematic diagram to show the center surround hypothesis of basal ganglia. Excitatory pathways are shown in pink arrows and inhibitory pathways are shown in red arrows. Str: striatum, SNr: substantia nigra pars reticulata, Th: Thalamus; otherwise the same abbreviations as in Figure 16-9 and Figure 16-10. Cited from Mink, 2007, with permission.

ii. Nigrostriatal system

The dopaminergic pathway from the substantia nigra pars compacta (SNc) to the striatum is crucial in modulating the basal ganglia. Its degeneration is clinically important in Parkinson disease. Metabolism of dopamine is shown in Figure 16-12a. It excites the D1 receptors of the GABA/SPergic neurons in the striatum (direct pathway) and inhibits the D2 receptors of the GABA/enkephalinergic neurons (indirect pathway) (see Figures 16-9 and 16-10). However, the distribution of these two receptors is not completely distinct, and many striatal neurons may have both types of receptors (though with predominance of one of them).

In Parkinson disease, the dopaminergic neurons in the SNc selectively degenerate, resulting in dopaminergic deficiency (Figures 16-13, 16-14, and 16-15). The GABA/enkephalinergic medium spiny neurons are disinhibited, and through the indirect pathway they increase activities of the STN and GPi, which then inhibits activities of the thalamic relay nuclei. This results in a decrease in the glutamatergic thalamocortical projection, thus lowering the excitability of motor cortices (see Figures 16-9 and 16-10). The direct pathway neurons are disfacilitated, also leading to increased activity of the GPi. In confirmation, if electrical activities are recorded at the time of deep brain stimulation in patients with Parkinson disease, neuronal discharge is abnormally increased in the STN and GPi. The projection from the basal ganglia to the motor cortex via the thalamus is believed to affect primarily the SMA.

The modern technique of molecular imaging has enabled us to visualize the dopamine transporter uptake at the terminal of the nigrostriatal pathway as well as the dopamine uptake at the D2 receptors of the striatal neurons in humans (Strafella et al, 2017).

A few cases of Parkinson disease are known to be familial, and the genes so far identified include *LRKK2*, *SNCA*, *Parkin*, *VPS35*, *PINK1*, *DJ-1*, *DNAJC13*, *GCH1*, *ATP13A2*, *PLA2G6*, *MPAN*, *BX07*, *RAB39B*, and *ATXN2* (*SCA2*) (Schneider & Alcalay, 2017). In addition, the gene beta-glucocerebrosidase 1 (GBA1) is known to serve as a risk factor for Parkinson disease (Greuel et al, 2020; Maple-Grødem et al, 2021; Sanyal et al, 2020). Autosomal recessive mutations of this gene cause the lysosomal storage disorder Gaucher disease.

iii. Deep brain stimulation in Parkinson disease

Deep brain stimulation (DBS) is commonly used for the symptomatic treatment of Parkinson disease (Katz et al, 2015). For DBS, microelectrodes are placed in the GPi or

STN and are continuously stimulated at a frequency of about 130 Hz. Because the neuronal discharge is increased in the two nuclei, it used to be believed that DBS has favorable effect by inhibiting the discharges, but its physiologic mechanism has not been clearly elucidated. In a positron emission tomography (PET) study using glucose metabolism, it has been reported that the metabolism is even increased in the STN and GPi by STN stimulation (Hilker et al, 2008). In fact, it has been shown that the neural network in the basal ganglia is not a simple circuit of excitatory and inhibitory pathways, either in series or parallel (see Figure 16-10). Moreover, the flow of impulses among different nuclei of the basal ganglia is mostly bidirectional. Therefore, it may not be reasonable to ascribe the effect of DBS simply to excitation or inhibition. MRI-guided laser thalamotomy has been applied in cases of medically intractable tremor disorders (Chang et al, 2018; Elias et al, 2016; Harris et al, 2019). Recently, focused ultrasound subthalamotomy was shown to be effective for tremor-dominant Parkinson disease (Martinez-Fernández et al, 2020; Meng et al, 2020) and also for essential tremor (Jones et al, 2020).

Based on the field potential recording from deep structures in patients with Parkinson disease, Brown et al. reported that oscillation in the β band at around 20 Hz shows high correlation among different basal ganglia nuclei as well as between the basal ganglia and the motor cortex (Hammond et al, 2007). Clinically, it has been reported that the higher the interregional correlation, the more severe the bradykinesia, and that the correlation is suppressed when the patient improves with L-dopa medication or DBS. This theory, which attaches importance to **synchronization** or **resonance** among neuronal activities, is called the **temporal information processing hypothesis**. For this functional interrelationship between the basal ganglia and the motor cortex, the direct pathway from the frontal cortex to the STN (hyperdirect pathway) seems to play an important role in addition to the corticostriatal input and the striato-thalamo-cortical output (see Figure 16-11). Recently, a closed-loop DBS system has been applied to treatment of Parkinson disease because it can deliver an automatically adapted stimulation by specifically blocking the β band and thus might be better than continuous DBS (Bouthour et al, 2019).

iv. Significance of raphe nucleus and locus coeruleus

The serotonergic raphe nucleus and the noradrenergic locus coeruleus are known to project to the SNc and striatum, and Lewy bodies have been shown to appear in both

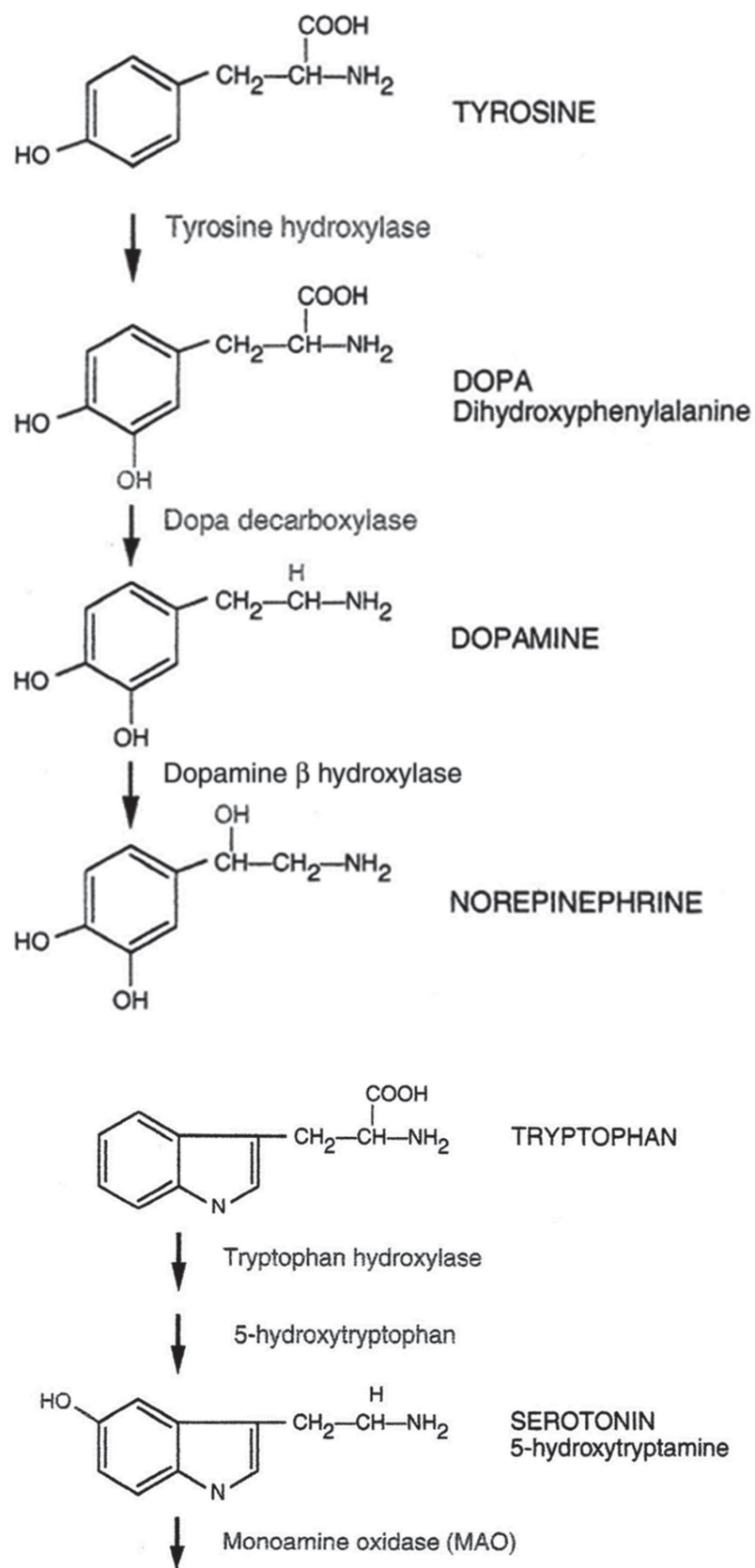


Figure 16-12 Synthesis of dopamine (a) and serotonin (b).
Cited from Brodal, 2010 with permission.

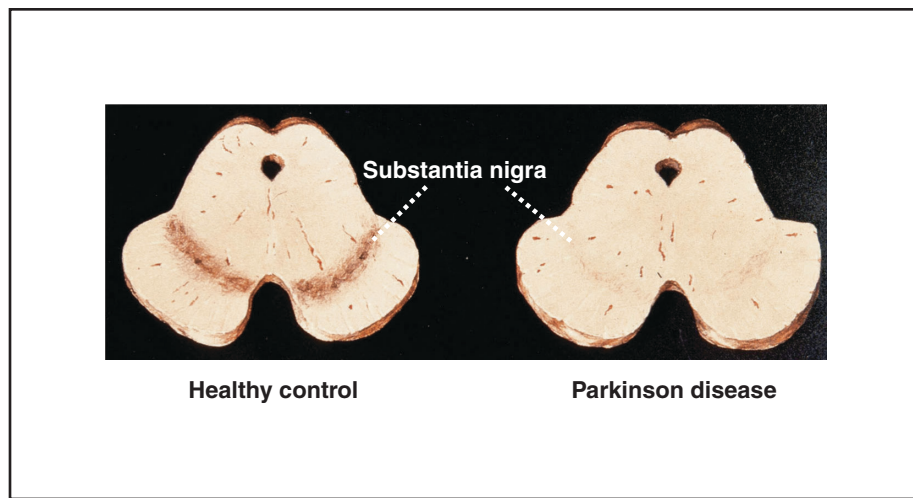


Figure 16-13 Transverse sections of midbrain from a patient with Parkinson disease and a healthy control. In Parkinson disease, melanin pigment is lost in the substantia nigra pars compacta.
By courtesy of Dr. Jun Tateishi, emeritus professor of Kyushu University.

nuclei at an early stage of Parkinson disease (Box 16-2). Impairment of the raphe nucleus might be related to the non-rapid eye movement sleep behavior disorders (RBD) and depression commonly encountered in patients with Parkinson disease, and that of the locus coeruleus might be related to sleepiness, REM sleep behavior disorder, and dysfunction of the autonomic nervous system, which are also common in this disease.

v. Parkinsonism

Clinical manifestations of basal ganglia disorders are diverse, but they can be largely divided into two groups. One group is characterized by an increase in muscle tone and a decrease in the speed, size, and frequency of movement (hypokinetic), and the other group is characterized by a decrease in muscle tone and frequent involuntary movements (hyperkinetic). A typical example of the former

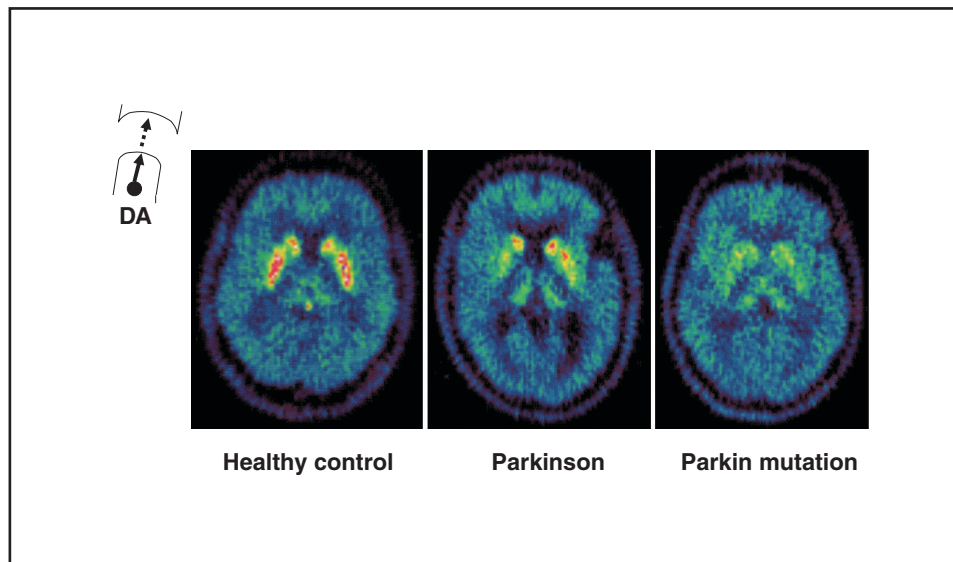


Figure 16-14 Dopamine (DA) uptake image in the terminal of nigrostriatal fibers in a patient with Parkinson disease and a patient with Parkin gene mutation in comparison with a healthy control. Positron emission tomographic (PET) study with ^{18}F -labeled DOPA. DA uptake is decreased in Parkinson disease and Parkin gene mutation.
By courtesy of Dr. David Brooks of Hammersmith Hospital, London, UK.

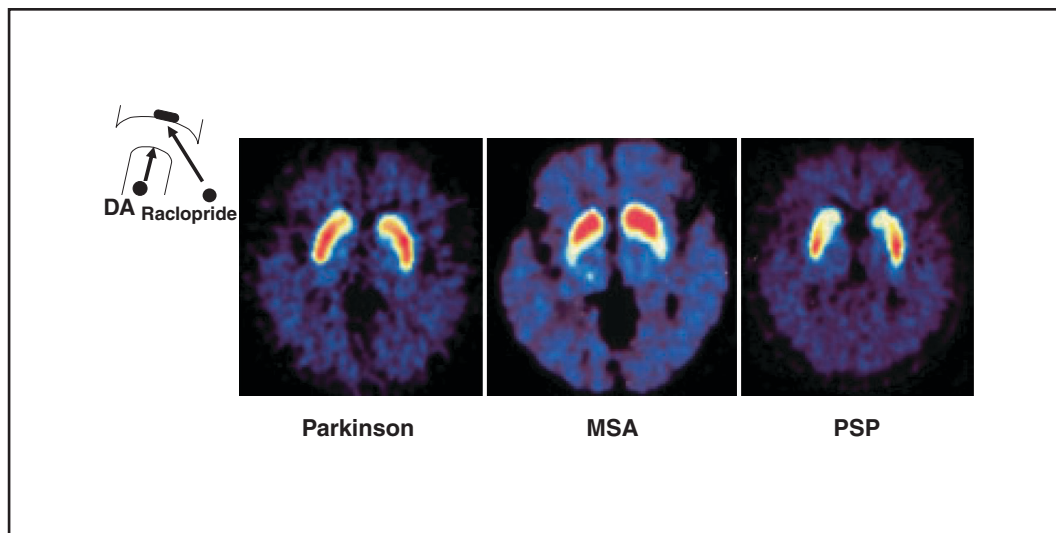


Figure 16-15 PET image of receptor binding with D2-selective ligand raclopride in a patient each with Parkinson disease, multiple system atrophy (MSA), and progressive supranuclear palsy (PSP). Binding of RP is partially decreased in the striatum of MSA and PSP but preserved in the striatum of Parkinson disease.

By courtesy of Dr. Hidenao Fukuyama of Kyoto University.

group is Parkinson disease or parkinsonism, in which movements are generally slow, small, and scanty. In particular, it takes time to initiate the intended movement, and the movement is slow throughout its execution (**bradykinesia**). In movements requiring repetition like walking and speaking, each movement becomes increasingly small (**sequence effect**) and the repetition rate increases until it finally stops. This phenomenon is called **festination: festinating gait** for small and quick steps and **oral festination** for quick

speech resembling stuttering (Moreau et al, 2007). Once the patient stops walking, it is extremely difficult to start walking again (**freezing phenomenon, freezing of gait, start hesitation**). Likewise, handwriting becomes smaller toward the end of the line (**micrographia**) (Figure 16-16), and the voice becomes softer while speaking (**hypophonia**).

Muscle tone is increased in Parkinson disease and parkinsonism. When a skeletal muscle is passively stretched by the examiner in a relaxed condition, resistance of various

BOX 16-2 NON-MOTOR SYMPTOMS AND SIGNS IN PARKINSON DISEASE

It had been believed until recently that motor disturbance is the sole manifestation of Parkinson disease. As the result of careful history taking and clinical evaluation, however, it has become clear that psychiatric symptoms, cognitive disturbance, impairment of the autonomic nervous system, sleep disorders, and sensory symptoms are not rare in this condition (Jagadeesan et al, 2017). Pathologically, Braak, a neuroanatomist from Frankfurt, and his colleagues studied a number of patients who died at various (including presumptive, presymptomatic) stages of Parkinson disease and found the appearance of Lewy bodies first in the dorsal motor nucleus of the vagus and the olfactory bulb, followed by the raphe nucleus, locus coeruleus, amygdala, pedunculopontine nucleus, and basal forebrain nucleus, and at a relatively advanced stage in SNc. Furthermore, in the more advanced stage, Lewy bodies were found even in the mesial aspect of the temporal lobe, insula, and cingulate gyrus, and finally in the neocortex of the temporal lobe (Braak & Del Tredici, 2008; Goedert et al, 2013). The insula plays an important role in integrating somatosensory, autonomic, and cognitive-affective information to guide behavior, and its potential role in non-motor symptoms of Parkinson disease has drawn recent attention (Christopher et al, 2014). Involvement of the brainstem respiratory center (see Figure 5-4) might be related to dyspnea and impaired respiratory function, which are encountered in some patients with Parkinson disease (Vijayan et al, 2020). Although the question of whether Lewy bodies or α -synuclein facilitates neuronal degeneration or protects neurons from degeneration is still controversial, at least as far as the distribution of Lewy bodies is concerned, it is postulated that the typical motor manifestation of Parkinson disease appears at a relatively advanced stage of its disease process.

Before treatment

今日は朝から震えています
ここは京都にある京大病院です。毎月1回付
科=内科へ通っています。糖尿病外来です。血糖値はコントロール
出来ています。

After treatment

平成10年11月26日から京大病院に入院治療しています。入院時より調子は良い状態に思っています。

Figure 16-16 Micrographia seen in a Japanese patient with Parkinson disease. Before treatment, the handwritten letters become smaller toward the end of each line, but this improves after treatment.

degrees is felt by the examiner's hand regardless of the speed of stretch (**rigidity**). Typically, this resistance is felt like trembling and is called the **cogwheel phenomenon**.

Given these symptoms, it is postulated that the basal ganglia keep controlling the state of muscles so that a voluntary movement can be executed smoothly and quickly. As the basal ganglia are closely connected to the ipsilateral motor cortices, the clinical manifestations due to a basal ganglia lesion appear in the contralateral extremities (see Figure 16-7). Furthermore, as the basal ganglia send their descending projections to the brainstem and spinal cord through the **pedunculopontine nucleus (PPN)** (Figure 16-17), a unilateral lesion of the basal ganglia is expected to cause motor disturbances bilaterally, as seen in postural abnormality and gait disturbance (Box 16-3).

vi. Causes of parkinsonism

Parkinsonism is characterized by bradykinesia, rigidity, resting tremor, and disturbance of postural control. The most common cause of parkinsonism is idiopathic **Parkinson disease**, but many of these cases are now recognized to be caused by genetic mutations even without an obvious family history. A number of important dominant and recessive mutations have been identified. Parkinsonism can also be caused by a variety of conditions, including (1) vascular disease, (2) side effects of drugs such as

first- and second-generation neuroleptic agents and other drugs that involve dopamine blocking, such as metoclopramide and sulpiride (Factor et al, 2019; Savica et al, 2017), (3) intoxication with manganese and carbon monoxide, and (4) sequela of acute encephalitis (postencephalitic parkinsonism, sequelae of encephalitis lethargica) (Lutters et al, 2018). Parkinsonism due to ischemic vascular disease (**vascular parkinsonism**) is commonly encountered, but its diagnostic criteria have not been fully established. By contrast with Parkinson disease, vascular parkinsonism commonly shows stepwise exacerbation of symptoms and responds poorly to L-dopa treatment. In addition, the presence of cognitive disorders, pyramidal signs, falls, pseudo-bulbar palsy, or urinary incontinence may also support its diagnosis (Kalra et al, 2010).

A variety of neurodegenerative disorders can cause parkinsonism; in particular, a group of hereditary diseases manifesting parkinsonism has drawn increasing attention. In addition to the diseases listed in Box 16-4, some patients with **Gaucher disease** have been shown to develop parkinsonism and Lewy body dementia (see Chapter 23-2A, p. 223) (Kraoua et al, 2009). In particular, patients with a heterozygous mutation of the glucocerebrosidase gene develop parkinsonism at a younger age and commonly have non-motor symptoms such as cognitive disorders and autonomic nervous disturbance (Brockmann et al, 2011; Seto-Salvia et al, 2012; Siebert et al, 2014).

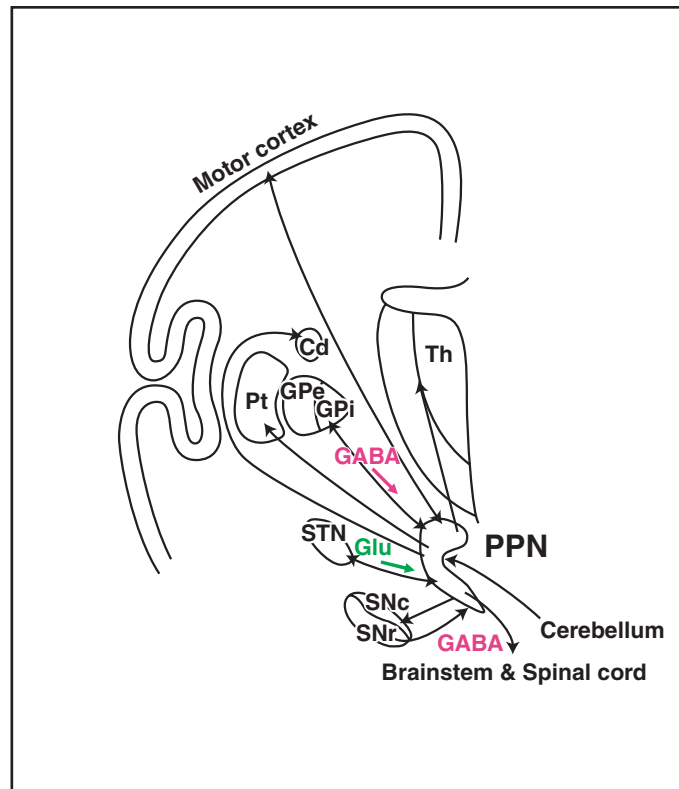


Figure 16-17 Schematic diagram showing the input and output of the pedunculopontine nucleus (PPN) in human. Cd: caudate nucleus, Pt: putamen; otherwise the same abbreviations as in Figures 16-9, 16-10, and 16-11. Modified from Jenkinson et al., 2009 with permission.

BOX 16-3 PEDUNCULOPONTINE NUCLEUS

It is considered that limb movement is influenced by basal ganglia mainly through the striato-thalamo-cortico-spinal circuit, but another pathway through the pedunculopontine nucleus (PPN) may be also important (see Figure 16-10). The PPN is located in the ventral tegmentum at the border zone between the midbrain and the pons, and it has fiber connections with many structures of the basal ganglia and thalamus. It projects cholinergic descending fibers to the brainstem and spinal cord (see Figure 16-17). Although the role of the PPN-related pathway in human has not been precisely elucidated, it is considered to be involved in the control of posture and gait as well as sleep (Jenkinson et al, 2009). It is noteworthy that Lewy bodies appear in the PPN at a relatively early stage of Parkinson disease (see Box 16-3). In fact, falling is commonly encountered in Parkinson disease, and a PPN lesion is considered to be one of its causes (Bohnen et al, 2009; Galea et al, 2017). DBS applied to PPN may improve gait freezing and falls in patients with Parkinson disease (Garcia-Rill et al, 2019; Snijders et al, 2016; Thevathasan et al, 2018; Garcia-Rill et al, 2019).

A syndrome characterized by parkinsonism of young onset and hypoventilation is known as **Perry syndrome**. In this condition, other various symptoms such as depression, behavioral abnormality, body weight loss, and vertical gaze palsy may be also seen. This is a rare autosomal dominant disease due to mutation of the DCTN1 gene, characterized by TDP43 pathology, and it responds to L-dopa treatment (Kim et al, 2021; Newsway et al, 2010). Recently several conditions characterized by parkinsonism and brain iron accumulation have drawn increasing attention (neurodegeneration with brain iron accumulation [NBIA]) (Box 16-5) (Figure 16-18).

L-3,4 dihydroxyphenylalanine (L-dopa) is known to be effective almost specifically for Parkinson disease, but it may improve the parkinsonian symptoms in other conditions as well, at least at a certain stage of the clinical course (Box 16-7).

vii. Disability rating score of Parkinson disease

On neurologic examination, it is quite important to rate the disability state objectively, which will serve as the basis for evaluating the effect of treatment. A commonly used

BOX 16-4 NEURODEGENERATIVE DISEASES PRESENTING WITH PARKINSONISM

Classical parkinsonism

Sporadic: idiopathic Parkinson disease

Hereditary: mutation of LRRK2, SNCA, Parkin, PINK1 and DJ-1 gene, and SCA2, etc.

Atypical parkinsonism

Sporadic: Lewy body disease, multiple system atrophy, progressive supranuclear palsy, corticobasal degeneration, etc.

Hereditary: mutation of FTDP-17 and SNCA, etc.

From Galpern & Lang, 2006.

disability scale for Parkinson disease is that of Hoehn and Yahr (Box 16-8). As the symptoms and signs of parkinsonism tend to fluctuate, especially over the course of L-dopa treatment (see Box 17-10, p. 151), the disability also varies according to the time when it is evaluated. For reporting the results of research or treatment of Parkinson disease internationally, the Unified Parkinson's Disease Rating Scale (UPDRS) is commonly used. The revised version of this scale (MDS-UPDRS) is made up of (1) evaluation of mentation, behavior, and mood, (2) self-evaluation of

activities of daily living (ADLs), including speech, swallowing, handwriting, dressing, hygiene, falling, salivating, turning in bed, walking, and cutting food, (3) clinician-scored monitored motor evaluation, (4) Hoehn and Yahr staging of severity of Parkinson disease, (5) Schwab and England ADL scale, and (6) complications of treatment (Goetz et al, 2008).

C. CEREBELLUM

The cerebellum, a relatively large symmetric structure located in the posterior cranial fossa, plays an important role in regulating the activity of motor cortices through the circuit with the cerebral cortex (see Figure 16-7, p. 116).

i. Cytoarchitecture of cerebellum

The cerebellum receives cerebral input mainly from the frontal cortex through the **corticopontine tract**, which descends through the medial and lateral borders of the cerebral peduncle to project to the ipsilateral **pontine nucleus** (see Figure 5-2, p. 32). The output projection from the pontine nucleus crosses at that level and reaches the contralateral cerebellar cortex through the **middle cerebellar peduncle** (brachium conjunctivum) (Figure 16-19).

BOX 16-5 NEURODEGENERATION WITH BRAIN IRON ACCUMULATION (NBIA)

Some neurodegenerative diseases are characterized by accumulation of iron in the brain, and most of them are hereditary. Those are **pantothenate kinase-associated neurodegeneration (PKAN)**, phospholipase A2G6-associated neurodegeneration, **Kufor-Rakeb disease**, fatty acid 2-hydroxylase-associated neurodegeneration, FA2H-associated neurodegeneration, **aceruloplasminemia**, **neuroferritinopathy**, and **static encephalopathy of childhood with neurodegeneration in adulthood (SENDA)**. Some investigators even include Friedreich ataxia in this group (Schipper, 2012; Schneider et al, 2012).

PKAN, previously known as Hallervorden-Spatz syndrome, is an autosomal recessive disease that presents with parkinsonism, dystonia, and cognitive disturbance in childhood or young adults (Kruer et al, 2011). Since iron accumulates in the GPi in this condition, the T2-weighted MRI shows a linear high-signal region in the middle of low-signal region in the globus pallidus bilaterally (the "eye of the tiger" sign) (see Figure 16-18) (Valentino et al, 2006).

Kufor-Rakeb disease is an autosomal recessive disease that presents with dopa-responsive parkinsonism, pyramidal signs, supranuclear upward gaze palsy, cognitive impairment in puberty (Schneider et al, 2012), and in some cases action myoclonus and seizures (Rohani et al, 2018).

Aceruloplasminemia is an autosomal recessive disease and presents with cerebellar ataxia, parkinsonism, and other neurological symptoms in addition to diabetes mellitus and retinal degeneration (Hida et al, 2010).

Neuroferritinopathy is an autosomal dominant disease and is an important entity when considering the differential diagnosis for extrapyramidal disorders.

Static encephalopathy of childhood with neurodegeneration in adulthood (SENDA) is an X-linked dominant disorder characterized by developmental delay in early childhood followed by progressive parkinsonism, dystonia, spasticity, and intellectual decline, and MRI findings suggestive of iron deposition in the globus pallidus and substantia nigra (Ozawa et al, 2014). This condition is now reclassified as **beta-propeller protein-associated neurodegeneration (BPAN)** (Morales-Briceño et al, 2018; Paudel et al, 2015).

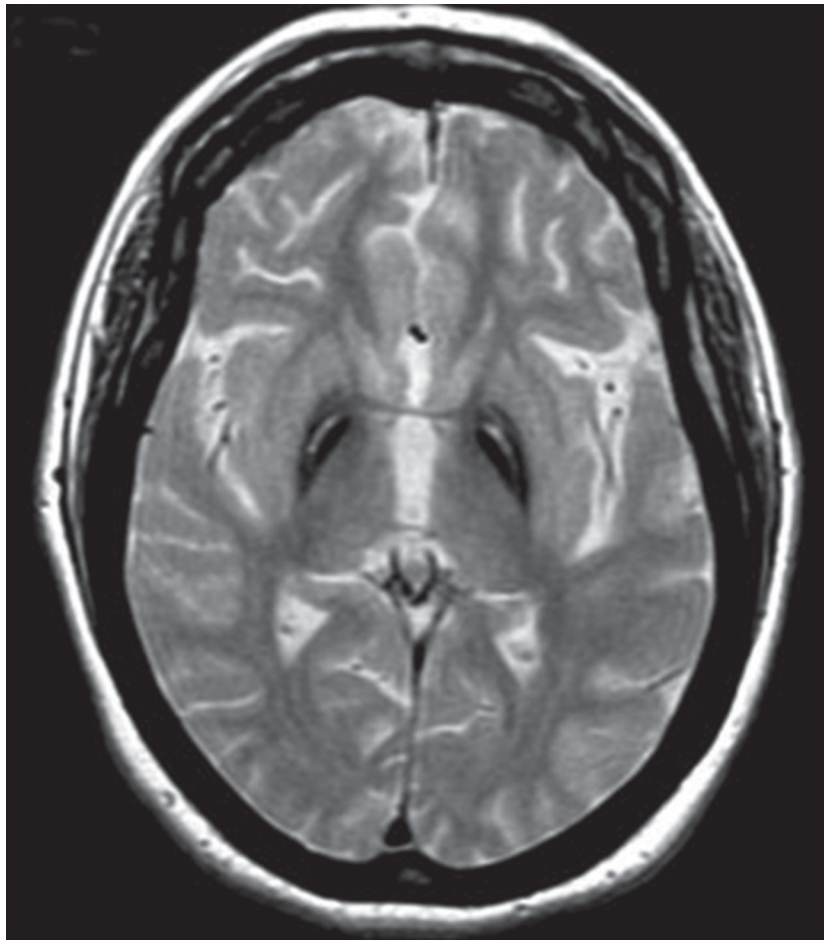


Figure 16-18 The “eye of the tiger” sign on MRI in a patient with pantothenate kinase–associated neurodegeneration (PKAN). Cited from Valentino et al., 2006, with permission.

BOX 16-6 LYSOSOMAL STORAGE DISEASE

Lysosomal storage diseases (LSDs) are a group of diseases characterized by lysosomal dysfunction, and most of them are inherited as autosomal recessive traits (Ebrahimi-Fakhari et al, 2018). The lysosome is the key cellular hub for macromolecule catabolism, recycling, and signaling, and defects that impair any of these functions cause the accumulation of undigested or partially digested macromolecules in lysosomes or impair the transport of molecules, which can result in cellular damage. Clinical symptoms appear in infancy and childhood, but adult-onset forms also occur. Many of them cause movement disorders such as ataxia, tremor, dystonia, and myoclonus. The most common forms associated with movement disorders are Niemann–Pick disease type C, neuronal ceroid lipofuscinosis, and different mucopolysaccharidoses (Ebrahimi-Fakhari et al, 2018).

In the cerebellar cortex, the middle cerebellar peduncle projects to the **granule cells** of the **granular layer** as **mossy fibers**, and axons of the granule cells project to the **molecular layer**, where they form the **parallel fibers**. Cerebellar granule cells are the most numerous cells in the brain. The parallel fibers make synaptic contact with the dendrites of **Purkinje cells** (Figure 16-20). Axons of Purkinje cells project to the deep cerebellar (intracerebellar) nuclei. The main output from the cerebellum to the cerebral cortex originates in the dentate nucleus and ascends as the **dentatothalamic tract (superior cerebellar peduncle)**, which crosses at the lower midbrain to pass through the red nucleus and projects to the **ventral lateral nucleus (VL)** of the contralateral thalamus (see Figures 16-7 and 16-19). This area of the thalamus corresponds to the **ventral intermediate nucleus (Vim)** of Hassler’s nomenclature (Krack et al, 2002; Molnar et al, 2005). The dentato-thalamo-cortical pathway projects primarily to the lateral premotor area and M1.

BOX 16-7 CONDITIONS OTHER THAN PARKINSON DISEASE IN WHICH L-DOPA MAY IMPROVE THE PARKINSONIAN SYMPTOMS

Parkinsonism in patients with neurodegenerative diseases other than Parkinson disease may also respond to L-dopa therapy (Martin et al, 2021). Patients with multiple system atrophy predominantly presenting with parkinsonism (MSA-P) (see Box 21-2, p. 209) may respond to L-dopa in the initial stage. Among spinocerebellar ataxias (SCAs) (see p. 129), patients with SCA2, SCA6, SCA8, and SCA17 may present with parkinsonism initially, and L-dopa may ameliorate the parkinsonism. In dystonia–parkinsonism syndrome (DYT3, Lubag syndrome), the initial symptoms of dystonia may be replaced by parkinsonism later and may respond to L-dopa (see p. 176). In the autosomal recessive dopamine-responsive dystonia with Segawa's syndrome (see Chapter 18-6, p. 176) and in the hereditary dystonia due to sepiapterin reductase deficiency, L-dopa is effective. Patients with tyrosine hydroxylase deficiency, an autosomal recessive disease, present with myoclonus–dystonia syndrome and may respond to L-dopa (Stamelou et al, 2012). Kufoor–Rakeb disease, an autosomal recessive neurodegenerative disease with brain iron accumulation (see Box 16-5), responds to L-dopa; and Perry syndrome with parkinsonism and hypoventilation (p. 124) may also respond to L-dopa. Furthermore, restless legs syndrome, although it may not belong to the neurodegenerative diseases in the narrow sense, responds to L-dopa, and so does the resting tremor due to mid-brain infarction (see Box 19-3, p. 170).

BOX 16-8 THE HOEHN AND YAHN SCALE FOR THE LEVEL OF DISABILITY IN PARKINSON DISEASE

Stage 0: no signs of disease
Stage 1: symptoms on one side only (unilateral)
Stage 1.5: symptoms unilateral and also involving the neck and spine
Stage 2: symptoms on both sides (bilateral) but no impairment of balance
Stage 2.5: mild bilateral symptoms with recovery when the “pull” test is given (the doctor stands behind the person and asks them to maintain their balance when pulled backwards)
Stage 3: balance impairment; mild to moderate disease; physically independent
Stage 4: severe disability but still able to walk or stand unassisted
Stage 5: needing a wheelchair or bedridden unless assisted

ii. Cerebellar circuit and clinical symptomatology

As the input from the cerebral cortex to the cerebellum crosses at the pons as the middle cerebellar peduncle, and as the output from the cerebellum to the motor cortex crosses at the midbrain as the superior cerebellar peduncle, the cerebellar hemisphere is functionally connected to the contralateral cerebral hemisphere (see Figure 16-7). Therefore, if a cerebellar hemisphere is unilaterally involved by a lesion, the motor disturbance is expected to occur in the ipsilateral extremities. This circuit is considered to regulate the motor cortex so that the limb movement can be executed skillfully, and its impairment will result in clumsy execution of voluntary movement (**ataxia**). Namely, all voluntary movements, including eye movements, speaking, and limb movements, are executed with large variability in size and time interval, and with unstable rhythm. For speech, the sounds become suddenly loud (**explosive speech**), and the intervals among syllables become slurred and variable (**scanning speech**, **slurred speech**). Limb movements become clumsy because the related muscles cannot contract appropriately (**incoordination**, decomposition of movement), and they often overshoot the target (**dysmetria**, **hypermetria**). Furthermore, repetitive alternating movements (diadochokinesis) such as pronation and supination of the hands are poorly executed (**adiadochokinesis**, **dyadiadochokinesis**).

For cerebellar control of limb movement, the neocerebellum (embryologically new cerebellum), which corresponds to the cerebellar hemispheres, plays an important role (Figure 16-21). In the cerebellar cortex, there is some somatotopic organization.

The outputs from the vestibular nerve and the medial and inferior vestibular nuclei are projected through the inferior cerebellar peduncle to the vermis, flocculus, and nodulus of the cerebellum, which corresponds to the archicerebellum (see Figure 16-21). If this region is involved by a lesion, equilibrium of the trunk is impaired (**truncal ataxia**). As was described in Chapter 13-1 (p. 93), a localized lesion in the cerebellum could cause vertigo in relation to this connection.

A large part of the cerebellar vermis and the medial part of the cerebellar hemisphere belong to the paleocerebellum (see Figure 16-21). The anterior lobe of the cerebellum receives projections from the proprioceptive receptors of muscles through the spinocerebellar tracts (see Figure 16-19). The spinocerebellar tracts ascend through the most lateral part of the spinal cord as the dorsal and ventral spinocerebellar tracts. The **dorsal spinocerebellar tract** originates in the column of Clarke in the posterior

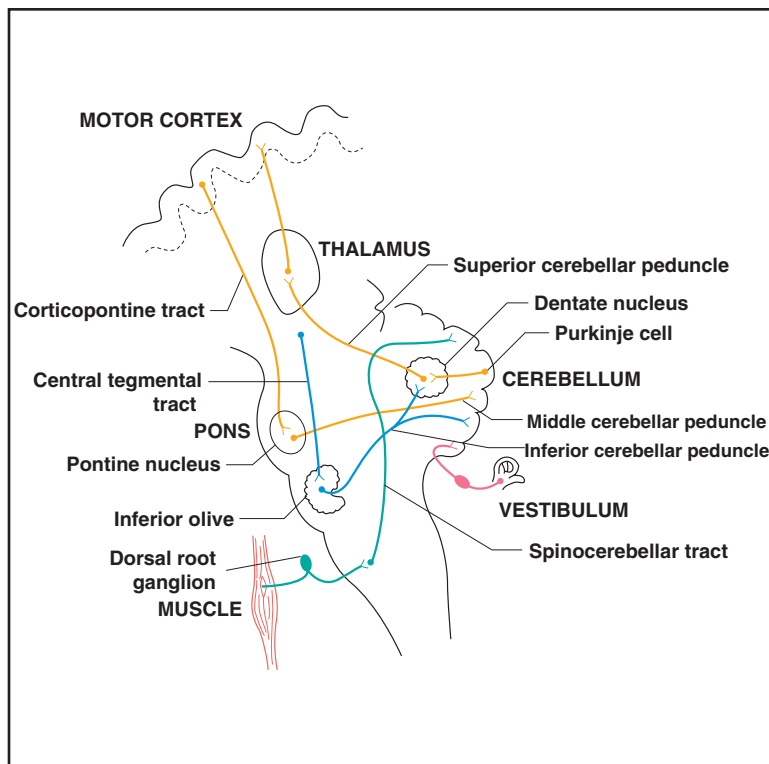


Figure 16-19 Input and output circuits of the cerebellum. In this figure, the input from the vestibular nuclei of the brainstem is not shown; only the direct input from the vestibulum is shown (see Chapter 13-1, p. 93).

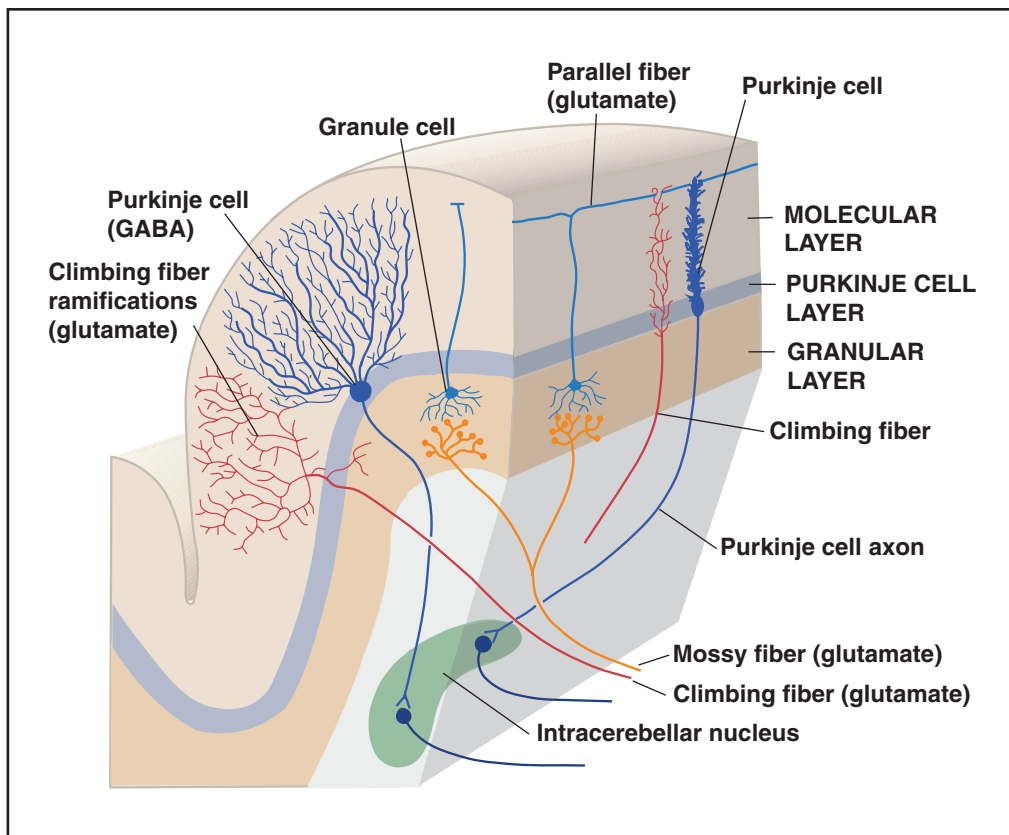


Figure 16-20 Schematic diagram showing cytoarchitecture of the cerebellar cortex. Cited from Brodal, 2010, with permission.

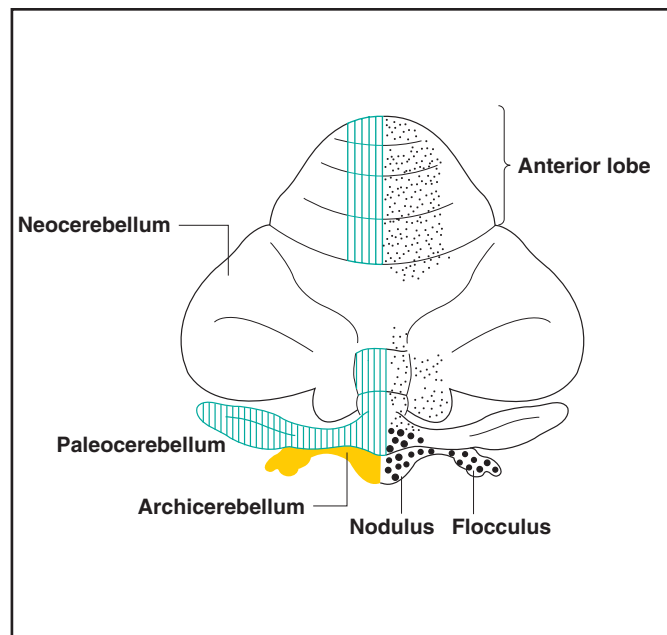


Figure 16-21 Embryologic division of cerebellum.

horn and ascends through the dorsal lateral cord to project to the ipsilateral cerebellum. The **ventral spinocerebellar tract** originates in the interneurons of layer 7 in the posterior horn caudal to the midthoracic level and crosses and ascends through the contralateral lateral column to project to the cerebellum through the superior cerebellar tract, and most fibers cross again in the cerebellum. The function of the spinocerebellar tracts has not been fully elucidated, but it is believed that the dorsal tract sends information about complex joint movements to the cerebellum while the ventral tract sends information about the activity of inhibitory interneurons (Brodal, 2010). Clinically, it is also believed to control muscle tone. In an acute cerebellar lesion, muscle tone is said to be decreased, but this is not always the case, and its underlying mechanism has not been fully clarified. It is known that section of the spinocerebellar tract at the time of spinal surgery does not cause ataxia.

Another input to the cerebellum comes from the inferior olive (see Figure 16-19). The inferior olive receives input from the ipsilateral red nucleus and probably also from the hypothalamus through the central tegmental tract. There may be also a direct pathway from the dentate nucleus of the contralateral cerebellum through the superior cerebellar peduncle and the central tegmental tract, forming the Guillain-Mollaret triangle (see Box 19-16, p. 182). Fibers from the inferior olive cross at that level and project through the inferior cerebellar peduncle directly to the contralateral deep cerebellar nuclei and the Purkinje

cells as the climbing fibers. The function of this circuit has not been clearly elucidated, but the inferior olive may be related to adaptation learning, which is an important function of the cerebellum.

iii. Spinocerebellar degeneration

A number of neurodegenerative disorders primarily involving the cerebellum are known, and recently genetic abnormalities have been identified in many of them. Among those disorders, a group of diseases of autosomal dominant inheritance is categorized into **spinocerebellar ataxia (SCA)**, and now more than 35 diseases have been reported in the category (Durr, 2010; Guo et al, 2014; Ikeda et al, 2012; Klockgether et al, 2019; Smets et al, 2014; Sugihara et al, 2012; Sullivan et al, 2019). Genetically SCA is classified into three groups. The first group is characterized by polyglutamine deposit in neurons due to CAG repeat expansion (**triplet repeat disease**) and is known as **polyglutamine disease**. To this group belong **SCA1, SCA2, SCA3, SCA6, SCA7, SCA17**, and **dentato-rubro-pallido-luysian atrophy (DRPLA)**. Besides SCA, Huntington disease (see Box 19-5, p. 171) and spinal and bulbar muscular atrophy (see Box 17-2, p. 137) belong to the triplet repeat disease.

The second group of SCA also belongs to the triplet repeat disease, but in this group the repeat is in the intron rather than the exon. The mechanism of damage of such mutations is not clear but is believed by some to be

mediated by RNA (Todd & Paulson, 2010). Members of this group are **SCA10** and **SCA12**, and probably **SCA8**. Other diseases, like myotonic dystrophy and **fragile X-associated tremor/ataxia syndrome** (see Box 19-4), also belong to the same genetic category.

The third group of SCA is due to the ordinary single base pair mutation, and **SCA5**, **SCA13**, **SCA14**, and **SCA27** belong to this group.

The prevalence of SCAs differs among ethnic groups. In Japan, for example, **SCA3** (Machado–Joseph disease), **SCA6**, chromosome 16q-linked dominant cerebellar ataxia, and **DRPLA** are common. Among others, **SCA6** is characterized by pure cerebellar ataxia of late onset and sporadic occurrence. Some types of SCA are associated with unique clinical features. **Machado–Joseph disease**, which typically begins between the ages of 20 and 50 years, is characterized by cerebellar ataxia, progressive external ophthalmoplegia, fasciculation in facial muscles, dysarthria, dysphagia, pyramidal signs, dystonia, rigidity, and distal muscle atrophy. **SCA36** is characterized by cerebellar ataxia of late onset associated with motor neuron involvement (Ikeda et al, 2012; Ohta et al, 2017).

In many triplet repeat diseases, it is known that the longer the repeat expansion, the greater the disease severity. Furthermore, the repeat expansion tends to be longer in the offspring; thus, the age of onset becomes younger in the succeeding generations (**anticipation**).

There is also a group of cerebellar degenerative disorders of autosomal recessive inheritance (Beaudin et al, 2019; Renaud et al, 2017; Rossi et al, 2018; Synofzik et al, 2019). **Friedreich ataxia** is its representative form, and in fact it is the most common cause of hereditary cerebellar ataxia in the world (Delatycki & Bidichandani, 2019; Pallardó et al, 2021). This is due to a GAA expansion in the first intron of the frataxin gene on chromosome 9q13. Clinically, ataxia of gait and limb movement, dysarthria, loss of vibration and joint position sense, loss of tendon reflexes, and pyramidal signs start at age 5 to 25 years. Pathologically, it is characterized by degeneration of large sensory neurons in the dorsal root ganglion, dentate nucleus of the cerebellum, and upper motor neurons. Friedreich ataxia is conventionally classified into spinal ataxia, but diagnostically it is important to note that there are also abnormalities in eye movements such as square-wave jerks and ocular flutter, although the saccadic speed has been reported to be normal (Fahey et al, 2008). Furthermore, systemic abnormalities such as cardiomyopathy, diabetes mellitus, scoliosis, and pes cavus are common, and in fact cardiomyopathy is the most common cause of death in this disease.

BOX 16-9 BASSEN–KORNZWEIG DISEASE

The authors saw a 36-year-old man with genetically proved Bassen–Kornzweig disease at NINDS, NIH. The patient was noted to have clumsy hand movement at age 5 years, and difficulty in writing and walking and tremor in head and voice at age 14. At age 16, loss of tendon reflexes, decrease in joint position sense and vibration sense, abetalipoproteinemia, hypcholesterolemia, acanthocytosis, decreased vitamin E level in serum, and steatorrhea were found. There was no similar disease in his family except for hand tremor. Neurologically, there were postural and action tremors in the head, limb and truncal ataxia, hypotonia of limbs, and ataxic-spastic gait. There was no pseudoathetosis, and Romberg sign was negative. This disease was reported by Bassen and Kornzweig of New York in 1950 as a case of atypical retinal pigment degeneration and acanthocytosis (Bassen & Kornzweig, 1950). This is inherited in an autosomal recessive trait, and mutation of the microsomal triglyceride transfer protein (MTP) gene has been reported (Zamel et al, 2008).

A variety of conditions may manifest with clinical symptomatology similar to Friedreich ataxia. Those include vitamin E deficiency, abetalipoproteinemia (Bassen-Kornzweig disease) (Box 16-9), Refsum disease, cerebrotendinous xanthomatosis (Box 16-10), mitochondrial encephalomyopathy (see Box 19-13, p. 181), ataxia–telangiectasia (Louis–Bar syndrome), early onset ataxia with ocular motor apraxia and hypoalbuminemia (Yokoseki et al, 2011), Charlevoix–Saquenay syndrome (see Box 17-6), and Marinesco–Sjögren syndrome (Box 16-10). **Ataxia–telangiectasia** is an autosomal recessive hereditary disease due to mutation of the ATM gene. The disease exhibits progressive cerebellar ataxia in early childhood associated with conjunctival telangiectasia, immunoglobulin deficiency leading to recurrent infections, endocrine abnormalities, and increased risk of cancer (van Os et al, 2019). When children with this condition grow up, they may show a wide spectrum of movement disorders (Levy & Lang, 2018).

iv. Acquired cerebellar ataxia

A number of acquired conditions due to toxic, ischemic vascular, inflammatory, metabolic, and autoimmune disorders may present with ataxia, and some of them may show clinical pictures similar to the neurodegenerative disorders.

BOX 16-10 HEREDITARY DISORDERS CHARACTERIZED BY CEREBELLAR ATAXIA, SPASTIC PARALYSIS, AND CATARACT

Cerebrotendinous xanthomatosis and Marinesco–Sjögren syndrome are clinically similar in that both are inherited as autosomal recessive traits and both manifest cataracts and cerebellar ataxia from childhood.

Cerebrotendinous xanthomatosis is caused by mutation of the CYP27A1 gene and is characterized by deposition of cholesterol in the brain, lens, and tendon due to lack of sterol-27-hydroxylase, which interferes with the synthesis of chenodeoxycholic acid (CDCA). Clinically, cataract and xanthoma in the tendon appear in the school-age youth or young adult, and psychomotor retardation, cognitive decline, cerebellar ataxia, and pyramidal signs are seen (Stelten et al, 2019). In addition, epileptic attacks, parkinsonism, psychiatric symptoms, and polyneuropathy may be also seen. The diagnosis is made by documenting enlargement of the Achilles tendon by visual inspection and MRI, and an elevated cholesterol level in the serum. It has been reported that supplementation of CDCA ameliorates the cholesterol deposit if treatment is initiated at the early clinical stage (Pilo-de-la-Fuente et al, 2011; Stelten et al, 2019).

By contrast, **Marinesco–Sjögren syndrome** is characterized by the onset of cataracts, cerebellar ataxia, psychomotor retardation, chronic myopathy, small stature, hypogonadotropic hypogonadism, and skeletal abnormalities in infancy, thus clinically resembling cerebrotendinous xanthomatosis. In a large proportion of patients with this condition, mutation of the SIL1 gene on chromosome 5q31 has been detected (Krieger et al, 2013; Yis et al, 2011).

As many of them are treatable if diagnosed in the early stage of illness, the diagnosis of those conditions is important (Ramirez-Zamora et al, 2015).

The cerebellum is especially vulnerable to ethanol, pharmaceutical drugs such as diphenylhydantoin and lithium, and some heavy metals (**toxic cerebellar diseases**). **Hunter–Russell syndrome (Minamata disease)** is a typical example of organic mercury poisoning (see Chapter 1-1, p. 2). The cerebellum also has a predilection for the demyelinating plaques in **multiple sclerosis**.

Transient cerebellar ataxia may be a sign of transient ischemic attack (TIA). It may present with dysarthria–clumsy hand syndrome (see Box 2-2) or paroxysmal dysarthria–ataxia syndrome (Gessani et al, 2019).

Purkinje cells are vulnerable to autoimmune pathology in relation to either viral infection or neoplasm of other organs. The latter condition, **subacute cerebellar degeneration**, is associated with elevated anti-Yo antibody in association with cancer of the uterus, ovary, or mammary gland or with elevated antibody against voltage-gated calcium channel (VGCC) in association with pulmonary small cell carcinoma (see Chapter 28-2, p. 264). Other antibodies such as anti-Hu, anti-Ri, anti-CV2, anti-Tr, anti-Ma, anti-Ta, anti-zic 4, and anti-mGluR1 might also be detected in paraneoplastic cerebellar degeneration.

Subacute combined degeneration of the spinal cord due to vitamin B₁₂ deficiency, although it does not directly involve the cerebellum, may show an ataxic-like gait due to proprioceptive loss in the lower extremities (sensory ataxia) in addition to hyperactive tendon reflexes (other than at the ankle) and Babinski signs. The ankle jerk may be lost because of the coexistence of polyneuropathy (see Chapter 18-2B, p. 160). Its most common cause is pernicious anemia, but it may be also caused by gastric resection and other forms of malnutrition. Pathologically, it is characterized by myelin loss of the posterior column and the lateral column of spinal cord. When it is severe, it may also damage cerebral white matter and cause cognitive impairment. Therefore, this condition is an important differential diagnosis for spinocerebellar degeneration.

Since 1998, when Hadjivassiliou et al. reported 28 patients with **gluten-sensitive ataxia**, this condition has drawn attention of many investigators (Hadjivassiliou et al, 1998; Lin et al, 2018; Vinagre-Aragón et al, 2018). Gluten ataxia is characterized by sporadic, progressive cerebellar ataxia of late onset and elevated antigliadin antibodies in the serum. The entity is highly controversial but is now considered by some to be a gluten-induced autoimmune disease. A gluten-free diet is recommended for treatment, although no controlled studies have been done (Zis & Hadjivassiliou, 2019). Other movement disorders like myoclonus, chorea, and tremor have also been reported in association with gluten sensitivity (Vinagre-Aragón et al, 2018).

v. Tumors of the cerebellum

Tumors of the cerebellum are more common in children than in adults, and most of them are **astrocytomas** or **medulloblastomas**. In adults, **hemangioblastoma** is the most common tumor of the cerebellum. Familial hemangioblastoma is known as **von Hippel–Lindau disease**, which, in addition to hemangioblastoma in the central nervous system and retina, is often complicated by pheochromocytoma



Figure 16-22 A striated mass lesion in the left cerebellar hemisphere on T1-weighted MRI (Lhermitte–Duclos disease) in a 24-year-old patient with Cowden syndrome.

and tumors in other visceral organs (Binderup et al, 2015; Frantzen et al, 2012). **Lhermitte–Duclos disease** is cerebellar dysplastic gangliocytoma, which is characterized by a striated appearance on MRI (Figure 16-22), and this tumor is commonly associated with an autosomal dominant hereditary cancer syndrome called Cowden syndrome (Abi Lahoud et al, 2014; Mester & Eng, 2015).

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EXAMINATION OF MOTOR FUNCTIONS

The motor system can be examined in the following order. In general, the examination may start with observation of visible findings, followed by manual examination of objective findings, and finally observation of the execution of the patient's voluntary movements, namely (1) posture, (2) the presence or absence of muscle atrophy and fasciculation, (3) muscle strength (muscle power), (4) muscle tone, (5) the presence or absence of muscle spasm and myotonia, and (6) coordination.

1. POSTURE

First it is useful to observe the **posture** of the upper limbs while they are kept in the natural **position** or in the forward-extended position. Any abnormal posture, especially of a twisted nature, suggests the possibility of **dystonia**. If it is simply an abnormal posture of either flexion or extension, the range of joint motion should be tested in order to exclude joint pathology. Generally speaking, postural abnormality due to the central nervous system pathology tends to disappear during sleep, whereas abnormality of the joint itself is unchanged during sleep. Patients with spastic hemiplegia show a characteristic posture with the upper limb flexed and with the lower limb extended and externally rotated (**Wernicke–Mann posture**). Postural abnormality of basal ganglia disorders, especially of Parkinson disease, is described in Chapter 16-2 (p. 121).

2. MUSCLE ATROPHY AND FASCICULATION

A. MUSCLE ATROPHY

If the history suggests the possibility of neuromuscular diseases, muscle volume should be carefully observed or palpated. If there is muscle atrophy, its distribution should be evaluated in detail. Generally speaking, in myopathy like **progressive muscular dystrophy**, atrophy is seen mainly

in the muscles of the shoulder and pelvic girdles, and of the proximal limbs. A typical form is **limb–girdle muscular dystrophy** (Narayanaswami et al, 2014), but **facioscapulohumeral muscular dystrophy** (FSMD) is not rare (Deenen et al, 2014). There are two types of FSMD, and FSMD type 1 results from deletion of a critical number of D4Z4 repeats on the A allele (Rieken et al, 2021; Tawil et al, 2015). In these conditions, the scapula may look protruded backward because of atrophy of the adjacent muscles (**scapula alata**, **winged scapula**) (Figure 17-1). **Oculopharyngeal muscular dystrophy** shows a characteristic distribution of muscle atrophy (Kroon et al, 2021). It is an autosomal dominant hereditary myopathy due to GCN repeat expansion of the PABPN1 gene and is associated with cognitive decline in some cases (Mizoi et al, 2011; Paul et al, 2020; Richard et al, 2017). The distribution of muscle atrophy in **Emery–Dreifuss syndrome** is the same as that of ordinary limb–girdle muscular dystrophy, but it is characterized by joint contracture and cardiomyopathy with conduction block (Liang et al, 2011).

The distal predominance of muscle atrophy usually suggests **neurogenic muscle atrophy**, but exceptions are not rare, like **distal myopathy** and **myotonic dystrophy**. Three groups of **distal muscular dystrophy** are known: (1) Welander type, with autosomal dominant inheritance, and (2) Nonaka type and (3) Miyoshi type, with autosomal recessive inheritance (Fam et al, 2019). In the Miyoshi type, the calf muscle is predominantly affected and the level of creatine kinase in serum is elevated, whereas in the Nonaka type there is little if any atrophy of the calf muscle. In **inclusion body myositis** (Box 17-1), the quadriceps femoris muscle is mainly involved in the lower limb; in the upper limb, the finger flexor and extensor muscles are primarily affected.

By contrast, **Kugelberg–Welander disease** is a progressive spinal muscular atrophy of juvenile onset and of autosomal recessive inheritance, but in this case atrophy is primarily seen in the proximal muscles even though it is a typical neurogenic muscle atrophy associated with fasciculation. In **bulbosplinal muscular**



Figure 17-1 Scapular winging (scapula alata) seen in a patient with facioscapulohumeral muscular dystrophy. By courtesy of Dr. Akio Ikeda of Kyoto University.

BOX 17-1 INCLUSION BODY MYOSITIS

This myopathy used to be considered a kind of chronic myositis because of the presence of a unique inclusion body in the cytoplasm as well as in the nucleus of muscle cells. It may be hereditary, but most cases are sporadic. The sporadic form mainly affects males older than 50 years. Muscle atrophy and weakness are mainly seen in the quadriceps femoris muscle and the finger flexors and extensors (Mavroudis et al, 2020). The clinical course is slowly progressive, and immunosuppressive treatments have no effect. As a result of the long-term follow-up study of many sporadic cases, it was shown that daily activity is significantly impaired due to progressive muscle weakness (Shelly et al, 2021). Deposits of β amyloid have been demonstrated in the muscle fibers of patients with this condition, and it has drawn special attention in relation to the mixture of an aging effect and inflammation. Tawara et al. (2017) found anti-cytosolic 5'-nucleotidase 1A antibodies in one-third of patients with sporadic inclusion body myositis. In relation to this, it should be noted that a similar term, "hereditary inclusion body myopathy," is applied to the condition histologically characterized by "distal myopathy with rimmed vacuoles."

atrophy, neurogenic muscle atrophy is seen in the bulbar muscles and proximal limb muscles (Box 17-2) (see Figure 14-8, p. 104).

Amyotrophic lateral sclerosis (ALS), one of the most incurable neurological diseases, is characterized by rapidly

BOX 17-2 SPINAL AND BULBAR MUSCULAR ATROPHY

Spinal and bulbar muscular atrophy (SBMA, Kennedy disease) is characterized by slowly progressive bulbar palsy and neurogenic muscle atrophy predominantly involving the proximal limbs in males over middle age and is often associated with gynecomastia and testicular atrophy. This clinical picture was initially described by Kawahara of Nagoya University at the end of the 19th century, and Kennedy, Alter, and Sung, of the University of Minnesota in 1968 coined the term "progressive proximal spinal and bulbar muscular atrophy of late onset" for this condition. This is a polyglutamine disease due to CAG repeats in the androgen receptor gene on chromosome Xq11-12 (see p. 129). The extraocular muscles are not involved, but unlike amyotrophic lateral sclerosis, the upper motor neuron is not affected and its clinical course is much slower (Chahin et al, 2008). There is a deposit of pathogenic androgen receptors in the nucleus of the remaining neurons, and the effect of androgen deprivation therapy has been experimentally shown by Sobue and his group. In a double-blind, long-term follow-up study in patients who were treated with subcutaneous injection of anti-androgen leuprorelin, it was shown to be effective for slowing progression of motor disabilities (Hashizume et al, 2020).

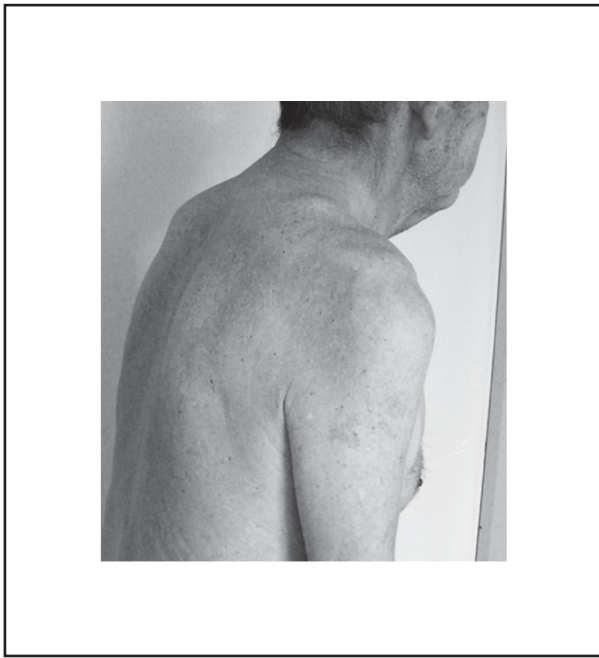


Figure 17-2 Muscle atrophy in the right shoulder girdle and proximal upper limb seen in a patient with clinical diagnosis of amyotrophic lateral sclerosis and progressive supranuclear palsy. By courtesy of Dr. Manabu Inoue, Takeda General Hospital, Kyoto.

progressive neurogenic muscle atrophy occurring in adults above middle age. Muscle atrophy and weakness of the limbs usually start distally, but they may also start from the proximal upper limb (Figure 17-2) or bulbar muscles. It is usually sporadic, but about 10% of cases are familial. In 1993, the superoxide dismutase 1 (SOD1) gene was reported as a responsible cause of familial cases, and it is found in about 20% of the familial cases. More recently, mutation of other genes, including TARDBP, FUS, optineurin, and VCP, has been discovered (Yamashita & Ando, 2015). Some of these gene mutations have also been found in other neurodegenerative diseases (Renton et al, 2011) (see Chapter 23-2A, p. 223). C9orf72 hexanucleotide repeat expansions are the most common cause of familial frontotemporal dementia and ALS, although patients with ALS who have these expansions more frequently show behavioral and cognitive symptoms than those with classical ALS (Cammack et al, 2019; Leko et al, 2019; Rohrer et al, 2015).

Multisystem proteinopathy is the name of a group of inherited disorders that cause neurodegeneration, myopathy, and bone disease and share common pathophysiology. The related genes include the gene encoding valosin-containing protein (VCP), heterogeneous nuclear ribonucleoprotein A2B1 and A1 (*hnRNPA2B1*, *hnRNPA1*), sequestosome 1 (*SQSTM1*), matrin 3 (*MATR3*), T-cell restricted intracellular antigen 1 (*TIA1*), and optineurin (*OPTN*), all of which share disruption of RNA stress granule function and autophagic degradation (Korb et al, 2021).

Spinal muscular atrophy type 1, previously called Werdnig–Hoffmann disease, is the leading genetic cause of infant death, and novel gene therapies are being applied to this condition (Farrar et al, 2017; Kolb et al, 2017).

In muscle atrophy following a nerve lesion, the speed of development of muscle atrophy and its severity heavily depend on the etiology (whether trauma or inflammation), the speed and severity of the nerve damage, and especially which of the two pathologies (axonal damage vs. demyelination) is predominant (p. 146).

B. FASCICULATION

Irregular twitches observed in the resting muscle are called fasciculations and are pathognomonic of neurogenic muscle disease. Therefore, if muscle atrophy is found, the presence or absence of fasciculation should be carefully sought. This phenomenon is caused by an increased spontaneous discharge of an enlarged motor unit as a result of reinnervation following chronic denervation. If fasciculation occurs in small muscles like the hand intrinsic muscles, it manifests as small finger movements resembling tremor. In this case, twitches may be palpable in individual interosseous muscles, which will exclude the possibility of tremor. Occasionally, fasciculation might be seen in healthy subjects with no muscle atrophy (benign fasciculation).

In active degeneration of anterior horn cells, the needle electromyogram (EMG) of the completely resting muscle might delineate small muscle discharges (**fibrillation**) and **positive sharp waves**, both of which are called **denervation potentials**. This represents the spontaneous activity of single muscle fibers.

Myokymia is a continuous, relatively slow undulation seen in the surface of muscle, and it is thought to be generated by an abnormal excitability of anterior horn cells like fasciculation. The needle EMG will show repetitive bursts of rhythmic discharges of a single motor unit potential associated with this phenomenon.

C. MUSCLE HYPERTROPHY

Muscle can be hypertrophic in some conditions. Calf muscle may look enlarged in patients with **Duchenne muscular dystrophy** and those with **Becker muscular dystrophy** of late onset, but this phenomenon is considered to be **pseudohypertrophy** because it is not real hypertrophy of the muscle fiber. In some patients with muscular dystrophy, a part of the biceps brachii muscle may look swollen like a ball (**boule musculaire**, **boule du biceps**) (El Sherif et al, 2020). Muscle hypertrophy may appear in patients with hypothyroid myopathy, a condition called **Hoffmann**

syndrome in adults and **Kocher–Debre–Semelaigne syndrome** in infants.

Firmness and tenderness of muscle can be checked by palpation and pressing with the examiner's fingers. In polymyositis, muscle is often painful spontaneously as well as on pressure and may feel firm, likely due to inflammatory changes.

3. MUSCLE STRENGTH

As muscle strength varies even among healthy subjects depending on the patient's sex, age, and body build, it is important to evaluate the strength of the muscle in question in comparison with normal muscles in each individual. How much in detail is needed in the muscle strength evaluation depends on the tentative clinical diagnosis derived from the history taking. If detailed information about muscle strength was judged to be unnecessary from the history, it is practical to test a few representative muscles from the proximal and distal limbs. It is important to pay attention to the symmetry between two sides, but there may be significant asymmetry even in healthy subjects depending on the handedness, occupation, and sports of the subject. In patients with mild hemiparesis, if both upper limbs are held in extension straight forward at the shoulder level with the palm up, the weaker arm might fall, depending on the degree of weakness. This phenomenon is called forearm drift or **Barré sign**. This phenomenon becomes manifest especially when the eyes are closed, but in this case the examiner must judge whether it is due to weakness or pseudo-athetosis as a result of proprioceptive loss (see Chapter 20-2B, p. 196). A similar phenomenon can be observed in the lower limbs if the lower legs are kept raised by flexing the knees with the patient placed in the prone position. Mild hemiparesis in an upper extremity might also manifest as asymmetry in the forearm rolling test, in which the patient rotates the fists and distal forearms around each other. The weaker arm will move less and the normal arm will rotate around it.

A. MANUAL MUSCLE TESTING

To express the degree of weakness quantitatively, manual muscle testing or manual strength testing is used. In the Medical Research Council (MRC) scale, the degree of weakness can be rated into five categories; normal strength is rated 5:

- 5: normal
- 4: good (weaker than normal, but the patient can move against the examiner's moderate resistance)
- 3: fair (can move well against gravity)

2: poor (can move well if gravity is removed)

1: trace (can contract the muscle but no joint movement)

0: zero (complete paralysis)

If a score of 4 is judged insufficiently precise, intermediate ratings of 4⁻ or 4⁺ can be used.

B. MUSCLE PAIN

In patients with severe muscle pain (**myalgia**), it may be difficult to evaluate the muscle strength precisely. Although myalgia is common in **polymyositis** and **dermatomyositis** (see Chapter 3-1, p. 11), myalgia is more severe in the condition called **polymyalgia rheumatica**. This is a systemic angiitis seen in aged subjects and shares pathogenesis with **temporal arteritis** (see Chapter 7-1C, p. 40). In fact, these two conditions commonly coincide or occur independently in the same patient. It is characterized by marked elevation of the erythrocyte sedimentation rate and a good response to corticosteroid treatment. By contrast, polymyositis is characterized by predominant muscle weakness with relatively mild myalgia, myogenic abnormality on the needle EMG, and an elevated creatine kinase level in the serum. Regarding the needle EMG, it should be kept in mind that, in polymyositis, the so-called active denervation potentials like fibrillation and positive sharp waves (see p. 138) can be observed, especially upon needle insertion. This is considered to be due to denervation, which could result either from segmental necrosis of muscle fibers separated from the end-plate region or from involvement of the terminal nerve endings (Kimura, 2013). It is important to keep in mind that dermatomyositis in aged subjects is often a manifestation of paraneoplastic syndrome (cancer-associated myositis) (Hida et al, 2016) (see Box 3-2, p. 17).

A condition similar to polymyalgia rheumatica, but more often associated with widespread musculoskeletal pain accompanied by fatigue, but no objective findings, is sometimes diagnosed as **fibromyalgia**. It is more common in females than in males, and there are often tender spots on the shoulder, low back, and proximal lower limbs. As there is no specific laboratory abnormality in this condition, and as the clinical course varies among patients, the diagnostic criteria are still controversial. Depression is common in these patients. There is a controversy as to its relationship with chronic fatigue syndrome.

C. SITES OF LESION FOR MONOPLEGIA

When muscle weakness is found in one hand or foot (**monoplegia**), the examiner must determine whether it

is due to disturbance of the central nervous system or the peripheral nervous system, and if it is peripheral, what part of the motor nerve is affected. In general, flaccid paralysis suggests a peripheral cause, while hypertonia or enhanced tendon reflex supports the diagnosis of central paralysis. The presence of muscle atrophy in the hand or foot suggests a peripheral pathology, but cases with no muscle atrophy are more often difficult. That is because muscle atrophy does not appear until several days after the onset of weakness of peripheral origin. Even in this case, however, muscle tone and tendon reflexes might give a clue to the diagnosis.

In a case with monoplegia of peripheral origin, it can be judged clinically whether the lesion is in the anterior spinal root, brachial plexus or lumbar plexus, or a single nerve or multiple nerves. In a case with monoplegia of the hand, for example, it is useful to start evaluating whether the muscle atrophy or weakness is localized to those muscles innervated by the median nerve, ulnar nerve, or radial nerve, or more than one of them. For this purpose, information about the nerve innervation and segment of each representative muscle is helpful (Table 17-1). **Juvenile muscular atrophy of the unilateral upper extremity**

TABLE 17-1 NERVE INNERVATION AND FUNCTION OF REPRESENTATIVE LIMB MUSCLES

Muscle	Nerve	Segment	Function
Upper limbs			
Supraspinatus	Suprascapular	C5	Initial abduction of arm from side of body
Infraspinatus	Suprascapular	C5	External rotation of arm at shoulder
Deltoides	Axillary	C5	Abduction of arm
Biceps brachii	Musculocutaneous	C6	Flexion of forearm at elbow
Triceps brachii	Radial	C7	Extension of forearm at elbow
Pronator teres	Median	C6	Pronation of forearm
Supinator	Posterior interosseus	C6	Supination of forearm
Flexor carpi radialis	Median	C7	Flexion of hand at wrist
Flexor carpi ulnaris	Ulnar	C8	Flexion and ulnar deviation of hand at wrist
Extensor carpi radialis	Radial	C7	Extension of hand at wrist
Extensor carpi ulnaris	Radial	C7	Extension and ulnar deviation of hand at wrist
Extensor digitorum	Radial	C7	Extension of fingers at MP joints
Flexor digitorum sublimis	Median	C8	Flexion of middle phalanges at proximal IP joints
Abductor pollicis brevis	Median	C8	Palmar abduction of thumb
Opponens pollicis brevis	Median	T1	Opposition of first metacarpal across palm
Interossei	Ulnar	T1	Abduction and adduction of fingers
Lower limbs			
Iliopsoas	Lumbar plexus	L2	Flexion of thigh at hip
Quadriceps femoris	Femoral	L3	Extension of leg at knee
Hamstrings	Sciatic	L5	Flexion of leg at knee
Tibialis anterior	Deep peroneal	L4	Dorsiflexion of foot
Gastrocnemius-soleus	Tibial	S2	Plantar flexion of foot
Peroneus longus, brevis	Superficial peroneal	L5-S1	Eversion of foot
Tibialis posterior	Posterior tibial	L5-S1	Inversion of foot
Extensor hallucis longus	Deep peroneal	S1	Extension of big toe
Flexor digitorum longus	Posterior tibial	L5-S1	Plantar flexion of toes at distal IP joints

MP: metacarpophalangeal, IP: interphalangeal, C: cervical, T: thoracic, L: lumbar, S: sacral

BOX 17-3 JUVENILE MUSCULAR ATROPHY OF UNILATERAL UPPER EXTREMITY (HIRAYAMA DISEASE)

This condition has been mainly reported among Asian populations, although it can be seen universally. It is characterized by slowly progressive muscle atrophy localized to the muscles innervated by the C7-T1 segments, namely the hand intrinsic muscles and the ulnar distal forearm, unilaterally (see Figure 17-3). Sensory symptoms are subtle if any. It is common in juvenile males, especially sports players. Dynamic flexion magnetic resonance imaging (MRI) shows anterior shifting of the posterior wall of the cervical dural sac. It is believed to be due to an ischemic lesion of the anterior horn as a result of repetitive neck flexion (Hirayama, 2008).

(**Hirayama disease**) is an example of segmental muscle atrophy (Box 17-3) (Figure 17-3).

A lesion of each nerve innervating the hand muscles shows a characteristic posture of the hand. In an ulnar nerve palsy, atrophy is seen in the interosseous and hypothenar muscles, and the metacarpal-phalangeal joints of the fourth and fifth fingers are hyperextended (**claw hand**) (Figure 17-4a). In a median nerve palsy, marked atrophy is seen in the thenar muscles, and the thumb is unopposed at the same plane as the palm (**ape hand**) (Figure 17-4b). In a radial nerve palsy, **wrist drop** is seen due to weakness of wrist extensors (Figure 17-4c).

Furthermore, in an ulnar nerve palsy, a maneuver of firm grasping of a piece of paper with the thumb and index finger of each hand and of strong lateral pulling causes flexion of the distal phalanx of the thumb on the affected side because of inability to adduct the thumb, which is compensated by the flexor pollicis longus muscle innervated by the intact median nerve (**Froment sign**) (Figure 17-5). In the lower limbs, **foot drop** occurs in a peroneal nerve palsy.

If a single nerve is severely damaged, muscles innervated by other nerves may also appear weak to some extent on manual muscle testing. That is because, to fully contract a particular muscle, that muscle has to be set ready in the most appropriate posture, which can only be done by other related muscles. For example, if the median nerve is markedly impaired due to **carpal tunnel syndrome** (see Box 20-3, p. 197), the muscles innervated by the ulnar nerve might be also judged slightly weak.

D. ROOT LESIONS

A lesion of the anterior spinal root causes atrophy in the muscles innervated by the corresponding segment (see Table 17-1). It is often associated with radicular pain and sensory symptoms in the segmental distribution due to simultaneous involvement of the posterior spinal root. A spread of metastatic carcinoma selectively involving anterior roots might cause rapidly expanding neurogenic muscle atrophy across multiple segments.

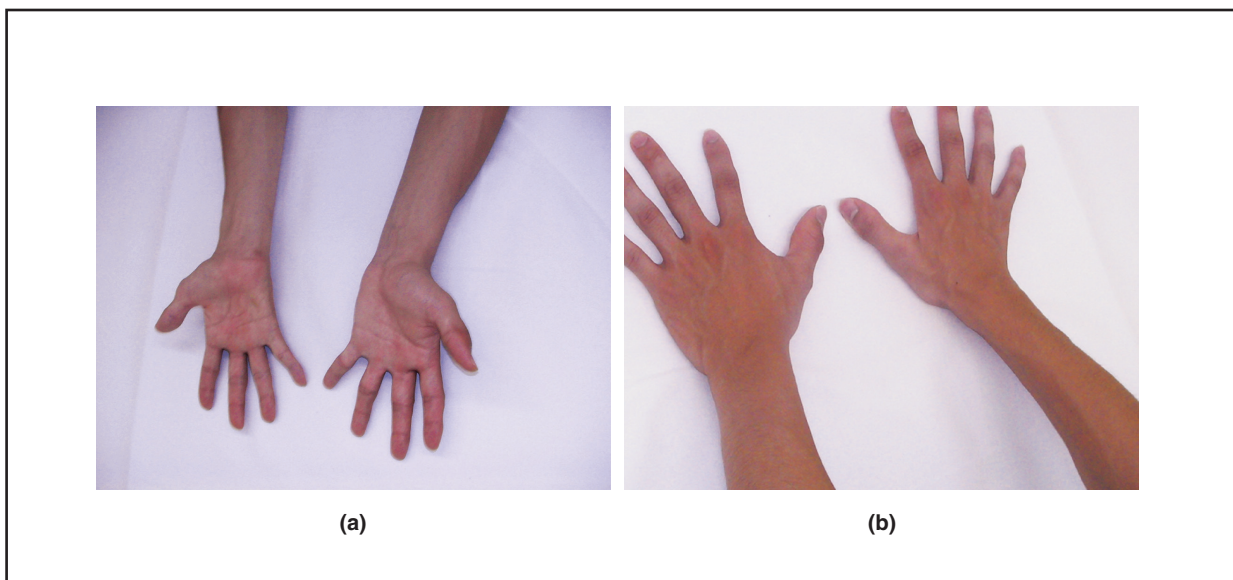


Figure 17-3 Juvenile muscular atrophy of unilateral upper extremity seen in a 23-year-old Japanese male with clinical diagnosis of Hirayama disease. (a) palmar view, (b) dorsal view.

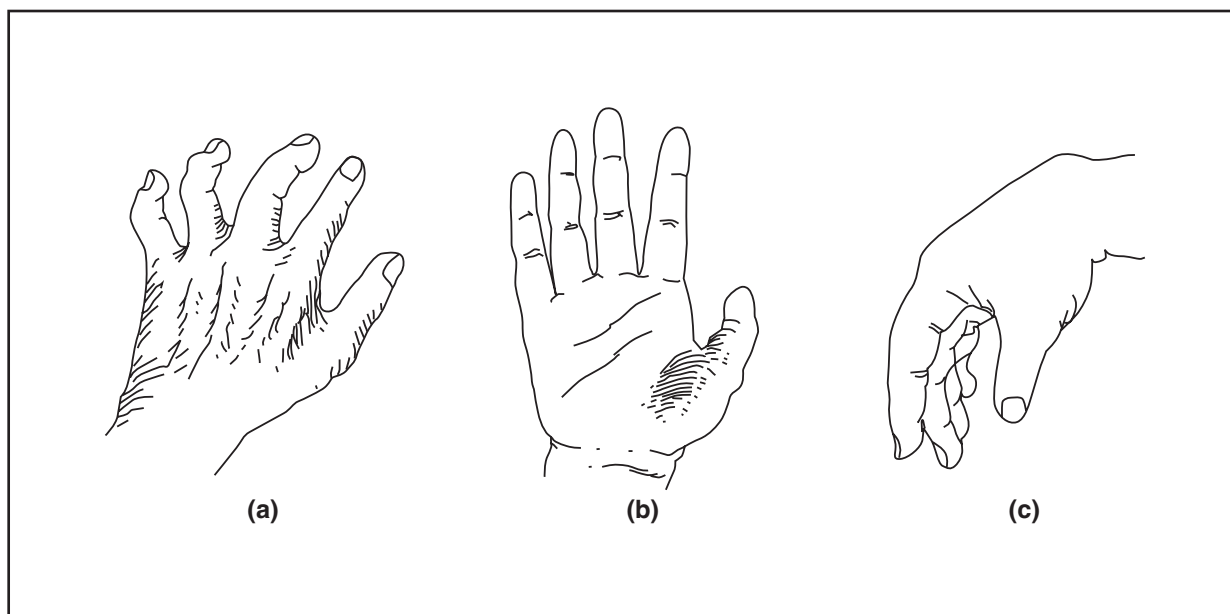


Figure 17-4 Claw hand due to ulnar nerve palsy (a), ape hand due to median nerve palsy (b), and wrist drop due to radial nerve palsy (c).

E. PLEXUS LESIONS

By contrast with a segmental lesion or a peripheral nerve lesion, lesions of the **brachial plexus** (Figure 17-6) and **lumbosacral plexus** (Figure 17-7) may involve multiple nerve roots or multiple peripheral nerves. Thus, if muscle weakness cannot be explained by a lesion of either a nerve root or of a peripheral nerve or nerves, it is advisable to consider the possibility of plexus lesion. Idiopathic inflammation of the brachial plexus can cause neuralgic pain and muscle atrophy in the unilateral upper limb. This condition, called **neuralgic amyotrophy** or idiopathic brachial plexopathy (Parsonage–Turner syndrome), is considered to be of autoimmune pathogenesis. In this condition, even though the clinical manifestation is limited to unilateral upper limb, the EMG study might disclose abnormality in the opposite



Figure 17-5 Froment sign in the right hand seen in ulnar nerve palsy. Cited from Dawson et al., 1999, with permission.

upper limb as well. Recently, this condition was reported in a patient with COVID-19 (Mitry et al, 2021).

Compression of the lower part of the brachial plexus at the thoracic outlet will cause pain and numbness in the upper limb. This condition, called **thoracic outlet syndrome**, is most commonly caused by a cervical rib, which is the rib coming out from the eighth cervical spine. Clinically, it is characterized by sensory loss on the ulnar side of the hand with atrophy of the thenar muscle due to a lesion of the lower trunk. The maneuver called the **Adson test** is often applied. In this test, with the patient placed in the sitting position, the examiner palpates the radial pulse bilaterally, and the patient is asked to take a rapid deep breath and rotate the extended neck to either side. If the pulsation diminishes on either side, the test is judged positive. In this case, a vascular bruit may be audible in the subclavian artery. However, the sensitivity and the specificity of this test are not high.

In patients with diabetes mellitus, **painful sensory polyneuropathy** is common (see Chapter 19-2B), but painless motor neuropathy may also appear in the territory of the lumbosacral plexus (**painless diabetic motor neuropathy**) (Garces-Sanchez et al, 2011).

F. ACUTE POLYNEURITIS

Diffuse inflammatory involvement of the peripheral nerves of acute onset is called **acute polyneuritis** or **acute polyradiculoneuritis**. This condition, generally called **Guillain–Barré syndrome**, is important because it is relatively

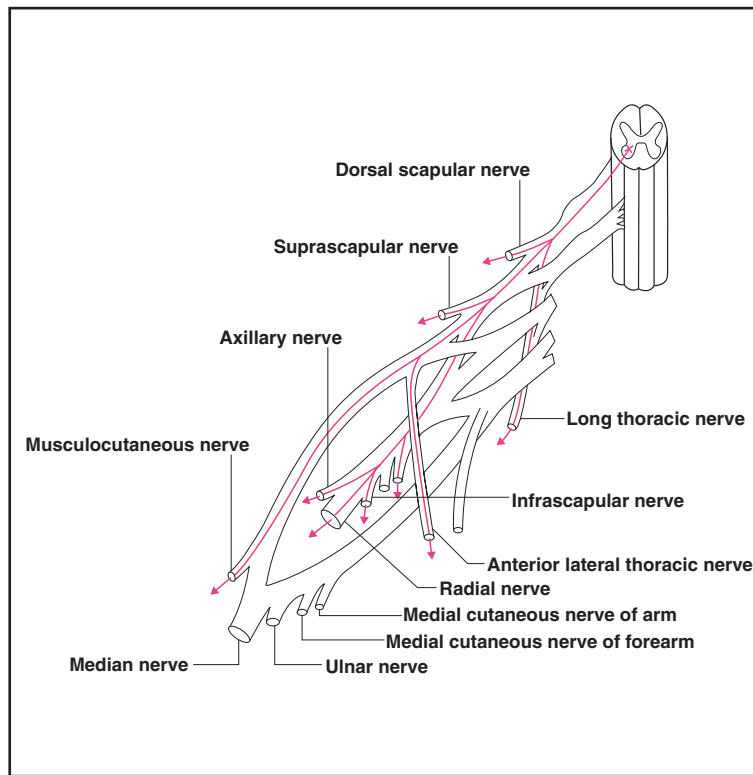


Figure 17-6 Schematic diagram of the brachial plexus.

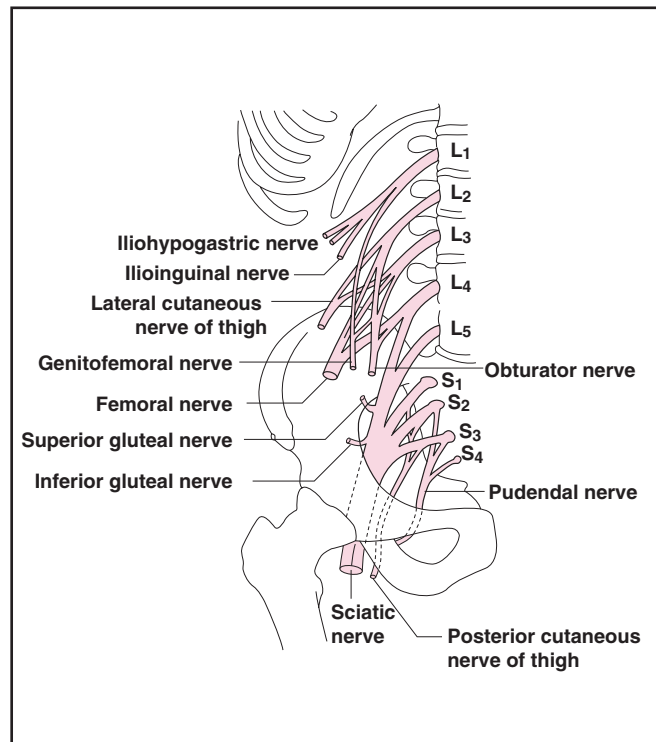


Figure 17-7 Schematic diagram of the lumbosacral plexus.
Modified from Mayo Clinic and Mayo Foundation, 1998.

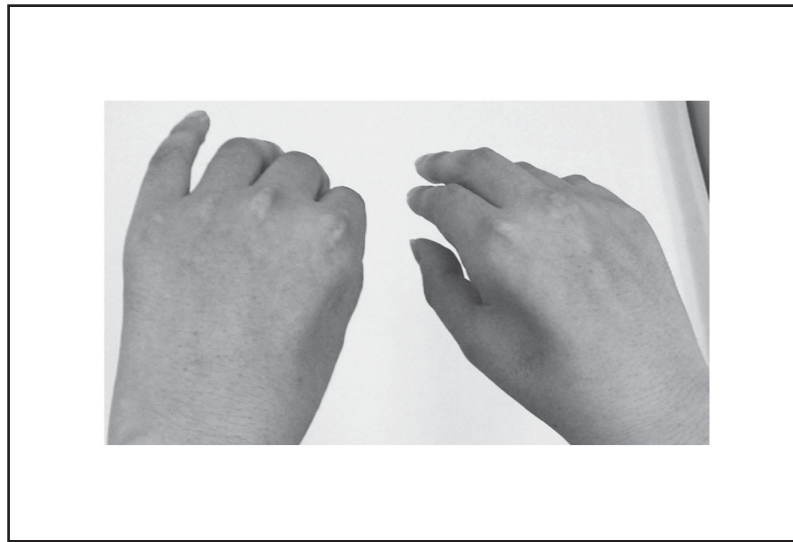


Figure 17-8 Muscle atrophy of hands seen 6 months after acute axonal polyneuritis.

common and mostly curable if appropriately treated in the early phase. In most cases, it is preceded by symptoms of upper respiratory or gastrointestinal infection either immediately or by a week or two. The common initial symptoms are weakness and numbness of the hands or feet, followed by rapid progression in the degree and distribution. Rapid ascending of the symptoms from the feet to the respiratory muscles is not infrequent. Occasionally the cranial nerves may also be involved. Tendon reflexes are diffusely lost from the acute phase. The signs are usually symmetric, but there may be asymmetry. An autoimmune pathogenesis is considered in most cases. After the acute phase, electrophysiologic studies will disclose its demyelinating nature, but in some cases axonal impairment may be a main feature. In the axonal type, muscle weakness tends to be severe, muscle atrophy will become apparent from about a week after the onset, and functional recovery is worse than with the demyelinating type (Figure 17-8). In patients with acute axonal polyneuritis who manifest gastrointestinal symptoms before the clinical onset, infection with *Campylobacter jejuni* may be found for its pathogenesis (Huizinga et al, 2015).

Guillain–Barré syndrome has to be differentiated from other conditions characterized by acute or subacute motor weakness, such as botulism (see Box 33-2, p. 286), acute anterior poliomyelitis (see later in the chapter), tic paralysis including Lyme disease (see later in the chapter), AIDS-related polyneuropathy, diphtheritic neuropathy (see later in the chapter), porphyric polyneuropathy (see p. 20), and polyneuropathies caused by toxins such as lead, hexane, thallium, acrylamide, and vincristine (see p. 145). These conditions usually show characteristic patterns of weakness

and associated systemic findings, and the mode of onset is usually different from typical Guillain–Barré syndrome.

In typical Guillain–Barré syndrome, motor paralysis predominates and sensory impairment is mild, if any. Furthermore, as motor paralysis is commonly seen in proximal muscles, differentiation from polymyositis of subacute onset may be necessary even in the acute phase. Even in cases with no sensory symptoms, if the straight leg raising test induces radiating pain over the posterior aspect of lower limb (positive Lasègue sign), it may suggest inflammatory involvement of the posterior roots.

In some patients with acute polyneuritis, only sensory symptoms may be present. In these cases, unlike typical Guillain–Barré syndrome, the posterior root ganglion cells are considered to be affected by an autoimmune pathogenesis (**polyganglioneuritis**). Moreover, impairment of the autonomic nervous system might predominate, suggesting involvement of the autonomic ganglion cells or autonomic nerve fibers. As the postganglionic fibers are either A- δ fibers or unmyelinated C fibers (see Chapter 21, p. 204), axons are primarily involved in these cases, resulting in greater resistance to treatment. In autoimmune polyneuritis, antibodies against various gangliosides have been demonstrated (Kusunoki et al, 2011).

G. ATYPICAL ACUTE POLYNEURITIS

If acute polyneuritis is preceded by symptoms of systemic infection and is associated with an atypical clinical picture, a possibility of infection with a variety of microorganisms has to be taken into account. Those include **Epstein–Barr**

virus, *Mycoplasma pneumoniae*, and *Borrelia burgdorferi*, a spirochete causing Lyme disease. *Mycoplasma pneumoniae* was shown to be associated with Guillain–Barré syndrome more commonly in children than in adults, and anti-GalC IgG antibody was shown to be related to its pathogenesis (Sauter et al, 2016).

Collagen diseases like **systemic lupus erythematosus (SLE)** and Sjögren syndrome may present with polyneuritis based on autoimmune angiitis. In **Sjögren syndrome**, sensory symptoms are especially common in the trigeminal nerve territory. By contrast, in **polyarteritis nodosa**, **mononeuritis multiplex** (multiple mononeuritis) is especially common. Mononeuritis multiplex associated with severe pain and marked eosinophilia characterizes **Churg–Strauss syndrome**.

Acute anterior poliomyelitis (polio, Heine–Medin disease) and **diphtheria** used to be known as common causes of acute focal motor paralysis and may be still prevalent in some regions of the world. Polio paralyzes predominantly unilateral upper or lower limb, or respiratory muscles, but other limbs may also be affected to a certain extent, and sensory impairment may also be present. By contrast, diphtheria is known to cause especially bulbar palsy. There have been some reports of vaccine-associated paralytic poliomyelitis following oral administration of live polio vaccine. Nowadays, therefore, subcutaneous injection of inactivated polio vaccine is being used in many parts of the world. Recently in the United States, a possible link between an outbreak of enterovirus D68 infection and **acute flaccid myelitis** has drawn attention, although their causative relationship has not been confirmed (Greninger et al, 2015; Lerner et al, 2021; Murphy & Pardo, 2020).

H. CHRONIC INFLAMMATORY DEMYELINATING POLYNEUROPATHY

A chronic form of polyneuritis is not uncommon. There are three groups: **chronic inflammatory demyelinating polyneuropathy (CIDP)**, **chronic inflammatory axonal polyneuropathy (CIAP)**, and **multifocal motor neuropathy (MMN) with conduction block**. In the latter condition, multiple nerves may not necessarily be affected all at one time; only mononeuritis may occur at a certain stage of illness, which is later followed by involvement of other nerves. This condition is occasionally called Lewis–Sumner syndrome, but in their original report of five cases published in 1982, there was also sensory impairment (Lewis et al, 1982). In many cases of this disease, levels of IgM antibody against GM1 are elevated in the serum, especially in severe

cases (Cats et al, 2010). Patients with these kinds of chronic polyneuritis are treated with intravenous immunoglobulin (IVIg). There might be some improvement, but the disease tends to show remissions and relapses, just like in multiple sclerosis of the central nervous system. A recent nationwide study of 100 cases of MMN in the Netherlands found that this disease is essentially a progressive one (Herraets et al, 2020).

Misdiagnosis of CIDP is common (Allen & Lewis, 2015). The initial symptoms of CIDP might appear acutely, and some cases of Guillain–Barré syndrome might show worsening of the symptoms after the acute phase, thus creating some diagnostic problem in the acute phase. It has been reported that if patients with an initial diagnosis of Guillain–Barré syndrome show worsening of the symptoms more than 8 weeks after the onset, or if they repeat more than three times, it is more likely CIDP (Ruts et al, 2010).

I. TOXIC POLYNEUROPATHY

As the peripheral nerves, like the cerebellum, are vulnerable to heavy metals and drugs, a possibility of toxic polyneuropathy should be considered in any polyneuropathy of subacute onset. Intoxication with lead, vincristine, and acrylamide presents primarily with motor paralysis. Toxins like arsenic, organic mercury, cisplatin, and *n*-hexane tend to affect the sensory ganglion, and organophosphate and thallium are known to affect the autonomic nervous system. Chemotherapy-induced peripheral neuropathy is of practical importance, because the symptoms persist after the termination of chemotherapy and may impair cancer survivors' quality of life (Gewandter et al, 2017; Staff et al, 2017).

J. METABOLIC POLYNEUROPATHY

Metabolic diseases such as diabetes mellitus, malnutrition, vitamin deficiency due to ethanol abuse, and porphyria are known to cause chronic polyneuropathy. Among others, **diabetic polyneuropathy** and **alcohol vitamin deficiency polyneuropathy** are notorious for causing severe dysesthesia with burning pain (see Chapter 20-2B, p. 197).

K. HEREDITARY POLYNEUROPATHY

There are a number of diseases in this category, but its most representative form is **hereditary motor sensory neuropathy (HMSN)**. HMSN is classified into three groups: type I (**Charcot–Marie–Tooth disease type I**),

which is characterized by a decrease in the motor nerve conduction velocity; type II (**Charcot–Marie–Tooth disease type II**), which is characterized by axonal damage; and type III (**Dejerine–Sottas disease**), which is characterized by nerve hypertrophy due to demyelination. The mode of inheritance is autosomal dominant for type I but not consistent for types II and III. Type II includes X-linked Charcot–Marie–Tooth disease (CMTX1) (Panosyan et al, 2017). Charcot–Marie–Tooth disease type IIA (CMT2A) is known to be caused by MFN2 mutations. A recent international multicenter prospective study of 196 patients with dominant and autosomal recessive CMT2A found that a childhood onset of autosomal dominant CMT2A is associated with higher rates of disability compared to adult-onset autosomal dominant CMT2A (Pipis et al, 2020).

In HSMN types I and III, the myelin sheath is primarily affected, and they are characterized by the presence of **onion bulb formation** in the histology of the peripheral nerve. In type III, there is prominent enlargement of the peripheral nerves, and the hypertrophy can be observed or palpated in the greater auricular nerve and peroneal nerve. An autosomal dominant hereditary polyneuropathy initially presenting with ataxic gait, pes cavus, and loss of tendon reflexes, and later manifesting with distal muscle atrophy, postural tremor, and sensory loss used to be called Roussy–Lévy syndrome, but this form is now considered to be a subtype of Charcot–Marie–Tooth disease type I (Planté-Bordeneuve et al, 1999). Genetic abnormalities have been elucidated for many subtypes of HSMN (Gutmann & Shy, 2015).

Other autosomal dominant forms of hereditary neuropathy include hereditary sensory neuropathy, hereditary sensory and autonomic neuropathy, **familial amyloid polyneuropathy** (see Box 21-1, p. 206), and hereditary neuropathy with liability to pressure palsy (pressure-sensitive neuropathy). **Pressure-sensitive neuropathy** is characterized by motor paralysis following mechanical compression of the corresponding nerve and a histologic “sausage-like” appearance due to the bead-shaped hypertrophy of nerve (**tomaculous neuropathy**). Clinically, this neuropathy is manifested as multiple mononeuropathy, but the tendon reflexes are lost not only in the compressed nerve but also diffusely. Another autosomal recessive form of hereditary polyneuropathy is **giant axonal neuropathy**, which is histologically characterized by the deposit of neurofilaments in the axon. Generally speaking, in these hereditary neuropathies, it is rare for the patients to be aware of numbness or pain even if the sensory nerves are involved.

L. SPASTIC PARALYSIS

Motor paralysis due to a lesion of the pyramidal tract is associated with increased muscle tone (spasticity) (see p. 147) and thus called **spastic paralysis**. As there is a clear somatotopic organization in the motor cortex (see Figure 11-2, p. 84, and Figure 16-1, p. 110), a localized lesion in the motor cortex causes monoplegia in the contralateral upper or lower limb. By contrast, a lesion in the internal capsule causes contralateral hemiplegia because all fibers of the corticospinal tract converge there. In this case, the paralysis also involves the contralateral lower face and the tongue, which will deviate to the hemiplegic side upon protrusion (see Box 14-2, p. 99).

In a transverse lesion of the spinal cord, spastic paralysis is seen bilaterally below the level of the lesion, namely paraplegia in a thoracic cord lesion and quadriplegia in a cervical cord lesion. If there is marked paraplegia while the upper limbs are intact or only mildly weak, if at all, the main lesion is usually in the thoracic cord. If a transverse lesion is just at the level of the Th10 segment, voluntary flexion of the neck in the supine position will shift the umbilicus rostrally (**Beevor sign**) (McCarter et al, 2018). This phenomenon occurs because the rostral half of the rectus abdominis muscle above the Th10 segment contracts normally but the caudal half is paralyzed.

M. GOWERS SIGN

Patients who have marked weakness in the pelvic girdle and the muscles of the proximal lower limb have great difficulty standing up from a squatting position, and they can barely climb themselves up with legs widely placed and by holding on to nearby objects like a chair. This phenomenon is called **Gowers sign**. Furthermore, they tend to swing the waist when walking (**waddling gait**). This condition is commonly seen in myopathy but may be also seen in patients with proximal neurogenic muscular atrophy like Kugelberg–Wielander disease (see p. 136).

N. DISUSE ATROPHY

In patients who have remained in bed for a long time for non-neurological diseases or in patients with neurological diseases without direct involvement of the neuromuscular system, muscles may lose their volume. This kind of muscle wasting is called **disuse atrophy**, and sometimes distinguishing it from “real” muscle disease becomes a practical problem. In disuse atrophy, muscle strength is relatively well preserved despite small muscle volume, and tendon reflexes

are preserved. The needle EMG study does not show any primary abnormality in this case.

Functional motor paralysis is not rare, and this will be described in the chapter on functional (psychogenic) disorders (Chapter 29, p. 268).

4. MUSCLE TONE

Muscle tone is tested by passively stretching the relaxed muscle and is judged by the degree of resistance against passive stretch. Therefore, it is important to place the corresponding muscle in complete relaxation. To distinguish between **spasticity** and **rigidity**, the passive stretch has to be given at two different speeds. Spasticity is speed-dependent and can be elicited only by quick stretch, whereas rigidity is independent of the stretch speed. Stretch of intermediate speed may not help distinguish one from the other.

A. SPASTICITY

Spasticity is characterized by a sudden increase in the resistance during passive stretch. If a flexion stretch is maintained, the resistance may suddenly disappear, leading to an extension movement. This phenomenon is called the

clasp-knife phenomenon and is explained based on the physiologic principle of **lengthening reaction**. Spasticity is commonly tested in the forearm flexor (biceps brachii muscle) for the upper limb and in the quadriceps femoris muscle for the lower limb. If we take passive stretch of the biceps brachii muscle as an example (Figure 17-9), an impulse produced by stretch of the stretch receptor (muscle spindle) is conducted through the Ia fibers to the anterior horn of the corresponding cervical cord segment where the α motor neurons are monosynaptically activated, causing reflex contraction of the muscle. This muscle contraction is excessively enhanced because the anterior horn cells are hyperexcitable due to a lesion of the pyramidal tract. Then the α motor neurons are inhibited by the inhibitory interneurons post-synaptically, causing the clasp-knife phenomenon. Regarding the afferent input for this phenomenon, it used to be believed that the Ib afferent fibers originating from the tension receptor (tendon organ) play a major role, but its precise mechanism in humans remains controversial (Burke et al, 2013).

B. CLONUS

When testing muscle tone in patients with marked spasticity, a single muscle stretch might induce repetitive brisk

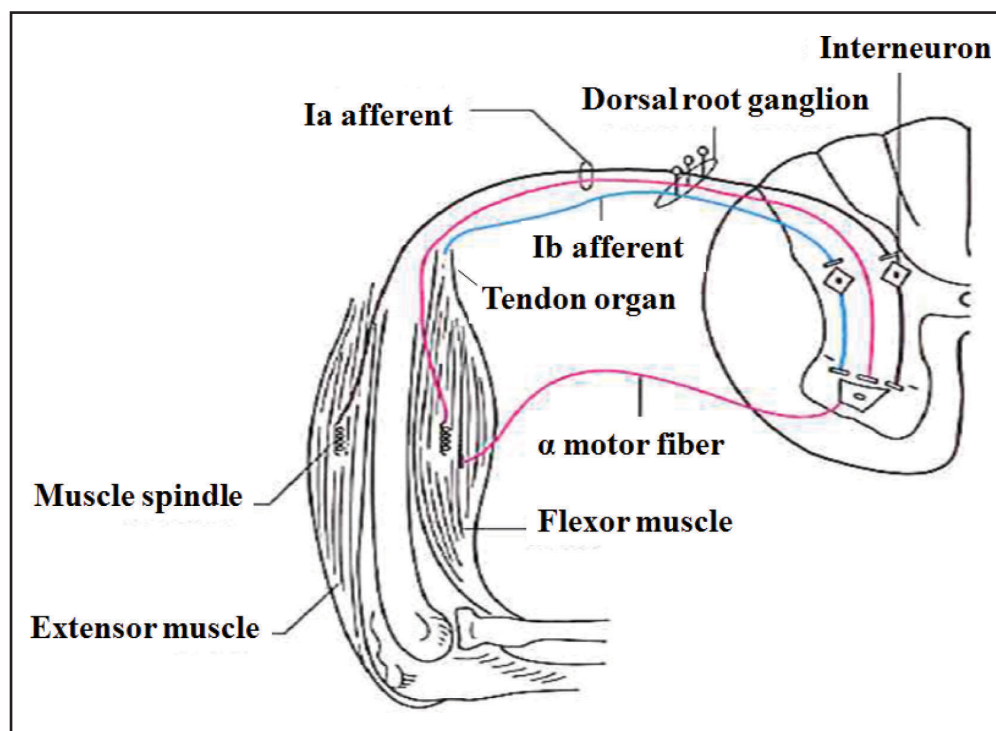


Figure 17-9 Schematic diagram showing the spinal reflex pathways. Interneurons are shown much more posterior from the real location for the sake of clarity.

BOX 17-4 NEUROMYELITIS OPTICA SPECTRUM DISORDERS (NMOSD)

After a case report by Devic in 1894, neuromyelitis optica (NMO) has been recognized as a special form of multiple sclerosis (MS). MS among Asian populations has drawn special attention because acute involvement of the optic nerve and spinal cord is seen in as many as 35% of all MS cases, and it has been called the optic spinal form of MS. Oligoclonal bands in the cerebrospinal fluid were found less commonly in the Asian MS cases compared with the Western patients. Even in those cases, however, pathologically periventricular demyelinating plaques have been found in addition to necrotic lesions of the optic nerve and spinal cord. It has now been identified that, in many patients with this condition, levels of an antibody against **aquaporin-4 (AQP4)**, which is a channel protein of the main water molecule in the brain, are elevated in the serum. Furthermore, spinal MRI in those cases showed the presence of a lesion longitudinally extending over more than three vertebral bodies.

The term **NMO spectrum disorders** was coined for the syndrome of bilateral optic nerve lesions, long spinal cord lesion, and positive anti-AQP4 antibody, and it is considered to be an entity independent of classical MS (Wingerchuk et al, 2015). Furthermore, antibodies against AQP4 were reported to be elevated in 20% of the patients with relapsing optic neuritis, and the prognosis of those cases was unfavorable and tended to shift to NMO (Matiello et al, 2008). It is especially noteworthy that corticosteroids and other immunosuppressive agents suppress relapse of NMO, whereas β interferon, which is used for MS, tend to cause acute severe exacerbation in NMO (Shimizu et al, 2010). It was reported that 7.4% of the patients with NMO spectrum disorders have antibodies against myelin oligodendrocyte glycoprotein (MOG) instead of AQP4 (Sato et al, 2014). The overall visual and motor outcome was shown to be better in patients with MOG-antibody-associated NMOSD compared with AQP4-antibody-positive patients (Cobo-Calvo et al, 2017). AQP4-antibody-positive NMOSD is now regarded as an immune astrocytopathy, and MOG-antibody-associated disease is known to target myelin rather than astrocytes, leading to an NMOSD syndrome with distinct clinical and radiologic features (Paul et al, 2021).

An international multicenter study showed that the postpartum period was a time of especially high risk for the initial presentation of NMOSD (Klawiter et al, 2017). A recent literature review showed variations in the prevalence and incidence among different geographic areas and ethnicities, highest in the Afro-Caribbean region and lowest in Australia and New Zealand (Papp et al, 2021).

Relapsing myelitis is also seen in **Sjögren syndrome**, which is characterized by thirst, antinuclear antibodies, and antibodies against SS-A and SS-B (SS derived from Sjögren syndrome) (Chahin et al, 2009). Intramedullary cord lesions resembling neuromyelitis optica may be also seen in association with sarcoidosis (Kwon et al, 2014).

BOX 17-5 ACUTE DISSEMINATED ENCEPHALOMYELITIS (ADEM) VERSUS MULTIPLE SCLEROSIS (MS)

The relationship between these two inflammatory demyelinating diseases of the central nervous system has been discussed for many years. Clinically, MS is characterized by frequent remissions and exacerbations followed by slow progression in general, whereas ADEM is an acute severe inflammatory disease, often associated with disturbance of consciousness. Pathologically, MS is characterized by multifocal, often confluent, demyelinating plaques, whereas ADEM is characterized by more disseminated demyelination with perivenous cell infiltration. A special form of demyelinating disease characterized by alternating concentric areas of increased FLAIR signal is called Baló concentric sclerosis (Hardy et al, 2017; Jolliffe et al, 2021). In some cases, however, there is some overlap of these clinical and pathologic features (Young et al, 2010). In fact, a multicenter study of 228 cases of ADEM in the United States showed that relapsing disease was fairly common (Koelman et al, 2016). The prevalence of MS in the United States is estimated to be 309.2 per 100,000 (Wallin et al, 2019).

contractions in the same muscle, which is called **clonus**. Clonus is interpreted as a repetition of the excessively enhanced stretch reflexes. Clonus is most easily demonstrated in the calf muscle (**ankle clonus**), but it can be also seen in the quadriceps femoris, wrist flexor, and masseter muscles. Clonus may occur spontaneously, and when it is vigorous, it may look like convulsions.

C. RIGIDITY

Rigidity is one of the most representative signs of parkinsonism. In rigidity, resistance against the passive muscle stretch appears independent of the speed of stretch, and typically it continues throughout the passive stretch in a cogwheel fashion (**cogwheel phenomenon**), which is due to action tremor superimposed upon the increased tone (Lance et al, 1963). Rigidity sometimes may continue throughout the passive stretch smoothly without cogwheel. In contrast with rigidity, **Gegenhalten** (paratonia) appears as a continuous resistance against the passive movement of a joint and is proportional to the effort with which the examiner tries to move. It is usually seen in association with a diffuse cerebral lesion.

BOX 17-6 DEGENERATIVE DISORDERS OF UPPER MOTOR NEURONS

There is a group of neurodegenerative disorders selectively involving the upper motor neurons. Clinically they present with progressive spastic paralysis in the lower limbs. Conventionally, they are called **primary lateral sclerosis (PLS)**. A retrospective study of 43 patients with pure PLS revealed that none of them converted to ALS after a median of 9 years (Hassan et al, 2021). Recently a large number of gene abnormalities have been identified in many of these conditions, and they are now called by the general term **hereditary spastic paraplegia** (Murala et al, 2021; Saputra & Kumar, 2021). The mode of inheritance is variable, ranging from the autosomal dominant or autosomal recessive to X-linked.

Charlevoix-Saguenay syndrome is an autosomal recessive disease manifesting with spastic paraplegia and cerebellar ataxia, in which mutation of the SACS gene was reported (Duquette et al, 2013; Gagnon et al, 2019; Pilliod et al, 2015). Nakamura et al. (1995) reported a family with **autosomal recessive spastic paraplegia associated with early baldness, mental retardation, and thinning of corpus callosum**. In this entity, mutation of the SPG11 gene on chromosome 15q21.1 has been reported from various countries (Samaranch et al, 2008).

D. HYPOTONIA

Decreased muscle tone is commonly seen in disorders of the peripheral nervous system or muscles. Among others, **tabes dorsalis** is known to cause marked hypotonia so that the toes may even come close to the face upon the straight leg raising test. Tabes dorsalis is a representative form of neurosyphilis with main involvement of the dorsal roots.

Hypotonia may be also seen in disorders of the central nervous system. In the acute stage of severe cerebral stroke, flaccid paralysis may be seen. This condition is called **spinal shock** even if the paralysis is caused by cerebral lesions. In newborns and infants with chronic diffuse cerebral lesions, flaccid paralysis is often seen (**floppy infant**). Hypotonia is also seen in acute cerebellar lesions, although its mechanism has not completely been elucidated. In relation to this, the cerebellar form of multiple system atrophy (MSA-C) may manifest hypotonia, whereas its parkinsonian form (MSA-P) commonly shows rigidity (see Box 21-2, p. 209).

BOX 17-7 ADRENOMYELONEUROPATHY

Adrenomyeloneuropathy (AMN) is considered to be a subtype of **adrenoleukodystrophy (ALD)**. ALD occurs in boys aged 5 to 9 years with behavioral abnormality, cognitive decline, cortical blindness, cortical deafness, optic atrophy, and convulsive seizures, followed by a progressive course. However, a few patients have a relatively steady course with a form of AMN characterized by asymmetric spastic paraparesis (Mallack et al, 2020). Like ALD, it is associated with adrenal insufficiency. It must be differentiated from degenerative diseases of the upper motor neurons (as listed in Box 17-6) and the chronic progressive form of MS. Cranial MRI shows abnormal signal intensity along the corticospinal pathway, and in fact cerebral demyelination is frequently seen in the later clinical stage (de Beer et al, 2014). Like ALD, the mode of inheritance is X-linked, but it can occur sporadically. Levels of **very long chain fatty acid** are elevated in the plasma. Improvement of clinical symptoms has been reported with bone marrow transplantation (Hitomi et al, 2003). Other entities in the differential diagnosis include **metachromatic leukodystrophy** due to arylsulfatase-A deficiency, the **adult form of Krabbe disease** due to β -galactosidase deficiency, and **adult polyglucosan body disease (APBD)**. APBD is characterized by progressive spastic paraplegia, sphincter disturbance, axonal sensorimotor polyneuropathy, and axonal enlargement due to deposit of polyglucosan (Lossos et al, 2009).

5. MUSCLE SPASM AND CRAMP

A continuous or intermittent state of spontaneous abnormal muscle contraction is called **muscle spasm**. Muscle spasm associated with pain is called muscle cramp or just **cramp**. Spasm is caused by abnormal electrical discharge in the nervous system or muscles. For neurogenic spasm, any part of the motor system may be responsible, including the motor cortex, corticospinal or corticobulbar tract, anterior horn cells or brainstem motor nuclei, α motor fibers, neuromuscular junction, and muscles. Therefore, the distribution of spasm varies depending on its source. A phasic repetitive form of spasm originating in the motor cortex may present with **epilepsia partialis continua (Kojevnikoff epilepsy or Kojevnikoff syndrome)**.

Convulsion rarely originates in the white matter, but **painful tonic spasm** can occur in multiple sclerosis, including neuromyelitis optica spectrum disorders, as a result of abnormal discharge and abnormal conduction of electrical